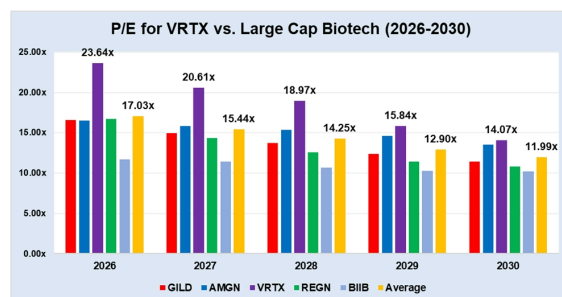
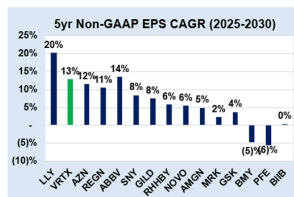
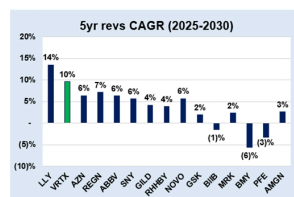


IgAN Data Looks In Line - We're Still Bullish on Oppty - Initiate BUY, PT \$580

We're Buy on VRTX given: 1) CF is a stable franchise for VRTX w/ limited competitive threats & supports floor valuation of ~\$422 (CF NPV + cash), 2) although pove's Ph.3 IgAN data came in line w/ sibe, IgAN market is large & pove auto-injector format could improve compliance, 3) we think Inaxaplin could show best-in-class data in AMKD, 4) while we're mixed on pain due to pbo risk in DPN trial, we still see Journavx used in acute pain. Buy, PT \$580

**** For our full report, please see the attached slide deck ****

We are initiating coverage on VRTX with a BUY rating & PT of \$580. VRTX's stock has underperformed its peers over the past 6 months (+16% vs +21-40% for other LC biotech's) following several setbacks last year. The stock trades at a forward P/E premium (~23.6x vs ~17.0x industry average), which is consistent w/ consensus modeling ~10% revenue & ~13% EPS CAGRs over the next five years - placing it among the higher-growth profiles in large-cap pharma/biotech.



Source: Jefferies Analysis, Factset

Notably, we see upside for VRTX on povetacept (anti-APRIL/BAFF). While the Ph.3 IgAN data came in line with sibe & our expectations ([HERE](#)), we think the IgAN market could be \$15-20Bn in size at peak & safety profile (no increase infections) de-risks the 80mg dose for autoimmune expansion trials. Overall, we think pove has the potential to diversify VRTX beyond CF, and we model pove revenues reaching ~50% of CF revenues longer term. Additionally, we think the stock cheap on a sum-of-the-parts basis, with a 'floor' valuation of ~\$422 (CF NPV + cash).

Let's go into the details below:

<i>FY (Dec)</i>	2024A	2025A	2026E	2027E
Rev. (MM)	11,020.1	12,001.3	13,007.2	14,428.3
Cons. Rev.	-	-	13,024.4	14,365.0
Cons. EPS Non-GAAP	-	-	19.11	21.83
EPS	0.29	18.40	18.74	22.14

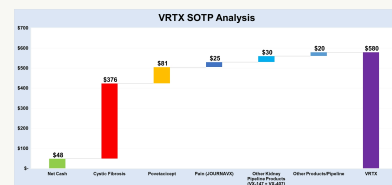
INITIATING COVERAGE

RATING	BUY
PRICE	\$460.87*
PRICE TARGET % TO PT	\$580.00 +26%
52W HIGH-LOW	\$519.68 - \$362.50
FLOAT (%) ADV MM (USD)	97.5% 655.08
MARKET CAP	\$119.6B
TICKER	VRTX

^aPrior trading day's closing price unless otherwise noted

FY (Dec)	CHANGE TO JEF _e		JEF vs CONS	
	2026	2027	2026	2027
REV	NA	NA	NM	NM
EPS	NA	NA	NA	NA

2026 (\$)	Q1	Q2	Q3	Q4	FY
EPS	4.45	4.59	4.90	4.80	18.74



Source: Jefferies Analysis

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Please see analyst certifications, important disclosure information, and information regarding the status of non-US analysts on pages 109 - 115 of this report.

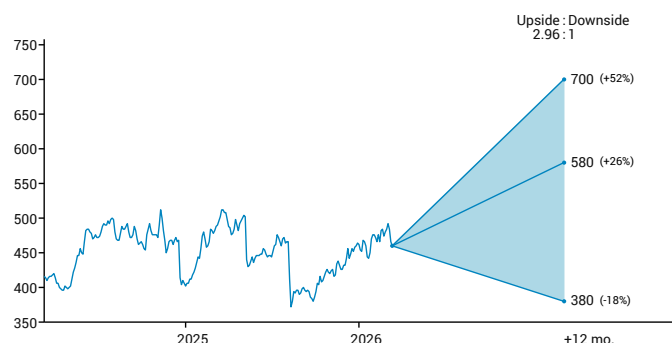
* Jefferies LLC / Jefferies Research Services, LLC

The Long View: Vertex

Investment Thesis / Where We Differ

We're Buy rated on VRTX given: 1) durable CF base business w/ limited competitive threat + patent out till late 2030s, 2) povetacept IgAN Ph.3 looks in line w/ sibe, but we think the market is large, pove will be conveniently administered via auto-injector, and competition has set a high starting price 3) we see add'l upside with w/ Inaxaplin being best/first in class for AMKD, 4) we're skeptical on rest of pipeline based on limited expansion opportunity, but our valuation does not rely on it.

Risk/Reward - 12 Month View



Base Case, \$580, +26%

- Continued strong CF business, with upside from the next gen VX-828 w/ PS \$16Bn.
- Pain gets continued uptake in acute and chronic w/ \$2.2Bn PS
- Pove gains market penetration across different indications, and we model \$8.5Bn PS.
- Inaxaplin Ph.3 hits in AMKD - we model \$3Bn PS.

Upside Scenario, \$700, +52%

- Pain gets meaningfully higher uptake in acute and chronic and Ph.3 DPN trials hit. We model \$5Bn PS.
- Pove shows best in class efficacy and is preferentially selected in IgAN. We model \$12Bn PS.
- Favorable efficacy and safety for the cells/device "combination" program and next-gen pain programs.

Downside Scenario, \$380, -18%

- Our \$380 downside scenario price target is based on a DCF that assumes a weaker-than-expected CF business and/or experiences macro risk (e.g. in the EU).
- Lower PoS and limited value from VRTX non-CF pipeline (eg. Pove shows worse data vs comps in IgAN + other autoimmune)
- Pain remains acute only. We model \$1.5Bn peak sales.

Sustainability Matters

Top Material Issue(s): 1) Access and Affordability: As the manufacturer of the most effective and safest cystic fibrosis medicine, VRTX plays a central role in improving the quality of life of patients suffering from CF. Affordability is key in lower-income countries for patients to obtain treatment. **2) Product Quality and Safety:** Ensuring the medicine patients obtain passes regulatory standards is critical for patients' health as well as VRTX's revenue stream and to ensure physician and patient confidence in its products.

Company Target(s): 1) Increase access to CF medicines around the world.

Qs to Mgmt: 1) Can you tell us more about your compassionate use programs and your continued efforts to increase equity in accessing CF medication? **2)** Can you tell us about your efforts in promoting clinical trial diversity when enrolling for diseases such as APOL-1-mediated kidney diseases?

ESG Sector Deep Dives: [US Biotech, Pharma, and Specialty Pharma](#)

Catalysts

- 1H26:** Ph.2a proof of concept data from SION
- 1H26:** Pivotal Phase III APOL1 kidney disease interim readout
- 2H26:** Potential Phase III data and regulatory filing for VX-880 (islet cell therapy) in T1D
- 1H27:** Phase III DPN chronic pain suzetrigine readout

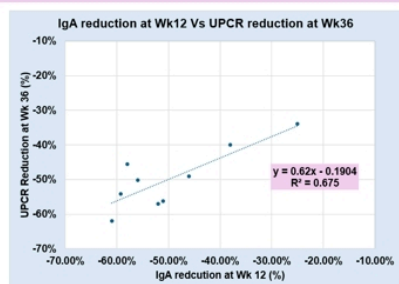
1) Povetacicept Ph3 RAINIER came in line with our expectations ([HERE](#)) at 52% abs UPCR reduction from baseline at wk 36 - while comparable to sibeprenlimab Ph.3's 50% UPCR reduction, we think povetacicept could be a larger oppt'y than expected & model \$8.5B total risk-adj PS (\$6B from IgAN) vs \$5.8B cons...*note cont'd pg.3*

Specifically for IgAN, we think a) market could be bigger over time, as pts need to be on drug for life & high pricing for the class (sibe's price was \$390k/yr vs \$200k/yr cons). b) povi offers a more convenient auto-injector (vs PFS for sibe) & has smaller volume of administration (0.46mL vs 2mL sibe) which could give it a compliance advantage.

Additionally, we're optimistic on pove's expansion opportunity into gMG, pMN, & wAIHA.

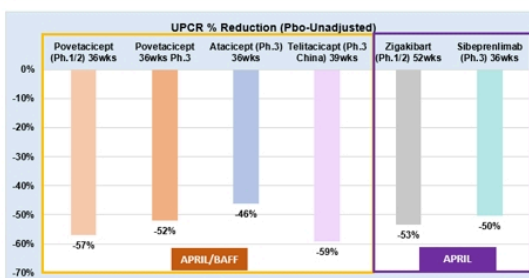
- For gMG, we think the pove's APRIL/BAFF works in this setting since Telitacicept (APRIL/BAFF) showed efficacy w/ -4.8 MG-ADL at 48wks, although it's tough to compare China-only data to global FcRn data – we think FcRns would still remain dominant here given faster drops in IgG, but do see value for Pove as 2L in FcRn refractory pts (~30% pts).
- For pMN, Pove showed -82% UPCR reduction and ~83% Anti-PLA2R reduction in Ph1/2 at 48wks (numerically similar UPCR to comps at 48wks, but caveats on sample size) – to us, Pove doesn't have to be best in pMN to still generate notable revs given pricing for the class is high
- Similarly for wAIHA, there are multiple tx being developed, which crowds the field for Pove, but pricing for the class still makes wAIHA an attractive opportunity

Our projections based on the Wk 12 IgA reduction –UPCR reduction relationship



Assumed Baseline UPCR for Pove Ph3	1.22
Assumed IgA reduction at Wk12	-51%
Proj Wk36 UPCR reduction with Pove in Ph3	-51.96%

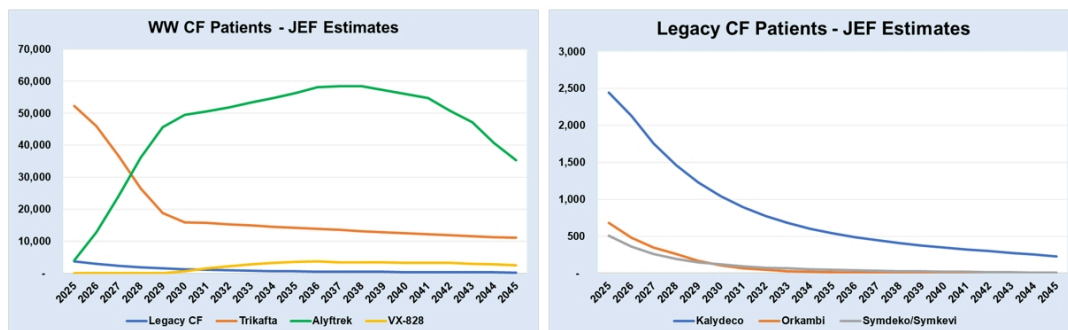
Interim analysis of Ph3 RAINIER reported 52% of abs 24-hr UPCR reduction from baseline at wk 36



Source: Jefferies, Company Presentations

2) Importantly, we think VRTX's CF base business looks stable w/ limited competitive threats (SION looks like an add-on tx) & we don't anticipate generic risk in the next decade – this supports a floor valuation ~\$422 (CF NPV + cash)

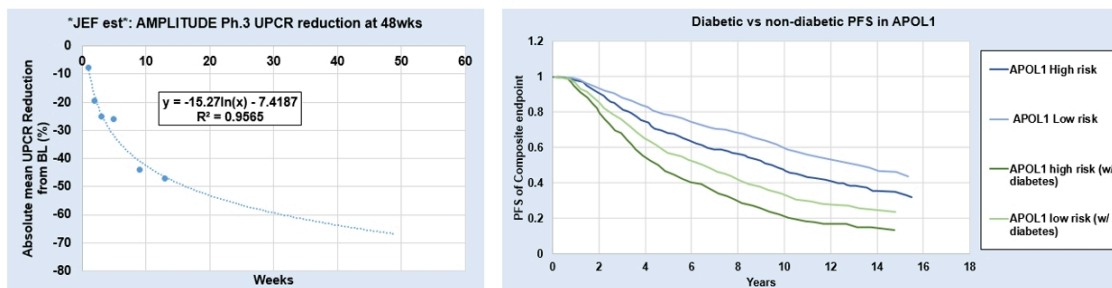
- Starting off, VRTX's CF franchise is the market leader (Trikafta in the US; Kaftrio ex-US; next-gen Alyftrek rollout ongoing) and continues to generate >\$10Bn annualized revenue w/ high margins & we don't see generic risk until post-2037. Note the CF market CAGR was ~17% over the past decade & cons projects cont. growth w/ ~3% CAGR over the next 5-7 yrs
- We see limited competitive threat from SION beyond as an add-on tx - ~90% of CF pts are eligible for CFTR modulators; ~2/3 of pts do not fully normalize CFTR function, but we see this as oppty for VRTX lifecycle mgmt rather than erosion. Addtl, current ex-US generics have not taken market share except in limited geographies & some states require step through via Trikafta.
- Importantly, mgmt expects most Alyftrek pts to come from Trikafta switch - in fact, our doc survey (n=25) suggests prescribers anticipate moderate long-term retention on Trikafta, a growing preference for Alyftrek, and gradual switching over 12-24mo. Ergo, we're modeling gradual Alyftrek expansion/switch with a ~15% switch rate from Trikafta in 2026, ramping up to 30% at peak.



Source: Jefferies Analysis

3) For the rest of VRTX's kidney pipeline, we think inaxaplin in AMKD could have first/best-in-class efficacy - our analysis suggests UPCR in inaxaplin's Ph.3 will trend to 64-70% reduction at 48wks in Ph.3 AMPLITUDE (note: currently no FDA approved targeted tx exist). **We're also not worried about competitor MZE829 & expect inaxaplin to show wider tx delta given a)** we think VRTX is reaching similar target coverage to MAZE and capping out on efficacy for the APOL1-targeted MOA, **b)** VRTX BL population does not including diabetics (therefore not being as pretreated for BL eGFR). We also think market could be large & estimate ~250K pts in AMKD in US & EU. That said, we currently have conservative penetration given the lower socio-economic status of these pts, though we expect dx rates to increase following tx approvals. We model 60% POS with risk-adj. PS of ~\$3Bn vs ~\$2.2Bn for cons.

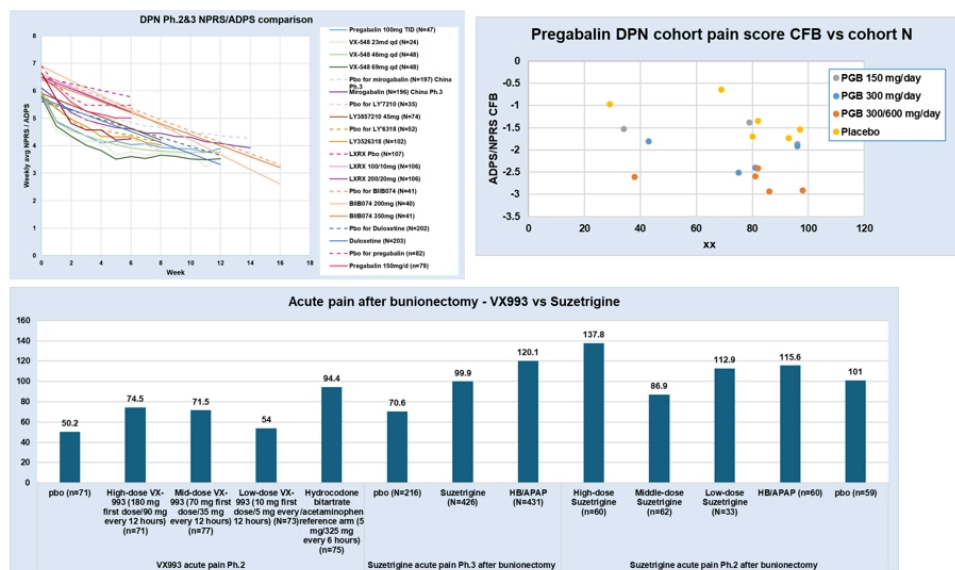
- For ADPKD, we think it's too early to say there will be differentiated efficacy for VX-407 from current SoC (Tolvaptan) since we've not yet seen data. That said, if it does in fact show similar safety to Trikafta (based on MOA similarity), pot'lly the black box warning won't be as demanding for liver function monitoring vs SOC (i.e no REMS) and this could further add value to kidney franchise. We model in line w/ cons at ~\$974M.



Source: Jefferies Analysis

4) We're mixed on VRTX's pain program but think it could still add \$25 NPV - we see risks to Ph.3 Diabetic Peripheral neuropathy (DPN) trials, but see room for addtl sales growth on Journavx (suzetrigine, Nav1.8 inhibitor) in acute pain.

- We model 30% POS for chronic pain DPN trials for suzetrigine & VX-993, given chronic neuropathic pain introduces greater heterogeneity & pbo variability - we think a combo-tx approach could offer greater efficacy. Commercial uncertainty (incl payer step edits, generic gabapentinoid use) could also be an overhang to Suzetrigine uptake in chronic pain.
- That said, we model acute pain as the bigger opportunity & expect ~3x Suzetrigine volume growth this year, based on recent VRTX mgmt comments). Our illustrative tracker analysis assuming ~50k patient adds per Q (in line with VRTX's expectation to 3x prescriptions this year) implies FY'26 US sales of ~\$242M (vs ~\$238M cons)



Source: Jefferies Analysis

5) Finally, on rest of pipeline, we're skeptical on Casgevy (slower ramp due to market uptake precedence) and Zimislecel (immunosuppression requirement & trial delays) and model below cons.

- In T1D, we think Zimislecel has a place as an islet cell replacement therapy. That said, the co-admin of chronic systemic immunosuppression to prevent rejection leaves high infection risk that limits TAM to those that have had 2+ severe hypoglycemic episodes per year and this will be challenging to overcome. On top of that, there are currently trial delays without visibility on resolution.
- For SCD, we think Casgevy is an efficacious & already approved (2023) tx. The TAM is 31-37k pts, but we've seen precedence that prior tx's had slower uptake/ under penetration due to patients/physician concerns and access barriers, hence we model SCD + TDT \$1.5Bn vs \$1.8Bn for cons

Income Statement

(In millions, except per share amounts)		Q126	Q226	Q326	Q426	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045
Jefferies	AP Total Revenues	\$ 3,108	\$ 3,198	\$ 3,344	\$ 3,357	\$ 13,007	\$ 14,428	\$ 16,205	\$ 18,295	\$ 20,216	\$ 22,422	\$ 24,238	\$ 26,262	\$ 28,410	\$ 30,183	\$ 31,560	\$ 31,861	\$ 32,097	\$ 27,384	\$ 25,294	\$ 23,634	\$ 21,645	\$ 19,817	\$ 17,749	\$ 15,904
	Non-GAAP Diluted EPS	\$ 4.46	\$ 4.59	\$ 4.90	\$ 4.80	\$ 18.74	\$ 22.14	\$ 26.45	\$ 31.85	\$ 36.63	\$ 42.52	\$ 47.59	\$ 53.39	\$ 59.93	\$ 65.67	\$ 70.34	\$ 72.56	\$ 74.46	\$ 62.63	\$ 57.45	\$ 53.78	\$ 49.05	\$ 44.91	\$ 39.97	\$ 35.77
Non-GAAP PNL																									
Product Revenue		3,108	3,198	3,344	3,357	13,007	14,428	16,205	18,295	20,216	22,422	24,238	26,262	28,410	30,183	31,560	31,861	32,097	27,384	25,294	23,634	21,645	19,817	17,749	15,904
Other Revenue		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total Revenue		3,108	3,198	3,344	3,357	13,007	14,428	16,205	18,295	20,216	22,422	24,238	26,262	28,410	30,183	31,560	31,861	32,097	27,384	25,294	23,634	21,645	19,817	17,749	15,904
COGS		430	440	462	462	1,784	1,932	2,136	2,341	2,635	2,955	3,215	3,476	3,854	4,197	4,363	4,310	3,579	2,548	2,303	2,108	1,919	1,745	1,587	1,407
Gross profit		2,678	2,758	2,883	2,895	11,214	12,496	14,069	15,955	17,582	19,467	21,023	22,785	24,747	26,386	27,698	28,250	28,518	24,836	22,991	21,526	19,726	18,072	16,162	14,497
R&D		941	964	987	987	3,849	4,080	4,284	4,498	4,678	4,818	4,915	5,013	5,113	5,164	5,216	5,268	5,163	5,008	4,808	4,567	4,293	3,993	3,673	3,343
Acquired IP R&D		25	25	25	25	100	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
SG&A		400	424	450	476	1,749	1,924	2,040	2,142	2,227	2,294	2,340	2,387	2,435	2,459	2,484	2,459	2,410	2,337	2,244	2,132	2,004	1,864	1,714	1,560
Other		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Operating income		1,313	1,352	1,443	1,407	5,515	6,402	7,745	9,315	10,676	12,354	13,768	15,386	17,199	18,763	19,998	20,523	20,946	17,491	15,939	14,827	13,429	12,216	10,795	9,594
Interest expense, net		(4)	(4)	(4)	(4)	(16)	(18)	(20)	(22)	(25)	(28)	(30)	(33)	(38)	(38)	(39)	(40)	(34)	(32)	(30)	(27)	(25)	(22)	(20)	(20)
Interest and other income, net		124	128	134	134	520	577	648	732	809	897	970	1,050	1,136	1,207	1,282	1,274	1,284	1,095	1,012	945	866	793	710	638
Pre-tax income (loss)		1,433	1,476	1,573	1,537	6,019	7,051	8,373	10,024	11,459	13,223	14,767	16,403	18,300	19,932	21,221	21,758	22,189	18,552	16,919	15,743	14,268	12,683	11,482	10,211
Income taxes		287	295	315	307	1,204	1,410	1,675	2,005	2,292	2,645	2,941	3,291	3,693	4,098	4,244	4,352	4,438	3,710	3,384	3,149	2,854	2,597	2,298	2,042
Non-GAAP Net Income		1,147	1,181	1,258	1,230	4,815	5,641	6,698	8,019	9,168	10,578	11,706	13,112	14,640	15,946	16,977	17,406	17,751	14,841	13,535	12,594	11,414	10,087	8,906	8,168
Income attributable to noncontrolling interest		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Comprehensive loss		1,147	1,181	1,258	1,230	4,815	5,641	6,698	8,019	9,168	10,578	11,706	13,112	14,640	15,946	16,977	17,406	17,751	14,841	13,535	12,594	11,414	10,087	8,906	8,168
Non-GAAP EPS																									
Non-GAAP Diluted EPS		\$ 4.46	\$ 4.59	\$ 4.90	\$ 4.80	\$ 18.74	\$ 22.14	\$ 26.45	\$ 31.85	\$ 36.63	\$ 42.52	\$ 47.59	\$ 53.39	\$ 59.93	\$ 65.67	\$ 70.34	\$ 72.56	\$ 74.46	\$ 62.63	\$ 57.45	\$ 53.78	\$ 49.05	\$ 44.91	\$ 39.97	\$ 35.77
Share Count (In millions)																									
Diluted		258	257	257	256	257	255	253	252	250	249	247	246	244	243	241	240	238	237	236	234	233	231	230	228

Source: Jefferies Analysis, Company data

Balance Sheet

(In millions, except per share amounts)		2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045	
Jefferies	Non-GAAP Total Revenues	\$ 12,001	\$ 13,007	\$ 14,428	\$ 16,205	\$ 18,295	\$ 20,216	\$ 22,422	\$ 24,238	\$ 26,262	\$ 28,410	\$ 30,183	\$ 31,560	\$ 31,861	\$ 32,097	\$ 27,384	\$ 25,294	\$ 23,634	\$ 21,645	\$ 19,817	\$ 17,749	\$ 15,904	
	Non-GAAP Diluted EPS	\$ 18.40	\$ 18.74	\$ 22.14	\$ 26.45	\$ 31.85	\$ 36.63	\$ 42.52	\$ 47.59	\$ 53.39	\$ 59.93	\$ 65.67	\$ 70.34	\$ 72.56	\$ 74.46	\$ 62.63	\$ 57.45	\$ 53.78	\$ 49.05	\$ 44.91	\$ 39.97	\$ 35.77	
Balance Sheet (Historicals from Company Filings)																							
Cash and cash equivalents		5,085	11,574	16,860	22,761	30,317	38,605	48,089	59,014	71,217	85,081	100,408	117,011	134,788	152,449	169,962	184,377	197,574	209,674	220,690	230,606	239,459	
Marketable securities		1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	1,523	
Accounts receivable, net		2,053	2,062	2,287	2,636	2,926	3,242	3,610	3,888	4,218	4,565	4,846	5,069	5,117	5,155	4,398	4,062	3,796	3,476	3,183	2,851	2,554	
Inventory		1,607	1,455	1,709	1,949	2,035	2,342	2,631	2,838	3,004	3,249	3,363	3,424	3,200	3,171	2,259	2,041	1,868	1,701	1,547	1,389	1,247	
Prepaid expenses and other current assets		853	844	945	1,088	1,205	1,337	1,488	1,603	1,739	1,862	1,998	2,090	2,110	2,125	1,813	1,675	1,565	1,433	1,312	1,175	1,053	
Total current assets		11,201	17,457	23,325	29,957	38,006	47,650	57,342	68,866	81,782	96,299	112,138	129,118	146,738	164,424	179,956	193,679	206,326	217,807	228,255	237,544	245,837	
Property and equipment, net		1,520	1,677	1,873	2,092	2,306	2,533	2,765	3,004	3,249	3,501	3,750	4,003	4,256	4,505	4,746	4,979	5,201	5,411	5,607	5,789	5,955	
Goodwill		1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	1,088	
Intangible assets, net		424	424	424	424	424	424	424	424	424	424	424	424	424	424	424	424	424	424	424	424	424	
Deferred tax asset		2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	2,886	
Operating lease assets		1,563	873	940	1,039	1,139	1,262	1,438	1,565	1,692	1,783	1,848	1,880	1,757	1,742	1,240	1,121	1,026	934	849	763	685	
Long-term marketable securities		5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	5,712	
Other long-term assets		1,237	1,260	1,442	1,595	1,814	1,997	2,219	2,387	2,588	2,810	2,988	3,122	3,152	3,175	2,709	2,502	2,338	2,141	1,960	1,766	1,573	
Total assets		25,643	31,389	37,702	44,806	53,388	62,985	73,887	85,954	99,443	114,516	130,945	148,245	166,026	183,967	198,774	212,403	225,014	236,416	246,794	255,974	264,172	
Current liabilities:																							
Accounts payable		462	494	551	617	698	771	855	924	1,002	1,084	1,151	1,204	1,215	1,224	1,044	965	901	826	756	677	607	
Accrued expenses		2,971	3,687	3,842	3,993	4,272	4,407	4,536	4,641	4,725	4,821	4,871	4,918	4,936	4,838	4,692	4,505	4,279	4,023	3,741	3,442	3,132	
Other current liabilities		420	587	555	597	645	654	679	695	705	721	728	735	736	722	701	673	640	601	559	514	469	
Total current liabilities		3,853	4,768	4,948	5,208	5,615	5,832	6,070	6,260	6,532	6,625	6,751	6,857	6,889	6,766	6,458	6,142	5,820	5,449	5,066	4,633	4,207	
Long-term operating lease liabilities		1,847	1,554	2,034	2,080	2,052	2,262	2,303	2,337	2,401	2,442	2,465	2,493	2,500	2,450	2,377	2,282	2,168	2,038	1,895	1,743	1,587	
Long-term finance lease liabilities		1,152	1,261	1,399	1,523	1,623	1,749	1,863	1,957	2,078	2,180	2,263	2,385	2,487	2,562	2,262	2,082	1,949	1,824	1,704	1,590	1,482	
Total liabilities		6,977	7,878	8,533	8,955	9,477	9,909	10,253	10,517	10,787	11,062	11,231	11,384	11,431	11,236	10,756	10,288	9,758	9,151	8,498	7,809	7,089	
Common stock and additional paid-in capital		5,122	5,548	5,961	6,341	6,777	7,172	7,547	7,980	8,473	8,928	9,338	10,304	11,028	11,809	12,650	13,609	14,551	15,542	16,582	17,671	18,809	
Accumulated other comprehensive gain		(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	(16)	
Retained earnings		13,560	17,979	23,224	29,526	37,148	45,921	56,103	67,472	80,199	94,442	109,962	126,543	143,963	160,930	175,383	188,202	200,720	211,739	221,720	231,599	238,291	
Total liabilities and stockholders' equity		25,643	31,389	37,702	44,806	53,388	62,985	73,887	85,954	99,443	114,516	130,945	148,245	166,026	183,967	198,774	212,403	225,014	236,416	246,794	255,974	264,172	

(In millions, except per share amounts)		2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045
Jefferies																					
Total Revenues		\$ 13,007	\$ 14,428	\$ 16,205	\$ 18,295	\$ 20,216	\$ 22,422	\$ 24,238	\$ 26,262	\$ 28,410	\$ 30,483	\$ 31,560	\$ 31,861	\$ 32,097	\$ 27,384	\$ 25,294	\$ 23,634	\$ 21,645	\$ 19,817	\$ 17,749	\$ 15,904
Diluted EPS		\$ 18.74	\$ 22.14	\$ 26.45	\$ 31.85	\$ 36.63	\$ 42.52	\$ 47.59	\$ 53.39	\$ 59.93	\$ 65.67	\$ 70.34	\$ 72.56	\$ 74.46	\$ 62.63	\$ 57.45	\$ 53.78	\$ 49.05	\$ 44.91	\$ 39.97	\$ 35.77
DCF Valuation																					
Operating Profit		4,906	5,883	7,136	8,705	10,066	11,744	13,159	14,778	16,589	18,153	19,388	19,914	20,336	16,881	15,330	14,217	12,820	11,606	10,185	8,985
Tax Provision		981	1,177	1,427	1,741	2,013	2,349	2,632	2,955	3,318	3,631	3,878	3,983	4,067	3,376	3,066	2,843	2,564	2,321	2,037	1,797
NOPAT		3,925	4,706	5,708	6,964	8,053	9,395	10,527	11,821	13,271	14,523	15,511	15,931	16,269	13,505	12,264	11,374	10,256	9,285	8,148	7,188
+ Stock Based Compensation		712	699	665	675	674	654	712	772	834	889	945	1,002	1,061	1,120	1,169	1,218	1,268	1,315	1,364	1,413
+ Depreciation & Amortization		256	275	289	304	316	325	332	339	345	349	352	353	346	336	323	306	288	268	246	224
- Capital Expenditures		(387)	(480)	(496)	(500)	(530)	(546)	(553)	(567)	(578)	(583)	(590)	(591)	(580)	(562)	(540)	(513)	(482)	(448)	(412)	(375)
(+/-) Change in Working Capital		1,947	(382)	(479)	(120)	(624)	(674)	(506)	(634)	(519)	(434)	(285)	318	(148)	2,084	449	242	248	160	184	102
Free Cash Flow		6,443	4,628	5,888	7,322	7,888	9,155	10,512	11,730	13,354	14,743	15,934	17,013	16,948	16,483	13,664	12,627	11,576	10,579	9,530	8,552
Terminal Value																					132,879
Discount Factor		1.00	2.00	2.99	4.00	5.00	6.00	6.99	8.01	9.01	10.01	10.98	12.01	13.01	14.01	14.97	16.01	17.01	18.01	18.96	20.01
PV Free Cash Flow		5,993	4,178	4,580	5,482	5,494	5,931	6,342	6,574	6,962	7,150	7,203	7,139	6,815	5,985	4,628	3,967	3,383	2,876	2,418	2,011
WACC		7.5%																			
Valuation Date		12/31/2025																			
Perpetuity Growth		1.0%																			
Growth Rate		1.0%																			
PV of Terminal Value		31,250																			
PV of Free Cash Flow		104,912																			
Enterprise Value		136,162																			
+ Cash		12,320																			
- Debt		-																			
Equity Value		148,483																			
Shares Outstanding		256																			
Fair Value		\$ 580																			

		Weighted Average Cost of Capital			
		6.5%	7.0%	7.5%	8.0%
Terminal Growth Rate	-	\$ 642.38	\$ 599.79	\$ 562.47	\$ 529.47
	0.5%	\$ 655.31	\$ 609.92	\$ 570.51	\$ 535.92
	1.0%	\$ 670.59	\$ 621.75	\$ 579.78	\$ 543.29
	1.5%	\$ 688.92	\$ 635.73	\$ 590.61	\$ 551.77
	2.0%	\$ 711.33	\$ 652.50	\$ 603.40	\$ 561.68

Source: Jefferies Analysis, company data

VRTX: IgAN Data Looks In Line – We’re Still Bullish on the Opportunity (BUY Rating, PT \$580)

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We're initiating coverage of VRTX with a Buy and a PT of \$580

INVESTMENT CONCLUSION:

We're initiating coverage of VRTX with a BUY rating and \$580 PT (~26% upside vs \$460.87 close on 3/9/26). VRTX's stock has underperformed its peers over the past 6 months (+16% vs +21-40% for other LC biotech's) following several setbacks last year. The stock trades at a forward P/E premium (~23.6x vs ~17.0x industry average), which we view as justified given consensus assumes 5-year revenue and EPS CAGRs of ~10% & ~13%, respectively - toward the high end of LC pharma and biotech peers. We like VRTX for several reasons: 1) We view Cystic Fibrosis (CF) as a stable franchise for VRTX which supports a floor valuation ~\$422 (CF NPV + cash), 2) while Ph.3 UPCR was in line w/ Sibe, to us what's more important is a) the IgAN market is larger than currently modelled and favorably priced (\$390K/yr), b) Pove is conveniently administered via 0.46ml autoinjector (vs Sibe's PFS) giving pot'l compliance advantage - we model \$8.5Bn vs \$5.8Bn cons., 3) We're expecting inaxaplin to produce 64-70% UPCR at 48wks in AMKD making it first/best in class & for VX-407 we think the lack of data makes it premature to make a differentiated call in ADPKD, hence we model ~\$974M in line w/ cons., 4) we're in line for acute pain (\$1.9Bn vs \$1.9Bn cons) based on expected volume growth, but we're below on chronic pain hitting (combo-tx likely needed) w/ \$245M vs \$1.1Bn cons, 5) that said, we're skeptical on rest pipeline (slower Casgevy ramp in SCD & Zimislecel immunosuppression requirement limiting TAM to severe T1D + trial delays)

REASONS BEHIND OUR RATING:

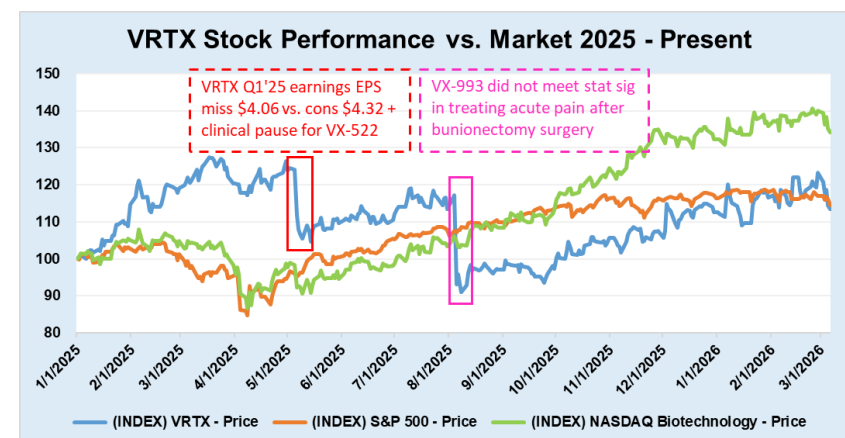
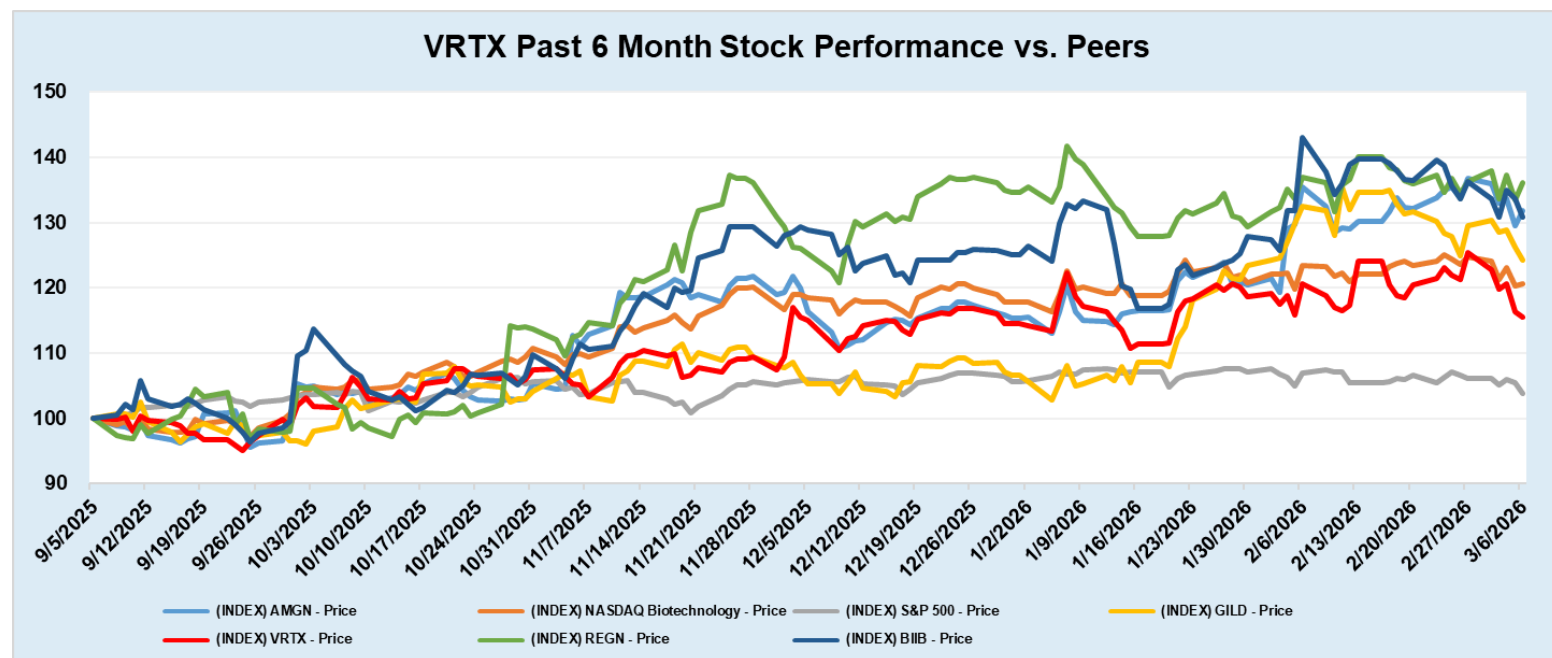
- We think VRTX's CF franchise (Alyftrek, Trikafta, legacy products) remains durable and expect gradual expansion and switch to Alyftrek
- IgAN market is big and Pove could be preferred as a more conveniently administered (autoinjector)
- APRIL/BAFF MOA is viable and Pove sees expansion in gMG, pMN, and/or wAIHA
- Long-term we see upside in AMKD w/ Inaxaplin being pot'ly first/best in class

RISKS TO OUR RATING:

- Competitor CF programs take share from VRTX or Alyftrek switch rate is minimal
- Povetacicept has poor market uptake in IgAN or high rates of chronic safety issues emerge (eg, infection, hypogammaglobulinemia)
- AMKD fails in its Ph.3 trial or competitor program shows better data

Source: Jefferies Analysis

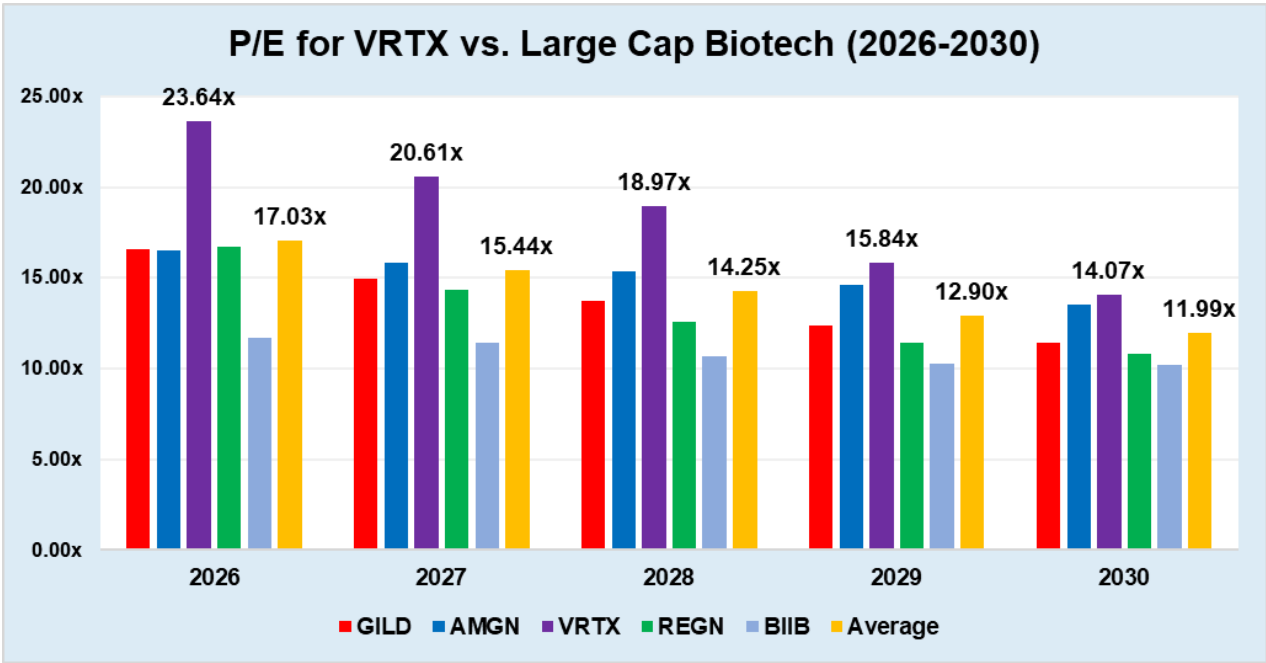
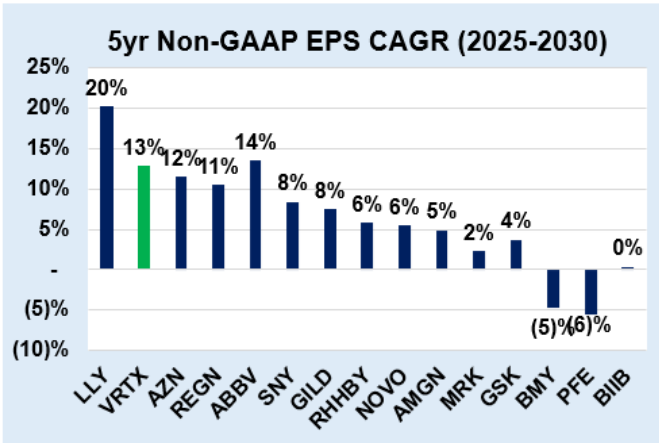
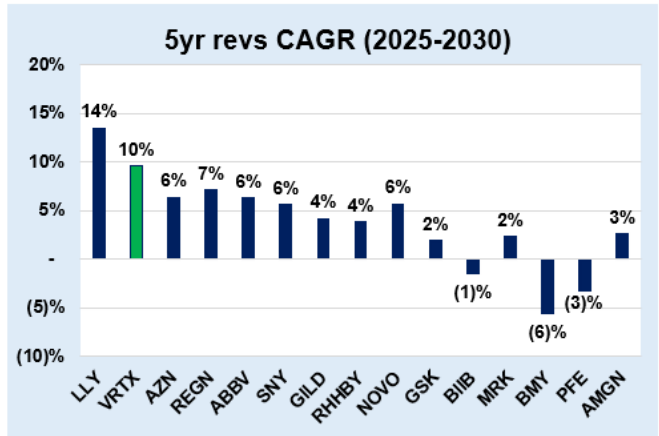
VRTX's stock has underperformed its peers over the past 6 months (+16% vs +21-40% for other LC biotech's) following several setbacks last year



VRTX underperformed vs peers last yr due to two major setbacks: 1) co missed Q1 on both top and bottom line (EPS \$4.06 vs. \$4.32 consensus - driven by rev declines for CF ex-US), 2) VX-993 failed to result in a stat. sig. improvement on the primary endpoint in its Ph.2 acute pain. VRTX has underperformed its peers following several setbacks last year (up 16% vs peers 21-36%). FY'26 guidance implies continued operating margin expansion (46~48% vs ~45% FY'25)

Source: Jefferies Analysis, Factset

Currently, the stock trades at a forward P/E premium (~23.6x vs ~17.0x industry average), which we view as justified given consensus assumes 5-year revenue and EPS CAGRs of ~10% & ~13%, respectively – toward the high end of LC pharma and biotech peers



Source: Jefferies Analysis, Factset

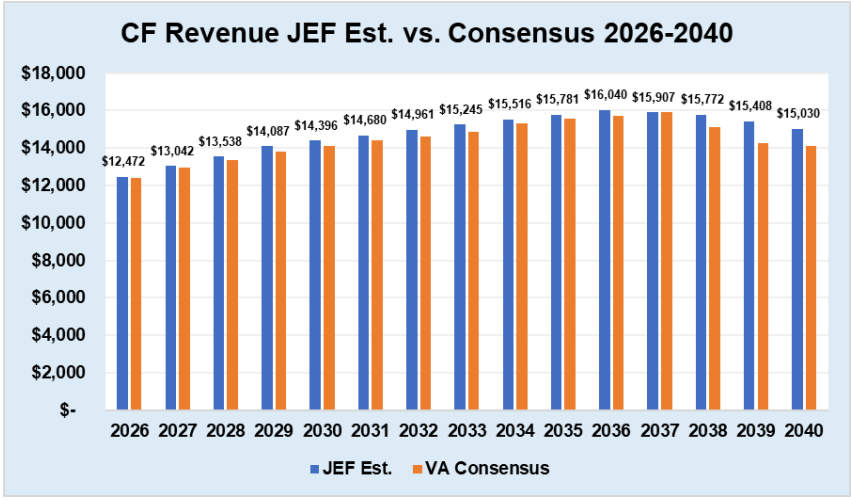
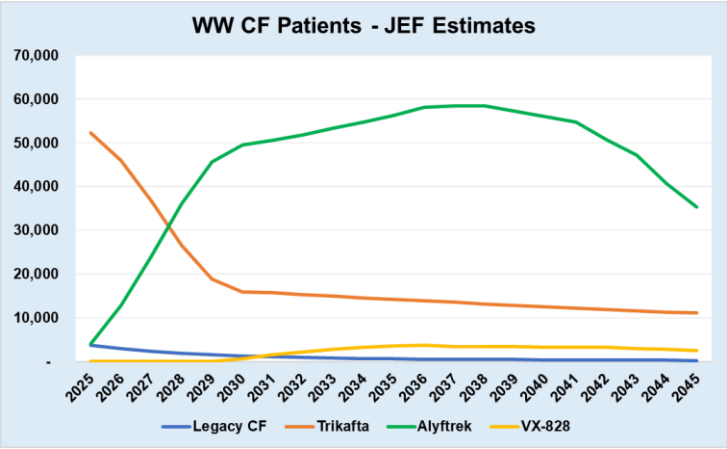
We're BUY rated on VRTX for a few reasons:

1) We view Cystic Fibrosis (CF) as a stable franchise for VRTX with limited competitive threats (SION looks like an add-on tx) & we don't anticipate generic risk in the next decade - this supports a 'floor' valuation ~\$422 (CF NPV + cash)

Cystic Fibrosis - Market Model	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045
Cystic Fibrosis - U.S. Build																					
US CF Market Model Assumptions	Kalydeco	Orkambi	Symdeko	Trikafta	Alyftrek	VX-828															
Launch Year					2025	2030															
Patent Expiry	2027	2031	2027	2037	2039	2042															
% Penetrator on new Pts	5%	-	1%	77%	17%		*2025 % NRx based on IQVIA														
Trikafta switch rate %	25%	20%	-	-	-																
Alyftrek switch rate %	25%	40%	40%	30%	-																
VX-828 switch rate %	-	-	-	10%	15%																
1+ yo Population Approval				2027	2029																
<1 yo Population Approval																					
US Cystic Fibrosis Patients	43,804	44,804	45,804	46,804	47,804	48,804	49,804	50,804	51,804	52,804	53,804	54,804	55,804	56,804	57,804	58,804	59,804	60,804	61,804	62,804	63,804
CF Incidence	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
US Modulator-Eligible %	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%
US Modulator-Eligible Pts	40,606	41,533	42,460	43,387	44,314	45,241	46,168	47,095	48,022	48,949	49,876	50,803	51,730	52,657	53,584	54,511	55,438	56,365	57,292	58,219	59,146
Treatment Penetration %	77%	79%	79%	80%	80%	80%	81%	81%	81%	81%	82%	82%	80%	78%	76%	73%	71%	64%	58%	52%	47%
Patients Treated	31,450	32,811	33,650	34,493	35,341	36,193	37,050	37,912	38,778	39,649	40,525	41,405	41,384	41,336	40,724	40,066	39,361	36,017	32,949	30,134	27,552

- VRTX's CF franchise (Trikafta in the US; Kaftrio ex-US; next-gen Alyftrek rollout ongoing) continues to generate >\$10Bn annualized revenue w/ high margins & we don't see generic risk until post-2037
- We see limited competitive threat from SION beyond as an add-on tx - ~90% of CF pts are eligible for CFTR modulators; ~2/3 of pts do not fully normalize CFTR function, but we see this as opportunity for VRTX lifecycle mgmt rather than erosion
- We expect gradual Alyftrek expansion/switch based on our survey results

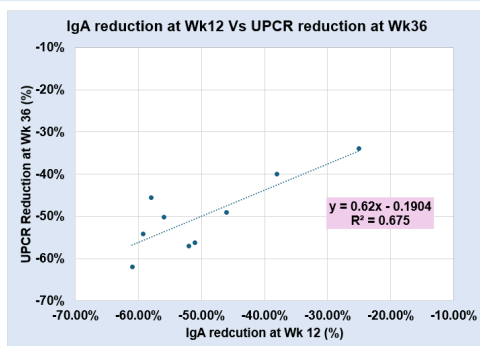
Source: Jefferies Analysis, Company data



2) While povetacicept Ph3 (RAINIER) interim data looks in line w/ sibeprenlimab Ph.3 (wk 36 tx arm UPCR reduction of 52% vs 50% w/ sibe), we think pove could be an \$8.5B drug over time given a) IgAN market could be larger than current cons. est b) pove offers a more convenient auto-injector, b) we're optimistic re: expansion opportunity into autoimmune indications incl. gMG, pMN, wAIHA

We had previously explored potential predictors of UPCR reduction - while limited by small number of data points, our illustrative regression analysis suggested wk12 IgA reduction could be a relevant predictor to wk36 UPCR reduction. Using our illustrative regression model, we expected Pove Ph3 to show ~52% absolute UPCR reduction. The reported interim data from Ph3 RAINIER came in line with our expectations.

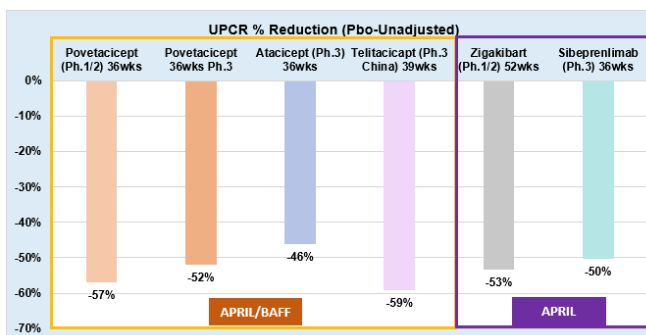
Our projections based on the Wk 12 IgA reduction –UPCR reduction relationship



Assumed Baseline UPCR for Pove Ph3	1.22
Assumed IgA reduction at Wk12	-51%
Proj Wk36 UPCR reduction with Pove in Ph3	-51.96%

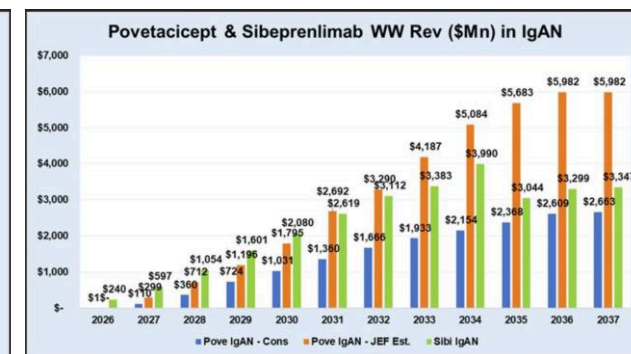
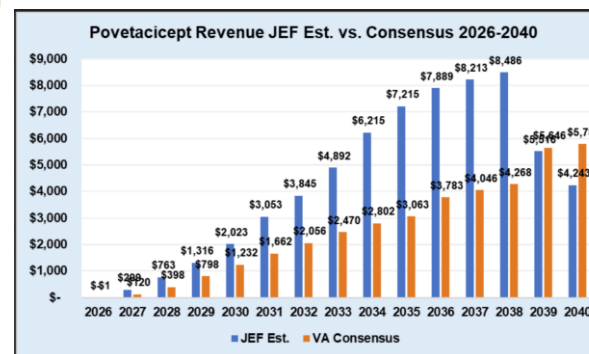
Source: Jefferies Analysis, company data

Interim analysis of Ph3 RAINIER reported 52% of abs 24-hr UPCR reduction from baseline at wk 36



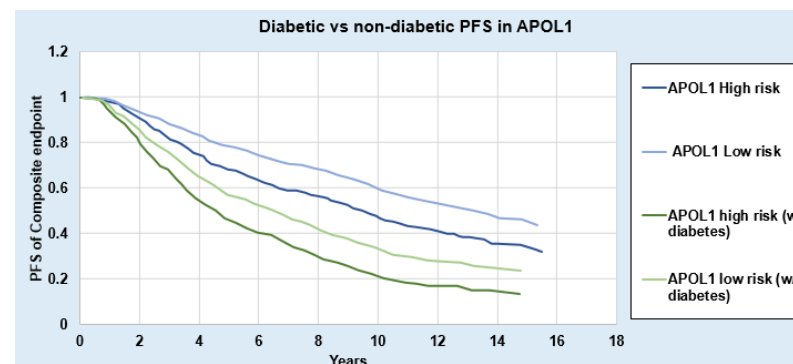
Our KOL Dr. Geoff Teehan previously noted ([HERE](#)) that, provided the APRL or APRIL/BAFF class show similar efficacy, **he believed dosing convenience would be a greater deciding factor for drug selection (~30-50% market share gained)**. He also felt **both classes were a better choice to SoC** given SoC does not effectively stabilize eGFR (*more on this later...*)

Despite Pove UPCR efficacy generally came broadly in-line with other APRIL & APRIL/BAFF approaches (eg. sibe, Atac, Teli, **caveats apply**) – we expect larger market opportunity for Pove, given - 1) Sibe reported ~500 pts starts after launch in late 2025 & price came in at \$390K/yr (vs \$200K/yr cons.) showing market is IgAN market is larger than anticipated and Sibe is under modelled, 2) Pove intends to use an autoinjector, but Sibe does not currently (we didn't see plans), 3) We're optimistic for indication expansion past IgAN - Telitacicept's data has helped derisk the MOA in gMG + we're also see viability in pMN and wAIHA given the orthogonal MOA with APRIL/BAFF. **Taken together, we model ~\$8.5Bn vs \$5.8Bn cons est.**

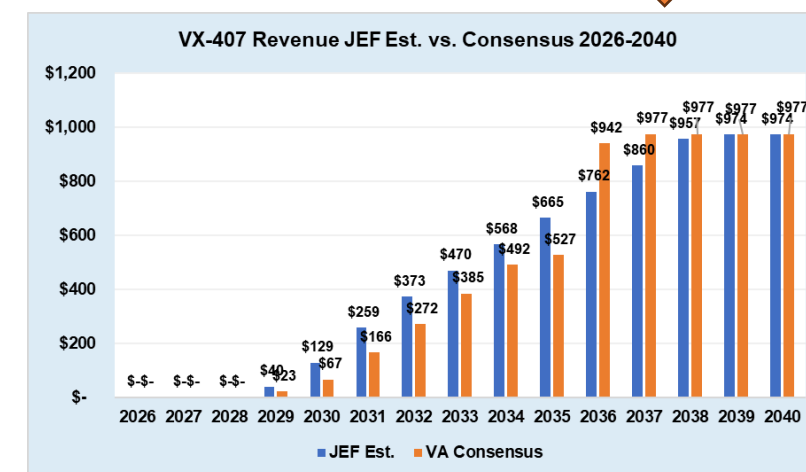
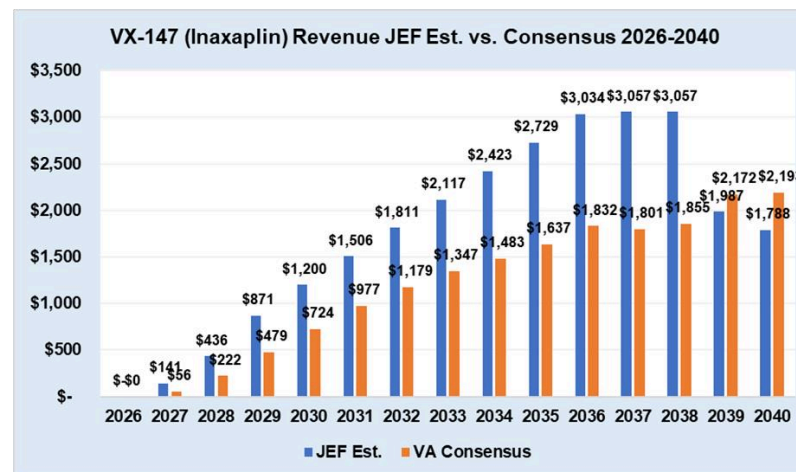
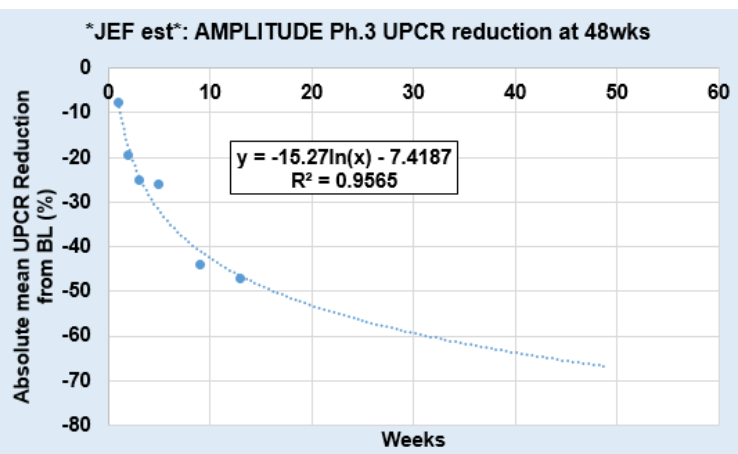


3) On other renal, we're also tracking inaxaplin in AMKD & could see LT upside as a potential first/best in class tx (TAM could increase w/ effective tx & more genetic testing awareness) – we model risk-adj. sale of ~\$3Bn (60% POS) vs ~\$2.2Bn for cons + we model \$974M risk-adj peak sales for ADPKD given data prematurity

- We see strong genetic validation of APOL1 as target & **our analysis suggests UPCR in inaxaplin's AMPK Ph.3 will trend to 64-70% reduction at 48wks in Ph.3 AMPLITUDE (note: currently no FDA approved targeted tx exist).**
- We're also not worried about competitors - we expect Inaxaplin to show a wider tx delta vs MZE829 Ph.2 due to VRTX BL population not including diabetics (therefore not being as pretreated for BL eGFR).
- We est. **~250K pts in AMKD in US & EU** but currently have conservative penetration given the lower socio-economic status of these pts. That said, a) we believe with the introduction of an effective tx there could be increased genetic testing, b) KOL believes with an increase in IgAN diagnosis rates from recent approval(s), there could be spillover detection for APOL1 detection in clinics.



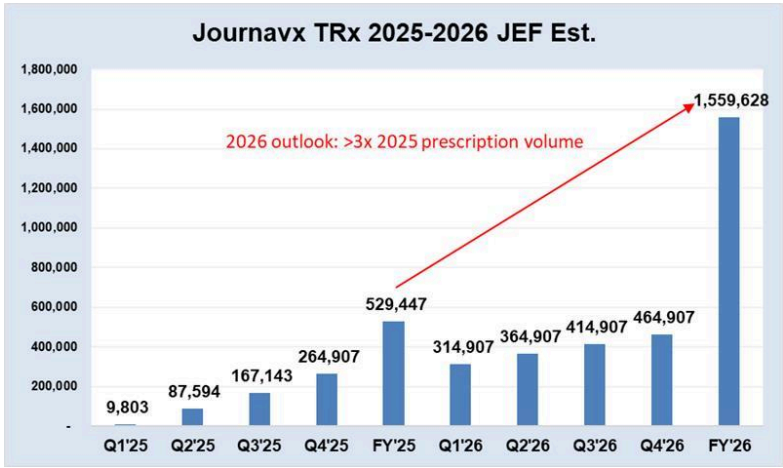
VX-407: We model in line w/ cons (~974M) - it's too early to claim differentiated efficacy for VX-407 vs tolvaptan without data. However, if safety mirrors Trikafta given MOA similarity, liver monitoring requirements could be less onerous.



Source: Jefferies Analysis, company data

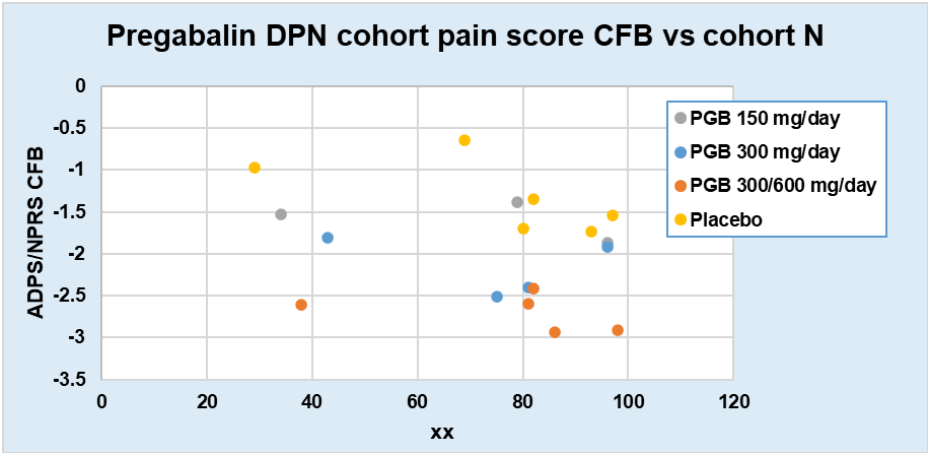
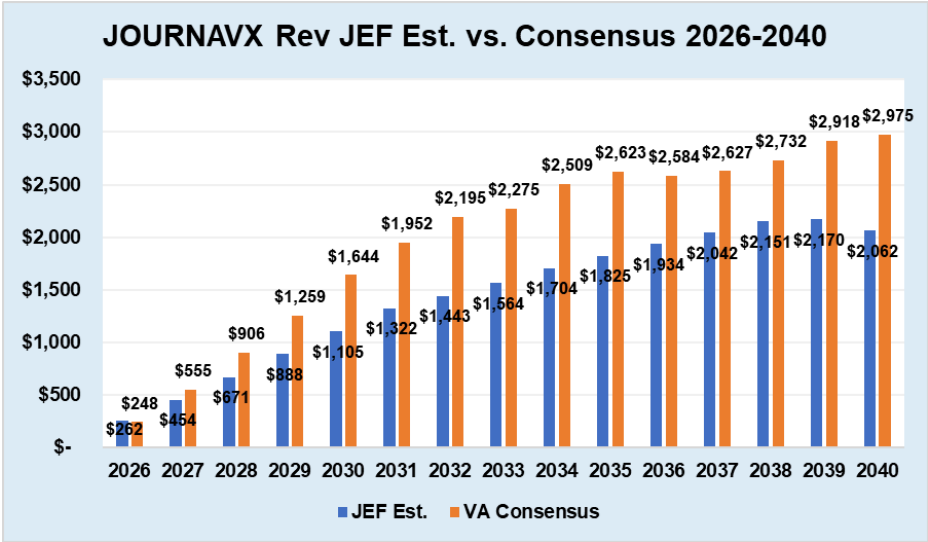
4) We assign \$25 NPV for the pain franchise – although we’re more bearish on VRTX’s DPN chronic pain trial (30% POS), we still see use for Journavx in acute pain, and model overall risk-adj PS \$2.2B JEF vs \$3B cons

- We see risks with the Ph.3 chronic pain DPN studies for suzetrigine & VX-993 given we've seen chronic neuropathic pain introduces greater heterogeneity and pbo variability - commercial uncertainty (inc. payer step edits, generic gabapentinoid use) could also be an overhang to Suzetrigine uptake in chronic pain
- On the other hand, we're slightly above/in line w/ cons on acute pain as we expect Suzetrigine volume growth this year (VRTX guided ~3x prescription in '26)



Journavx Summary	Q1'25	Q2'25	Q3'25	Q4'25	FY'25	Q1'26	Q2'26	Q3'26	Q4'26	FY'26
Total Scripts	9,803	87,594	167,143	264,907	529,447	314,907	364,907	414,907	464,907	1,559,628
New Pt Adds		77,791	79,549	97,764		50,000	50,000	50,000	50,000	
JEF Est. Rev	\$ 1	\$ 12	\$ 20	\$ 27	\$ 60	\$ 39	\$ 52	\$ 67	\$ 84	\$ 242
VA Consensus	\$ 1	\$ 12	\$ 20	\$ 27	\$ 60	\$ 36	\$ 46	\$ 62	\$ 91	\$ 234
Beat (Miss) vs. Consensus						3	6	6	(7)	8

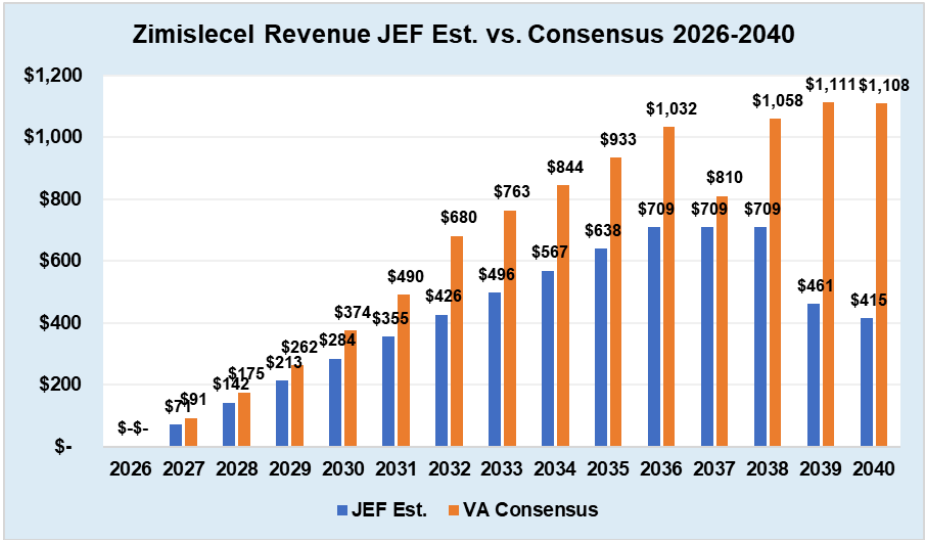
Source: Jefferies Analysis, company data



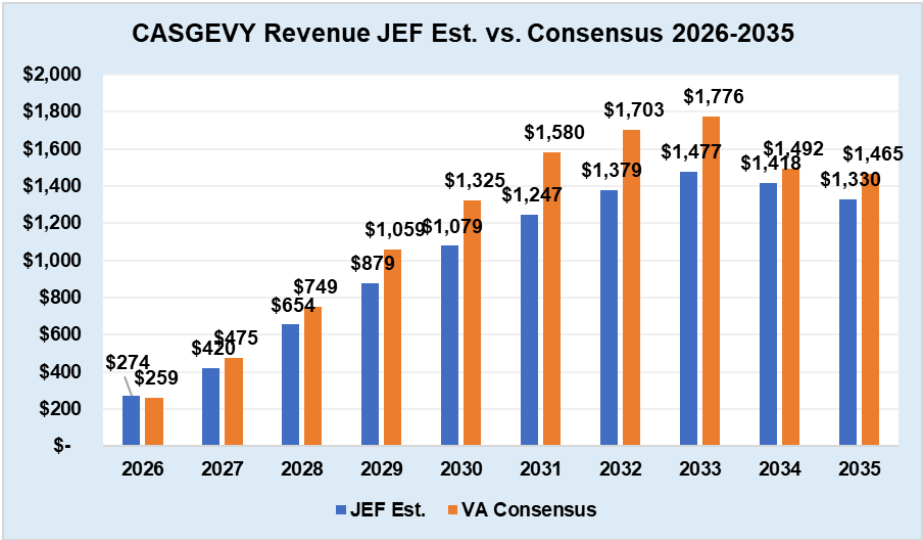
5) On rest of pipeline, we're skeptical on Casgevy (slower ramp based on market uptake precedence) and Zimislecel (immunosuppression requirement & trial delays) and model below cons

- For T1D, we think Zimislecel is an effective treatment for islet cell replacement in pts that can't otherwise produce insulin. That said, the co-admin of chronic systemic immunosuppression to prevent rejection leaves high infection risk that limits TAM to those that have had 2+ severe hypoglycemic episodes per year and this will be challenging to overcome. On top of that, there are trial delays without visibility on resolution.
- For SCD, we think Casgevy is an efficacious & already approved (2023) tx. The TAM is 31-37k pts, but we've seen precedence that prior tx had slower uptake/ under penetration due to patients/physician concerns and access barriers.

Zimislecel: we model \$709M risk-adj peak sales

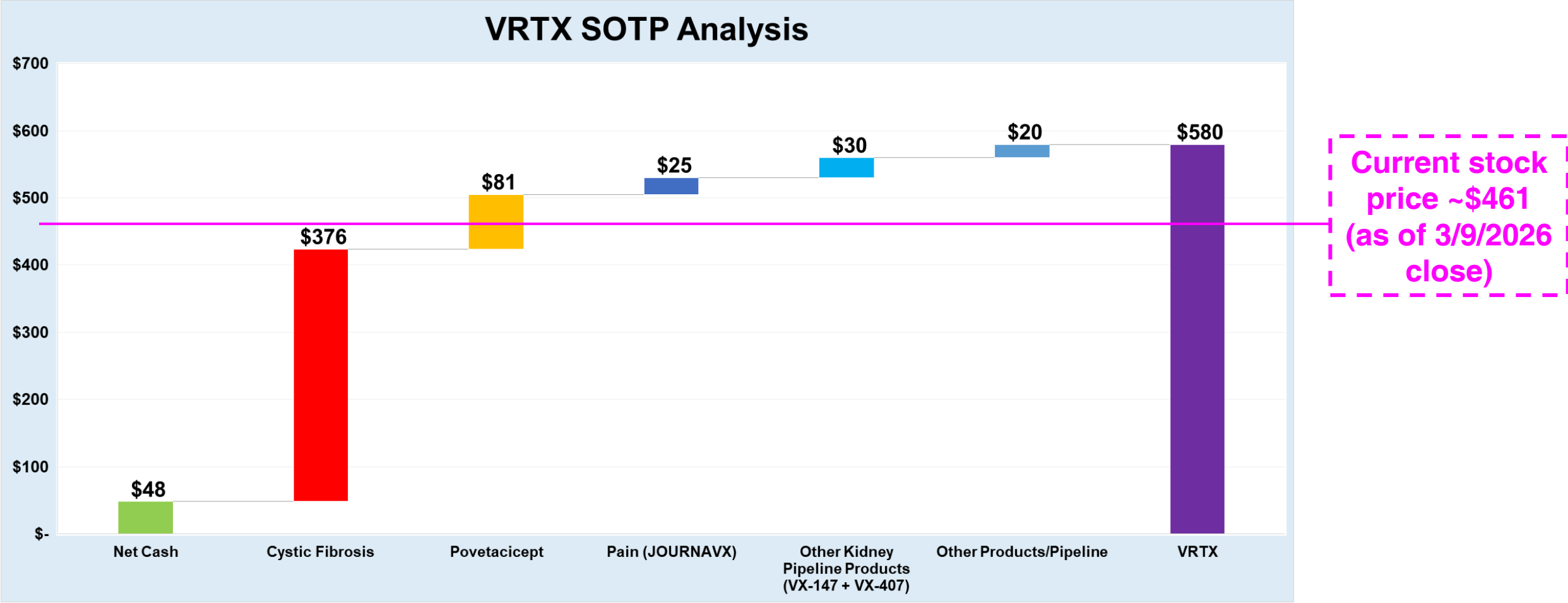


Casgevy: we model \$1.5Bn risk-adj peak sales



Source: Jefferies Analysis

Taken together, our valuation supports a \$580 price target, implying upside from the current \$461 share price



Source: Jefferies Analysis

Let's start with VRTX's cystic fibrosis business

Cystic Fibrosis

We expect VRTX's CF franchise to remain durable

- We're expecting CF to remain a durable revenue base for VRTX and expect gradual switch & new starts for pts onto Alyftrek
- We expect upcoming comps will work as add-on tx to VRTX, but we're not seeing evidence as a mono-tx they could produce greater efficacy
- We expect minimal pt adds from VRTX mRNA CF tx, but limited ability for generics to gain market share

IgAN & autoimmune

We see versatility for APRIL/BAFF across autoimmune indications

In IgAN, we saw 49.8% Pbo adj. UPCR reduction at 9mo (clinically in line w/ Sibe):

- To us, the impt point is: a) IgAN is a bigger market with favorable pricing (\$390K/yr), b) Pove has a convenience advantage with 1.5sec admin autoinjector
- We're optimistic on the indication expansion opportunity into gMG, pMN, wAIHA

AMKD & ADPKD

We see LT upside from Inaxaplin becoming first/best in class

- We're expecting Inaxaplin to be first & best in class in the AMKD market (currently no tx)
- We expect 64-70% UPCR reduction at 48wks + we think VRTX has the more favorable BL population vs comps
- We think the AMKD market could be bigger than initially expected given a) an approved tx will likely increase dx rates, b) our KOL expects spillover dx from the IgAN market growing from Povetacicept
- In ADPKD, we think the data is too early for VX-407 but could see pot'l reduced LFTs based on Trikafta read across (similar MOA)

Pain

We're mixed on Ph.3 DPN hitting, but are more confident in the acute pain opportunity

- We're mixed on whether chronic DPN can hit for Suze, given the high variability of previous data and Pbo (combo approach is more likely here).
- In acute pain, we expect 3x growth for Journavx based on recent access expansion

Rest of Pipeline

Rest of pipeline is interesting but we're skeptical on cons valuation

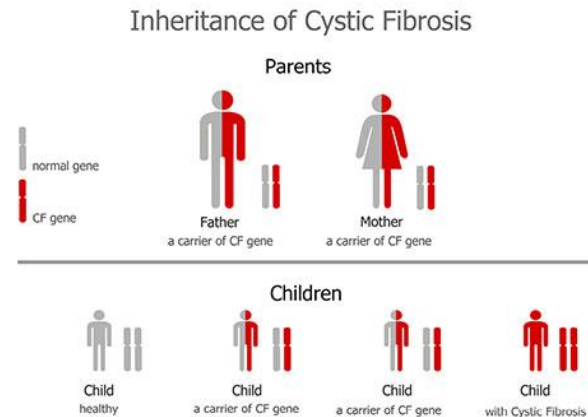
- We're skeptical on expansion opportunity for rest of pipeline:
- For Casgevy, we've seen precedence that viable tx's in SCD launch slower / underpenetrate the market due to patient/physician concerns & access barriers
- Zimislecel appears limited to severe T1D population based on immunosuppression requirement & clinical trial delays

Source: Jefferies analysis

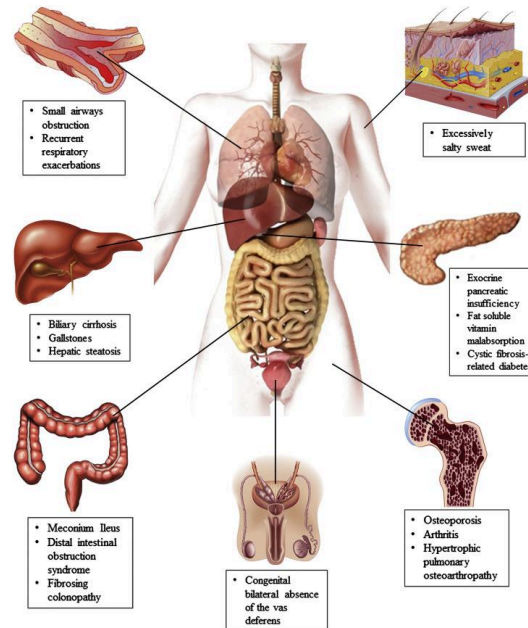
As a background, cystic fibrosis is caused by mutations in the CFTR gene which encodes a channel protein that transports chloride – the disease is autosomal recessive and affects the lungs, pancreas & other organs

Dysfunction of the CFTR protein leads to imbalance of ions and fluid in the cells of the sweat glands, airways, intestine, etc. If CFTR proteins don't open correctly, chloride ions don't move out of cells and this leads to the buildup of thick mucus in organs, like the lungs.

Autosomal Recessive Inheritance



CF is a multi-organ disease; the respiratory disorder represents the major cause of morbidity and mortality



There are 6 functional classes of CFTR mutations:

- **Class I** – No protein made (nonsense/frameshift/splice): CFTR isn't synthesized.
- **Class II** – Trafficking defect (e.g., F508del): CFTR misfolds and is degraded, so it never reaches the cell surface.
- **Class III** – Gating defect: CFTR gets to the membrane, but the channel doesn't open properly (low ion flow).
- **Class IV** – Conductance defect: CFTR opens but ions flow poorly through the channel pore (partially working channel).
- **Class V** – Reduced synthesis: Less normal CFTR is produced
- **Class VI** – Decreased stability: CFTR reaches the surface and works but is unstable and removed too quickly.

Class I–III are usually considered more severe vs. Class IV–VI tend to retain residual functions and are often milder.

Correctors (lumacaftor, tezacaftor, elexacaftor, vanzacaftor) help the misfolded CFTR protein fold properly and reach the cell surface. The **potentiators (ivacaftor or deuterivacaftor)** make the CFTR channels that are already at the surface open more often to allow chloride to flow.

Source: Jefferies Analysis, Cleveland Clinic, H. Li et al, S. Molina et al, PARI

As a background, VRTX has a robust CF franchise (5 approved US tx) w/ ~90% of CF pop eligible for its CFTR modulators

Legacy modulators (pre-2019): Prior to Trikafta, CFTR modulators delivered incremental efficacy with limited population coverage. Kalydeco (2012) was the first potentiator and demonstrated large ppFEV1 improvements but was restricted to gating mutations (~5% of CF pts). Orkambi (2015) expanded use to F508del homozygotes but showed modest efficacy (ppFEV1 +2.6–3 pts) and was associated with tolerability issues, including respiratory AEs, BP increases, and DDIs. Symdeko (2018) improved tolerability and efficacy (~+4-7 pts ppFEV1) and broadened eligibility, though uptake was limited following the introduction of Trikafta (2019).

Current modulators (post-2019): Trikafta is a triple-combination CFTR modulator initially approved in 2019 for pts with ≥ 1 F508del mutation (~90% of the CF population). In pivotal studies, Trikafta improved ppFEV1 by +14.3 pts vs placebo in F508del/MinFx pts and reduced pulmonary exacerbations by ~63%. The label has expanded to include pts ≥ 2 y, supporting broad use across age groups. Reported safety findings include transaminase elevations, rash, and modest BP increases, w/ recommended monitoring q3mo during the first year and annually thereafter.

CF Drug label comp

Drug	Active Ingredient(s)	Dosage/Format	Indicated Population	Warnings	Liver Monitoring	Key Efficacy (label, incl. trial size/duration)	Safety (common AEs $\geq 5\%$)
Kalydeco	Ivacaftor (CFTR potentiator)	150 mg tablet q12h (≥ 6 y), granules for ages ≥ 4 mo; always with fat-containing food	CF pts ≥ 4 mo with ≥ 1 responsive gating/residual function mutation	Transaminase elevations, cataracts (pediatric), CYP3A interactions	Baseline LFTs; q3mo in first year; annually thereafter	Pivotal trials in G551D and other gating/residual fx mutations (N $\approx$ 200 across multiple 24–48wk RCTs). Improved ppFEV1 ~10%, sweat Cl $\downarrow$ ~ -48 to -55 mmol/L, fewer exacerbations.	Headache, URTI, abdominal pain, diarrhea, rash, $\uparrow$ ALT/AST, nasal congestion, dizziness
Orkambi	Lumacaftor (corrector) + Ivacaftor	≥ 12 y: 2 tabs (lumacaftor 200mg/ivacaftor 125mg) q12h with food; granules for ages 2–5	CF pts ≥ 2 y homozygous for F508del	Respiratory events, liver-related events, blood pressure $\uparrow$, cataracts, CYP3A induction	Baseline LFTs; q3mo in first year; annually thereafter; more frequent if history of liver disease	Trials 1 & 2 (N=1108, 24wks, ppFEV1 40–90%). ppFEV1 $\uparrow$ 2.6–3.0 points vs placebo; pulmonary exacerbations $\downarrow$ 30–40% (rate ratio 0.6–0.7).	Dyspnea, nasopharyngitis, nausea, diarrhea, URTI, fatigue, rash, flatulence, $\uparrow$ CK, influenza
Symdeko	Tezacaftor (corrector) + Ivacaftor	1 tab (tezacaftor 100mg/ivacaftor 150mg) AM + 1 tab ivacaftor 150mg PM, with fat-containing food	CF pts ≥ 12 y (later expanded to ≥ 6 y) with homozygous F508del or ≥ 1 responsive mutation	Transaminase elevations, cataracts, CYP3A interactions	Baseline LFTs; q3mo in first year; annually thereafter	Trial 1 (N=504, 24wks, F508del homozygotes) ppFEV1 $\uparrow$ 4.0 pts; Trial 2 (N=244, 8wks, heterozygotes w/ responsive mut) ppFEV1 $\uparrow$ 6.8 pts. $\downarrow$ sweat Cl ~ -9 to -10 mmol/L.	Headache, nausea, sinus congestion, dizziness; generally similar to placebo in trials
Trikafta	Elexacaftor (corrector) + Tezacaftor + Ivacaftor	≥ 12 y: 2 tabs AM (ELX 100mg/TEZ 50mg/IVA 75mg each) + 1 IVA 150mg PM with fat-containing food; weight/age-based adjustments for 2–11y	CF pts ≥ 2 y with ≥ 1 F508del or another responsive mutation	Elevated transaminases, hepatic injury, cataracts, CYP3A interactions	Baseline LFTs; every month for the first 6mo, then every 3mo for the next 12mo, then annually thereafter	Trial 1 (N=403, 24wks, F508del/MinFx) ppFEV1 $\uparrow$ 14.3 pts vs placebo; Trial 2 (N=107, 4wks, F508del homozygotes) ppFEV1 $\uparrow$ 10 pts vs TEZ/IVA. Pulm exacerbations $\downarrow$ 63% (RR 0.37). Sweat Cl $\downarrow$ ~ -41 mmol/L.	Headache, URTI, abdominal pain, diarrhea, rash, $\uparrow$ ALT/AST, nasal congestion, $\uparrow$ CPK, influenza
Alyftrek	Vanzacaftor (corrector) + Tezacaftor + Deutivacaftor	≥ 12 y: 2 tabs once daily (VNZ 10mg/TEZ 50mg/D-IVA 125mg); pediatric weight-based dosing (6–11y)	CF pts ≥ 6 y with ≥ 1 F508del or another responsive mutation	Boxed warning: Drug-induced liver injury/liver failure; hypersensitivity; cataracts; CYP3A interactions	Baseline LFTs; every month for the first 6mo, then every 3mo for the next 12mo, then annually thereafter	Trial 1 (N=398, 52wks, F508del/MinFx) ppFEV1 non-inferior to Trikafta; Trial 2 (N=573, 52wks, F508del homozygous or responsive) ppFEV1 non-inferior. Sweat Cl $\downarrow$ -8.4 mmol/L (Trial 1), -2.8 mmol/L (Trial 2).	Cough, nasopharyngitis, URTI, headache, oropharyngeal pain, influenza, fatigue, rash, $\uparrow$ ALT/AST, sinus congestion

Source: Jefferies Analysis, Company Publications

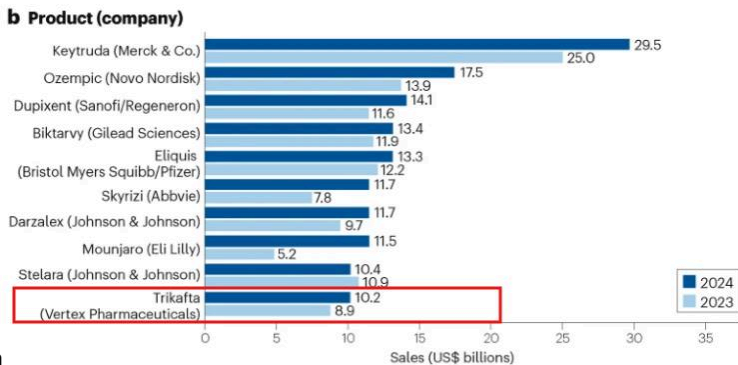
Importantly, the co is transitioning its pts from Trikafta (which currently takes majority market share) to Alyftrek, which supports CF margin expansion given its lower royalty burden

“Unsurprisingly, the uptake has been fastest in those who are new to medicine. ALYFTREK not only has greater sweat chloride benefit but it has 31 more mutations than TRIKAFTA. Those 31 mutations translates to something like 400, 500 patients in the US. ...And on the switches, I would say it's steady and it continues to be so.” – VRTX Nov’25

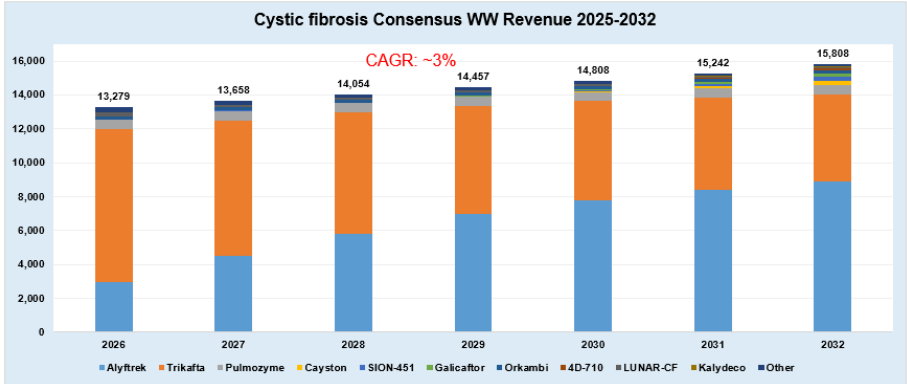
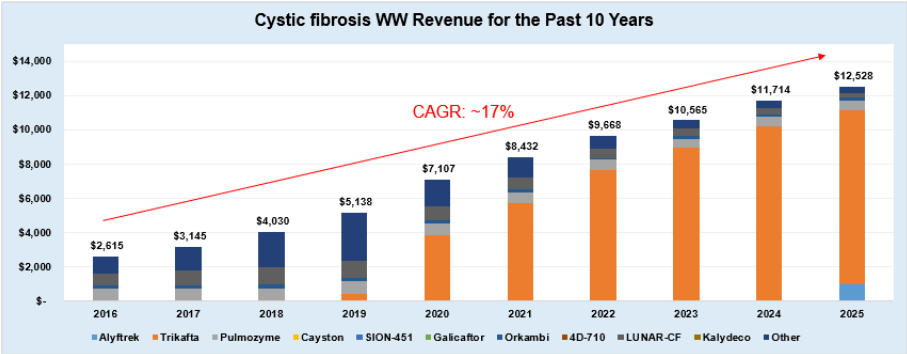
“The majority of prescriptions, the biggest population and thus, the majority of prescriptions of ALYFTREK in the second quarter, though, are patients who switch from TRIKAFTA to ALYFTREK, and we've been pleased with that pace. In the U.S., there is additional liver monitoring requirement, right? So that has taken sort of an extra step of conversation between physicians and their patients to weigh the 6 months of the monthly monitoring that's required versus the lifetime of benefit of lower sweat chloride levels and once-daily dosing.” -VRTX Sep’25

“...the most rapid uptake has been among those rare mutations that did not have access to a therapy before, followed by patients who discontinued a previous CFTR modulator and were no longer on treatment.” – VRTX mgmt. Sep ‘25

VRTX pays a 9.33% royalty on Trikafta (now assigned to Royalty Pharma), while Alyftrek carries a 4.0% royalty burden (note: Royalty Pharma initiated confidential arbitration Oct. ’25 alleging royalty burden is ~8% for Alyftrek). Overall, switching to Alyftrek would improve overall CF franchise margins.



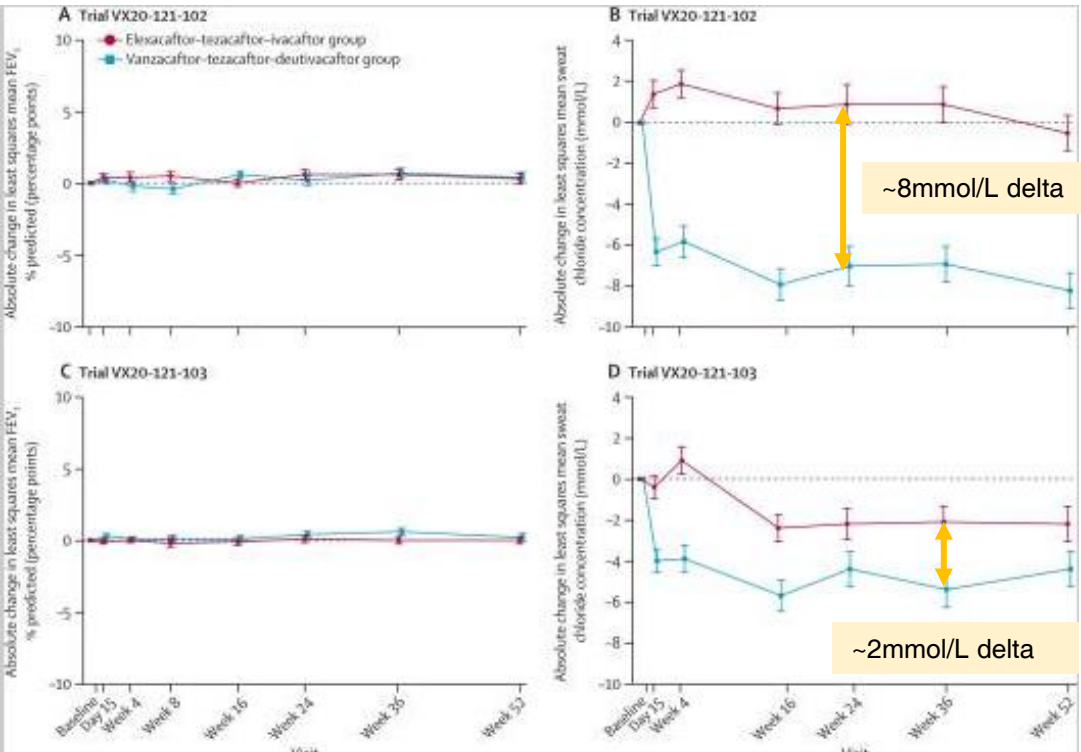
The CF market has shown consistent growth over the past decade w/ a CAGR ~17%. VRTX’s triple combination therapy, Trikafta, now takes majority market share and ranks among the top ten drugs globally by total sales. Consensus expects continued market expansion at CAGR ~3% for the next 7 years, with VRTX maintaining its leadership position through Trikafta and Alyftrek.



Source: Jefferies Analysis, Nature, Evaluate Pharma

Stepping back, Alyftrek is a 1x daily triple modular tx that was FDA approved Dec ‘24 after showing non-inferiority to 2x daily Trikafta (+0.2pp difference in ppFEV1) & a numerically greater sweat chloride reduction (2-8 mmol/L benefit vs Trikafta)

Alyftrek Non-inferiority study (Keating et al. 2023)

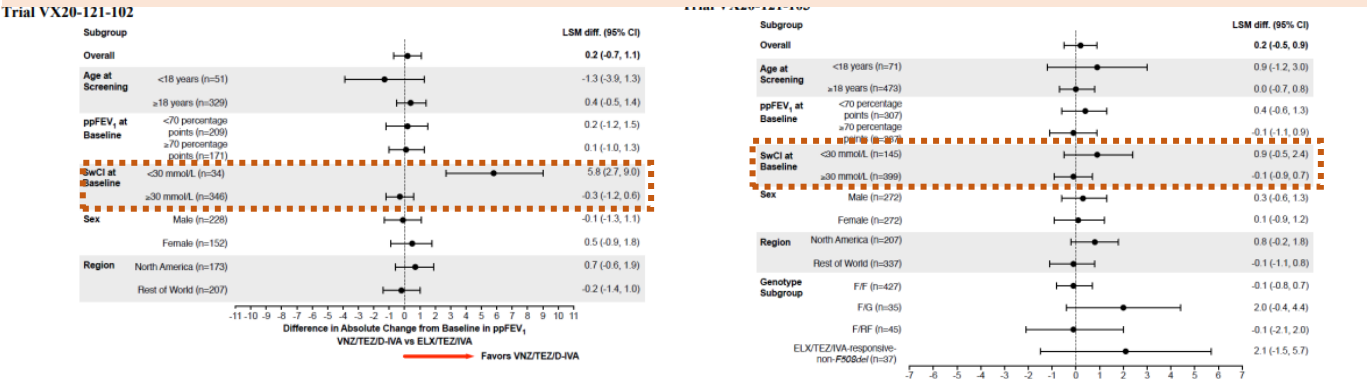


The primary endpoint for both trials was absolute change in FEV₁ % predicted from baseline. In Trial VX20-121-102, the least-squares mean change in ppFEV₁ through week 24 was +0.5% with Alyftrek vs +0.3% with Trikafta, for a tx difference of +0.2 pp (95% CI -0.7 to 1.1; one-sided p<0.0001). In Trial VX20-121-103, the change was +0.2% vs 0.0%, also a +0.2-point difference (95% CI -0.5 to 0.9; one-sided p<0.0001), showing virtually identical lung-function effects while statistically showing non-inferiority.

Source: Jefferies Analysis, Company Publications

We noted that Alyftrek showed greater reduction than Trikafta in sweat chloride in both Ph.3 studies (~2-8mmol/L delta). That said, Alyftrek showed noninferiority in ppFEV₁ reduction vs Trikafta (0.2 percentage points - meeting the non-inferiority criteria) likely b/c the pts in both studies were already on a stable dose of Trikafta prior to the trial so there was little room for their ppFEV₁ to increase further. There were no significant differences in quality of life in CFQ-R respiratory domain (-1.7 vs 0.5 for trial 102 and -1.2 for both arms in trial 103).

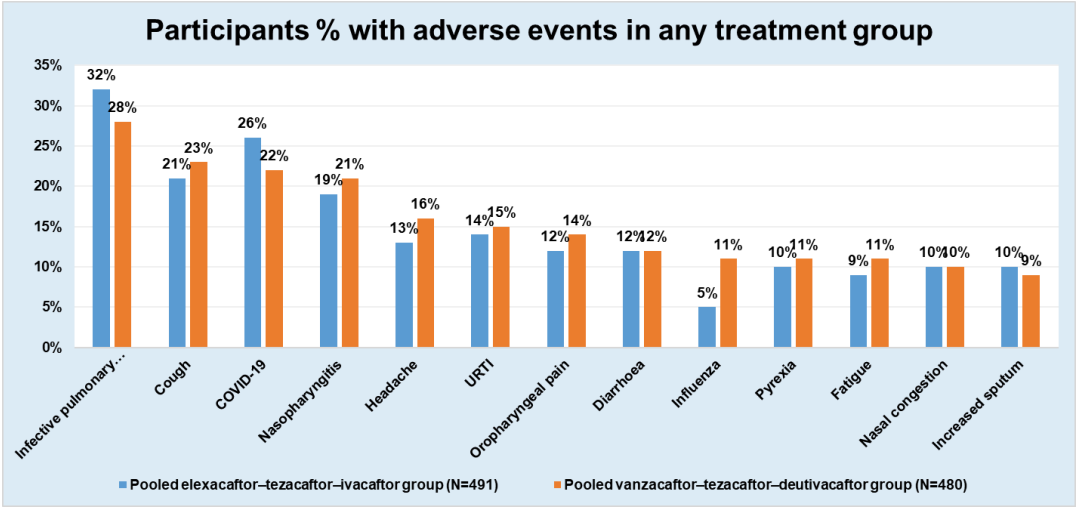
In study 102 there was stat. sig greater efficacy for Alyftrek w/ pts starting at a <30mmol/L sweat chloride, but this was only directionally present in the 103 study, but not stat. sig. This was initially conducted in ages 12+ but there are currently various Ph.3 label expansion trials for pts ranging 1-11y/o in total.



VRTX is looking to expand their label to 1/2y/o and completed a Ph.3 OL study. We'll flag in Ph.3 OL for children 2-5, cough (~99%), pyrexia (~35%), and rhinorrhea (~33%) were the most common AEs. ALT increases occurred in ~11% of children, but no children had ALT and/or AST levels >3x the upper limit of normal concurrent with total bilirubin level >2x the upper limit of normal. Furthermore, none had d/c as a result of increased ALT.

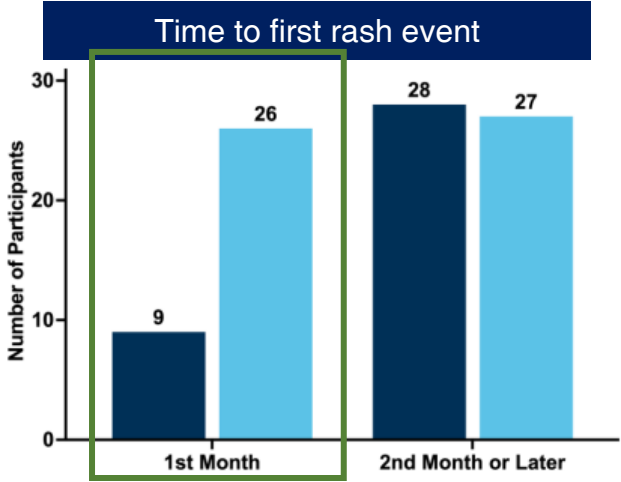
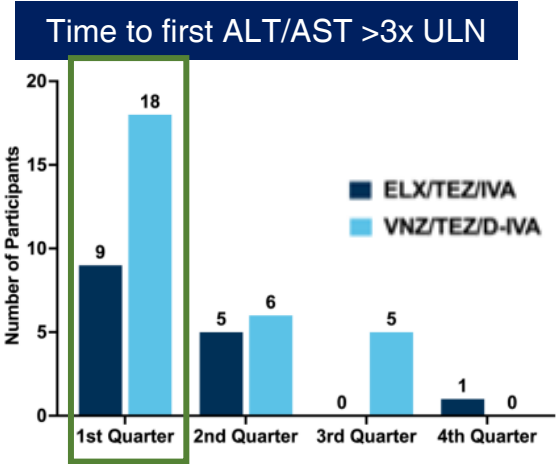
Safety profiles look similar btwn the 2, w/ majority of AEs mild/mod – Alyftrek had slightly higher transaminase elevations and d/c due to liver enzyme elevations vs Trikafta, but rates remain low (~9% vs ~7% any grade; d/c ~1.5% vs ~0.6%)

Alyftrek and Trikafta showed similar safety: ~96% had AEs (mostly mild/moderate), with comparable common AEs, serious AEs (14% vs 16%), and AE-related discontinuations (4% vs 4%, mainly liver enzyme elevations). No on-treatment deaths occurred (2 post-treatment deaths in Trikafta arm were deemed unrelated).



Before 2025, Trikafta required only periodic LFTs (q3 months in year 1, then annually). The Dec '24 update added a stricter liver monitoring. **Now both Alyftrek and Trikafta carry boxed warnings for liver injury and require LFTs before start, then monthly x6 months, q3 months x12 months, and annually.** Legacy modulators (Kalydeco, Orkambi, Symdeko) still require only baseline ALT/AST and q3mo checks in year 1, with no Boxed Warning.

- That said we noted the following safety differences:
1. **AST/ALT >3x ULN occurred more frequently in the Alyftrek arm vs Trikafta arm (18 vs 9 pts) in the 1st Q. D/c due to liver enzyme elevation as slightly more frequent than w/ Alyftrek vs Trikafta (1.5% vs 0.6%)**
 2. **Any-grade transaminases elevations were slightly higher w/ Alyftrek vs Trikafta (9% vs 7.1%)**
 3. **Rash more common w/ Alyftrek vs Trikafta (11% vs 7.7%). Most were mild/moderate; 1 event led to d/c in the Alyftrek arm. Rash events occurred more commonly during the 1st month w/ Alyftrek (26 vs 9 pts)**
- That said, we haven't seen these lead to meaningful dose limitations for Alyftrek



Source: Jefferies Analysis, Keating et al , 2025

On access, Alyftrek’s prior auth generally looks in line w/ other CFTR modulators (confirmed CF diagnosis, responsive genotype, specialist attestation), but select states (e.g. LA) require step-through, which could slow Alyftrek uptake

Most commercial payers (e.g. UHC, Cigna, MedMutual, and Kaiser) use CFTR modulator prior auth templates similar to Trikafta, requiring age ≥6, confirmed CF diagnosis with a responsive genotype, and specialist attestation. For most covered CF pts, approvals are processed within 3-5 days once genotype documentation filed, and Alyftrek does not add new biomarker requirements beyond standard genotype verification. However, some formularies (e.g. Ventegra, UHC’s July 2025 update, and Louisiana Medicaid) are introducing step-therapy rules requiring prior failure, intolerance, or contraindication to Trikafta before Alyftrek approval. This can create a tier-2 positioning similar to early Symdeko/Orkambi transitions, adding delays of about 1-2 weeks. To us, this is an overhang in transitioning pts from Trikafta to Alyftrek

Re. Medicaid, states like MS, LA, and FL imposing stricter rules such as step-therapy (LA) or quantitative renewal criteria (MS). Commercial Blues and large PBMs generally follow UHC/Cigna frameworks but differ in documentation requirements, (e.g. mutation reports or sweat chloride, which could slow about 15-20% of early treatment starts). Access is supported by VRTX’s limited-distribution specialty pharmacy network (CVS Specialty, Accredo, Optum), which handles PA submissions, documentation, and refills. For pts switching from Trikafta, many PAs are expected to auto-convert or renew using existing genotype and lab data.

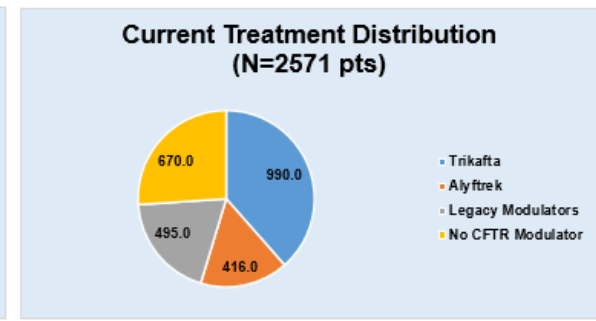
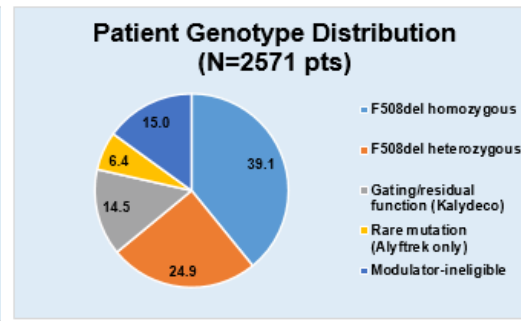
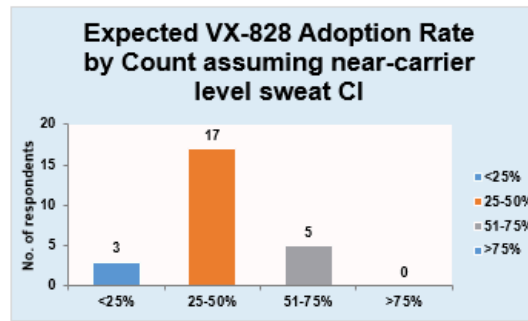
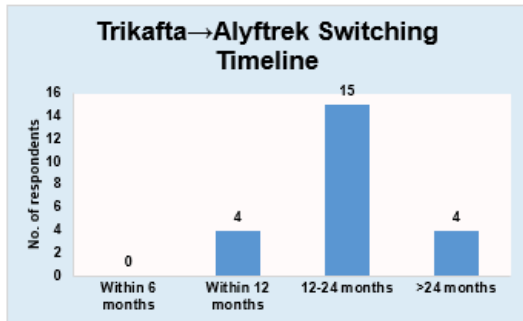
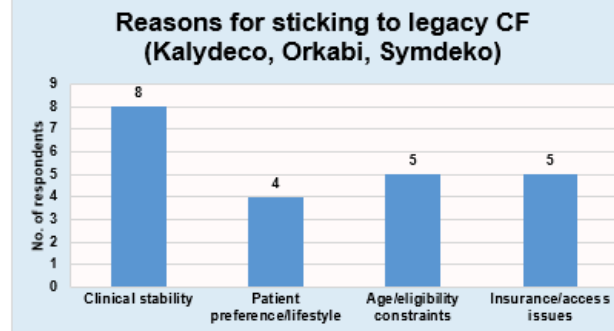
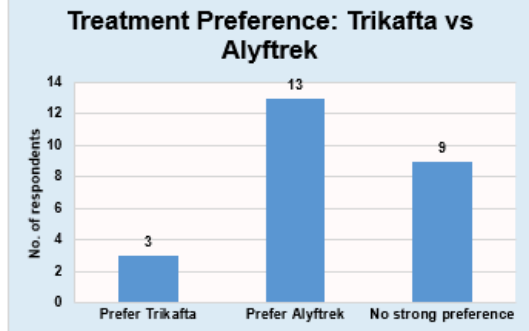
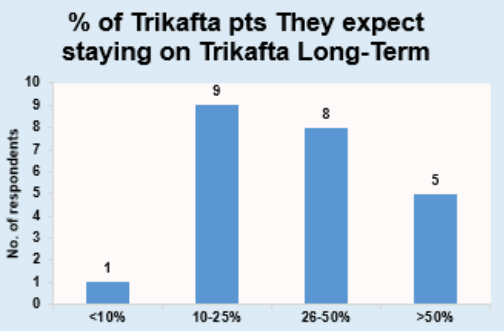
Alyftrek prior auth criteria								
Payer / Policy	Age Requirement	Prescriber Requirement	Genotype Criteria	Baseline / Monitoring Requirements	Exclusions / Step Therapy	Renewal Criteria	Auth Duration	Distinctive Features
Kaiser Permanente NW	≥6 yrs	CF specialist	≥1 responsive CFTR mutation (FDA or CLIA lab)	Must exclude severe hepatic impairment, prior transplant unless transplant center approves	No mention of step therapy	Not specified; for new starts only	Not specified	Streamlined criteria; limited to CF specialists; no explicit monitoring requirement
Florida Medicaid (AHCA)	≥6 yrs	CF specialist or consulted	≥1 F508del or responsive mutation	LFTs at baseline and ongoing; no Child-Pugh C	No CFTR modulator co-use	Clinical improvement or stabilization (FEV1, sweat chloride, BMI, CFQ-R, exacerbations)	Initial 1 yr	Comprehensive renewal endpoints; requires evidence of clinical response
NC Marketplace (CareSource)	≥6 yrs	Pulmonologist / ID specialist	≥1 F508del or listed responsive mutation	Baseline LFTs; attestation of no severe hepatic impairment	No mention of step therapy	Improved/stable FEV1, sweat chloride, fewer exacerbations, fewer hospitalizations	6 mo + 12 mo renewal	Mutation list explicit; detailed reauth; 6 mo initial term
UnitedHealthcare (May 2025 version)	≥6 yrs	Any CF-treating provider	≥1 F508del or responsive mutation (clinical/in vitro/extrapolated)	None specified beyond genotype verification	None noted	Documented positive clinical response	12 mo	Uses automated re-approval; minimal burden
UnitedHealthcare (Jul 2025 updated)	≥6 yrs	CF specialist	≥1 responsive mutation	Requires documentation of CFTR mutation via lab results	Must fail Trikafta OR mutation not responsive to Trikafta	Documented improvement/stable function	12 mo	Adds mandatory Trikafta failure or non-responsive genotype; excludes manufacturer sample starters
Louisiana Medicaid	≥6 yrs	CF-treating prescriber	≥1 F508del or responsive mutation	None specific; attestation of positive response for continuation	Step through preferred agents (e.g. Trikafta)	Positive response to therapy	12 mo	Explicit step therapy to preferred CFTRs; ties to LA PDL
Cigna	≥6 yrs	Pulmonologist / CF specialist	≥1 F508del or large responsive mutation list	Must confirm CF diagnosis (genotype + sweat test + clinical presentation)	Disallows CFTR combination use	Meets full criteria + evidence of efficacy	12 mo	Extremely detailed; integrates diagnostic hierarchy and clinical confirmation
Ventegra	≥6 yrs	Pulmonologist / CF specialist	≥1 F508del or responsive mutation (Table 5)	LFTs baseline + monthly ×6, q3mo ×12, then yearly	Requires Trikafta failure or contraindication	Stable lung function and tolerability	6 mo + reauth	Most stringent → requires Trikafta failure, frequent LFTs
Mississippi Medicaid	≥6 yrs	CF specialist/pulmonologist	≥1 F508del or responsive mutation (lab proof required)	Baseline FEV1, BMI, exacerbations, LFTs	Excludes other CFTR modulator co-use	Improved/stable FEV1, BMI, exacerbations, LFTs	6 mo + 12 mo	Data-driven reauth; quantitative clinical metrics (FEV1/BMI)
Medical Mutual of Ohio	≥6 yrs	Pulmonologist / CF specialist	≥1 F508del or responsive mutation	Diagnostic confirmation (sweat chloride ≥ 60 mmol/L or 2 CFTR mutations)	Excludes other CFTR modulators	Improved lung function, fewer exacerbations, weight gain	6 mo + 12 mo	Nearly identical to Cigna; explicit physiologic criteria

Source: Jefferies Analysis, Company Publications

Notably, our physician survey (N=25) indicates that on average, the surveyed prescribers expect to switch 15~30% of their Trikafta pts to Alyftrek between 2026-2030

Our survey (n=25 prescribers, conducted ~Oct'25) showed that **39.1%** of patients were F508del homozygous and **24.9%** were F508del heterozygous, but when including gating/residual function (**14.5%**) and rare mutations (**6.4%**), then only about **15%** of patients with CF are not eligible for a VRTX therapy. Currently, **38.5%** are on Trikafta (990/2571), **16.2%** on Alyftrek (416/2571), and **19.2%** on legacy modulators (495/2571), with the remaining **26%** receiving no CFTR modulator (670/2571).

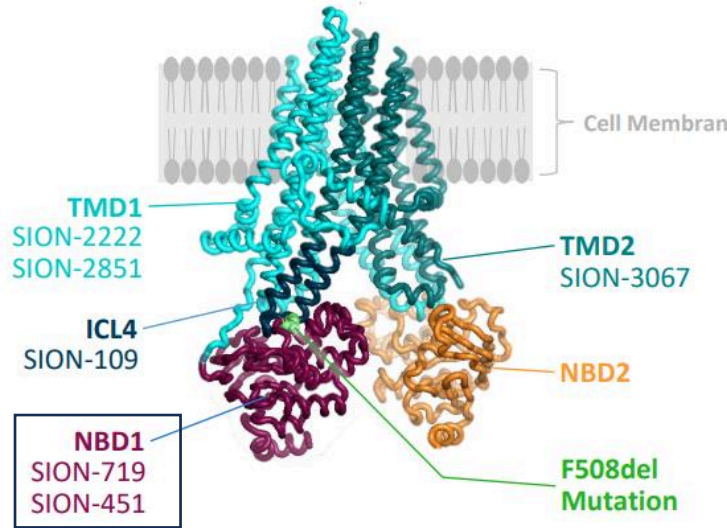
- Overall, most prescribers expect meaningful Trikafta retention (~4% predict <10%; ~39% expect 10–25%, ~35% expect 26–50%, ~22% expect >50%)
- Alyftrek is preferred by ~52% of prescribers vs ~12% for Trikafta (~36% neutral) & legacy modulators are mainly continued for clinical stability (n=8), with lifestyle, eligibility, and access each cited by 4–5
- Switching from Trikafta to Alyftrek is expected mostly within 12–24 months (~65% of prescribers). ~17% would switch within 12 months & ~17% in >24 months
- For VX-828, assuming near-carrier sweat chloride levels, projected uptake is moderate: ~68% expect 25–50%, ~12% expect <25%, ~20% expect 51–75%, and 0% expect >75%



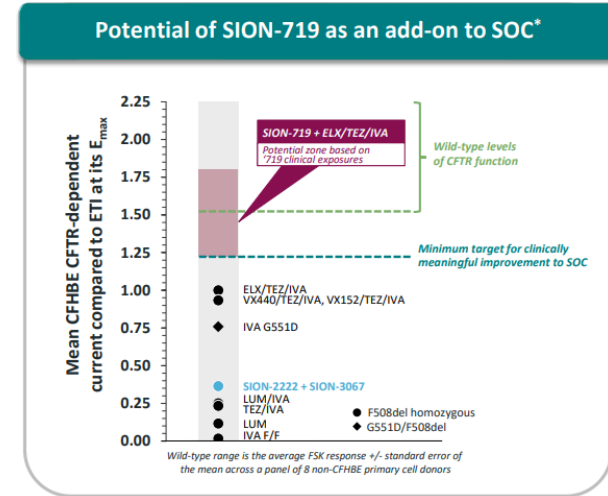
Source: Jefferies Analysis

LT, we expect the CF franchise to be stable, given VRTX's next-gen txs & limited competitors – a) SION is running a trial for their NBD1 stabilizer (SION-719) as add-on to Trikafta in Ph2a (data ~mid-'26), & in vitro data suggests SION-719 can improve Trikafta's efficacy in HBE assays (+116% vs Trikafta mono-tx)

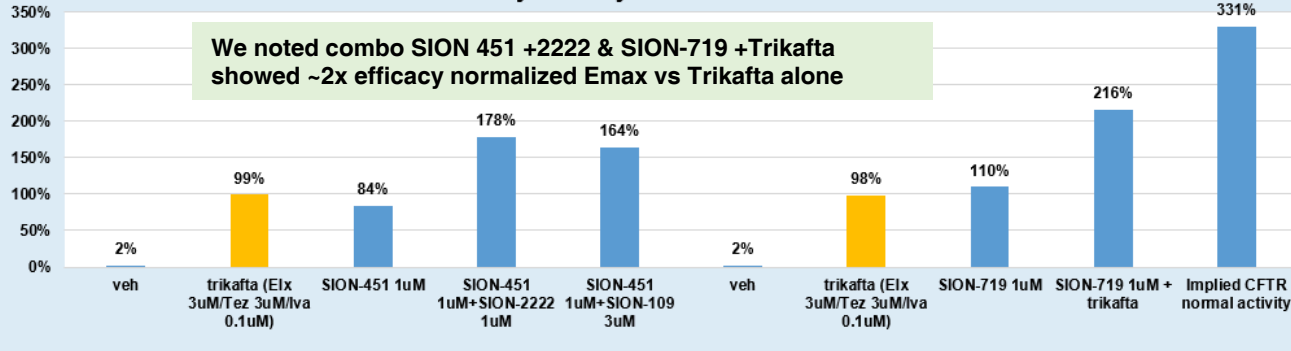
SION reports direct NBD1 stabilization as the core differentiating mechanism from VRTX. SION-719 is an NBD1 stabilizer being advanced initially as an add-on to SOC (Trikafta) in the PreciSION CF Ph.2a trial (PCD: July 2026), with the goal of driving incremental CFTR function as measured by sweat chloride (≥ 10 mmol/L superiority threshold). Ph.2 primary endpoint is safety/tolerability and secondary endpoints inc. change from BL in SwCl. We'll be paying attention to the expected SwCl data.



SION-451 (and other combos) are being developed as the second component of a proprietary dual-combination regimen pairing NBD1 stabilization w/ complementary domain correction, designed to approach or achieve near-WT CFTR function using two agents vs VRTX's triple therapy. Ph.1 PCD: June 2026. Primary endpoint is safety/tolerability & secondary endpoints are PK parameters in mono/combo tx.

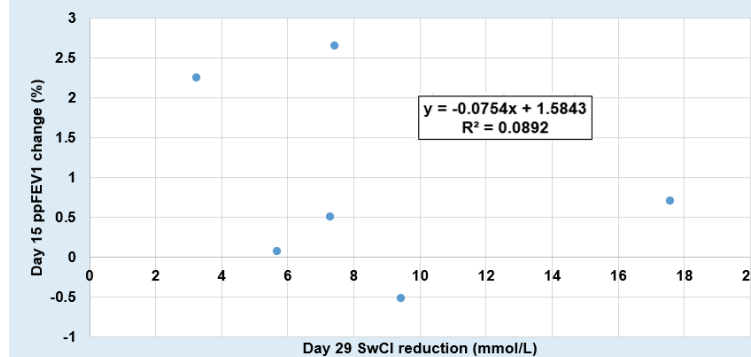


HBE assay Efficacy normalized to Trikafta Emax



Source: Jefferies Analysis, Company Publications

GLPG2222: SwCl reduction at day 29 vs ppFEV1 % change at day 15

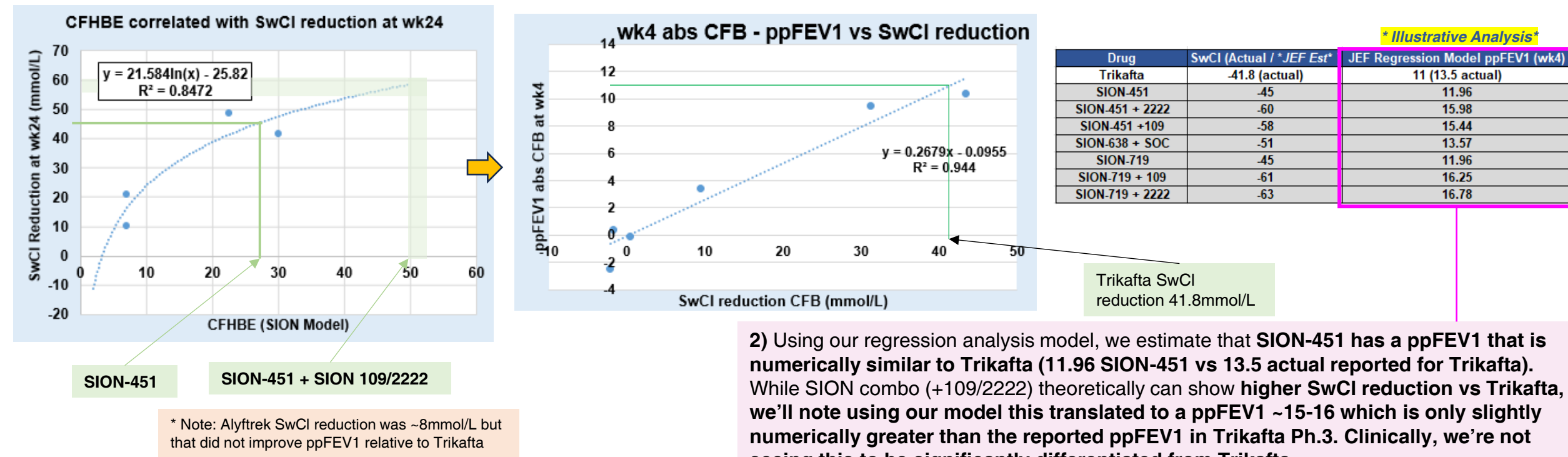


From the GLPG data, we saw as mono-tx GLPG2222 did not show a clear response in SwCl reduction and % improvement in ppFEV1 ($R^2=0.089$). To us, this suggests GLPG2222 as mono-tx is unlikely to be effective tx for ppFEV1 reduction.

In fact, our illustrative analysis based on CFHBE suggests that SION-451 would look similar on ppFEV1 to Trikafta if administered as mono-tx (~12 SION-451 vs 13.5 Trikafta) & the add-on combo could show ppFEV1 of ~15-16 at week 4

1) We plotted the wk24 SwCl reduction for Orkambi, Symdeko, Kaldeco and Trikafta vs the reported CFHBE model value from SION and found a log correlation fit best (R²=0.8). We used this to translate the CFHBE to SwCl at wk24 that we might anticipate for SION drugs

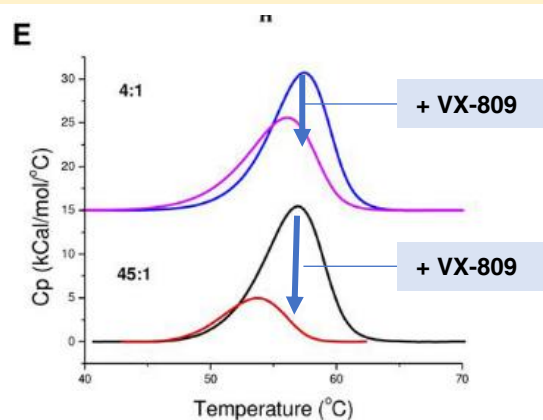
Caveat: We're assuming there is in fact a linear correlation btw SwCl reduction and ppFEV1 improvement. While our analysis shows there is, there are few datapoints in our correlation.



2) Using our regression analysis model, we estimate that SION-451 has a ppFEV1 that is numerically similar to Trikafta (11.96 SION-451 vs 13.5 actual reported for Trikafta). While SION combo (+109/2222) theoretically can show higher SwCl reduction vs Trikafta, we'll note using our model this translated to a ppFEV1 ~15-16 which is only slightly numerically greater than the reported ppFEV1 in Trikafta Ph.3. Clinically, we're not seeing this to be significantly differentiated from Trikafta.

Further, while SION is positioning their drug as stabilizing NBD1, we've seen hints that VRTX modulators also hit NBD1, potentially making SION less differentiated – preclinical papers showing lumacaftor (VX-809) also binds the NBD1 region + VX445 (elexacaftor) directly stabilizes NBD1 as a type 3 corrector

Biophysical NMR study (Hudson et al. 2017) showed that binding of lumacaftor (VX-809) perturbs contacts between the C-terminal helices and the NBD1 core, evidenced by chemical shift changes in residues along helices and β -strands S3, S9, and S10 on the NBD1 surface. **Notably, VX-809 binding was associated with a decrease in NBD1 thermal melting temperature**, suggesting an interaction that shifts the conformational equilibrium of NBD1 (i.e VRTX binds NBD1). Furthermore, Baroni et al. 2025 reported that VX-445 directly stabilized NBD1: *“Other pivotal results of this study showed that VX-445 binds directly to isolated F508del NBD1, preventing its unfolding and promoting CFTR stabilization.”* – Baroni et al. 2025. **Taken together, we're seeing that VRTX modulators are likely also binding the NBD1 regional and at least partially stabilize it.**



Hudson et al. showed temperature stabilization for VRTX drug

Source: Jefferies Analysis, Company Publications

We caught up with SION mgmt. here are our takeaways:

- On VRTX's impact, mgmt noted modulators have transformed QoL + disease trajectory in CF, but SOC does not normalize CFTR for most pts. ~1/3 on Trikafta/Alyftrek reach normal CFTR function, leaving ~2/3 w/ residual deficit -the core rationale for a next-gen entrant.
- On NBD1 as the MOA, mgmt emphasized NBD1 is central to F508del (~90% of pts), w/ mutation causing misfolding and impaired trafficking/function. They argue no approved modulator directly stabilizes NBD1, which explains persistent residual dysfunction. They have seen incremental efficacy from adding SION on top of SOC. On CFHBE translation, mgmt highlighted it as the gold-standard predictive assay and noted they reproduced VRTX clinical translation in-house. They argue direct NBD1 stabilization can push CFTR beyond Trikafta/Alyftrek ceilings given SOC lacks this mechanism.
- On VX-828, mgmt hypothesize it is another ICL4-like corrector (refined “triple”) vs direct NBD1 stabilization. While both co's see headroom in sweat chloride, they diverge on FEV1 translation; mgmt contend $\geq \sim 10$ mmol sweat chloride reduction should translate, citing Alyftrek's non-inferior FEV1 + modest sweat chloride delta as insufficient CFTR movement.
- On 719 vs 451, mgmt reported no meaningful tolerability split in HVs; 719 viewed as more potent low-dose add-on, while 451 supports higher exposure dual-combo strategy. Add-on path is viable if PoC positive, but dual likely preferred; running both is capital-intensive. Split-asset approach provides optionality + pricing flexibility.
- On market durability, mgmt see CF market ~\$11-12B today, growing to ~\$15B by decade-end, w/ limited generic risk in IP-protected geographies.

b) VRTX has their next-gen CFTR corrector VX-828 (Ph1 HV data 2H'26), which mgmt. has called their most efficacious CFTR corrector based on in vitro data

- VX-828 is the next-gen molecule for improving folding, trafficking and expression of CFTR in CF pts. Mgmt has commented it is the most efficacious CFTR corrector they have tested, although we haven't seen any data
- "The backbone of this N.G 3.0 combination is VX-828, the most efficacious CFTR corrector that we have ever studied in vitro. It is completing Phase 1 development, and we remain on track to initiate a study with VX-828 in patients with CF before the end of this year" – VRTX mgmt. 5 May 2025*

- The Ph.1 study of VX-828 trial is structured in multiple parts, beginning with single-ascending dose (SAD) cohorts in healthy volunteers, then multiple-ascending dose (MAD). The study will also evaluate VX-828 in combo with other CFTR modulators (EX, D-IVA) to assess drug-drug interactions and combo PK.
- Primary endpoints focus on safety and tolerability (AEs) & PK parameters (Cmax, AUC, t½). Change in SwCl at day 28 is listed as a secondary endpoint. The study is designed to inform dose selection and pot'l combo strategy for subsequent Ph.2 development.

Recruiting ⓘ

Evaluation of VX-828 in Healthy Participants and in Participants With Cystic Fibrosis

ClinicalTrials.gov ID ⓘ NCT06154447

Sponsor ⓘ Vertex Pharmaceuticals Incorporated

Information provided by ⓘ Vertex Pharmaceuticals Incorporated (Responsible Party)

Last Update Posted ⓘ 2026-02-24

VRTX 10-K 2025:

We estimate that nearly 95% of people with CF could benefit from our five approved medicines, and, in connection with our serial innovation approach, we continue to identify and develop additional CFTR modulators with the goal of developing best-in-class medicines that can treat more people with CF. We have advanced several next-generation, 3.0 CFTR modulators into the clinic. VX-828 is the first of these and is being evaluated in a proof-of-concept clinical trial of people with CF. We expect to complete enrollment and dosing in the first half of 2026.

Study Start (Actual) ⓘ

2023-12-12

Primary Completion (Estimated) ⓘ

2026-04-23

Study Completion (Estimated) ⓘ

2026-04-23

Enrollment (Estimated) ⓘ

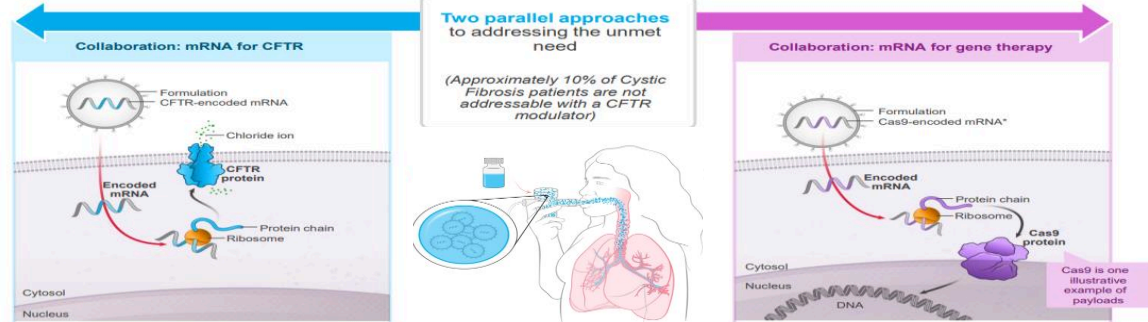
255

PCD is April 2026 and mgmt. has guided data disclosure expect 2H26

Source: Jefferies Analysis, CT.gov

c) VRTX is also developing inhaled mRNA CF tx VX-522 w/ Moderna, which could target the ~5k pts that don't respond to Alyftrek/Trikafta (Ph.1 PCD Dec 2026) – we currently don't model this as the program is still early

VX-522 is a genetic inhaled LNP-mRNA (mRNA-3692) that aims to deliver full-length CFTR to airway epithelium (collaboration with Moderna). The target pop are the ~5k pts whose CF genotypes are not responsive to CFTR modulators (i.e Trikafta/Alyftrek). VX-522 is being tested in adults 18+ with non-responsive CFTR genotypes, including nonsense/"no protein" mutations. **We'll flag VX-522 is not a replacement for Trikafta/Alyftrek, but a mutation-agnostic add-on to VRTX's CF franchise.**



VX-522 is being tested in Ph1/2 NCT05668741 SAD/MAD trial with concomitant IVA, which keeps CFTR channels open longer (PCD Dec '26). IVA may boost efficacy via a different MOA and may also prevent any early regression in function before VX-522 takes effect, keeping net benefit stable. **We'll flag VX-522's Ph1/2 program was temporarily halted after two pts in MAD Cohort 1, Tx Arm 2 (VX-522 + IVA) developed fever, dyspnea, a drop in ppFEV₁ and CT evidence of ground-glass opacities; 1 case was classified as a SUSAR, the other as non-serious but moderate, with both improving rapidly after VX-522 d/c plus antibiotics & corticosteroids.** At the time, VRTX & IDMC concluded these events likely reflected a pulmonary inflammatory response to VX-522 and that existing mitigation (inhaled steroids/bronchodilators) was insufficient, leading the sponsor pause in Apr 2025, but regulators did not deem this a permanent change in benefit-risk, and VRTX resumed the MAD dosing (w/ protocol amendments) ~Aug 2025.

Source: Jefferies Analysis, Company Publications

While there are many tx in clinic using the mRNA/AAV modality to treat CF Class I pts, we haven't seen clear differentiation w/ efficacy endpoints varying across early-stage clinical trials. We'll flag a few that reported prelim efficacy data:

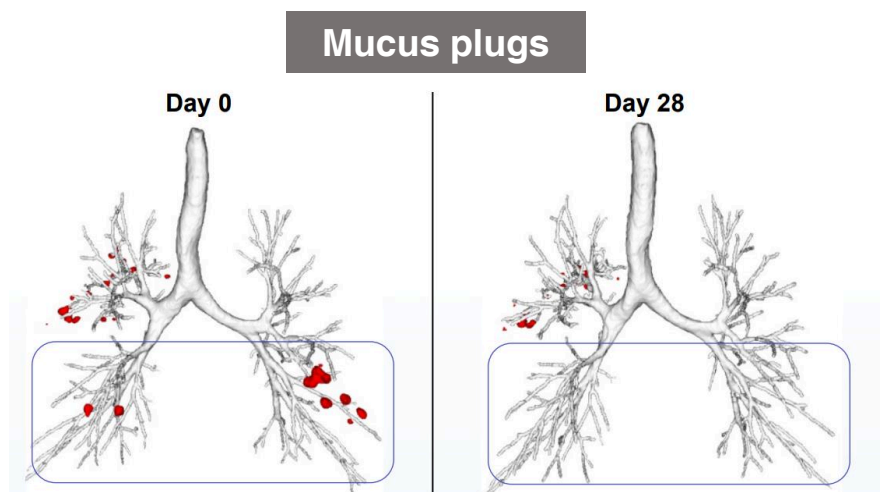
- **4D-710:** In lower-dose cohorts (C3/C4; n=9), lung biopsies showed dose-dependent CFTRΔR mRNA expression w/ the selected Ph2 dose (2.5E14 vg) achieving the target 17% mean of airway epithelial cells and CFTR protein expression in ~91% of epithelial cells, approaching non-CF levels (normal lung ~99.9% vs CF lung ~18.4%). In 3-12mo F/U, pts at the Ph2 dose had ppFEV₁ +3 to +10 pts, LCI_{2.5} improvement -0.9 to -1.9, and CFQ-R-R +17 to +50 pts.
- **KB407:** Ph1 CORAL-1 interim - highest dose (4x daily 10⁹ PFU) achieved wild-type CFTR expression. Per-pt transduction ranged 29.4%-42.1% across airway cells regardless of modulator status or genotype (incl class I null). In modulator-ineligible pts (n=4), positive-cell rates were 33.1%-62.0%; in modulator-eligible pts (n=2), rates were 36.8% and 31.4%.
- **SP-101:** preclin data suggest low-dose doxorubicin can enhance AAV transduction by improving intracellular processing of the vector (10⁴ vs ~10⁶ log₁₀ mRNA copies)

Drug	Company	Modality	Mutation Coverage	Status
ARCT-032	Arcturus Therapeutics	mRNA therapy (LUNAR LNP)	Mutation agnostic	Phase 2
RCT2100	ReCode Therapeutics	mRNA therapy (SORT LNP)	Class I mutations	Phase 1b
4D-710	4D Molecular Therapeutics	Gene therapy (AAV)	Mutation agnostic	Phase 1/2
SP-101	Spirovant Sciences	Gene therapy (AAV + augmentor)	Mutation agnostic	Phase 1/2
KB407	Krystal Biotech	Gene therapy (HSV-1)	Class I mutations	Phase 1
Gensaic CF Program	Gensaic	Bacteriophage vector	NR	Preclinical
Prime Editors (CFTR)	Prime Medicine	Gene editing	Class I mutations	Preclinical
SalioGen CF Program	SalioGen Therapeutics	Gene coding platform	Mutation agnostic	Preclinical
Carbon CF Program	Carbon Biosciences	Gene therapy (Parvovirus vector)	Class I mutations	Preclinical
SION-638/451/2222/719	Sionna Therapeutics	Oral small-molecule CFTR NBD1 stabilizer	Primarily ΔF508 (NBD1 defect)	Phase 1 completed (SAD/MAD); ongoing formulation/food-effect studies
Necafter	Fair Therapeutics (Fair TX)	Oral triple CFTR modulator with organoid-based companion diagnostic	Ultra-rare CFTR variants (organoid-selected responders)	Phase 1b (CHOICES) – enrollment completed
ETD001	Enterprise Therapeutics	Inhaled ENaC blocker (mucociliary clearance; CFTR-independent)	Mutation-agnostic	Phase 2 in CF ongoing; Phase 1 in healthy volunteers published
GDC-6988	Genentech / Roche	Inhaled TMEM16A potentiator (CFTR-independent chloride secretion)	Mutation-agnostic	Phase 1 safety published; ongoing early clinical in muco-obstructive disease cohorts
Brensocatib	Insmed	Oral DPP1 (Cathepsin C) inhibitor – anti-inflammatory adjunct	Mutation-agnostic	Phase 2a in CF completed (PK/PD and safety characterized)
Verducatib (BI-1291583)	Boehringer Ingelheim	Oral Cathepsin C/DPP1 inhibitor – anti-inflammatory (bronchiectasis, incl. CF bronchiectasis)	Mutation-agnostic	Phase 3 (AIRTIVITY) recruiting; Phase 2 CF bronchiectasis (Clairafly) completed

We'll also call out ARCT-032 as another mRNA tx for Class I CF pts to keep an eye on – while it didn't hit on ppFEV1 reduction, the co is attempting to get approval on other endpoints (ie, drug showed mucus plug reduction)

In Oct 2025, ARCT-032 did not demonstrate meaningful improvement in FEV₁, but noted a 5.1% relative increase in ppFEV₁ in 4/6 Class I CF pts. That said, there was efficacy demonstrated for improvement in % change in # mucus plugs and mucus volume.

Initial analysis comparing FEV₁ values from Day 1 to Day 28 did not demonstrate meaningful improvement. However, a post hoc exploratory analysis comparing the average of two pre-treatment FEV₁ measurements (screening and Day 1) for the baseline with the Day 42 post-treatment value, assuming an extended activity of functional CFTR protein triggered by daily ARCT-032 administration, suggests improvements in lung function, in four of six Class I CF participants with an average absolute increase of 3.8% and a relative increase of 5.1% in percent predicted FEV₁ (ppFEV₁). While the magnitude of these changes falls within the



Source: Jefferies Analysis, Company Publications

We caught up with ARCT mgmt:

- On ppFEV₁, mgmt commented while FEV₁ has traditionally been used as the primary efficacy signal, they **do not believe regulatory success for mRNA therapies in CF must hinge solely on FEV₁ improvements** - particularly in Class I pts, where unmet need is highest and lung damage may be less advanced; **they view LCI and mucus plug reduction as potentially approvable endpoints pending alignment with FDA.** Mgmt highlighted that in pediatric populations LCI is often more informative than FEV₁, while in infants neither metric is practical, reinforcing the need for a broader endpoint framework.
- On mucus plug imaging, mgmt highlighted the FDA itself requested the 28-day CT scans, signaling regulatory interest in lung-remodeling biomarkers; mgmt added that mucus plug reduction is known to precede functional improvement in lung disease. Mgmt stated that while they still intend to demonstrate lung function improvement, they view the regulatory “trigger” as establishing an endpoint framework (FEV₁, LCI, mucus plug reduction) acceptable to FDA, and expect additional clarity following an upcoming Type C meeting.
- On competitive differentiation vs VX-522, mgmt argued that their LUNAR delivery system **offers superior endosomal escape, improved stability through the liposomal process, engineered biodegradability, and proprietary RNA purification that minimizes inflammatory impurities.** They suspect reduced impurities helps reduce the acute inflammatory reaction in the body.
- On longer-term outlook, mgmt added that **mRNA therapies are not intended to replace CFTR modulators but should reduce overall treatment burden,** and once proof-of-concept is firmly established, they expect strong demand for these therapies.

d) Finally, we’re not worried about generics near-term – While generics exist in select countries (e.g. Argentina & Bangladesh, Russia) where VRTX lacks enforceable patent protection, none have taken notable market share + Trikafta has IP until 2037 in US & Alyftrek till 2039

Lupin’s ivacaftor ANDA has received tentative FDA approval, but VRTX is suing Lupin in Delaware based on patent infringement and a 30mo stay was placed till 2026. That said, in certain ex-U.S. markets (e.g., Argentina), VRTX lacks enforceable patent protection, allowing legally manufactured ivacaftor generics to be sold locally. These products are sometimes imported by patient groups, illustrating manufacturing feasibility and demand, but they lack FDA or EMA approval and have not translated into meaningful penetration in regulated western markets.

There are currently no ANDAs in the U.S. for lumacaftor/ivacaftor or any triple-combination CFTR modulators. **We’ll flag high burden of proving bioequivalence, and small pt n, has deterred generic entry beyond single-agent ivacaftor.** While some emerging markets have petitioned for compulsory licensing due to affordability concerns, these efforts have been slow.

Russia introduced an unauthorized Argentina-made generic Trikafta called Trilexa, and regional physicians have shifted some CF patients away from Trikafta despite the gx version lacks clinical evidence. This stems from the economic strain caused by war, combined with the country’s shift toward domestic manufacturing.

Patient advocacy groups (CF Buyers Club) are partnering with Beximco Pharmaceuticals in Bangladesh to produce a generic version of Trikafta, in response to its high price and limited availability. **The generic will cost roughly \$6,375 annually for children and \$12,750 for adults.**

VRTX CF franchise gx activity			
Product	Generic-development activity	Status & timing	Key evidence (public sources)
Kalydeco (ivacaftor)	- Lupin (India) filed an ANDA for 150 mg tablets (2022) and for pediatric oral granules (2024). - FDA has issued tentative approvals for both forms; Lupin holds 180-day first-to-file exclusivity.	Final U.S. launch blocked by Vertex patents through Feb 2030–Feb 2033 (composition & formulation). - Vertex is suing Lupin in Delaware; 30-month stay runs to mid-2025/26.	- FDA tentative-approval letters & patent-block notice. - Vertex v. Lupin patent litigation docket (C.A. No. 1:22-cv-00966-RGA).
Orkambi (lumacaftor / ivacaftor)	- No U.S./EU ANDA filings public yet (patents expire 2030 US, 2026 EU). - Gador (Argentina) manufactures a generic (Lucaftor) because Vertex never patented Orkambi there. “Orkambi Buyers Club” in the UK/NL imports the Argentine generic under personal-import rules.	Legal in Argentina; import permitted for personal use in UK (private prescriptions). - No Western regulatory filing yet; market size & patent barriers deter.	- UK press on Buyers Club & Argentine source. - DrugPatentWatch shows Orkambi patents & DMF entries, but zero ANDAs so far.
Symdeko / Symkevi (tezacaftor / ivacaftor)	- No ANDA filings disclosed. - Earliest U.S. composition patents run to 2034; fewer countries without coverage vs Orkambi.	Nothing in FDA queue yet	Patent-expiry listings in SEC filings & DrugPatentWatch (no ANDAs posted).
Trikafta (elexacaftor / tezacaftor / ivacaftor)	- No public ANDAs (too early; COM patents to 2037). - Activist compulsory-license petitions filed/under discussion in South Africa, Brazil, Ukraine, India and Chile; litigation in South Africa led by patient Cheri Nel.	- Petitions still unresolved; no local license granted yet. - If granted, would allow state import of Argentine copy or local production for public-sector use only.	- Guardian coverage of SA lawsuit & compulsory-license push (The Guardian). - Advocacy letters urging USTR not to retaliate against CL petitions (STAT Pharamlot).
Cross-product supply efforts	Multiple LMIC humanitarian campaigns (JustTreatment, CF Buyers Club) sourcing Argentine generics to bypass high prices.	Operate under “named-patient” import rules; volumes remain tiny (<100 pts) and outside regulated supply chains.	CF Buyers Club blog & pharma press explaining Argentine sourcing model.

Source: Jefferies Analysis, Company Publications, Press Sources Note: no company comments

Putting it together, we built a rolling model to capture market/switch dynamic across VRTX's cystic fibrosis portfolio: **1)** We model ~44k CF patients in US in 2025, ~93% eligible for modulator therapy + penetration growing from ~77% w/ modest decline after Trikafta and Alyftrek LOE w/ a bigger impact post VX-828's assumed 2042 LOE

Cystic Fibrosis - Market Model	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045
Cystic Fibrosis - U.S. Build																					
US CF Market Model Assumptions	Kalydeco	Orkambi	Symdeko	Trikafta	Alyftrek	VX-828															
Launch Year					2025	2030															
Patent Expiry	2027	2031	2027	2037	2039	2042															
% Penetration on new Pts	5%	-	1%	77%	17%		*2025 % NRx based on IQVIA														
Trikafta switch rate %	25%	20%	-	-	-																
Alyftrek switch rate %	25%	40%	40%	30%	-																
VX-828 switch rate %	-	-	-	10%	15%																
1+ yo Population Approval				2027	2029																
<1 yo Population Approval																					
US Cystic Fibrosis Patients	43,804	44,804	45,804	46,804	47,804	48,804	49,804	50,804	51,804	52,804	53,804	54,804	55,804	56,804	57,804	58,804	59,804	60,804	61,804	62,804	63,804
CF Incidence	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000	1,000
US Modulator-Eligible %	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%	93%
US Modulator-Eligible Pts	40,806	41,533	42,460	43,387	44,314	45,241	46,168	47,095	48,022	48,949	49,876	50,803	51,730	52,657	53,584	54,511	55,438	56,365	57,292	58,219	59,146
Treatment Penetration %	77%	79%	79%	80%	80%	80%	80%	81%	81%	81%	81%	82%	80%	78%	76%	73%	71%	64%	58%	52%	47%
Patients Treated	31,450	32,811	33,650	34,493	35,341	36,193	37,050	37,912	38,778	39,649	40,525	41,405	41,384	41,336	40,724	40,066	39,361	36,017	32,949	30,134	27,552

We estimated ~44k patients are affected by cystic fibrosis in the US as of 2025. In our model, we also assume ~1k new patents add yoy. We also estimated ~58k patients in the EU, bringing the total to ~101k as of 2025

We define the addressable patient population as patients with CF who are eligible for modulators. According to the Cystic Fibrosis Foundation, ~7.3% of CF patients remain ineligible for any modulator therapy due to age or genotype. Therefore, we estimated that ~93% of the CF population can be treated w/ VRTX's CF products.

We also assume market penetration starting at ~77% in the US (in line w/ VRTX's comments to match pts adds) and increases over time. We estimated a modest decline in penetration following the LOE for Trikafta and Alyftrek, the current triple combo therapy. In our model, we also assume VX-828 will reach LOE in 2042 (12 yrs post launch in 2030) w/ more impactful decline in market share.

Source: Jefferies Analysis, Cystic Fibrosis Foundation, Company Filing, Cromwell et. al, 2024

2) We model continued switch from the legacy portfolio to next gen products – we expect Kalydeco to maintain a portion of the new patient uptake (5%) since it's the only CF tx approved for ≥ 1 mo for VRTX & demonstrated strong persistence in TRx w/ more switches post approval in ages 1+ to Alyftrek

	Q1 2024	Q2 2024	Q3 2024	Q4 2024
KALYDECO WAC	86,275	86,275	86,275	86,275
GTN	30%	30%	30%	30%
Compliance	85%	85%	85%	85%
Net Price	51,333	51,333	51,333	51,333
Est Rev	70,891,325	72,636,658	71,712,659	68,221,992
Reported Est	74,152,749	75,754,185	74,901,035	71,202,569
Capture Rate	96%	96%	96%	96%
Implied Pts	1,445	1,476	1,459	1,387

	Q1 2024	Q2 2024	Q3 2024	Q4 2024
ORKAMBI WAC	86,275	86,275	86,275	86,275
GTN	30%	30%	30%	30%
Compliance	85%	85%	85%	85%
Net Price	51,333	51,333	51,333	51,333
Est Rev	667,333	1,437,333	1,899,333	1,437,333
Reported Est	9,924,445	24,452,067	32,311,660	26,208,314
Capture Rate	7%	6%	6%	5%
Implied Pts	193	476	629	511

	Q1 2024	Q2 2024	Q3 2024	Q4 2024
SYMDEKO WAC	80,873	80,873	80,873	80,873
GTN	30%	30%	30%	30%
Compliance	85%	85%	85%	85%
Net Price	48,119	48,119	48,119	48,119
Est Rev	15,061,359	15,157,598	15,879,388	13,665,898
Reported Est	20,458,680	19,759,054	21,125,957	17,995,953
Capture Rate	74%	77%	75%	76%
Implied Pts	425	411	439	374

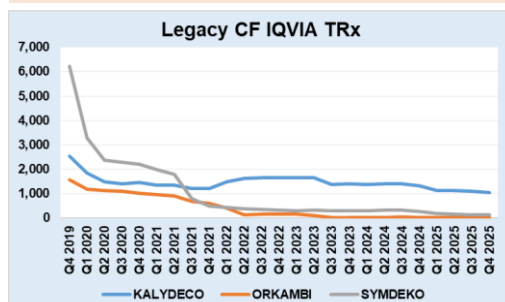
	2025	2026	2027	2028	2029	2030
Cystic Fibrosis - Market Model						
CF US Patient Build						
Kalydeco						
Beginning Patients Balance	1,387	1,216	1,070	839	665	534
% Penetration on new Pts	5%	5%	5%	5%	5%	5%
New Patients Adds (Loss)	37	37	36	36	36	35
Switches to Trikafta	208	182	267	210	-	-
Trikafta switch rate %	15%	15%	25%	25%	-	-
Switches to Alyftrek	-	-	-	-	166	134
Alyftrek switch rate %	-	-	-	-	25%	25%
Ending Patients Balance	1,216	1,070	839	665	534	436
Orkambi						
Beginning Patients Balance	511	307	184	110	66	40
% Penetration on new Pts	-	-	-	-	-	-
New Patients Adds (Loss)	-	-	-	-	-	-
Switches to Trikafta	102	61	37	22	-	-
Trikafta switch rate %	20%	20%	20%	20%	-	-
Switches to Alyftrek	102	61	37	22	26	16
Alyftrek switch rate %	20%	20%	20%	20%	40%	40%
Ending Patients Balance	307	184	110	66	40	24
Symdeko						
Beginning Patients Balance	374	232	146	94	63	44
% Penetration on new Pts	1%	1%	1%	1%	1%	1%
New Patients Adds (Loss)	7	7	7	6	6	6
Switches to Trikafta	-	-	-	-	-	-
Trikafta switch rate %	-	-	-	-	-	-
Switches to Alyftrek	150	93	59	38	25	17
Alyftrek switch rate %	40%	40%	40%	40%	40%	40%
Ending Patients Balance	232	146	94	63	44	32

We expect continued switching from the legacy products to concentrate in Trikafta in the NT, given it's approved for patients 2yo+ (vs. 6yo+ for Alyftrek). That said, VRTX is conducting trials in the earlier age group for both Alyftrek and Trikafta. Following the approval in younger cohorts (e.g. 1yo+), we assume switches will mostly towards Alyftrek based on its better drug profile.

Our assumptions for switch %:

- Kalydeco:** TRx declined ~20% year over year, while NRx ~13% year over year. Following approval of the 1–11 yo patient population for Alyftrek and Trikafta, we expect an increased switch rate to Alyftrek, assuming patients remain on Kalydeco until age one. Mgmt plans to initiate studies in patients ages 1–2 in 2026 for Alyftrek (Trikafta's 1–2-year-old trial took ~ 3 years, we model Alyftrek approval in 2029). **Overall, we model a 15-25% patient switch rate from Kalydeco to Alyftrek.**
- Orkambi:** TRx ~46% decrease YoY on avg; NRx 38% decrease YoY on avg – we assume 20% switch to trikafta (mostly 2-6 age group) and 20% for alyftrek. Once Alyftrek is approved in 1yo+(assuming 2029), we model all switches to Alyftrek. **Overall, we model ~40% switch rate in aggregate.**
- Symdeko:** TRx ~37% decrease YoY; NRx 40% decrease YoY. Since the eligible age group overlaps w. Alyftrek, we assume all patients switch to Alyftrek.

We estimate revenue and patients on legacy products by applying VRTX company-level revenue mix to IQVIA TRx capture rates and net price assumptions, estimating ~ 2.3k patients on the legacy portfolio



We expect to see continued uptake of Kalydeco among newly initiated patients (5% based on NRx data from IQVIA), since it is the only therapy approved for patients aged ≥ 1 mo+. IQVIA scripts data also indicate stickiness in TRx for Kalydeco compared to other legacy CF products. Additionally, we model an acceleration in switches from Kalydeco to Trikafta and Alyftrek once approved in 1yo + population in 2027 and 2029 respectively.

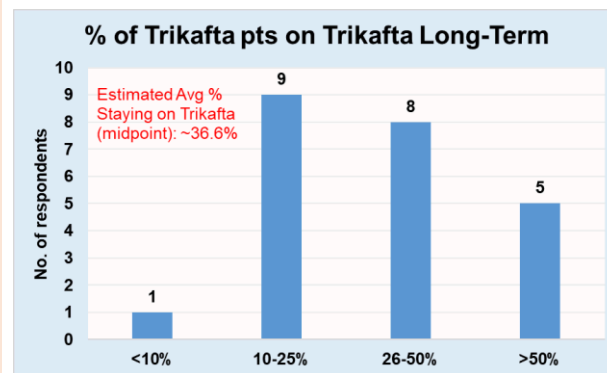
Source: Jefferies Analysis, Cystic Fibrosis Foundation, Company Filing

US CF Market Model Assumptions	Kalydeco	Orkambi	Symdeko	Trikafta	Alyftrek	VX-828
Launch Year					2025	2030
Patent Expiry	2027	2031	2027	2037	2039	2042
% Penetration on new Pts	5%	-	1%	77%	17%	
Trikafta switch rate %	25%	20%	-	-	-	
Alyftrek switch rate %	25%	40%	40%	30%	-	
VX-828 switch rate %	-	-	-	10%	15%	
1+ yo Population Approval				2027	2029	
<1 yo Population Approval						

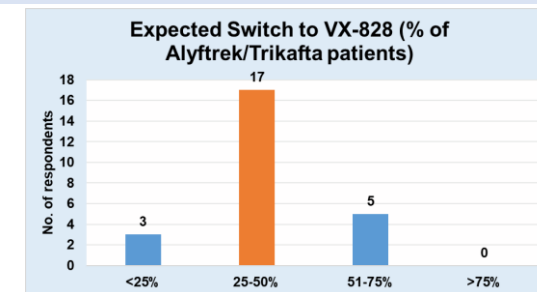
3) Based on our survey results, we assume a ~15% Trikafta->Alyftrek switch rate in 2026 ramping to ~30% peak & model ~36% of Trikafta patients remaining on Trikafta through 2029 – for the switch to next gen, we assume a ~23-31% switch from Trikafta/Alyftrek to VX-828 post launch starting in 2030

Cystic Fibrosis - Market Model	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045
Trikafta																					
Beginning Patients Balance	28,460	27,288	23,851	18,979	13,753	9,753	9,631	9,450	9,214	8,984	8,759	8,540	8,326	8,118	7,915	7,717	7,525	7,336	7,153	6,974	6,800
% Penetration on new Pts	77%	62%	47%	32%	17%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
New Patients Adds (Loss)	553	454	345	236	126	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Switches from Legacy CF	310	244	304	232	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Switches to Alyftrek	2,035	4,134	5,522	5,694	4,126	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Alyftrek switch rate %	7%	15%	23%	30%	30%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Switches to VX-828	-	-	-	-	-	488	722	945	921	898	876	854	833	812	792	772	752	734	715	697	680
VX-828 switch rate %	-	-	-	-	-	5%	8%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
Risk adjustment from VX-828	-	-	-	-	-	366	542	709	691	674	657	640	624	609	594	579	564	550	536	523	510
Ending Patients Balance	27,288	23,851	18,979	13,753	9,753	9,631	9,450	9,214	8,984	8,759	8,540	8,326	8,118	7,915	7,717	7,525	7,336	7,153	6,974	6,800	6,630
																					-31%
Alyftrek																					
Beginning Patients Balance	-	2,409	7,561	13,628	19,947	24,971	25,434	26,179	27,013	27,890	28,758	29,785	30,842	31,208	31,408	31,011	30,577	30,096	26,976	24,225	21,735
New Patients Adds (Loss)	122	864	451	565	681	609	613	617	621	624	628	632	(56)	(82)	(644)	(689)	(734)	(3,369)	(3,091)	(2,835)	(2,599)
Switches from Legacy CF	252	154	95	60	218	167	131	106	88	74	65	59	52	47	44	41	38	36	33	30	29
Switches from Trikafta	2,035	4,134	5,522	5,694	4,126	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Switches to VX-828	-	-	-	-	-	1,249	1,908	2,618	3,377	4,184	4,314	4,468	4,626	4,681	4,711	4,652	4,587	4,514	4,046	3,634	3,260
VX-828 switch rate %	-	-	-	-	-	5%	8%	10%	13%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%	15%
Risk adjustment from VX-828	-	-	-	-	-	936	1,908	2,729	3,546	4,352	4,648	4,834	4,997	4,916	4,915	4,866	4,801	4,727	4,354	3,948	3,563
Ending Patients Balance	2,409	7,561	13,628	19,947	24,971	25,434	26,179	27,013	27,890	28,758	29,785	30,842	31,208	31,408	31,011	30,577	30,096	26,976	24,225	21,735	19,467
																					-23%
VX-828																					
Beginning Patients Balance	-	-	-	-	-	-	637	1,021	1,351	1,619	1,883	1,978	2,036	1,874	1,842	1,836	1,815	1,788	1,759	1,630	1,490
New Patients Adds (Loss)	-	-	-	-	-	812	817	822	827	833	838	843	-	-	-	-	-	-	-	-	-
Switches from Trikafta	-	-	-	-	-	488	722	945	921	898	876	854	833	812	792	772	752	734	715	697	680
Switches from Alyftrek	-	-	-	-	-	1,249	1,908	2,618	3,377	4,184	4,314	4,468	4,626	4,681	4,711	4,652	4,587	4,514	4,046	3,634	3,260
POS	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%	25%
Ending Patients Balance	-	-	-	-	-	637	1,021	1,351	1,619	1,883	1,978	2,036	1,874	1,842	1,836	1,815	1,788	1,759	1,630	1,490	1,358
Cystic Fibrosis US Patients	31,450	32,811	33,650	34,493	35,341	36,193	37,050	37,912	38,778	39,649	40,525	41,405	41,384	41,336	40,724	40,066	39,361	36,017	32,949	30,134	27,552

We assumed ~15% switch rate in 2026 w/ a peak switch rate of ~30% from Trikafta to Alyftrek based on IQVIA and our survey results. Our survey suggests ~36% pts on Trikafta are expected to stay over time prior to the launch of VX-828. In our rolling build, we model ~36% of patients will continue on Trikafta through 2029 (w/ VX-828 assumed to launch in 2030), after which we expect these patients to switch to VX-828 post approval.

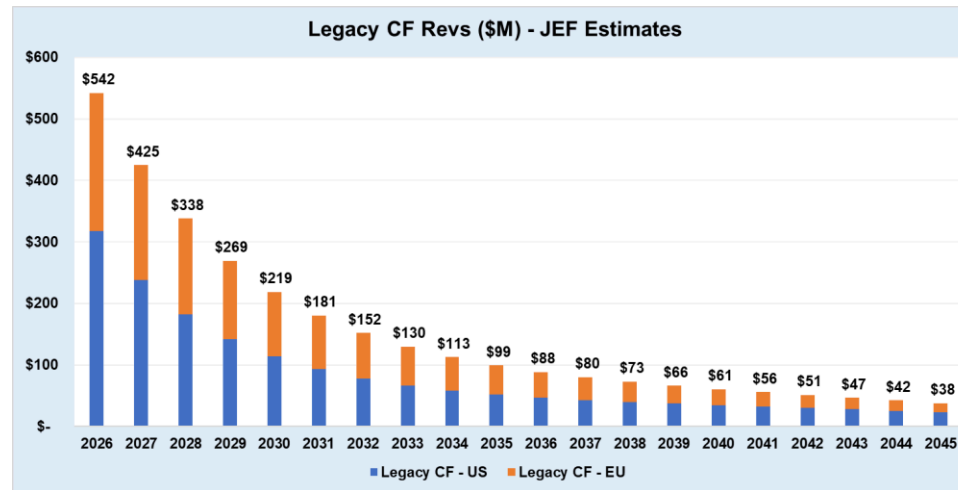
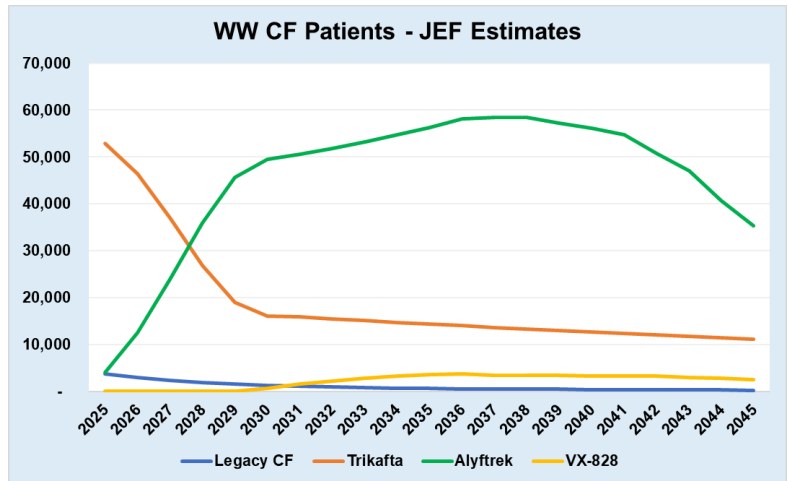
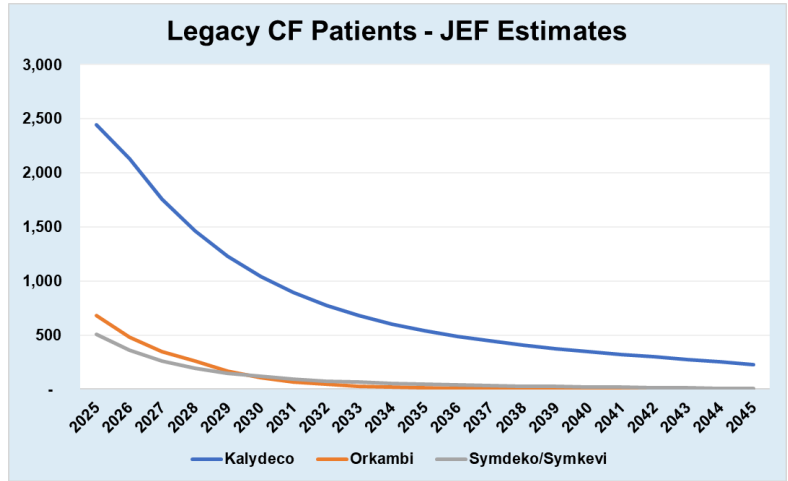


Our survey suggest most respondent think the switch rate to VX-828 (% of Alyftrek/Trikafta patients) falls in 25-50%. In our model, we assume ~23-31% from Trikafta and Alyftrek to switch to VX-828 overtime.

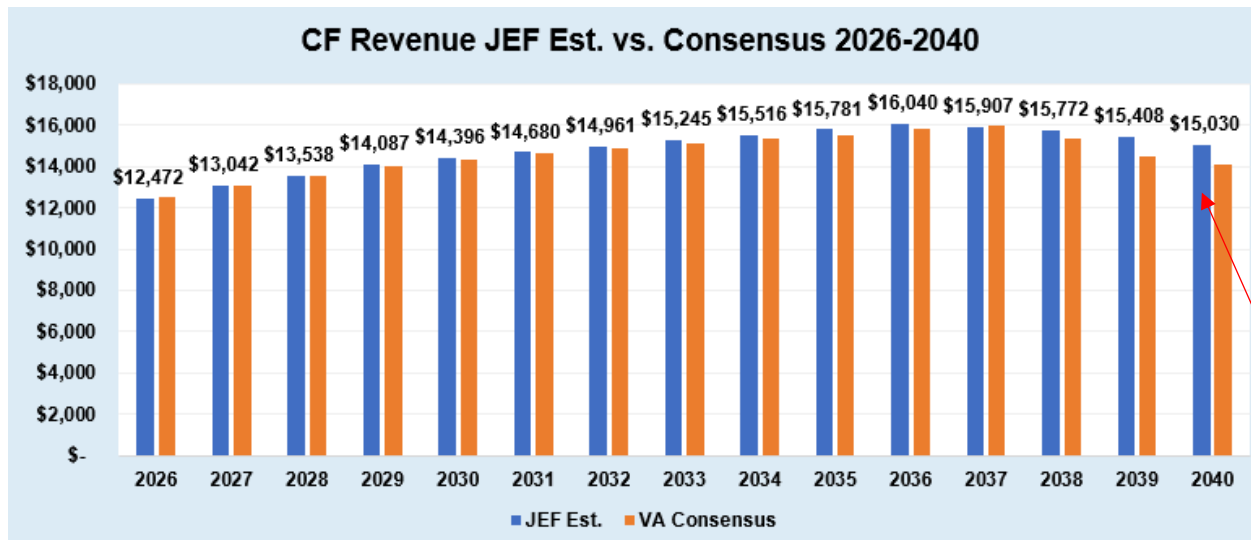


Source: Jefferies Analysis

Overall, we model ~\$542M sales for the legacy CF portfolio in '26 (trending down as the next-gen portfolio gets approved in younger pts) + PS of ~\$16B for VRTX CF franchise overall (in line with cons) – this supports a VRTX floor valuation ~\$422 (CF NPV + cash)



On switch, rate, we model a higher rate vs cons, hence we model \$10.7Bn sales for Alyftrek by 2030 vs. \$7Bn cons.



We model slightly less erosion in CF due to us modeling \$900M for VX-828, which cons does not include but we haven't seen data yet

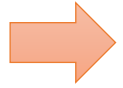
Source: Jefferies Analysis

Moving on, we'll discuss Povetacicept

Cystic Fibrosis

We expect VRTX's CF franchise to remain durable

- We're expecting CF to remain a durable revenue base for VRTX and expect gradual switch & new starts for pts onto Alyftrek
- We expect upcoming comps will work as add-on tx to VRTX, but we're not seeing evidence as a mono-tx they could produce greater efficacy
- We expect minimal pt adds from VRTX mRNA CF tx, but limited ability for generics to gain market share



IgAN & autoimmune

We see versatility for APRIL/BAFF across autoimmune indications

In IgAN, we saw 49.8% Pbo adj. UPCR reduction at 9mo (clinically in line w/ Sibe):

- To us, the impt point is: a) IgAN is a bigger market with favorable pricing (\$390K/yr), b) Pove has a convenience advantage with 1.5sec admin autoinjector
- We're optimistic on the indication expansion opportunity into gMG, pMN, wAIHA

AMKD & ADPKD

We see LT upside from Inaxaplin becoming first/best in class

- We're expecting Inaxaplin to be first & best in class in the AMKD market (currently no tx)
- We expect 64-70% UPCR reduction at 48wks + we think VRTX has the more favorable BL population vs comps
- We think the AMKD market could be bigger than initially expected given a) an approved tx will likely increase dx rates, b) our KOL expects spillover dx from the IgAN market growing from Povetacicept
- In ADPKD, we think the data is too early for VX-407 but could see pot'l reduced LFTs based on Trikafta read across (similar MOA)

Pain

We're mixed on Ph.3 DPN hitting, but are more confident in the acute pain opportunity

- We're mixed on whether chronic DPN can hit for Suze, given the high variability of previous data and Pbo (combo approach is more likely here).
- In acute pain, we expect 3x growth for Journavx based on recent access expansion

Rest of Pipeline

Rest of pipeline is interesting but we're skeptical on cons valuation

- We're skeptical on expansion opportunity for rest of pipeline:
- For Casgevy, we've seen precedence that viable tx's in SCD launch slower / underpenetrate the market due to patient/physician concerns & access barriers
- Zimislecel appears limited to severe T1D population based on immunosuppression requirement & clinical trial delays

Source: Jefferies analysis

Starting off, Povetacicept is a TACI receptor fusion protein targeting APRIL/BAFF - it's currently in pivotal trials for IgAN (RAINIER) and pMN (OLYMPUS) & will also be studied in gMG and wAIHA

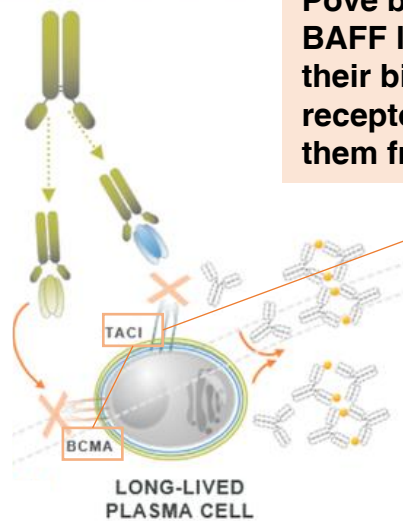
Pove is a TACI receptor fusion protein (engineered) that binds to both APRIL and BAFF (which then bind to TACI and BCMA receptors). **APRIL is an upstream cytokine that help with survival of mature plasma cells. BAFF supports B-cell maturation into plasma cells.**

Pove recently read out it's Ph.3 topline data. As a reminder, the Ph.3 go-forward dose is 80mg and mgmt. commented they are currently also running trials to collect data on at-home autoinjector administration at monthly dosing intervals to administered as a 0.46ml at home.

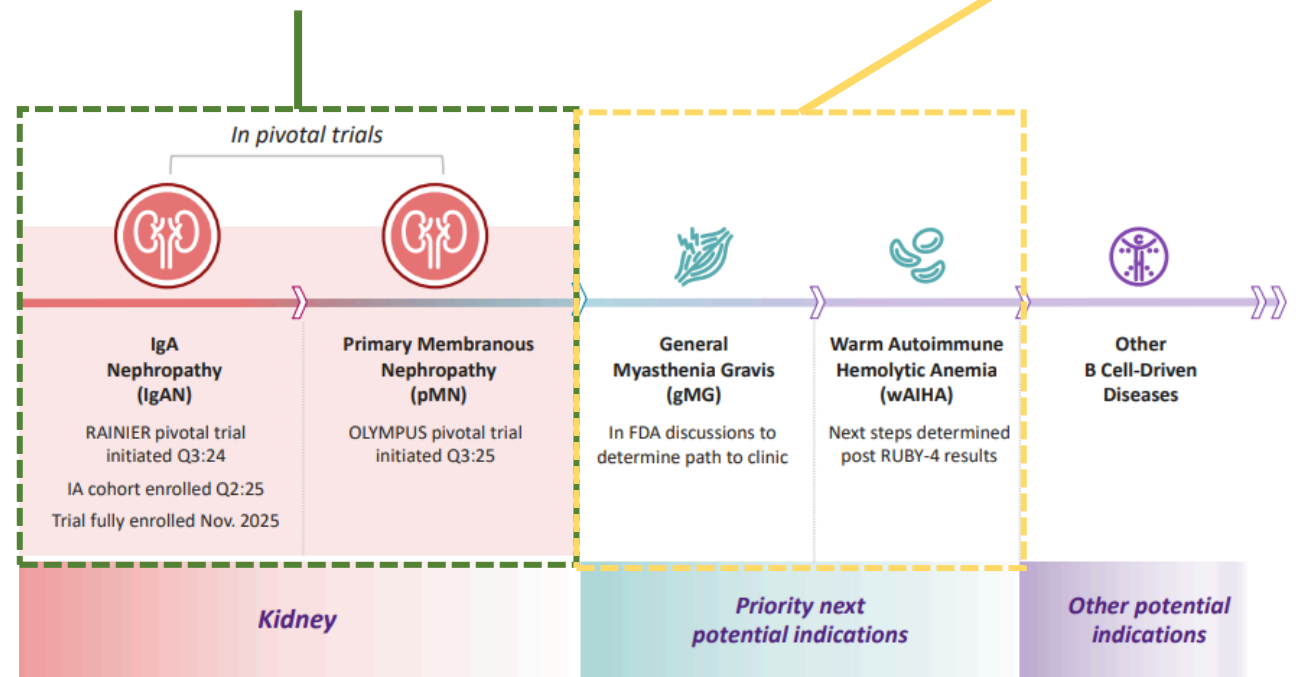
While mgmt. has commented they intend to run trials in gMG and wAIHA, **since neither program has started yet, we model lower PoS vs IgAN for each.** That said, we like the commercial optionality of Pove given its upstream MOA.

Povetacicept

Neutralizes both BAFF and APRIL



Pove binds to both APRIL & BAFF ligands thus preventing their binding to TACI and BCMA receptors therefore preventing them from binding to BCMA



Source: Jefferies Analysis, Company Publications

We see meaningful opportunity for VRTX to diversify beyond CF through pove:

#1) Starting with IgAN interim analysis data from Pove Ph3 (RAINIER) reported **52% abs 24-hr UPCR reduction from baseline at wk 36 (49.8% pbo-adj),**

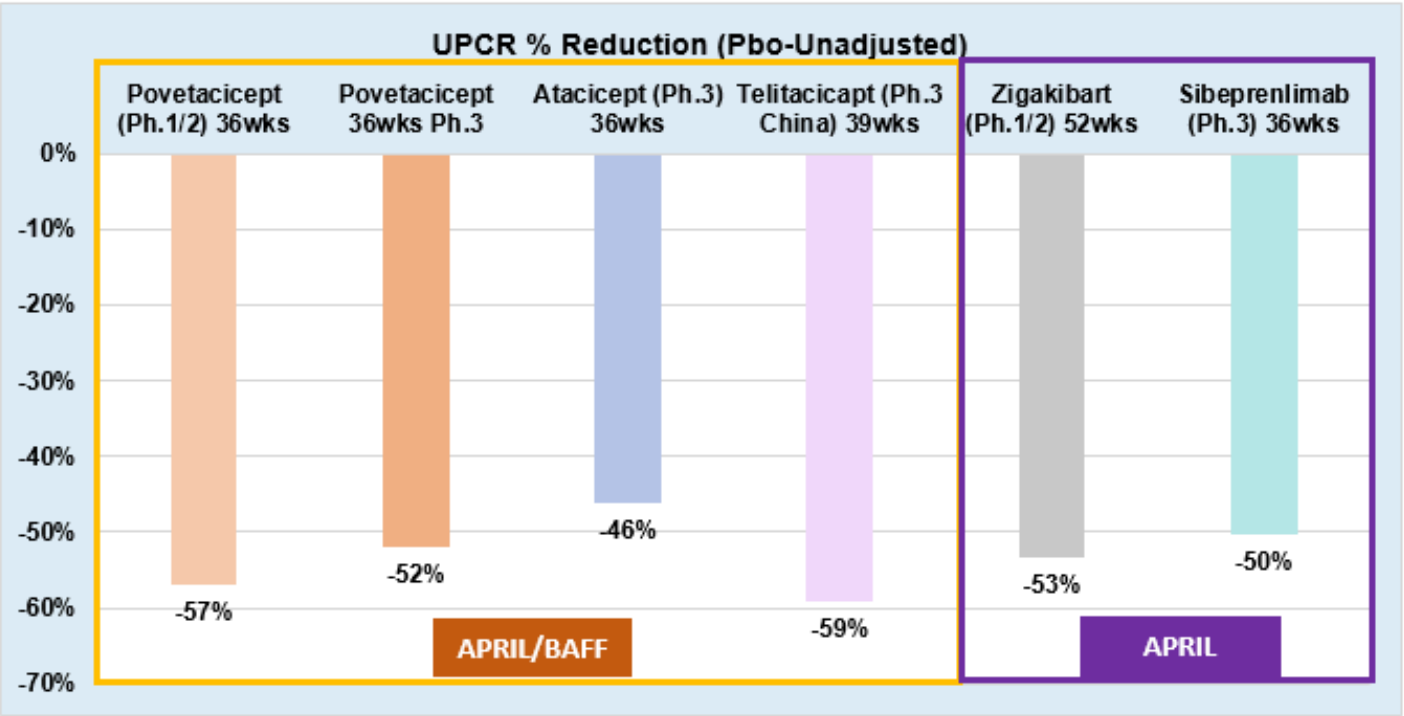
g 36wk interim data of RAINIER showed 52% abs 24-hr UPCR reduction from baseline (49.8% Pbo adjusted UPCR reduction). Overall, the UPCR reduction with Povi appears to be comparable/incrementally higher vs the ~50% abd UPCR reduction reported with Sibe Ph3

Efficacy Results for the RAINIER Interim Analysis Population at Week 36		
	Placebo N=68	Povetacicept N=131
Primary Endpoint		
% change from baseline in 24-hr UPCR	-4.3	-52.0
% change versus placebo		-49.8
P value vs. placebo ^a		<0.0001
Secondary Endpoint: Gd-IgA1		
% change from baseline in serum Gd-IgA1	+9.1	-77.4
% change versus placebo		-79.3
P value vs. placebo ^a		<0.0001
Secondary Endpoint: Hematuria Resolution ^b		
proportion of patients achieving resolution (%)	23.4	85.1
difference in proportion versus placebo (%)		61.7
P value vs. placebo ^a		<0.0001
Select Exploratory Endpoint		
proportion of patients achieving 24-hour UPCR <0.5 g/g (%)	6.2	42.2

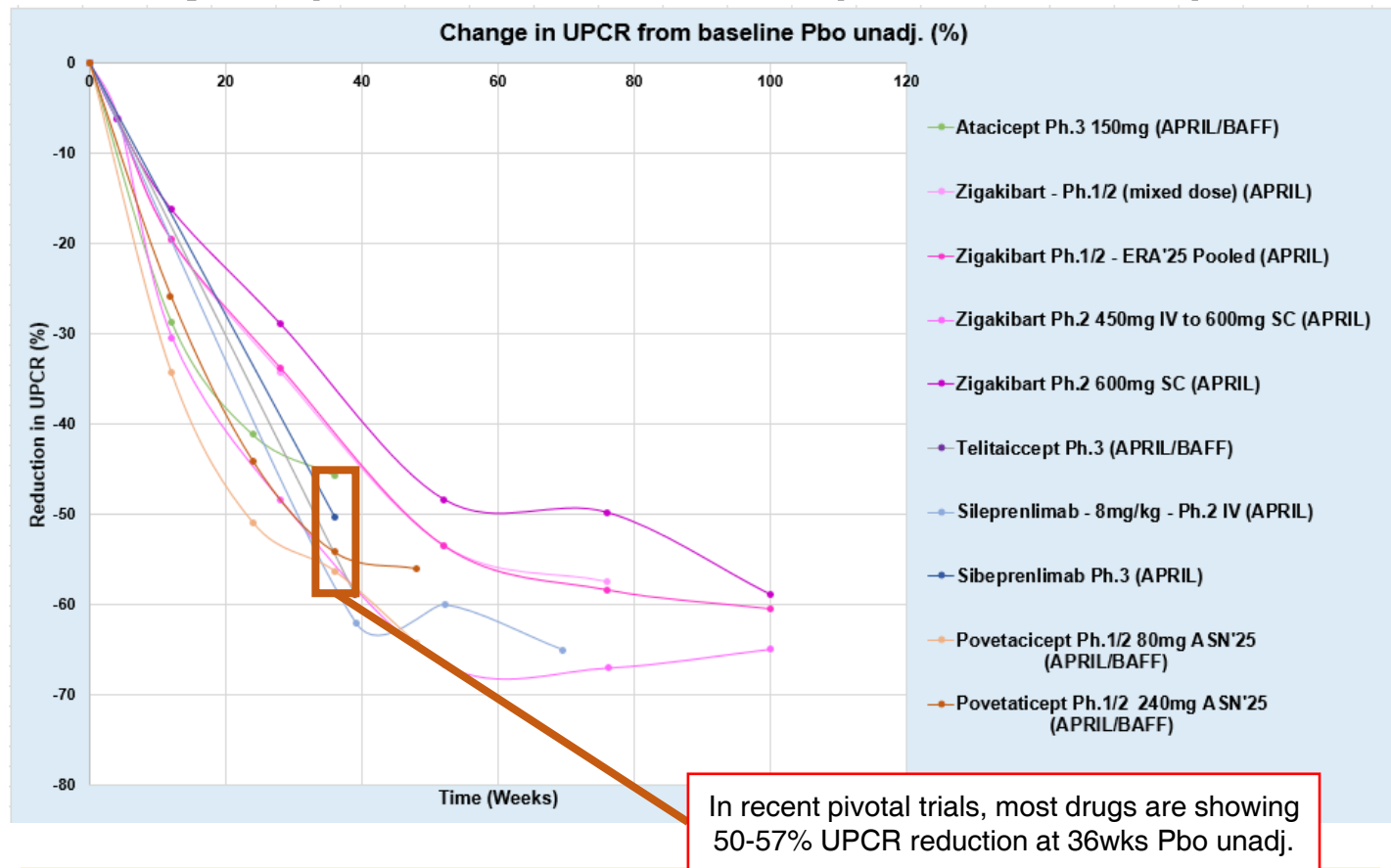
^aTwo-sided P value.

^bHematuria resolution in patients who had hematuria at baseline (N=36 for placebo and N=67 for povetacicept groups).

Source: Jefferies Analysis, company presentations



This was in line w/ our expectations ([HERE](#)) – recall that in prior Ph.2, pove showed ~56-57% UPCR reduction (vs 46-55% for comps) & the Ph.3 recruited more pts (n=~300 in tx arm) vs Ph.1/2 (n=17 at 80mg dose)



Criteria	Phase 2 (RUBY-3)	Phase 3 (RAINIER)
Dose	80mg, 240mg	80mg
Study Design	Multi-indication, open label, phase 1b/2 study	Randomized, double-blind, placebo-controlled trial
Sample Size	72 (54 IgAN patients) per CT.gov	~605 patients per CT.gov
Biopsy Status	Adults with biopsy ≤ 10 years	Biopsy-proven IgAN
Baseline UPCR	≥ 0.5 g/g	24-hr proteinuria excretion ≥1.0 g/day or 24-hour uPCR ≥0.75 g/g
Baseline eGFR	≥ 30 mL/min/1.73m ²	≥ 30 mL/min/1.73m ²
Prior Medications	Maximal ACE inhibitor or ARB therapy required	Stable angiotensin converting enzyme inhibitor (ACEi) or angiotensin II receptor blocker (ARBs)
Key exclusion criteria	eGFR <30 or rapidly progressive GN; recent or ongoing serious infection; prior B cell, BAFF, or APRIL-targeted therapy within protocol-defined windows.	Received immunosuppressive treatment or procedures (e.g., steroids, B/plasma cell or complement therapies, herbal meds, tonsillectomy) within protocol-defined washout; or had rapidly progressive GN with >50% eGFR drop within 12 weeks of screening.
Geographic Scope	Global (U.S., Europe, and Asia)	Global (U.S., Europe, and Asia)
Primary Endpoint	Safety (AE & SAEs) secondary: Proteinuria, eGFR, renal response	Interim: UPCR change at 36 weeks; Final: total eGFR slope through Week 104

- We'll call out two key difference in inclusion/exclusion criteria btw Pove Ph.2 and Ph.3: 1) the baseline threshold for UPCR increased from 0.5g/g to 0.75g/g minimum, 2) the number of patients is increasing from ~21 in the treatment arm to ~300 in the treatment arm. We'll go into them more later...

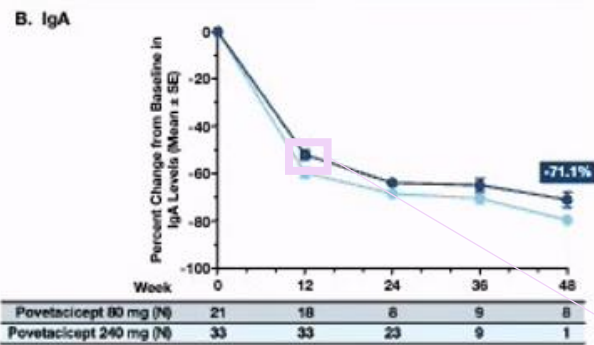
Note: Sibeprenlimab 8mg/kg IV (APRIL) showed -63% UPCR reduction at ~9 mo in Ph.2 but go-forward Ph.3 dose (400mg subQ dose), likely due to limitations around subQ volume. In Ph.3 Sibe showed -51.2% (Pbo adj.) UPCR reduction.

Source: Jefferies Analysis, company publications

Our illustrative regression analysis suggested wk12 IgA reduction could be a relevant predictor to wk36 UPCR reduction & we had estimated that **Pove Ph3 would show ~52% absolute UPCR reduction** *caveats apply*

	N	BL UPCR*	IgA reduction at Wk 12 (%)	UPCR reduction at Wk 36 (%)
Sibeprenlimab Ph2 - 2mg/Kg Q4W	38	1.46	-46.00%	-49%
Sibeprenlimab Ph2 - 4mg/Kg Q4W	41	1.53	-52%	-57%
Sibeprenlimab Ph2 - 8mg/Kg Q4W	38	1.44	-61%	-63%
Sibeprenlimab Ph2 - Pbo	38	1.68	-0.36%	-13%
Sibeprenlimab Ph3 - 400mg SC Q4W	152	1.22	-56%	-50.20%
Sibeprenlimab Ph3 - Pbo Q4W	168	1.28	0.40%	2.10%
Atacicept Ph2b 75mg QW	33	1.4	-44%	-34%
Atacicept Ph2b 150mg QW	33	1.4	-56%	-40%
Atacicept Ph2b Pbo	34	1.5	0.90%	-2%
Atacicept Ph3 150mg QW	106	1.7	-58%	-45.60%
Atacicept Ph3 Pbo	97	1.8	0%	-6.80%
Povitacicept Ph2 80mg Q4W	21	1.3	-51%	-56.20%
Povitacicept Ph2 240mg Q4W	33	1.2	-59.30%	-54.10%

*mean/median/geo.mean; Teli & Ziga excluded due to data comparability



(in line with Sibe Ph3, given the similar inclusion criteria)
(conservatively assuming similar W12 reduction as in Pove Ph2)

Assumed Baseline UPCR for Pove Ph3
Assumed IgA reduction at Wk12

1.22
-51%

Proj Wk36 UPCR reduction with Pove in Ph3

-51.96%

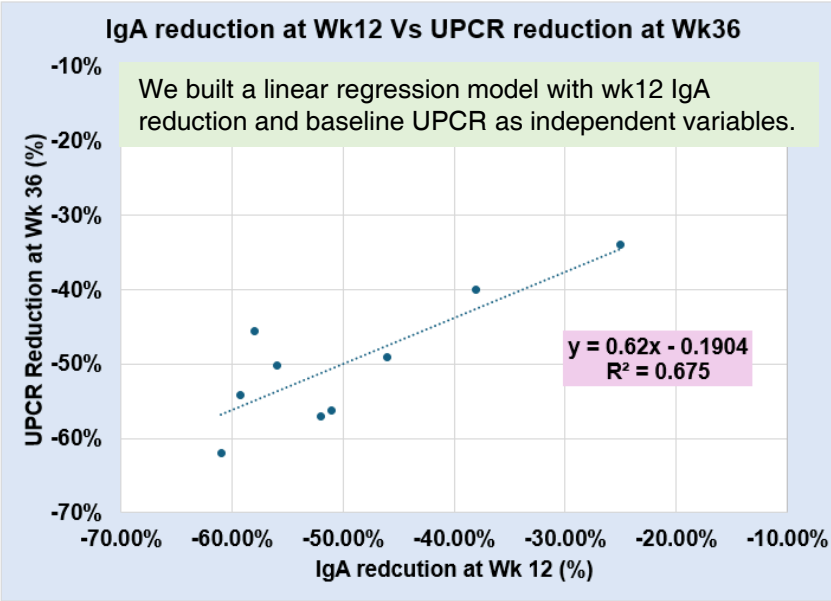
Assuming a baseline UPCR of ~1.2 in Pove Ph3 similar to Sibe Ph3 (given Pove Ph3 has similar inclusion criteria) & IgA reduction from baseline of ~51% at wk12 for the 80mg dose (as in Pove Ph.1/2 80mg dose, we are not seeing a Ph3 to Ph3 deterioration in Wk12 IgA reduction efficacy), our regression model suggest Pove Ph3 could show ~52% UPCR reduction from baseline at Wk 36
(significant caveats apply: our regression model is built from minimal data points affecting the reliability of projection, differences in baseline UPCR reporting methods, which adds further uncertainty)

Regression Statistics	
Multiple R	0.830753874
R Square	0.690151998
Adjusted R Square	0.586869331
Standard Error	0.056777676
Observations	9

ANOVA					
	df	SS	MS	F	Significance F
Regression	2	0.043082662	0.021541331	6.682166692	0.0297472
Residual	6	0.019342227	0.003223705		
Total	8	0.062424889			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	-0.289105352	0.201800814	-1.432627282	0.201939905	-0.782894157	0.204683452
BL UPCR*	0.069855236	0.128841001	0.542181725	0.607226584	-0.245407336	0.385117808
IgA reduction at Wk 12 (%)	0.619027687	0.17151592	3.60915586	0.011242319	0.19934335	1.038712023

Interestingly we saw a moderately strong ($R^2=0.68$) relationship between IgA reduction at wk12 and the UPCR reduction at wk36



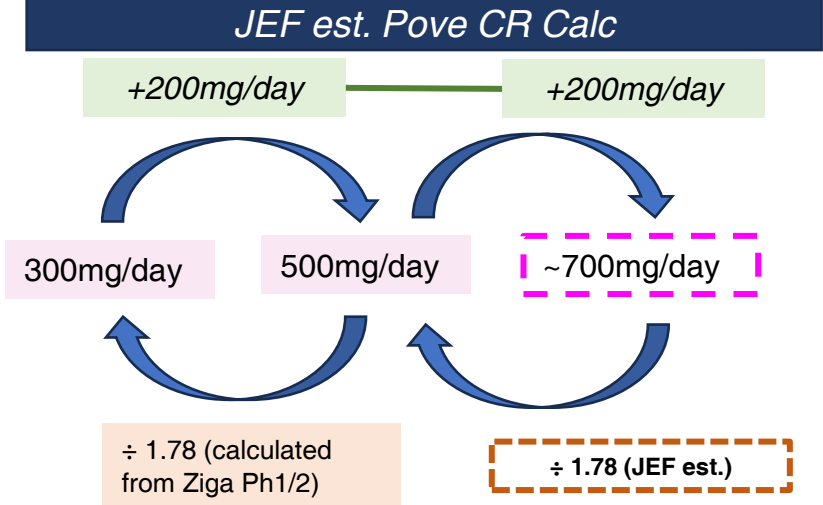
Source: Jefferies Analysis, company presentations

For clinical remission (not reported in VRTX PR), while difficult to compare, our illustrative adjustment suggested pove could show ~25% CR, which could be similar to Sibe's 29% CR at 4mg/kg Ph.2 **Caveat: Different definitions of clinical remission, assay variability, & variable durations make direct/standardized comparisons difficult**

1) We'll call out that the definition of clinical remission varied by study, but the **KDIGO definition is now "preferably below 0.3g/day"** of proteinuria. Pove (APRIL/BAFF), Ziga (APRIL) and Sibe (APRIL) were the 3 drugs that reported CR, but had different definitions

VRTX mgmt. commented that they have not specified a CR efficacy bar for Pove.

3) We'll also flag that duration of the study was different (and therefore data collection). Telitacicept had the shortest study (24w), but most studies passed 36w.



This is illustrative JEF analysis and cannot be directly be compared w/ other drug CRs

We'll flag that at 36wks pove 80mg reported ~44% CR (their definition) in Ph.1/2 and we found this could translate to KDIGO definition ~13.8% (0.3g/day) or ~24.7% (0.5g/day) using our calculated ÷ 1.78 multiple. Note: at 48wks this would be 29.7% and 16.7% respectively – aligning even closer to Sibe Ph.2.

As a reminder, Sibe showed a CR of 29.3% at the 4mg/kg dose, 28.9% at the 8mg/kg dose, in Ph.2 at 12mo using the 0.5g/day threshold and 34.3% Pbo adjusted remission in Ph.3. **To us, Pove looks numerically similar to Sibe Ph.2, so we're not expecting Pove to differentiate in CR.**

Drug	Study	Definition of Clinical Remission
Povetacicept	Ph.2 (RUBY-3)	UPCR <0.5g/g, negative hematuria, <25% reduction in eGFR vs baseline
Zigakibart	Ph.2 extended	UPE of <0.3g/day and <0.5g/day at 100w
Sibeprenlimab	Ph.2 ENVISION	UPCR <0.3g/day (matching KDIGO 2024 guidelines)

2) The LLoQ for Sibe and Pove studies was 50pg/ml, but the LLoQ for BAFF in Pove's study was 391pg/ml, so we cannot accurately say BAFF had similar levels of suppression as APRIL in the 80mg dose. We'll also call out that Telitacicept reported 100% APRIL suppression, but we'll caveat that the LLoQ for its APRIL assay was 780pg/ml (much higher than Sibe and Pove), so we cannot confirm whether APRIL was inhibited to the same extent.

APRIL Assay LLoQ			BAFF Assay LLoQ		
Company	Drug	APRIL assay	Company	Drug	BAFF assay
Otsuka	Sibeprenlimab	50 pg/ml	Otsuka	Sibeprenlimab	-
Alpine	Povetacicept	50pg/mL	Alpine	Povetacicept	391 pg/mL
RemeGen	Telitacicept	780pg/ml	RemeGen	Telitacicept	62.5pg/ml

Source: Jefferies Analysis, company data

On safety, while there's concern that BAFF inhibition may lead to increased infection rates, we haven't seen this translate clearly into trials, as severe infection rates looked similar to pbo for both APRIL-only & APRIL/BAFF (although some diffs in mild infection rates)

	Ph2 ENVISION Study - Sibe anti-APRIL					Ph3 VISIONARY study - Sibe anti-APRIL		Ph1/2 Zigakibart anti-APRIL	Ph2b ORIGIN Atacicept Study anti-APRIL/BAFF						Ph3 Atacicept (ORIGIN 3) anti-APRIL/BAFF		Telitacicept Ph2 anti-APRIL/BAFF		Povetacicept Ph1/2 RUBY anti-APRIL/BAFF		
Safety	Placebo	Sibe 2mg/kg Q4W	Sibe 4mg/kg Q4W	Sibe 8mg/kg Q4W	Sibe Pooled	Placebo	Sibe 400mg SC Q4W	Zigakibart All pts	Placebo	Atacicept t 25mg	Atacicept t 75mg	Atacicept 150mg	Atacicept 150mg & 75mg	Atacicept All doses	Placebo	Atacicept 150mg QW	Pbo	Teli 160 mg	Teli 240 mg	Pove 80mg SC Q4W	Pove 240mg SC Q4W
N	38	38	41	38	117	251	259	40	34	16	33	33	66	82	214	214	14	16	14	21	33
Any adverse event	71.1%	73.7%	80.5%	81.6%	78.6%	82.1%	74.1%	85%	79%	69%	73%	76%	74%	73%	50%	59%	85.7%	93.8%	92.9%	76%	82%
Gr1																				29%	21%
Gr2																				43%	58%
Gr3																				5%	3%
Gr4																				0%	0%
Maximum severity of any adverse event																					
Mild	60.5%	50.0%	53.7%	57.9%	53.8%										34.60%	42.10%					
Moderate	7.9%	18.4%	22.0%	21.1%	20.5%	3.2%	1.5%								11.2	15.9					
Severe	2.6%	5.3%	4.9%	2.6%	4.3%										4.2	1.4					
Adverse events related to drug or placebo	13.2%	18.4%	17.1%	10.5%	15.4%	0.0%	0.4%	27.5%	41%	38%	52%	58%	55%	51%	10%	29%				29%	30%
Serious adverse events	5.3%	5.3%	4.9%	2.6%	4.3%	3.2%	1.5%	2.5%	9%	0%	3%	3%	3%	2%	5%	1%	7.1%	6.3%	14.3%	5%	3%
AEs that led to interruption of tx or placebo	0%	13.2%	2.4%	7.9%	7.7%										2.3	2.3					
AEs resulting in reduction or temporary d/o															3.7	0.9					
AEs that led to discontinuation of tx or placebo	0%	2.6%	0.0%	0.0%	0.9%	1.6%	0.4%		3%	0%	0%	3%	2%	1%			7.1%	6.3%	21.4%	0%	3%
AEs that resulted in death	2.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%		0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%		
Specific AEs																					
COVID-19	42.1%	28.9%	26.8%	34.2%	29.9%			45%	18%	25%	27%	24%	26%	26%						10%	9%
Asymptomatic COVID-19								5%													
Pneumonia	15.8%	13.2%	12.2%	15.8%	13.7%	4.0%	5.4%													14%	15%
Nasopharyngitis	7.9%	10.5%	12.2%	15.8%	12.8%	10.0%	12.4%	10%	3%	0%	3%	9%	6%	5%	6%	8%				29%	27%
Upper respiratory tract infection	0.0%	7.9%	12.2%	5.3%	8.5%	13.9%	14.7%	32.5%	0%	0%	9%	6%	8%	6%	9%	12%	42.9%	31.3%	42.9%	10%	6%
Viral upper respiratory tract infection								5%												0	6%
Headache	10.5%	2.6%	12.2%	7.9%	7.7%	6.0%	5.3%														
Hypertension	2.6%	10.5%	7.3%	0.0%	6.0%															5%	12%
Diarrhea	2.6%	0.0%	9.8%	5.3%	5.1%																
Muscle spasm	2.6%	2.6%	9.8%	2.6%	5.1%																
Adverse events related to infections and infestations system organ class	55.3%	39.5%	56.1%	52.6%	49.6%			77.5%							1%	0%					
Urinary tract infection								7.5%	0%	13%	3%	3%	3%	5%						0	9%
Viral infection									6%	0%	6%	0%	2%	2%						10%	9%
Influenza						6.4%	8.1%	15%	3%	0%	3%	0%	2%	1%							
Tonsillitis									0%	6%	3%	0%	2%	2%							
Injection site reactions						12.0%	13.1%								5	29	0%	56.3%	71.4%	14	9
Injection site erythema						14.3%	11.8%	12.5% (tx related)							0.5	5.6					
Injection site pain						9.2%	10.0%	5% (tx related)													
Injection site swelling						5.2%	6.2%														
Injection site induration																					
Blood IgG decreased																	7.1%	31.3%	28.6%	5%	6%
Blood IgM decreased																	0%	31.3%	35.7%		
Hyperuricemia (elevated uric acid in blood)																	14.3%	18.8%	7.1%	5%	9%
Bronchitis								12.5%													
Sinusitis								10.0%													
Pharyngitis streptococcal								7.5%													
Ear infection								5.0%													
Gastroenteritis viral								5.0%												5%	6%
Rhinitis								5.0%													
Fatigue								5.0%													
Back pain						6.0%	6.6%	7.5% (tx related)												5%	0%
Malignancy																				43%	64%
Any infection																					
Infection Grade 1																					
Infection Grade 2																					
Infection Grade 3																					
Cough						4.2%	5.3%													0%	3%
Peripheral edema						7.1%	1.3%														

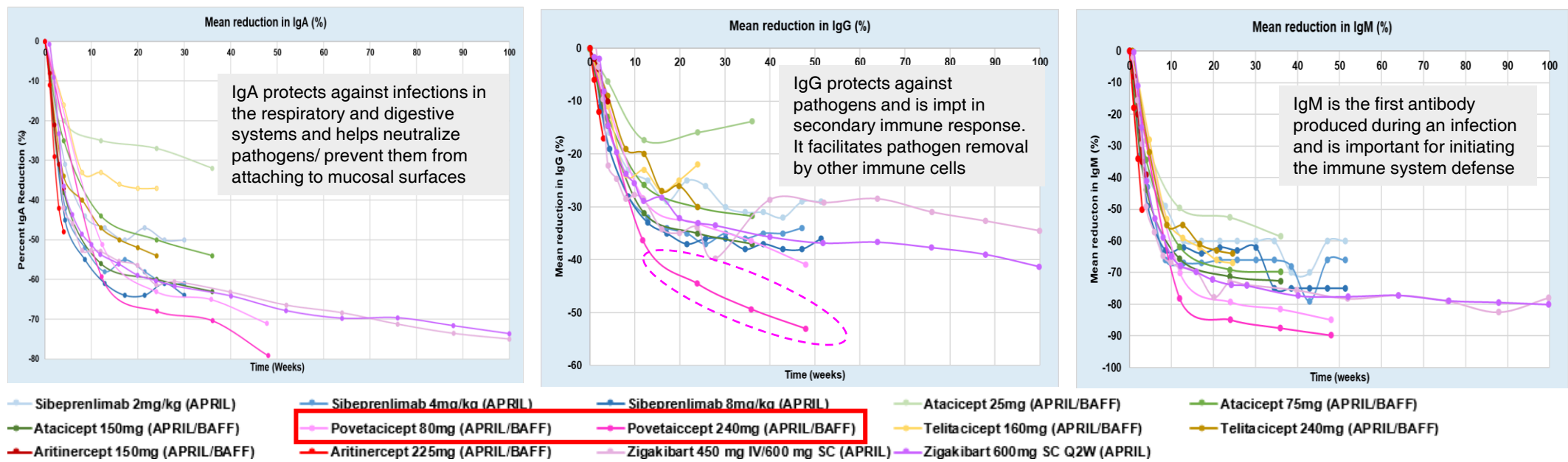
There was no increased risk of infection seen with sipeprelimab (49.6%) vs pbo (55.3%) and majority of AEs were mild or moderate in severity. Incidence of SAEs was equally distributed across sibe and pbo groups w/ none considered tx-related

VRTX mgmt. commented 80mg was well tolerated & most AEs were mild-moderate.

Source: Jefferies Analysis, Company Publications

Generally speaking, we've seen dose dependent decreases in IgA, IgG and IgM w/ both APRIL and APRIL/BAFF inhibitors, which are both important for fighting infections – this could be why pove's 240mg had numerically higher % infections...

We generally saw more significant decreases from baseline in IgM and IgA, followed by lower decreases in IgG. Notably, we see that Povetacicept's 240mg dose reduces these immunoglobulin levels the most vs other anti-APRIL+BAFF mAbs (caveat – there are differences in baseline char), which could explain the more significant infection risk seen with this dose of pove



We flag three points from Sibeprenlimab's Nov 2025 FDA approval: 1) a live-vaccine warning we expect could also apply to Pove, 2) 8% ADA rates that reduced efficacy (pot'l differentiation if Pove maintains its no-ADA profile), and 3) approval for at-home monthly pre-filled syringe use, which sets precedent for Pove's lower-volume (0.46 mL) at-home autoinjector dosing.

Source: Jefferies Analysis, Company Publications

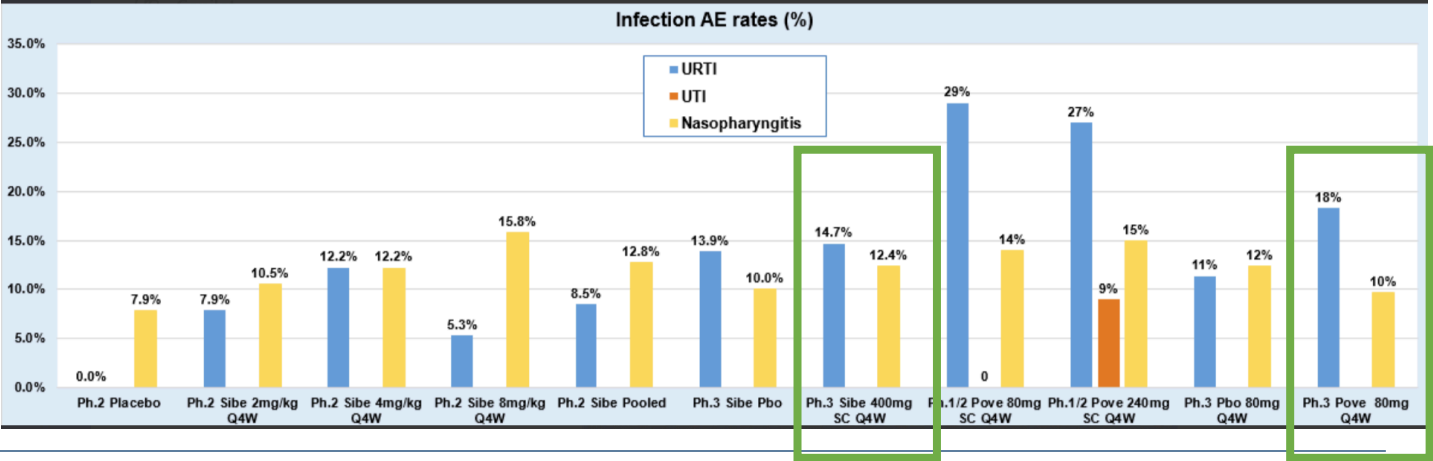
Consistent with our expectations, RAINIER Ph3 interim analysis reported Pove safety profile generally comparable with Sibe Ph3 numbers

Serious AEs occurred in 4.3% (8 out of 185 patients) of the pbo group and 3.0% (11 out of 372 patients) of the pove group. The same incidence of SAEs of infection (0.5%) occurred in both the placebo and pove groups. There were no serious adverse events (SAEs) related to pove and no deaths in the trial. There were no opportunistic infections, and no discontinuations due to infections. Treatment discontinuations (for any reason) were 8.8% in the placebo group and 3.8% in the pove group, and trial discontinuations (for any reason) were 1.5% in the placebo group and 0.8% in the pove group in this interim analysis population

Adverse Events Occurring During the Treatment Period in the Safety Population		
	Placebo N=185 n (%)	Povetacicept N=372 n (%)
Patients with any adverse event during the treatment emergent period	145 (78.4)	279 (75.0)
Adverse events occurring in ≥10% of patients in either group		
Upper respiratory tract infection	21 (11.4)	68 (18.3)
Nasopharyngitis	23 (12.4)	36 (9.7)
Injection site reactions ^a	4 (2.2)	54 (14.5)

^aInjection site reactions (ISR) were identified using a pre-specified set of Preferred Terms (including, for example, injection site reaction and injection site erythema). All ISRs were non-serious and mild or moderate in severity.

Overall safety events reported in the RAINIER Ph3 interim with Pove looks comparable to the rates reported in Sibe Ph3



Source: Jefferies Analysis, Company Publications

The current RAINIER interim PR didn't mention any cases of hypogammaglobulinemia (in pove's prior IgAN Ph.2, 1 severe case was reported w/ pove 80mg & 4 severe cases w/ pove 240mg)...that said, we've seen this reported w/ other drugs w/ this MOA

	Dose	N=	No. Hypo-g Cases	% of cases	Definition	FDA label immune-related monitoring requirements:
Povetacicept Ph.1/2	80mg	21	1	5%	Defined as single read of IgG < 300 mg/dl for participants with IgAN	Not approved
	240mg	33	4	12%		
	80mg pMN	10	1	10%	IgG < 150 mg/dl for participants with pMN.	
Povetacicept Ph.3	80mg	~605 total	-	-	If serum IgG is < 3 g/L on two consecutive timepoints at least 28 days apart, representing severe hypogammaglobulinemia, the study drug will be discontinued. Dose hold if <300 mg/dL until above 300mg/dL. Permanent d/c severe IgG hypogammaglobulinemia (<150 mg/dL) if confirmed by re-testing	
Atacicept Ph.3	150mg	214 (safety analysis)	0	0%	Serum IgG is < 3 g/L on two consecutive timepoints at least 28 days apart, representing severe hypogammaglobulinemia, the study drug will be discontinued.	Not approved
Atacicept Ph.2	25-150mg	82 (all doses)	0	0%	Serum IgG < 3 g/L (300 mg/dL) on two consecutive timepoints at least 28 days apart	
Atacicept SLE pooled (Gordon et al. 2019)	150mg	572	4	0.7%	Severe hypogammaglobulinaemia (IgG <3 g/l)	
Telitacicept Ph.2	160mg	16	5	31% *	Hypo: below the lower limit of normal. On our end, we've seen the NIH define mild Hypo-g is <7g/L and for severe <2g/L, but we can't confirm these were the thresholds used in this study	Not approved
	240mg	14	7	50% *		
Sibeprenlimab Ph.3	400mg	259 (safety)	NR	NR	NR	General infection monitoring language
Rituxan (GPA/MPA)	375mg/m2/wk IV	99 in tx arm	-	58%	Hypogammaglobulinemia (IgA, IgG, or IgM below the lower limit of normal)	Infections: Withhold RITUXAN and institute appropriate anti-infective therapy
Uplizna (NMOSD, gMG, IgG4-RD)	300mg IV day 1 & 15	174	-	4% in NMSOD long-term followup, but 0% in initial trial	Severe Hypogammaglobulinaemia <300mg/dL	Immunoglobulin levels: Monitor the level of immunoglobulins at the beginning, during, and after discontinuation of treatment with UPLIZNA until B-cell repletion. Consider discontinuing UPLIZNA if a patient develops a serious opportunistic infection or recurrent infections if immunoglobulin levels indicate immune compromise.

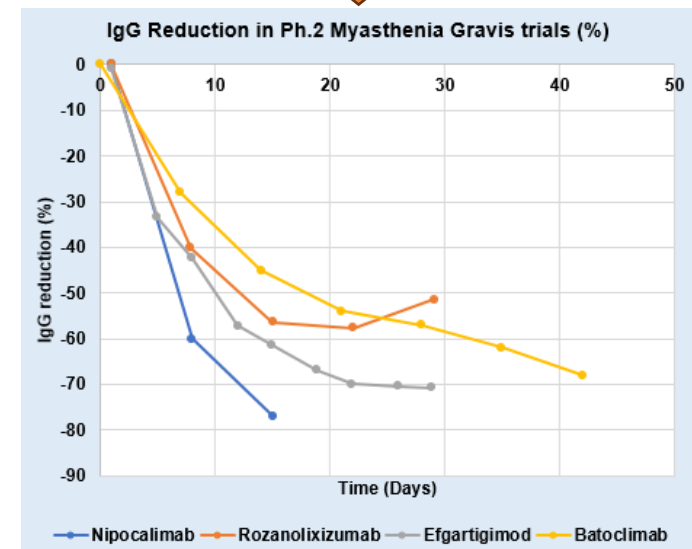
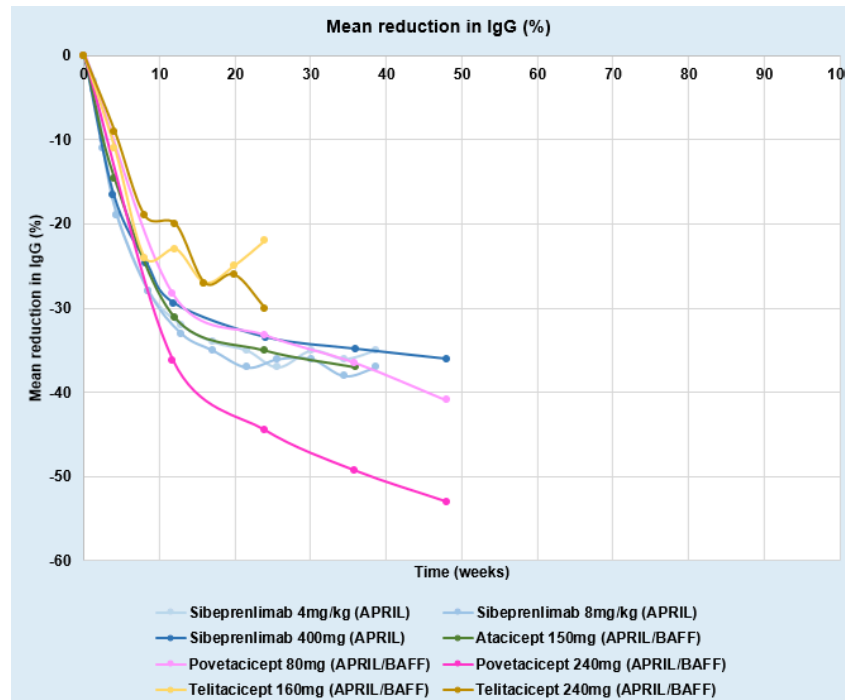
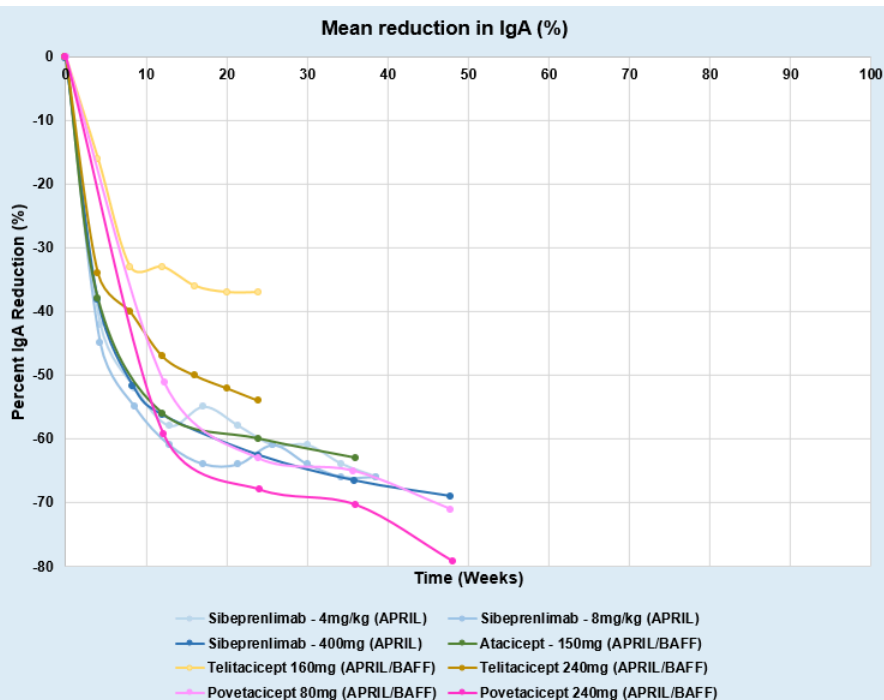
Source: Jefferies Analysis, Company Publications

We'll note that pove 80mg drops IgG generally in line w/ other anti-APRIL/BAFFs or anti-APRIL, including sibe

We noted that at Pove, Ata, and Sibe showed numerically similar reductions of IgA (~62%-67%), and numerically similar IgG reductions (~35-38%) at 36wks. That said, we'll note the IgG reduction trajectory for Pove appears to have it continuing to decline whereas Sibe and Ata appear to be stabilized. Conversely, Teli has numerically lower reductions at its 240mg go-forward dose (~53 for IgA and ~30 for IgG), but we'll caveat their IgG is measured to a 24wk timepoint. To us, Hypo-g is not clearly associated with reduction in IgG or IgA, or appears to be a trajectory-based phenomenon. That said, we can't exclude other mechanisms that may also be contributing to Hypo-g.

We'll flag that Pove has a IC_{50} for APRIL of 3.8nM, Atacicept 20.4nM, and Sibe 5.1nM – making Pove the most potent APRIL inhibitor. The higher potency could be a pot'l explanation why there was a greater Hypo-g response seen in Pove Ph.1/2 so far.

We'll also flag that FcRns (e.g. Efgart, Nipo) lowered IgG above 60% at a faster trajectory (~2wks) vs APRIL/BAFF and this did not lead to reported Hypo-g in pivotal trials (but recall IgG reduction is on target for them)



Source: Jefferies Analysis, Company Publications

Notably, for FDA review/label for Sibe, there weren't stringent infection/Ig monitoring requirements despite a) 31-38% of pts having IgG drop <600mg/dL, b) 9 pts had IgG drop <400mg/dL in Ph.3 – we think Hypo-g is something to track for Pove, but for now, we're not worried

We've seen in the FDA clinical review of Sibe that the FDA expected a higher proportion of pts to have a decline in Ig's below the LLN vs Pbo. Sibe Ph.3 didn't include d/c criteria based on low Ig and no pts d/c'ed due to low Ig. There were ~31-38% pts that had IgG < 600mg/dL.

Sibe FDA Label:	
WARNINGS AND PRECAUTIONS	
- Immunosuppression and Increased Risk of Infections: Before initiating VOYXACT, assess for active infections. During treatment, monitor for signs and symptoms of infection. (5.1) - Immunosuppression and Immunization Risks: Live vaccines not recommended within 30 days prior to initiation of VOYXACT or during treatment. (5.2)	
ADVERSE REACTIONS	
Most common adverse reactions are upper respiratory tract infection and injection site erythema. (6.1)	

We'll flag on label, FDA merely requires infection symptom monitoring, but without specific schedule or biomarker testing.

In Sibe Ph.3 there were 9 pts that had IgG <400mg/dL. While the study did not provide a definition for Hypo-g, we've typically seen in other IgAN studies it was <300mg/dL (e.g. Ata Ph.3). While this was not classified as Hypo-g for Sibe, we'll flag directionally it could suggest there were subjects close to the <300mg/dL threshold and the FDA did not flag this as a concern in their medical review or labelling.

Sibe FDA review

Given its mechanism of action, sibeprenlimab is expected to decrease immunoglobulin levels. As expected, a higher proportion of subjects in the sibeprenlimab group had postbaseline immunoglobulin levels below the lower limit of normal compared with those in the placebo group in the individual studies and the double-blind pool (Table 32). The protocol for Trial 417-201-00007 did not include discontinuation criteria based on low immunoglobulin levels and no subjects discontinued sibeprenlimab due to low immunoglobulin levels in Trial 417-201-00007. For details on sibeprenlimab's pharmacodynamic effects on IgA, see Section 6.1.

Table 32. Subjects With IgA, IgM, and IgG Values Below the Lower Limit of Normal, 9-Month Interim Analysis Safety Set^a, Trial 417-201-0007, Trial VIS649-201, and Double-Blind Pool

Immunoglobulin Antibodies ^b	Trial 417-201-00007		Trial VIS649-201		Double-Blind Pool	
	Sibeprenlimab N=152 n/Ne (%)	Placebo N=168 n/Ne (%)	Sibeprenlimab N=117 n/Ne (%)	Placebo N=38 n/Ne (%)	Sibeprenlimab N=269 n/Ne (%)	Placebo N=206 n/Ne (%)
IgA < LLN (<70 mg/dL)	37/152 (24.3)	0/167 (0)	18/117 (15.4)	0/38 (0)	55/269 (20.4)	0/205 (0)
IgM < LLN (<40 mg/dL)	126/144 (87.5)	5/160 (3.1)	98/116 (84.5)	1/36 (2.8)	224/260 (86.2)	6/196 (3.1)
IgG < LLN (<600 mg/dL)	57/152 (37.5)	3/164 (1.8)	36/117 (30.8)	0/38 (0)	93/269 (34.6)	3/202 (1.5)

Source: Summary of Clinical Safety, Table 2.7.4.4.4-1
^aTrial 417-201-00007 duration is 104 weeks, 26 doses administered once every 4 weeks. Trial VIS649-201 duration is 12 months of monthly doses. Only data in the first 9 months of double-blind treatment from both trials are included.
^bThe subjects' IgA, IgM, and IgG values were ≥ the LLN at baseline.
Abbreviations: Ig, immunoglobulin antibodies (IgA, IgM, and IgG); IV, intravenous; LLN, lower limit of normal; n, number of subjects with ≥1 postbaseline Ig value < the LLN; Ne, total number of subjects with baseline IgG values ≥ the LLN

Sibe NEJM Ph.3

reported in 8.1% and 6.4%, respectively. Nine patients in the sibeprenlimab group had post-

baseline IgG levels of less than 400 mg per deciliter; two of those patients had adverse events (moderate Covid-19 in one and mild urinary tract infection in the other) that occurred during treatment and resolved without dose modification.

Baseline (g/L):	IgG	IgA	IgM
Telitacept Ph.2 240mg	10.65	3.61	0.77
Povetacept Ph.1/2	NR	NR	NR
Atacept Ph.3 150mg	11.65	3.367	1.118
Atacept Ph.2 150mg	10.59	3.2	0.94
Sibeprenlimab Ph.3 400mg	NR	NR	NR
Sibeprenlimab Ph.2 4mg/kg	11.03	3.1	0.9
Sibeprenlimab Ph.2 8mg/kg	10.5	3.27	0.8

	IgG Exclusion Criteria
Telitacept Ph.2	NR
Povetacept Ph.3	IgG < 700mg/dL
Povetacept Ph.1/2	IgG < 700mg/dL or IgA < 10mg/dL
Atacept Ph.3	IgG less than 7g/L
Atacept Ph.2	IgG below 7 g/L
Sibeprenlimab Ph.3	IgG < 600mg/dL
Sibeprenlimab Ph.2	IgG: ≥ 700 mg/dL (inclusion)

We'll also flag the BL IgG, IgA, IgM and exclusion criteria were similar amongst Ph.2 and Ph.3 trials.

Source: Jefferies Analysis, FDA

Moving to the market, we see sig. opportunity for Pove as a) Pove has smaller volume of administration vs Sibe (0.46ml vs 2ml) & is dosed as autoinjector (vs PFS dosing for Sibe) which could give it a compliance advantage

Recall both Pove (0.46ml) and Sibe (2ml) are Q4W dosing, but we're expecting Pove to have autoinjector administration vs Sibe that is given via PFS. Currently, we've not yet seen plans from Sibe to use an autoinjector

	Povetacicept	Sibeprenlimab
MOA	APRIL/BAFF	APRIL
Frequency	Q4W	Q4W
Volume	0.46mL	2mL
Syringe Type	Autoinjector (likely)	Pre-filled Syringe
Admininstration	At home admin (likely)	At home admin (likely)
Time of Administration	1.5 sec	We've not seen reported by Otsuka but precedence suggests ~ 10 sec per ml for PFS

Push the plunger as far as it will go and it stops, and all the medicine is injected (see Figure I).

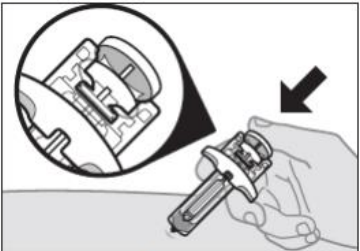


Figure I

Sibe's label shows the Pre-filled syringe is already filled, but still requires manual effort to physically give the injection

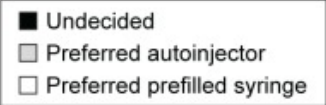
We've seen in a study of Golimumab in UC pts (Vermeire et al. 2018) comparing autoinjector use vs PFS that ~77% of pts preferred the autoinjector admin immediately after first use and then 71% still preferred autoinjector 2 wks later. To us, this suggest so long as efficacy is similar to Sibe, Pove's autoinjector could making it slightly more convenient for pts.

Sibe has self-admin on label as a single dose 2ml PFS Syringe. We haven't seen a reported injection time from Otsuka, but VRTX reports a 1.5 second autoinjector time

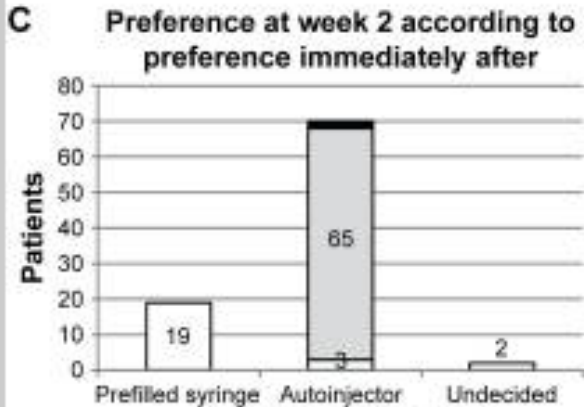
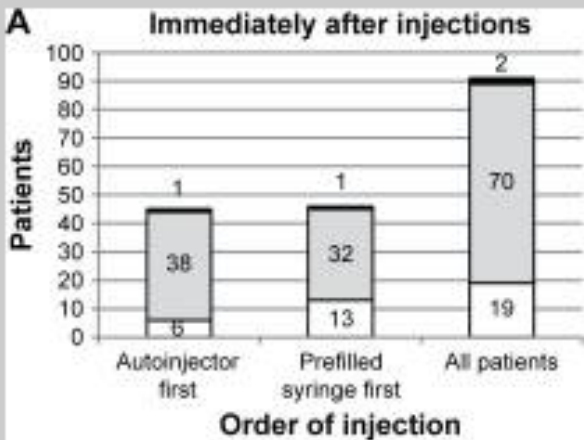
DOSAGE FORMS AND STRENGTHS
Injection: 400 mg/2 mL (200 mg/mL) in a single-dose prefilled syringe.
(3)

2.3 Preparation and Administration

VOYXACT is intended for patient self-administration or for administration by a caregiver. Provide proper training to patients and/or caregivers on the administration of VOYXACT prior to use according to the "Instructions for Use".



Golimumab: PFS vs Autoinjector



Source: Jefferies Analysis, company publications

b) the IgAN market is large (could be as big as ~408k pts vs 330k VRTX est.), Sibe's price was higher vs cons (\$390k/yr vs \$200k/yr cons), and pts are on tx for their lifetime

We've seen from DeCongelio et al. that **the US population estimate for IgAN prevalence is 54.2-62.7/100,000** therefore the high end could suggest ~214k pts in the US alone (assuming 342M ppl). Buisker et al. estimated the IgAN prevalence in **the EU is 4.31/10,000** (wide country variation) suggesting a population of ~194k pts (assuming 450M ppl) in the EU. Taken together, this illustrative analysis could suggest the IgAN market is 408k pts vs ~330k that VRTX is currently estimating.



Source: Jefferies analysis, Company Publication, KOL comments

Initially, cons modeled the price of **Sibe to be ~\$200k/yr, but the launch price was actually set at 390k/yr**. To us, this makes the IgAN market more attractive since Sibe's higher price now paves the way for Pove to match their price assuming efficacy is similar (*which we anticipate it will be*).

Given this, we suspect that Sibe might be under modelled based on TAM.

We'll also note that the majority of IgAN pts are located ex-US & EU. This supports VRTX's thesis that global expansion of Pove's could lead to a +1.5M TAM.

Barbour et al. 2024 reported that the proportion of kidney biopsies that are due to IgAN is highest in Asia (**39.5%**) vs **22.1%** in EU and **11.8%** in North America

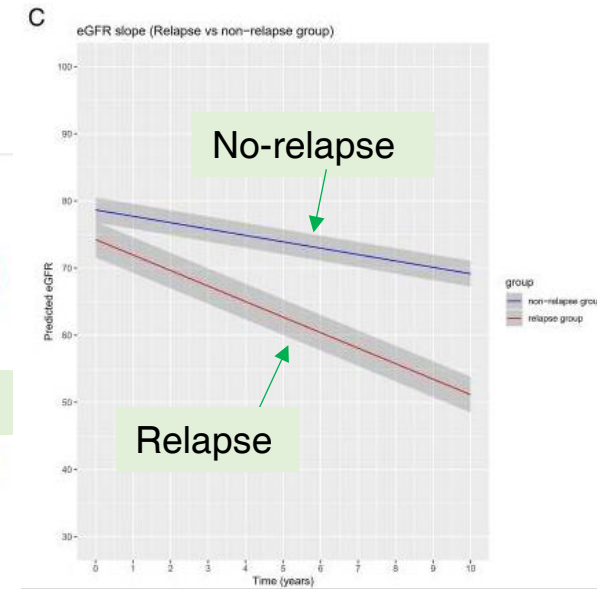
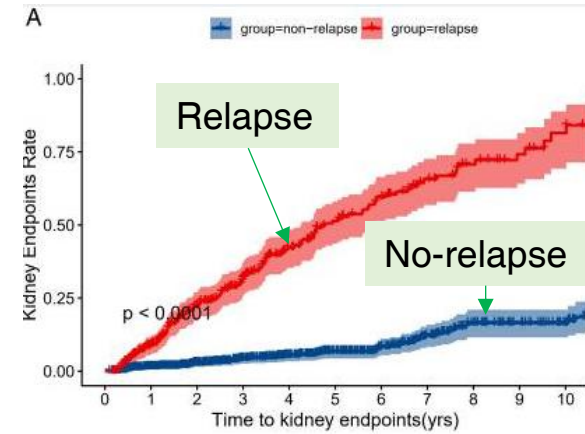
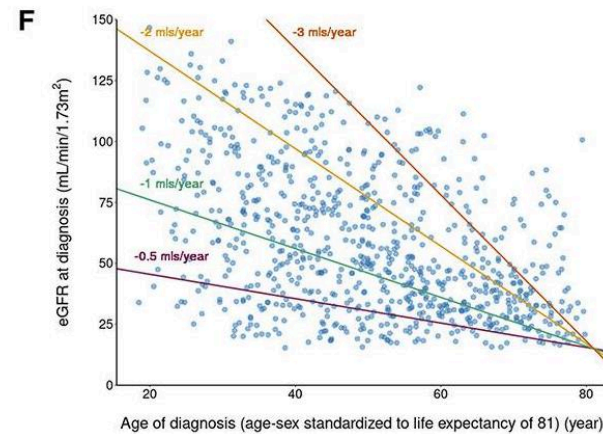
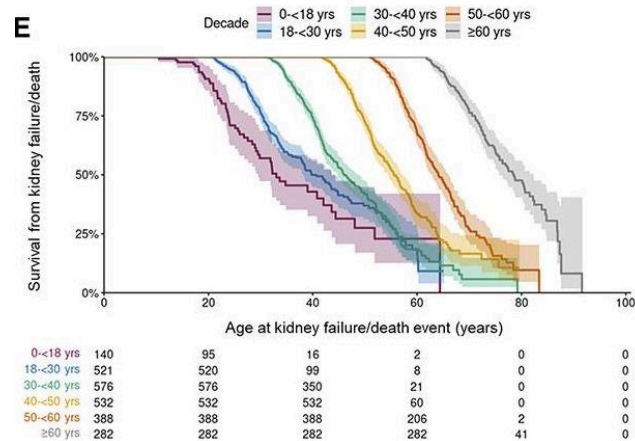
Our KOL previously noted ([HERE](#)) that **IgAN pts are otherwise healthy and typically diagnosed at a young age**. Hence, we expect that pts will require tx for a lifetime. To us, this suggests the market is durable.

At ASN'25, VRTX mgmt. commented that trial recruitment occurred fast (~600pts in 15mo). To us, this is +ve suggesting that there is a high unmet need and high demand for Pove. This also suggests VRTX has large network of pts.

c) We see durability in the market b/c IgAN is a lifelong disease with current median lifespan of ~66yrs (we suspect could be longer with a disease modifying tx like pove) & pove's monthly convenience could make tx easier to adhere to and prevent relapses

a) IgAN is a lifelong disease with a slow disease progression. Hastings et al 2017 reported the median age of diagnosis was ~37y and the median age of death was ~66y with current SoC. Pitcher et al 2019 reported that patients reach kidney failure with variable eGFR decline depending on age of diagnosis (younger age = lower rate of decline). To us, this suggests a drug that is effective at stabilizing eGFR decline (recall SoC does not achieve this) would likely increase life expectancy & also durable use in the market.

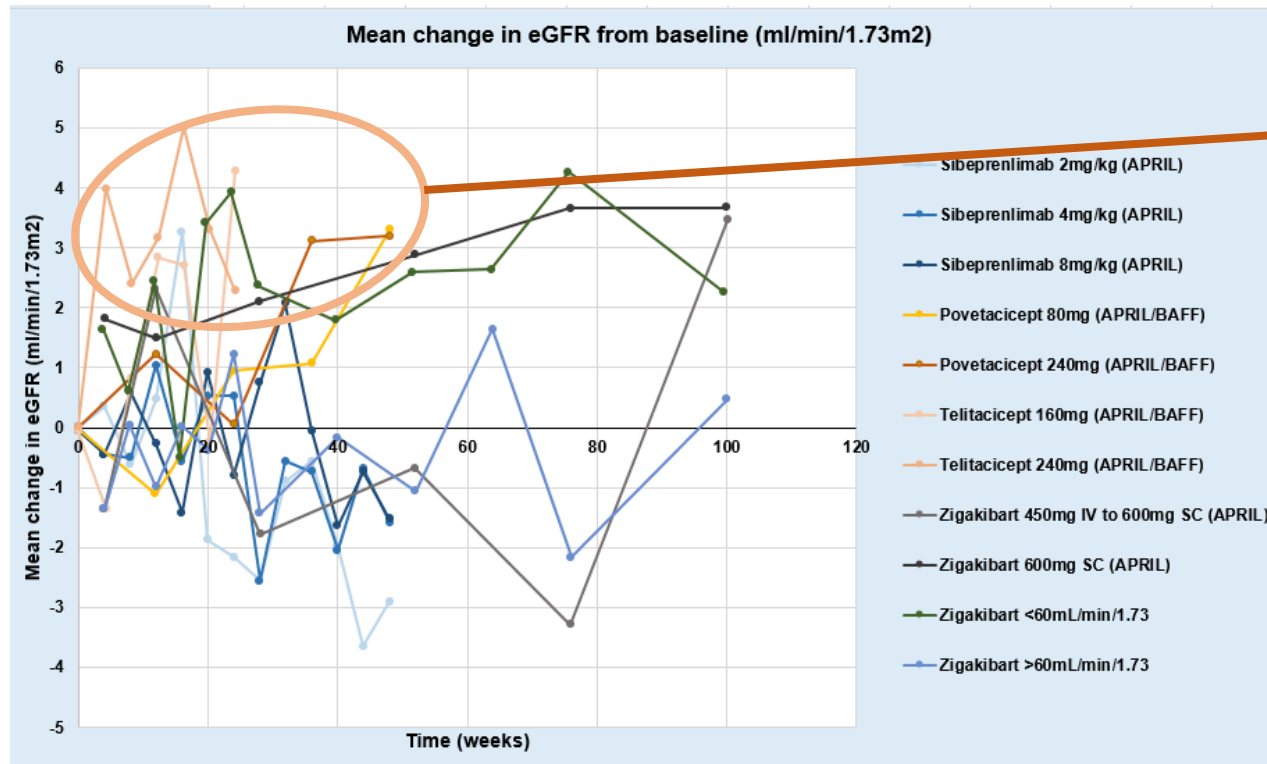
b) We've also seen in Shen et al. 2025 that without consistent tx, relapses can occur and the rate of eGFR decline increases. To us, this suggests that a convenient monthly dosed drug could reduce relapse rates by helping to stabilize eGFR and therefore could improve to durability of tx for Pove.



Source: Jefferies analysis, Company Publications

d) APRIL/BAFF MoA is disease modifying and stabilizes eGFR & hematuria (which we don't see w/ current SoC ACEi & steroids)

We looked at the eGFR mean change from baseline and noted that both MOAs led to stable eGFRs, apart from Sibe 2mg/kg (likely b/c too low dose for efficacy) – any numerical differences are hard to cross-compare given differences in baseline, small n, etc



That said, we noted Pove 80mg and Teli 240mg had the highest increases in mean eGFR, however we'll flag that these results are theoretically improbable b/c, nephrons cannot regenerate and these results suggest that nephrons are regenerating to increase eGFR. Hence, **we expect Pove to achieve stabilized eGFR in line w/ comps, but this won't be a differentiating factor since other drugs can achieve this as well. Acute improvement may be related to reduced local inflammation.**

From ASN'25, we also noted that Pove showed a +3.3ml/min/1.73m2 eGFR at 80mg. While KOLs are unsure exactly why the eGFR is +ve, we suspect it could be due to a reduction in acute glomerular inflammation leading to more vasodilation, but long-term, we're not expecting to see continued eGFR improvement. We'll also note our KOL flagged all the APRIL and APRIL/BAFF therapies have shown stabilized eGFR better than standard of care (ACE/ARB & Steroids)

On hematuria, while Pove, Sibe, Ata, Ziga all showed reductions in hematuria, we can't directly compare due to 1) different sensitivity levels for dipstick detection of hematuria, 2) different definitions of hematuria (e.g. 'trace' vs '1+2+3+'). That said, we're seeing at 36wks, Atacicept showed -81% hematuria reduction and Sibe showed ~-80% hematuria reduction. Pove showed 91% hematuria resolution at 36wks. While Pove may have numerically higher hematuria reduction, in the real-world, we expect this may be clinically similar to comps, therefore not a differentiating factor.

Source: Jefferies Analysis, company data

We prev. spoke w/ KOL Dr. Geoff Teehan ([HERE](#)), Chief of Nephrology at Kidney Care Specialists LLC – he emphasized there is a high unmet need in IgAN & believes novel agents that could stabilize eGFR could gain market share (30~50%+) – he would feel comfortable putting pts on APRIL or APRIL/BAFF provided there wasn't sig. safety risk

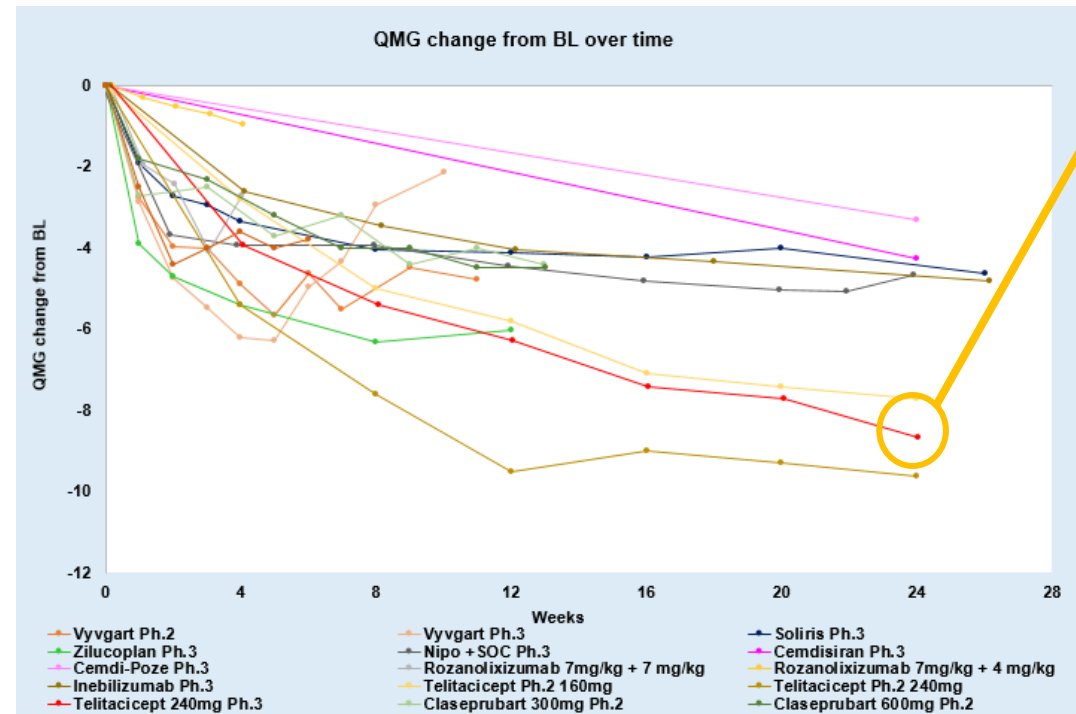
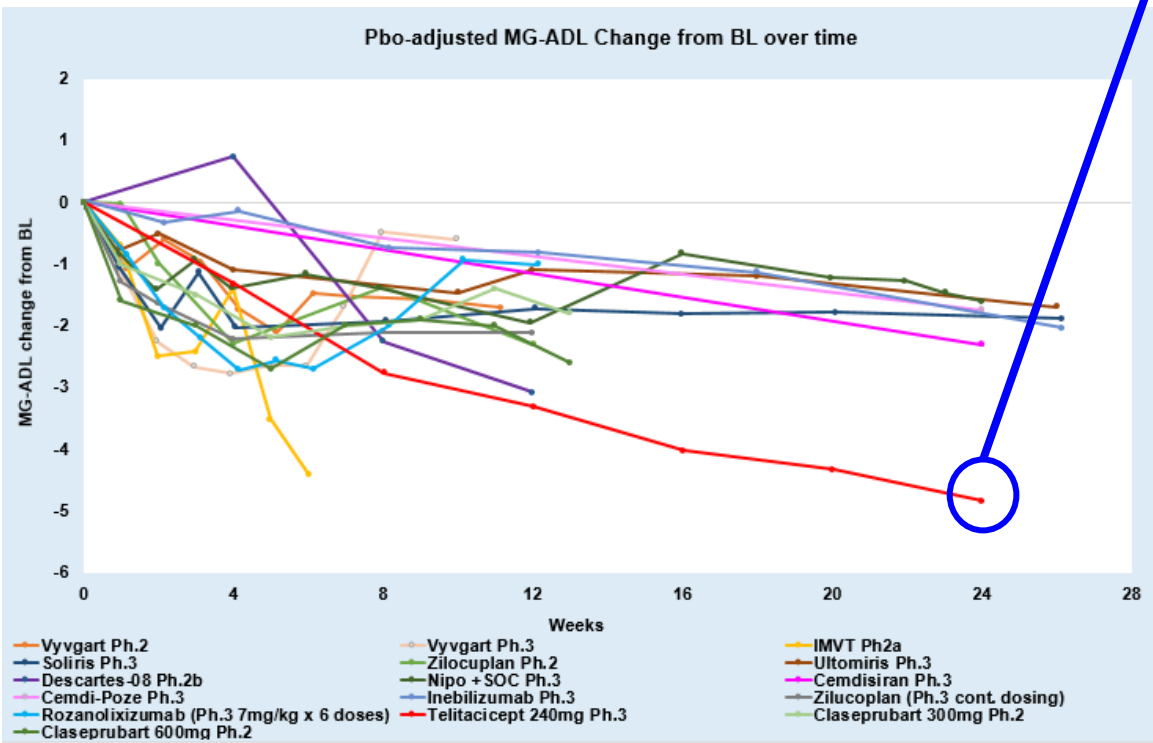
- **On timing of doctor's visits**, our KOL explained that pts with eGFR of <20 typically see their HCP every 1-2 months since they are at risk for needing dialysis or transplant. Further, our KOL stated that any patient with poor biopsy findings must visit their doctor frequently, particularly in the first 6-12 months when experimenting with therapies, regardless of kidney function
- **Regarding the current unmet need**, our KOL believes novel agents seem favorable – specifically, he is interested in the combination of available and novel agents (eg APRIL & dual APRIL/BAFF inhibitors) and noted that he's seen novel combo agents showing 96 weeks of eGFR preservation.
- **On disease treatment**, our KOL also flagged that current drugs on market are costly and that there is a lot of paperwork required to put patients on medications. As a result, he believes there is high unmet need in this indication.
- **On our KOL's patient breakdown**, our KOL stated that he's seen ~50 patients with IgAN in his career & ~10-14 of those were post the development of new medications (patients were prev. tx with ACE inhibitors, statins, fish oil etc.). He believes that 50 is a large number of patients, as IgAN is not a common condition (Sibe trial had ~300 patient in it) – he noted that the patient/doctor ratio may be more meaningful at a long-term GM center where doctors have quaternary referrals vs he is tertiary care and gets referrals from private practice nephrologists, which is divided up among partners.
- **On clinical endpoints**, our KOL stated that eGFR stabilization is most important to him vs proteinuria decline – specifically, he stated that he'd rather have kidney function rather than reduced proteinuria (preserving renal function is the goal). Further, our KOL intends to focus on agents that primarily show eGFR stabilization, though he noted that current trials likely are not long enough to show true signal – therefore, he believes that if UPCR is declining then the drug is probably slowing kidney progression
- **Re: dosing interval**, our KOL believes that a drug with similar/better efficacy vs comps would be favored by prescribers and pts if it had a longer dosing interval. That said, he also noted that typically nephrologist would change a drug if it was not showing adequate efficacy in 9mo b/c nephron loss is permanent so there should not be a large delay in switching to a more effective treatment.
- **On home administration**, our KOL believes that post - Ozempic it is normalized to be taking injections at home w/o attending a clinic. That said, the method to self inject must be taught to pts for optimum adherence.
- **Regarding APRIL/BAFF vs APRIL MOA**, our KOL flagged he was not well versed in the data specifically
- **On safety**, our KOL flagged all the drugs in development are showing generally favorable safety profiles. He could not imagine a scenario where a pts would not opt for a medication that stabilized eGFR based on fear of a mild infection. On balance, there are other IgAN tx (e.g. mycophenolate) that produce worse efficacy with worse safety profiles and those are still prescribed to pts.
- **For market share**, our KOL believes there could be 33% market share for new agents, and this could grow further over time b/c the therapies are targeted for IgAN. That said, a key headwind is IgAN must be diagnosed by biopsy and, since it is an uncommon condition, he did not think that there would be a significant increase in biopsies being performed. He also thought the market would be affected by IgAN severity type – he sees the split as 25% slow progression, 70% medium progression, 5% rapid progression. He believed APRIL/BAFF MOA could gain 30-50%+ of the market.

Source: Jefferies Analysis, KOL

#2) We think MOA also works in gMG since Telitacicept (APRIL/BAFF) had efficacy w/ -4.8 MG-ADL at 48wks, but we'll caveat that this was China-only data, which is tough to compare vs global FcRn data – we think FcRns are still dominant here given faster drops in IgG, but do see value for Pove as 2L in FcRn refractory pts (~30% pts)

Vyvgart's Phase 3 ADAPT trial showed the most rapid decline in MG-ADL of -4.6 (placebo-adjusted -2.8) at week 4. Vyvgart's trial had a larger sample size (n=167). We'll call out that Telitacicept Ph.3 showed a numerically greater MG-ADL decline of -4.83 Pbo adj., but this occurred at 24wks. That said, cross-trial comparisons have limitations due to differences in pt populations, geographies (China-only vs global for Vyvgart), dosing frequencies (Vyvgart Ph.3 allowed for repeat infusion cycles provided there was improvement, whereas Teli had standard weekly 240mg dosing).

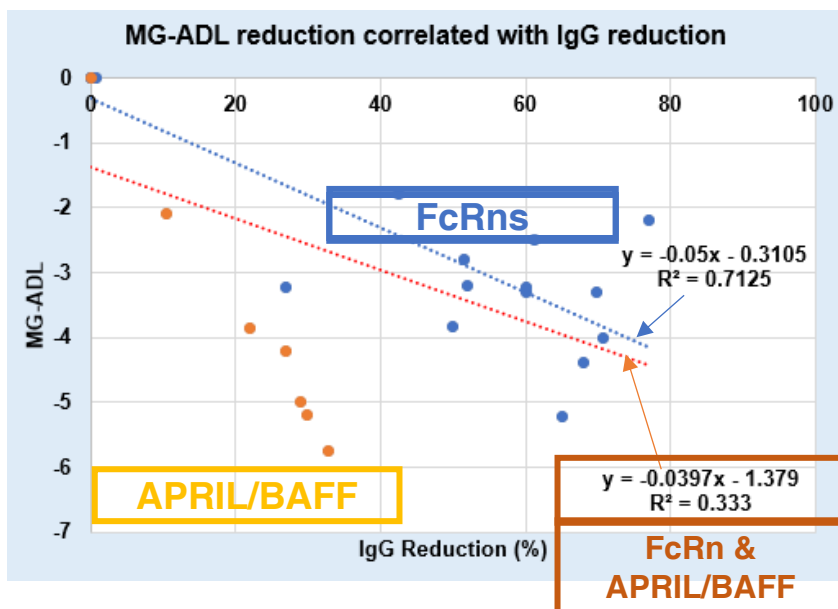
Vyvgart's Ph.3 ADAPT trial showed a week-4 QMG change of -6.2 (-5.2 placebo-adjusted), with dosing ending at 4 weeks. Telitacicept reached ~-8 to -10 in Ph.2 at 160mg/240mg, but in Ph.3 the 240mg dose regressed slightly to -8.66 (Pbo -2.27). Overall, Teli only matched Vyvgart's week-4 effect by week 12, indicating slower onset.



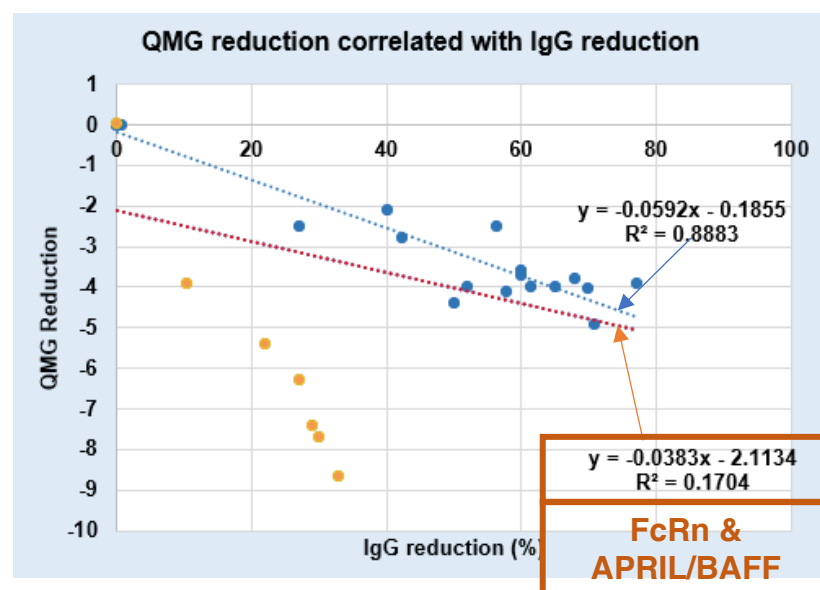
Source: Jefferies Analysis, Company Publications

For FcRN's, IgG reductions are a good predictor of efficacy...that said, we think the relationship scales differently for APRIL/BAFF - Teli had numerically lower IgG reductions than FcRns, but efficacy was generally in line

We noted that w/ the incorporation of Telitacept Ph.3 data into the correlation along with FcRns (red line; orange pts) the strength of the correlation decreased from $R^2=0.71$ & $R^2=0.88$ to $R^2=0.33$ for MG-ADL and $R^2=0.17$ for QMG respectively. To us, this suggests that IgG reduction is not the sole contributor to efficacy.



Telitacept showed numerically lower IgG reduction, but numerically similar MG-ADL reductions thus lowering the correlation strength from $R^2=0.71$ to $R^2=0.33$



Telitacept showed numerically lower IgG reduction, but numerically similar MG-ADL reductions thus lowering the correlation strength from $R^2=0.89$ to $R^2=0.17$

We've seen that the APRIL/BAFF class (inc. Pove & Teli) binds to the APRIL + BAFF ligand, preventing their binding to the BCMA receptor (as well as BAFF-R and TACI). We've also seen that Descartes-08 is a CAR-T that binds and blocks BCMA as well. Therefore, we used the Descartes-08 data to base our correlation.

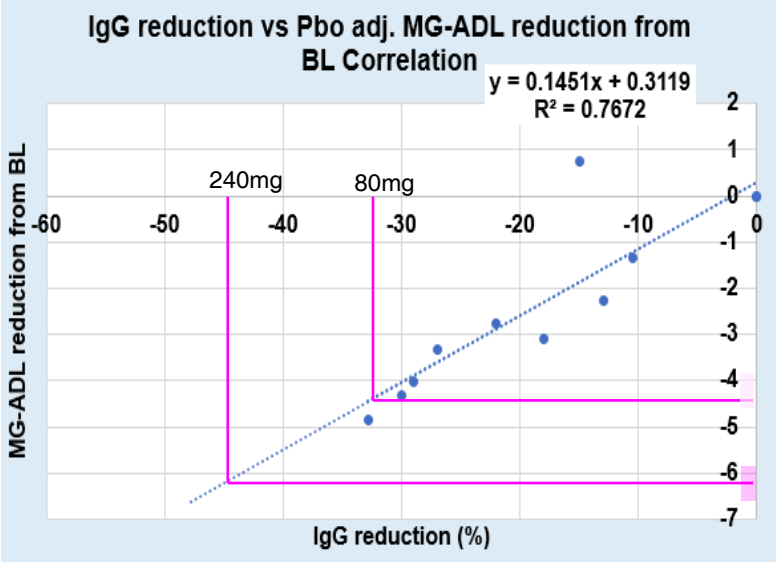
We'll also call out that Teli's data was collected exclusively from Chinese population and we'll flag that, generally speaking, we've seen subjective endpoints (e.g. MG-ADL + QMG) can vary between studies conducted in China vs ex-China.

Telitacept received approval in China for SLE, RA, and gMG based in Ph.3 data. It also received orphan drug designation from the EMA.

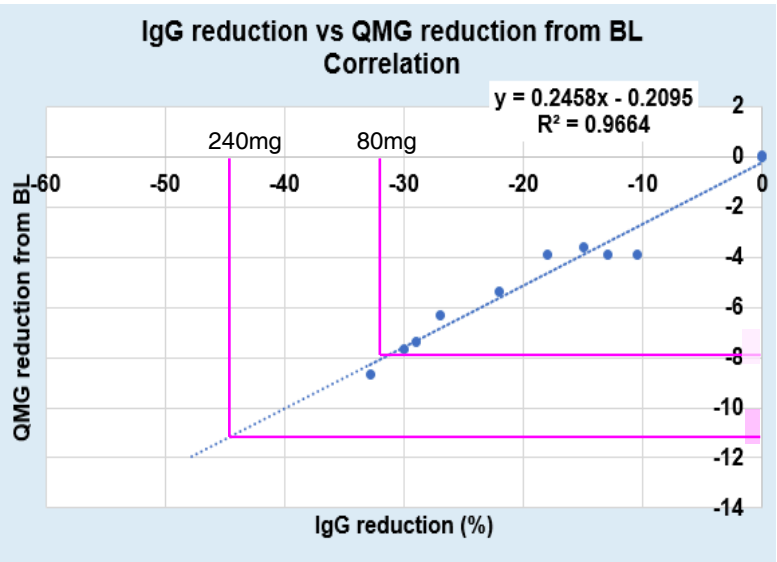
Source: Jefferies Analysis, Company Publications

To estimate what Pove could show in Ph.3, we used Descartes-08 (BCMA targeted CAR-T) & Teli's (APRIL/BAFF) gMG data to correlate IgG reduction and efficacy – we think 80mg Pove could show pbo-adj. MG-ADL reduction of -4.5 to -5 & QMG reduction of -5 to -6 at 24wks, which would look in line w/ Vyvgart at a time point ~6x later

We used the data from Descartes-08 Ph.2b and Teli Ph.3 to produce a correlation for IgG reduction and **MG-ADL reduction ($R^2=0.77$) and QMG reduction ($R^2=0.97$)**. We also noted that Telitacicept showed greater efficacy at higher doses, which would suggest the 240mg dose for pove would be the ideal go-forward dose (but we'll explain why VRTX may select the 80mg dose later...)



We estimate **-4 to -4.5 reduction at 80mg** and **-6 to -6.5 reduction at 240mg** in MG-ADL score at 24wks Pbo adj.



We estimate **-7.5 to -8 reduction at 80mg** and **-10 to -11 reduction at 240mg** in QMG score at 24wks Pbo unadj.

JEF Case for Pove	MG-ADL Change	QMG Change
Bear	< -4.5	< -5
Base	-4.5 to -5	-5 to -6
Bull	> -5	> -6

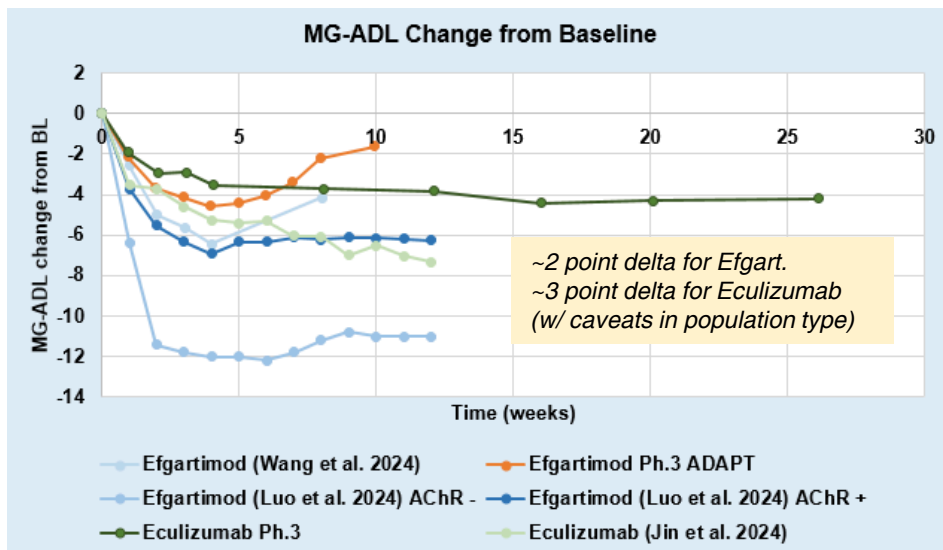
We also caught up w/ VOR mgmt:

- On their Ph.3 trial, mgmt reiterated the China study was protocol-compliant and that Chinese pts are not known to have genetic differences in the disease. They noted Teli controls hyperactivated B-cell clones (and likely tempers T cells, though this is assumed).
- They attributed greater-than-expected efficacy vs FcRn data to the BAFF component: FcRns mainly lower IgG, whereas BAFF reduces IgG plus other inflammatory markers relevant in gMG. Teli is described as closest to “wild-type,” enabling more natural inhibition.
- Mgmt sees Teli’s key advantage vs FcRns as durability: IgG reduction may be slower, but it avoids the rapid rebound seen w/ FcRn discontinuation. They also believe there is room in gMG and data need not exceed Vyvgart to be viable. On affinity, they position APRIL/BAFF closer to atacicept, vs higher-affinity povetacicept.

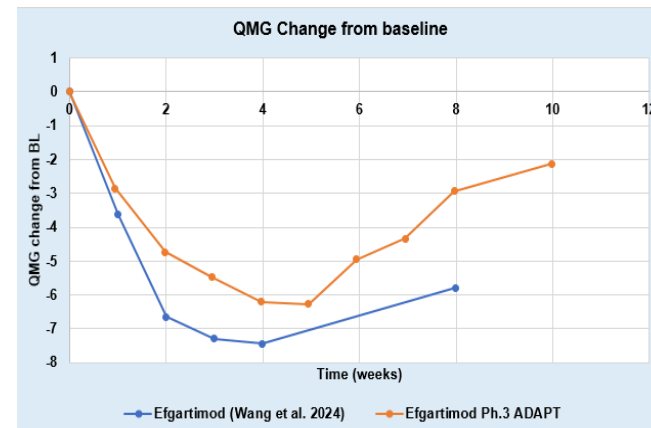
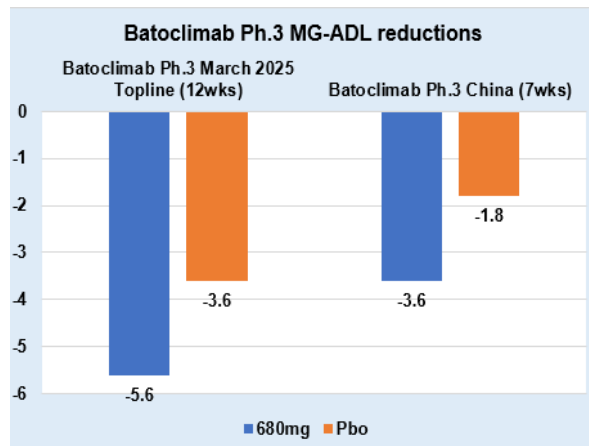
Source: Jefferies Analysis, Company Publications

Keep in mind, we've also seen numerically higher MG-ADL reductions and QMG reductions in Vyvgart studies conducted in China vs ex-China (~2 point delta for MG-ADL & QMG), so we performed similar adjustments to a western population for Teli

In order to determine the read across for Teli 240mg from a Chinese population to a global population (and by doing so, get an estimate for Pove), we compared the MG-ADL for Efgartigimod Ph.3 ADAPT w/ Efgart studies at the same dose in Chinese populations (10mg/kg IV). **We saw that Efgart had the numerically lowest MG-ADL change in their global Ph.3 ADAPT trial (max decline ~4.4).** Likewise, **Wang et al. 2024 showed a numerically greater reduction in QMG score vs Efgartigimod Ph.3 ADAPT (~2 additional pts).** We noted that compared to Chinese studies, there was ~2pt reduction in MG-ADL and QMG scores. (Note: we compared the AChR+ datasets given that the Ph.3 ADAPT was ~69% AChR +pts). For Eculizumab, we noted there was a ~3pt delta, but we don't think this is a good read across given that the trial population was acute exacerbation, impending crisis, or crisis and we don't expect this to be the trial population for Pove.



Bato's Ph.3 topline suggested MG-ADL scores ran ~2 points lower in China vs ex-China for both treatment and placebo at 680 mg, but the timepoints weren't comparable, so we aren't relying on this difference.



Key Trial Design Differences:

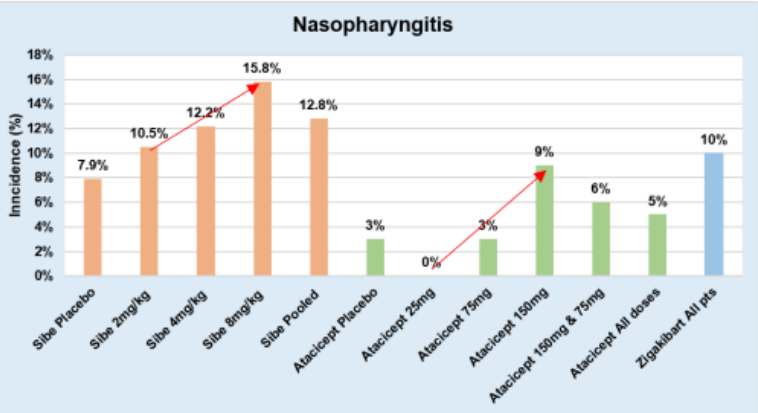
All studies used 10 mg/kg IV, but cycle frequency differed: global Ph.3 allowed up to 3 cycles, Wang et al. had 1, and Luo et al. used a flexible cycle design. ADAPT had the largest sample (n=129; 84 tx, 65 AChR+), vs Wang et al. (n=14, all AChR+) and Luo et al. (n=61; 54 AChR+). Wang et al. allowed concomitant therapy in the treatment arm, while ADAPT and Luo et al. did not.

Source: Jefferies Analysis, Company Publications

For Pove's APRIL/BAFF MOA, we see a dose dependent increase in infection risk (overall infection 43% at 80mg, 64% at 240mg) suggesting BAFF could pot'l limit higher dosing when it's needed to push efficacy

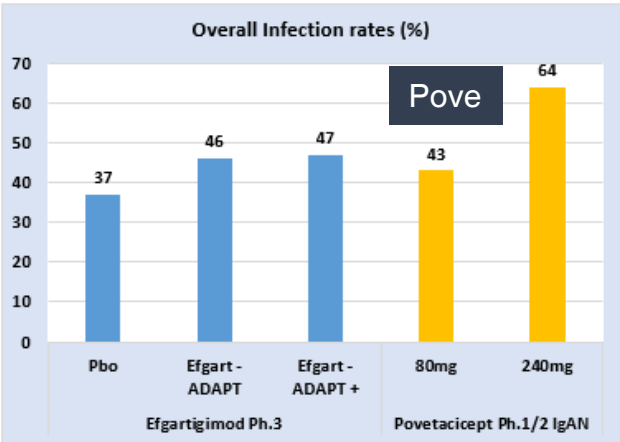
For Teli, we've not seen a notable increase in infection rates (e.g. URTI: ~29% at 160mg vs ~20% at 240mg in gMG), and similarly for IgAN at 160-240mg (~25-50% UPCR reduction, respectively; URTI: ~0% pbo adj). However, for pove (VRTX's APRIL/BAFF), we've seen dose-dependent overall infection increase in IgAN (~43% at 80mg vs ~64% at 240mg). Nasopharyngitis was dose dependent for Sibe (APRIL) and Ata (APRIL/BAFF). So far, we're not seeing infections as limiting for Teli, given we haven't seen a dose-dependent risk, which could be due to pove's higher APRIL/BAFF potency vs Teli (Kd ~59pM vs ~491pM for BAFF; 1pM vs undetectable for APRIL; Evans et al) making it more difficult for pove to thread the needle on efficacy/safety.

We'll flag Teli suppressed APRIL 100% btw 80-240mg in Ph.1, but had lower BAFF suppression than VRTX's APRIL/BAFF - povetacicept (45-60% Teli vs 78% Pove at 80mg). Interestingly, Otsuka's APRIL-only (sibe) inhibited BAFF 35% at 6mg/kg, but APRIL still drove efficacy. To us, it appears efficacy of Teli is largely driven by APRIL suppression – further suggesting you don't need BAFF



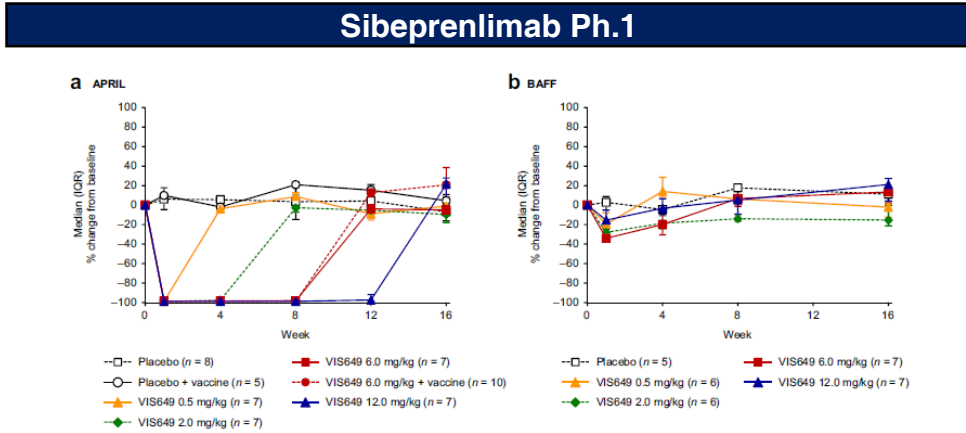
We caught up w/ JBIO mgmt. who reiterated that Nasopharyngitis is a mild infection, not dose limiting.

Source: Jefferies analysis, Company Publication



Telitacicept Ph.1					
Check Index Visit	Statistics	80 mg Group (N = 12)	160 mg Group (N = 12)	240 mg Group (N = 12)	Total (N = 36)
BlyS (pg/mL)	Mean ± SD	73736.6 ± 18724.2	117440.7 ± 14059.2	158662.8 ± 25707.2	116613.3 ± 40209.2
	Median	70208.3	116465.4	165223.1	115683.8
Day 57	Mean ± SD	33156.0 ± 14757.2	62312.2 ± 21308.8	95978.2 ± 16855.1	63815.5 ± 31275.6
	Median	30348.5	56976.6	90596.5	61344.5
APRIL (ng/mL)	Mean ± SD	0 ± 0	0 ± 0	0 ± 0	0 ± 0
	Median	0	0	0	0
Day 57	Mean ± SD	0 ± 0	0 ± 0	0 ± 0	0 ± 0
	Median	0	0	0	0

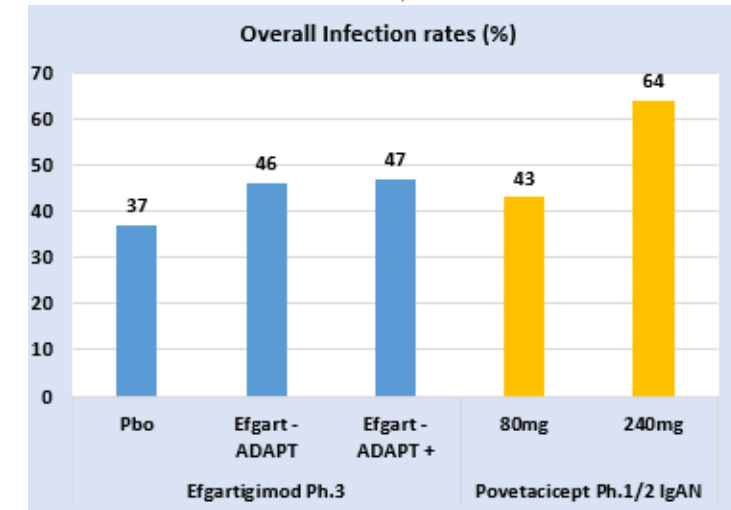
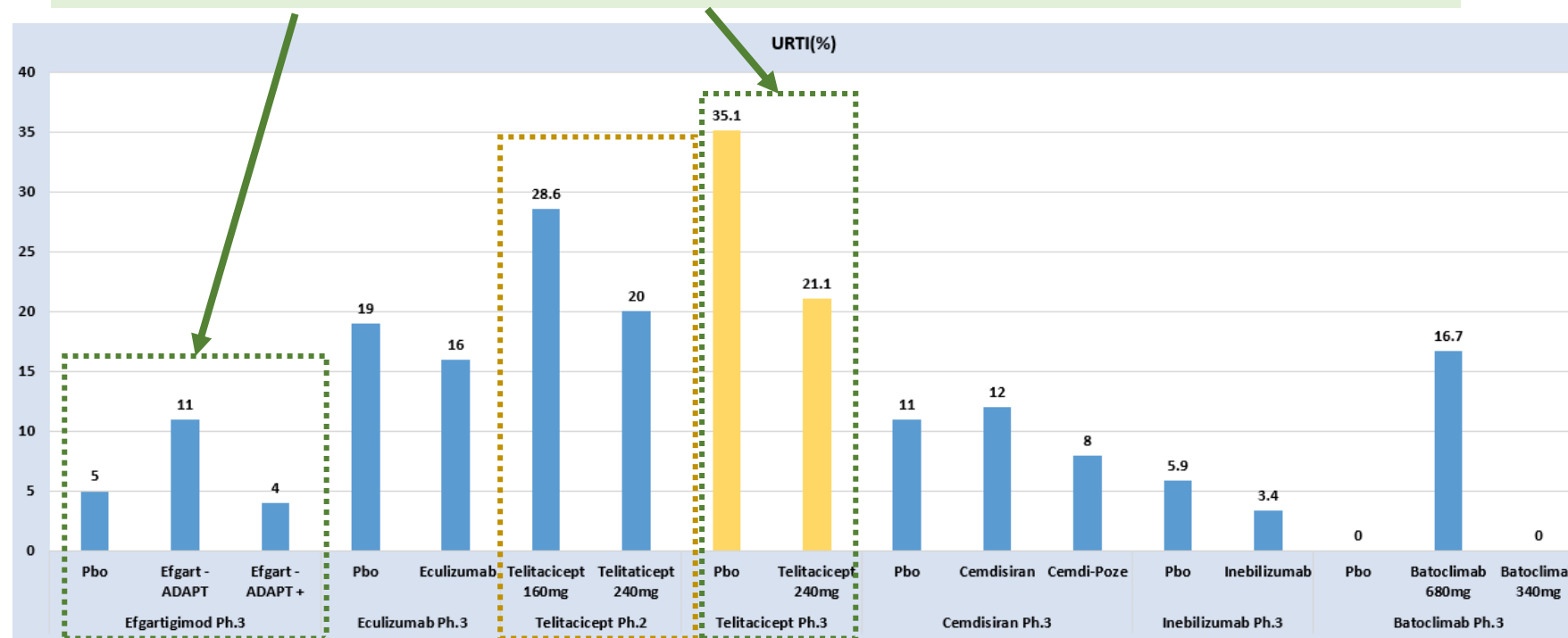
Change value = postdose value – baseline value.



To us, since Teli also showed numerically higher URTI rates vs Vyvgart (21% vs 11% tx arms), it wouldn't make sense for Pove to be used 1L, but infection rates still look manageable such that there's still opportunity in post-Vyvgart refractory setting

We'll call out that Teli had the numerically higher rates of URTI (21.1%) vs Vyvgart (11% Ph.3 adapt), however this is difficult to interpret given the Pbo AE rate was higher than Teli 240mg (35.1%).

Unlike Teli in gMG, Pove had a dose dependent AE response with infection (43% at 80mg vs 64% at 240mg. We'll also flag that Pove's overall infection was numerically similar to Vyvgart (caveat different indications), but it was not Pbo controlled.



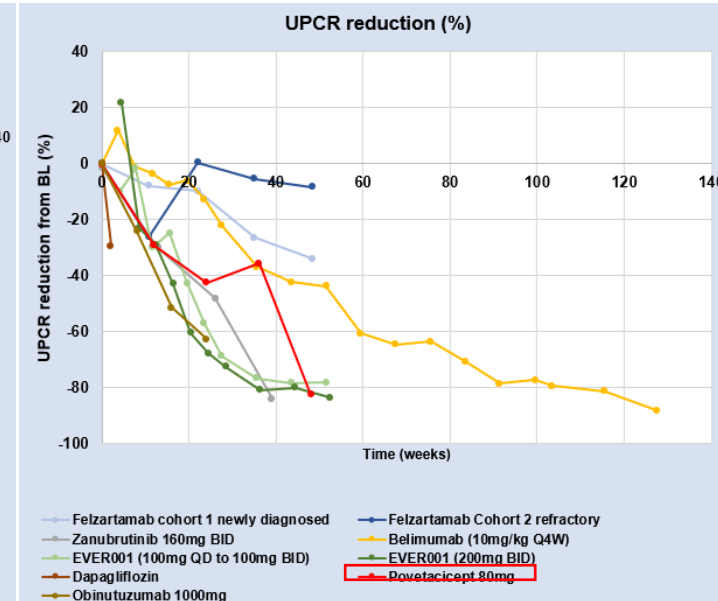
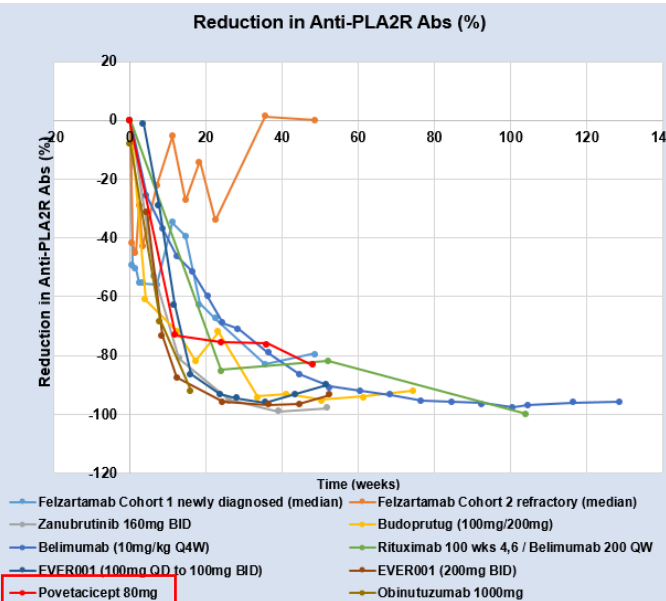
Ph.2 gMG, Teli 240mg dose had numerically lower URTI rate vs 160mg (28.6% vs 20%) meaning the high dose could be selected as go-forward since it did not increase the infection risk. **We suspect the APRIL-only MOA could also avoid infections seen w/ pove's APRIL/BAFF MOA while pot'l getting to similar efficacy in some indications (but it's TBD)** - for sibeprenlimab (APRIL-only), we've seen ~53% AE's related to infections at the highest tested 8mg/kg dose in Ph.2 IgAN which was lower vs ~64% infections for pove 240mg (caveat differences in AE reporting).

Source: Jefferies Analysis, Company Publications

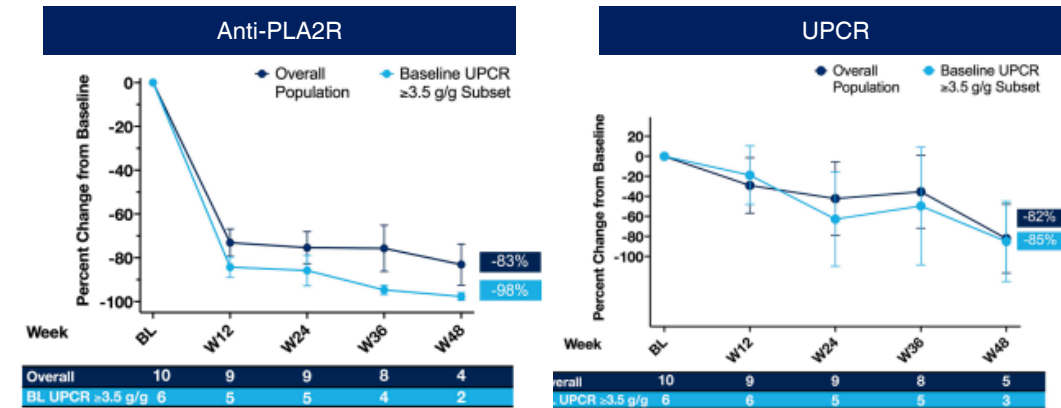
#3) For pMN, Pove showed -82% UPCR reduction and ~83% Anti-PLA2R reduction in Ph1/2 at 48wks (numerically similar UPCR to comps at 48wks, but caveats on sample size) – to us, Pove doesn't have to be best in pMN to still generate notable revs given pricing for the class is high

Primary membranous nephropathy (PMN) is an autoimmune kidney disease in which antibodies (most often against PLA2R) damage the glomerular basement membrane. This leads to immune complex deposition, impaired filtration, and significant protein loss in the urine. PMN mainly affects adults and can cause swelling, high cholesterol, and progression to kidney failure if untreated.

We noted that **Zanubrutinib** and **EVER001** showed numerically the greatest percent reductions in **Anti-PLA2R Abs** (~95% reduction by 36wks, but caveats in sample size). Pove showed a reduction in PLA2R antibodies of 83% at week 48 which puts it in numerically lower than the efficacy leaders (*Caveat: n=4 for Pove*). However, pove also showed ~82% UPCR reduction at 48w, which we're seeing is in line w/ Zanubrutinib and EVER001. To us, **Pove is positioned to be a joint leader for pMN efficacy, but we're still watching to see if the decline in both biomarkers continues with time given small sample size in Ph.1/2. Recall, Sibe's price was \$390K/yr in IgAN so to us, pMN still works based on the price favorability.**



In Ph.1/2 Pove showed 83% Anti-PLA2R reduction and 82% UPCR reduction at wk48 (caveat n=9 starting sample size)



At 48 weeks, all pts reached immunologic remission (n=4 at wk48), and UPCR continued to decline. Pts showed **100% partial and 40% complete remission; despite the small sample (n=5 at wk48)**

VRTX estimates the addressable pMN market is ~150K in US and EU (4Q25 earnings).

Povetacipt – pMN

~150K
(>600K globally)

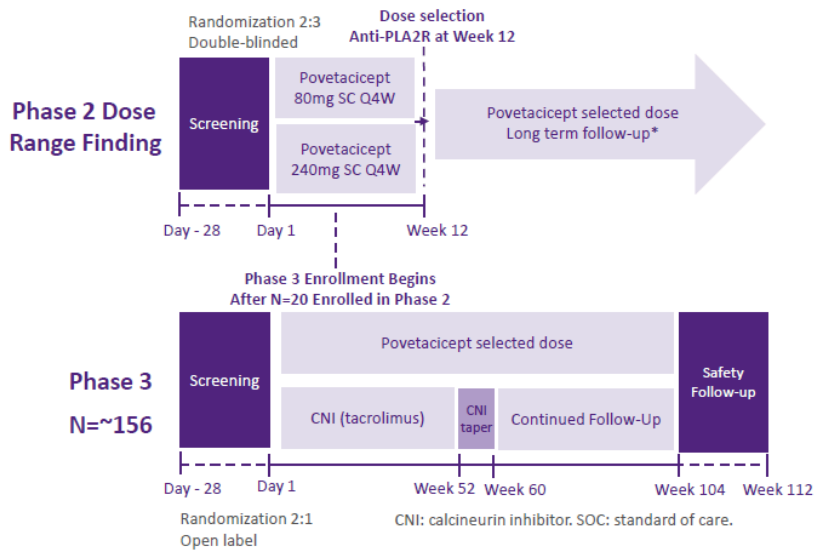
Phase 2/3
pivotal trial ongoing

Source: Jefferies Analysis, Company Publications

We model ~119K pts in US & EU, but Zhang et al. 2018 reported 1.2/100,000 incidence rate in US & EU (work out to ~9400 cases per year), which directionally support VRTX's ~150K prevalence assuming 10-15 yr disease duration.

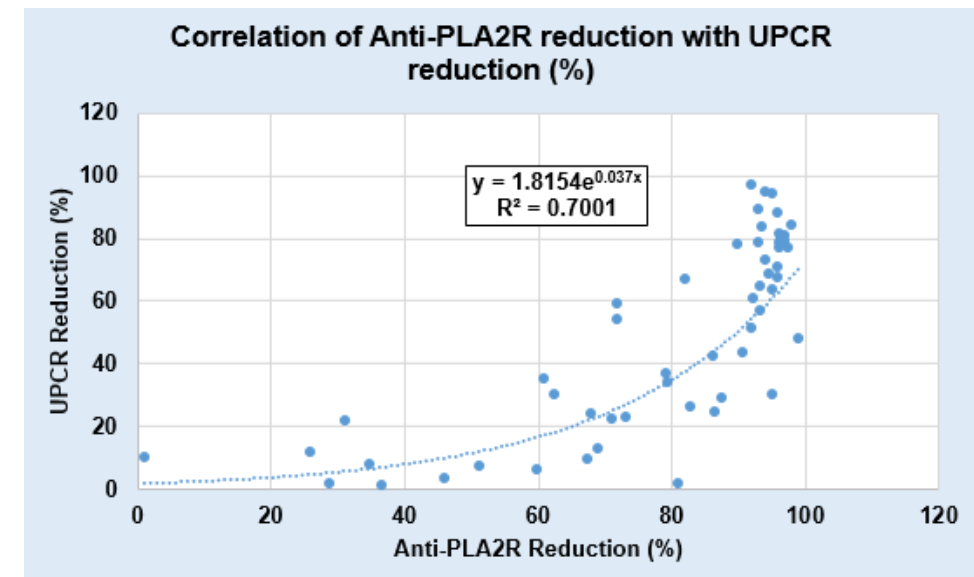
Our illustrative Anti-PLA2R to UPCR correlation ($R^2=0.7$) suggests Pove could achieve ~40% UPCR reduction at 104wks for OLYMPUS Ph.3, but Ph.1/2 data suggests it could be higher, as the correlation fit is not as strong past 80% Anti-PLA2R reduction

We'll flag the OLYMPUS trial (Ph.2/3 is VRTX's pivotal study to evaluate Pove vs Tacrolimus. PCD is Dec 2028. As a background, OLYMPUS is a Phase 2/3 (n~176). In the Phase 2 portion, pts will be randomized to pove 80mg vs 240mg. In the Phase 3 portion, pts will be randomized to receive either the selected dose of pove or a calcineurin inhibitor (tacrolimus). Final analysis will occur at two years of treatment, with a primary endpoint of proportion of participants with complete clinical remission at Week 104.



Atacicept (VERA's APRIL/BAFF) is also being tested in pMN (PIONEER trial; PCD Nov 2027). We est. Pove could show greater efficacy based on greater Ig reductions (IgA +5%, IgG similar, and IgM +10% reductions) in IgAN

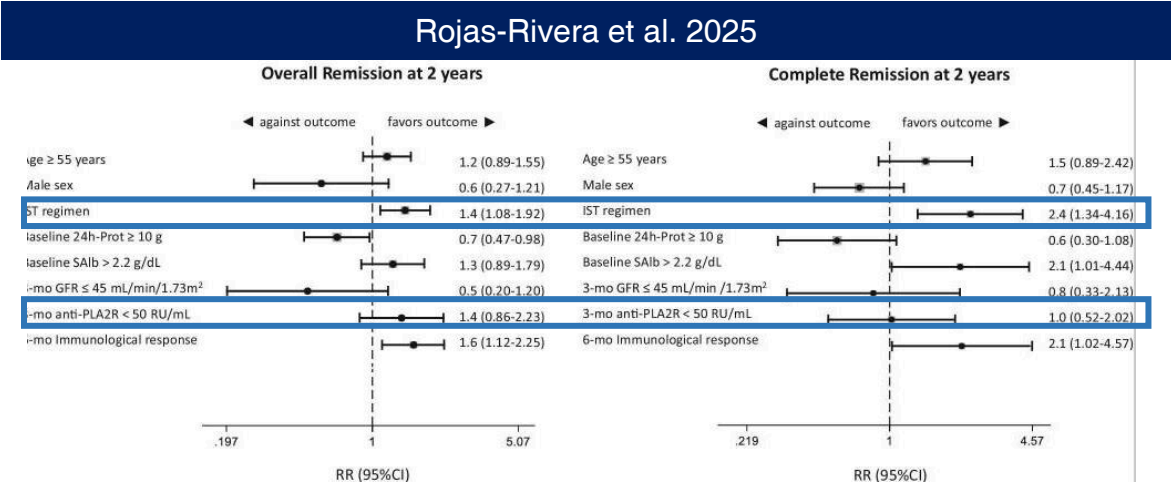
We plotted the Anti-PLA2R reduction for current comps in pMN with UPCR reduction and found a +ve correlation ($R^2=0.7$). That said, we think the high variability of data btw drugs makes this correlation not accurate at predicting UPCR reduction especially above 85% reduction where the line fit does not appear as strong. Given that -83% Anti-PLA2R was achieved at 48wks in Ph.1/2 and it continues to trend down, our correlation would suggest ~40% UPCR reduction in OLYMPUS by 104wks. That said, we've already seen in RUBY-3 that the highest achieved was ~80% UPCR reduction (n=4 at 48wks) so the expected Ph.3 correlation might be higher than 40% - to us the small sample size makes it too early to tell based on noise.



Source: Jefferies Analysis, Company Publications

Baseline characteristics for Pove looked in line w/ comps (age=58.2, duration=15.6mo, proteinuria=3.8g/g, eGFR=78.1mL/min/1.73²), but we’ve seen evidence prior IST use was associated w/ better clinical outcome (Pove had 10% prior use)

Pove’s had BL age=58.3, which was in line w/ comps, but disease duration=15.6mo which was numerically higher vs comps. Proteinuria=3.8g/g which was lower than most comps, but eGFR=78.1ml/min/1.73m² which was clinically in line). We’ll flag that the Anti-PLA2R titer was numerically in line w/ comps (139.5RU/mL) but we’ve seen evidence from Rojas-Rivera et al. 2025 that lower Anti-PLA2R levels are directionally associated with higher rates of overall/clinical remission (but not stat. sig). Therefore, we don’t think this would have favorably skewed the results for Pove. We also noted that prior immunosuppressive therapy (IST) was associated with favorable outcomes for the drug, but Pove reported numerically the lowest (10%) prior use.



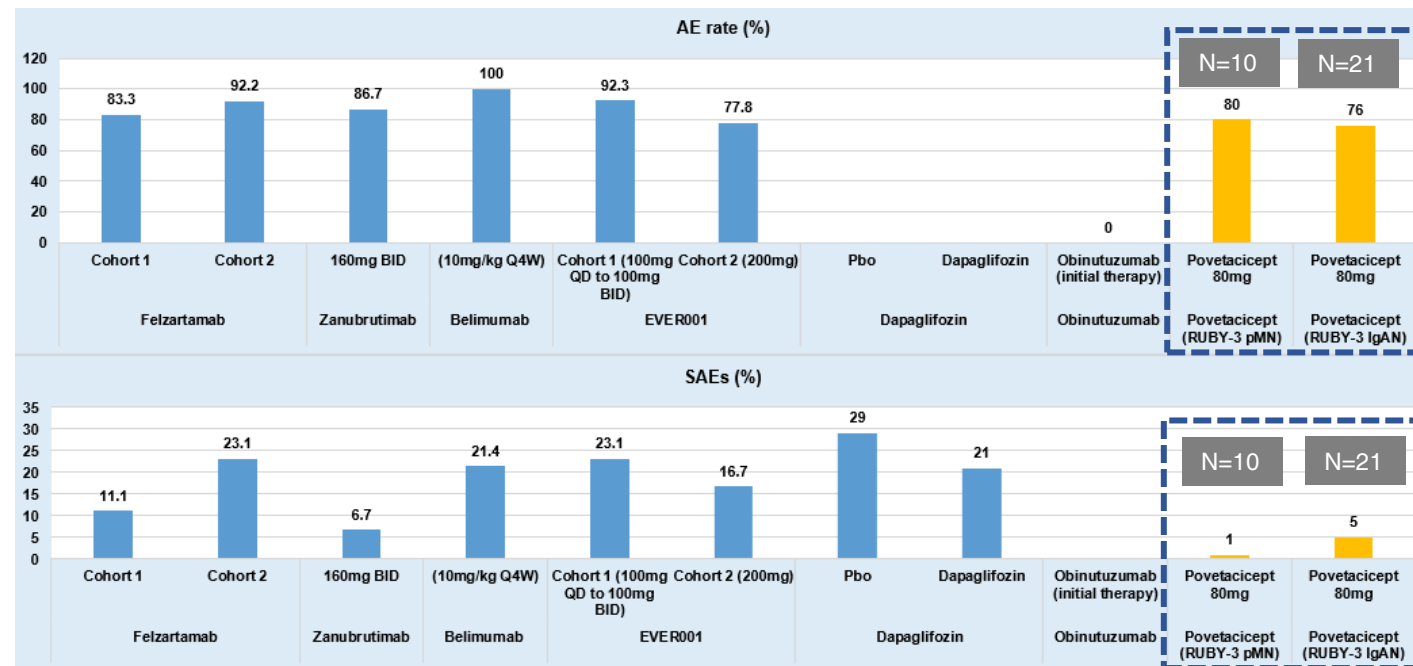
Drug	Sample Size in tx arm	Age (yr)	Sex (% Male)	Ethnicity	Disease Duration (months)	Prior Immunosuppressive Therapy (%)	Anti-PLA2R Titer (RU/mL)	Proteinuria	eGFR (mL/min/1.73m ²)
Belimumab	14	48	79	UK-based	6	21	168	11.7 g/day	78
EVER001	31			Chinese		0	>20	~6.4 g/day	Stable
Felzartamab	31	57.5	77	Mixed (EU, US, Korea)	11.9	48	161	6.1 g/g	61
Zanubrutinib	30	46.5	66.7	93.3% Asian	3.1		161	7.5 g/g	85.2
Dapagliflozin	19	61.5	68.4	47% White 42% Asian				1.6g/g	46.9
Budoprutug	n=5 for protenuria; n=3 for Anti-PLA2R						~71		
Obinutuzimab (initial tx)	20	50.1	55.1			0	81	5.2g/day	98.4
Povetacicept	10	58.2	80	20% Asian, 40% white, 40% Black	15.6	10	139.5	3.8g/g	78.1

Source: Jefferies Analysis, Company Publications

On safety, Ph.1/2 AE rates were in line w/ comps (n=10) - we suspect there could be slight regression to ~76% overall + Pove's overall infection rate (43% in IgAN) could remain numerically lower than Belimumab (86%)

Pove showed 80% AE rate and 1% SAE rate, but we'll flag a low n=10. We'll also call out that Belimumab (BAFF-only) had numerically high AE rate (100%) and an SAE rate of (21.4%). Like w/ efficacy, we think the question is whether this holds for Pove with a larger n. To scale the AEs, we looked at the AE rate from Pove in IgAN pts which showed AE rates of 76% and 5% SAE rate at 80mg. We're seeing the AE rates were in line w/ Pove in IgAN, so we're not expecting them to change much. Nonetheless in both pMN and IgAN the majority of AEs were mild therefore we are not concerned by the safety profile should it be numerically higher than the competition.

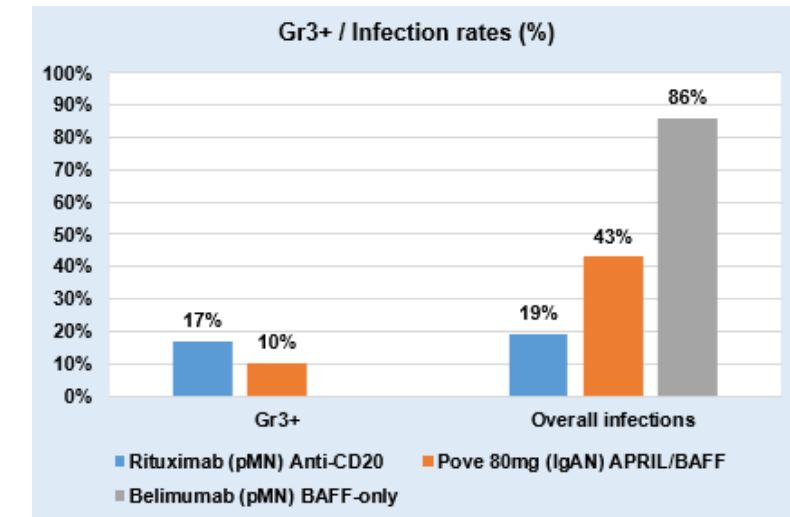
Aside from Pove, AEs looked in line across the drugs btw 80-100% (apart from EVER001 cohort 2 that showed 77.8% AE rate). SAEs were also similar across studies w/ 10-25%, but we'll flag that Zanubrutinib had the lowest numerical SAE rate (6.7%), aside from Pove 80mg (1%).



**Caveat: different MOAs are associated with different AE rates and different individual AEs so cross trial comparison is difficult is due to lack of standard reporting*

Source: Jefferies Analysis, Company Publications

We noted that Pove's 80mg IgAN data (n=21) showed lower rates of Gr3+ AEs and (17 vs 5%) and lower overall infection rates vs Belimumab (43% vs 86%) but higher than Rituximab (19%). We're watching to see what the infection rates are for other comps to determine if this is effective.



#4) wAIHA: Pove is being tested in RUBY-4 (basket trial inc. wAIHA), but we're still not seeing a strong R=0.41-0.53 w/ IgG (though we expect it contributes to disease), making it difficult to correlate pove's MOA to efficacy in wAIHA

We'll call out the RUBY-4 trial is a basket Ph.1/2 trial that will include pts with wAIHA. It has a PCD of July 2026 and we expect the data to be used to support a label expansion of Pove into wAIHA. We'll also flag that it has a small n (n=30 pts total) and we expect that wAIHA would not make up the majority. VRTX mgmt. commented they anticipate ~35K pts with wAIHA in US & EU. We suspect this may be conservative.

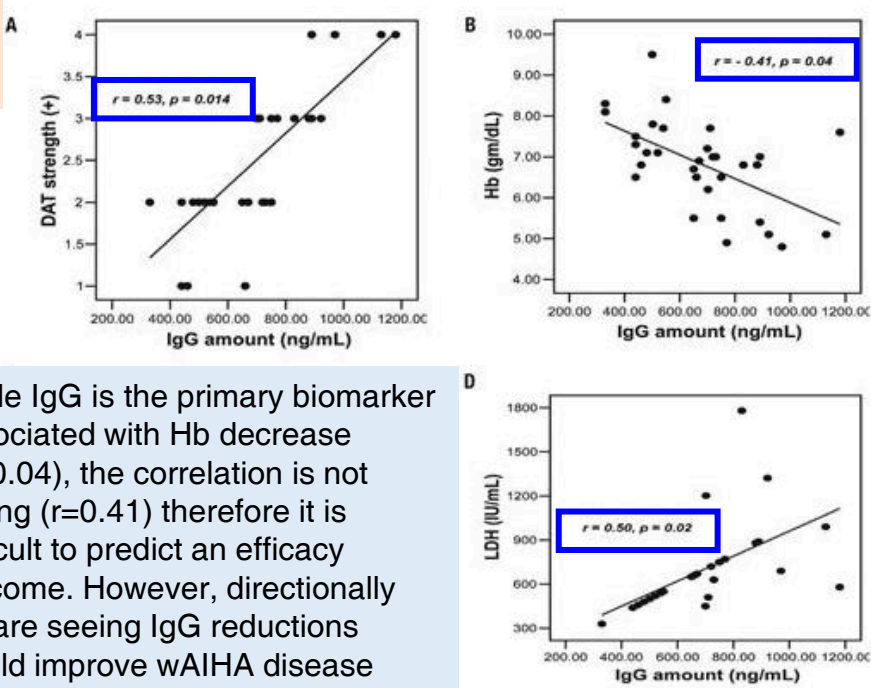
RUBY-4 Ph.1/2 Basket trial

Eligibility Criteria	
Description	
Key Inclusion Criteria:	
1. Indication-specific Criteria	
1. Immune Thrombocytopenia (ITP)	
• Documented persistent or chronic primary ITP of at least 12 weeks duration from diagnosis to Cycle 1 Day 1	
• History of failure or relapse to at least 2 treatment regimens for ITP	
• History of exposure to a TPO-RA unless otherwise contraindicated or unavailable	
• Documented history of platelets <30 x 10 ⁹ /L	
2. Warm Autoimmune Hemolytic Anemia (wAIHA)	
• Diagnosis of primary wAIHA of at least 12 weeks duration documented with a current or prior positive direct antiglobulin test (DAT) for anti-IgG (±C3d)	
• Documented history of anemia with hemoglobin ≤10 g/dL	
• At least one of the following: (i) haptoglobin < lower limit of normal (LLN) (ii) indirect bilirubin > upper limit of normal (ULN) (iii) lactate dehydrogenase>ULN	
• History of failure or relapse to at least 2 treatment regimens for wAIHA	
3. Cold Agglutinin Disease (CAD)	
• Diagnosis of primary CAD of at least 12 weeks duration with all of the following: (i) chronic hemolysis (ii) polyspecific DAT positive (iii) monospecific DAT strongly positive for C3d (iv) cold agglutinin titer ≥64 at 4°C (v) IgG DAT ≤1+ (vi) no overt malignant disease	
• Documented history of anemia with hemoglobin ≤10 g/dL	
• Evidence of hemolysis during screening: (i) indirect bilirubin >ULN and (ii) lactate dehydrogenase>ULN or haptoglobin <LLN	
• History of failure or relapse to at least 1 treatment regimen for CAD	
2. (All indications) If receiving protocol-specified standard-of-care medications, doses must be stable for protocol-specified durations	
Ages Eligible for Study	
18 Years and older (Adult, Older Adult)	
Sexes Eligible for Study	
All	
Accepts Healthy Volunteers	
No	
Study Start (Actual)	
2023-07-03	
Primary Completion (Estimated)	
2026-07	
Study Completion (Estimated)	
2026-07	
Enrollment (Actual)	
30	
Study Type	
Interventional	
Phase	
Phase 1	
Phase 2	

As a background, the literature typically reports the Direct Coombs Test (DAT) as the primary endpoint in wAIHA. However, since there is hemolysis, LDH is raised and hemoglobin is reduced. All three of these are stat-sig associated with IgG levels.

We think Pove uses an effective MOA for treating wAIHA for three reasons: 1) Xu et al. 2015 showed APRIL and BAFF levels are elevated in wAIHA, 2) Lewis et al. 2013, who demonstrated a stat sig improvement in DAT, Anti-RRBC, and LDH (p=0.0011, <0.0001, and 0.0410) in mouse models from Pove. 3) Go et al. 2017 showed raised IgG, IgA, and warm-reacting IgM are all associated with wAIHA and Pove has show reduction in all 3 of these Abs

Chaudhary et al. 2009.



While IgG is the primary biomarker associated with Hb decrease (p=0.04), the correlation is not strong (r=0.41) therefore it is difficult to predict an efficacy outcome. However, directionally we are seeing IgG reductions would improve wAIHA disease course.

Parameters	No./Median	%/Range
WAIHA type (N)		
• Primary	15	44.1
• Secondary	19	55.9

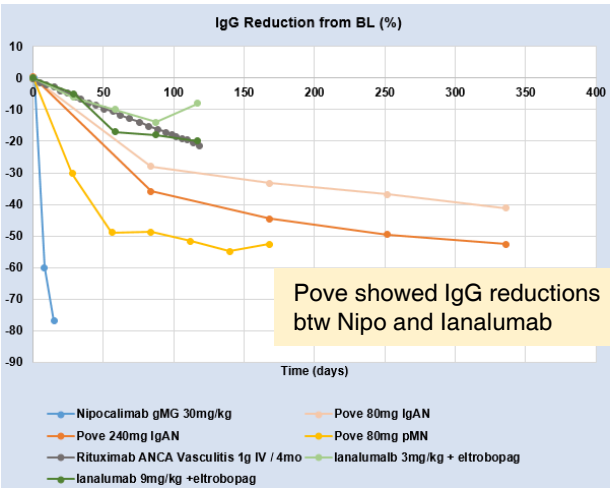
Source: Jefferies Analysis, Company Publications

We're watching 1) ENERGY Ph.2/3 (Nipocalimab), 2) VAYHIA Ph.3 (lanalumab) as both have pot'l to become efficacy leaders – that said, we've already seen Syntimmune's FcRn data show efficacy in wAIHA, which we think helps derisk Pove's MOA

We'll flag there are 2 main trials that we are watching: 1) ENERGY (Nipocalimab) which reported a hemoglobin level above 10 g/dL and an increase of at least 2 g/dL for at least 28 days for sBLA filing, 2) VAYHIA (lanalumab). Both are label-expansion Ph.3's with PCDs within 1H26 and use durability of Hb response as the primary endpoint.

3) Both sovleplenib (SYK) and rilzabrutinib (BTK) have demonstrated sustained hemoglobin responses in wAIHA (67% and 64%, respectively), supporting B-cell-targeted modulation as a valid MOA in wAIHA

4) Zhang et al. reported a China-only refractory/relapsed wAIHA study (n = 44) using sirolimus that achieved a stat. sig ~40 g/L hemoglobin improvement, also supporting upstream B-cell targeting as a valid MOA.



We'll highlight syntimmune also advanced its FcRn inhibitor SYNT001 into a n=8 Phase 1b/2a study. The study reported -0.044×10^{12} mean reticulocyte count reduction and 0.66g/dl reduction in Hb at day 33 – to us this suggests the Pove's MOA is viable in wAIHA. That said overall AEs were 100% AEs & 25% SAEs, hence program was d/c'd after Alexion's acquisition due to safety concerns & weak commercial potential. We're not expecting the same level of AE given we've seen a safety profile generally in line w/ other APRIL/BAFFs from Pove in IgAN already.

	RUBY-4 (Povetacicept)	ENERGY (Nipocalimab)	VAYHIA (lanalumab)
ClinicalTrials.gov ID	NCT05757570	NCT04119050	NCT05648968
Phase	Phase 1b	Phase 2/3	Phase 2/3 (double-blind)
Target Enrollment	7-14 per cohort (wAIHA, ITP, CAD)	~111 patients	~90 patients
Treatment Duration	24 weeks (+ optional 24-week extension)	24-week DB period + 144-week OLE	16 weeks (4 doses Q4W)
Dosing	240 mg SC Q4W	IV at 2 different dose schedules	High dose, low dose, or placebo IV Q4W
Primary Endpoint	Safety/tolerability	Durable Hgb response (≥ 10 g/dL and ≥ 2 g/dL increase at 3 consecutive visits)	Durable Hgb response (≥ 10 g/dL and ≥ 2 g/dL increase for ≥ 8 weeks, Week 9-25)
INCLUSION CRITERIA			
Age	≥ 18 years	≥ 18 years	≥ 18 years
Diagnosis	Primary wAIHA documented with current/prior positive DAT for anti-IgG ($\pm$ c3d)	Primary or secondary wAIHA with positive DAT	Primary or secondary wAIHA documented by positive DAT for anti-IgG or anti-IgA
Disease Duration	≥ 12 weeks	≥ 3 months	Not specified
Hemoglobin Requirement	Hgb ≤ 10 g/dL on 2 occasions ≥ 7 days apart during screening	Hgb < 10 g/dL with signs of hemolysis (high LDH, low haptoglobin, or high indirect bilirubin)	Hgb ≥ 5 g/dL and < 10 g/dL at screening and Week 1, with anemia symptoms
Prior Treatment Required	Failure or relapse to ≥ 2 treatment regimens	Prior treatment required (treatment-naïve not eligible); insufficient response or intolerance to available treatments	Insufficient response to or relapse after ≥ 1 line of treatment (includes steroid resistance/dependence/intolerance)
EXCLUSION CRITERIA			
AIHA Type	Secondary autoimmune cytopenias excluded	Cold antibody AIHA, cold agglutinin syndrome, mixed (warm+cold) AIHA, paroxysmal cold hemoglobinuria excluded	Cold or intermediate AIHA forms excluded
Secondary wAIHA Restrictions	Secondary autoimmune cytopenias excluded	Secondary wAIHA allowed	Secondary wAIHA with bone marrow involvement (e.g., CLL) or requiring prohibited meds excluded; autoimmune diseases after wash-out allowed
Prior Rituximab/B-cell Therapy	Rituximab must be stopped ≥ 12 weeks before enrollment	Not specified	B-cell depleting therapy (e.g., rituximab) within 12 weeks excluded, or no hematological response to last course
Prior Splenectomy	Splenectomy within 12 weeks excluded	Not specified	Not specified
Prior IVIg	IVIg must be stopped ≥ 4 weeks before	Not specified	Not specified
Infections	Active infections requiring treatment excluded	Not specified (clinically relevant abnormalities excluded)	Active viral/bacterial/other infections (including TB, SARS-CoV-2) requiring systemic treatment excluded; history of recurrent significant infection excluded
Plasmapheresis	Blood transfusions or plasmapheresis shortly before trial excluded	Not specified	Not specified
Other Clinical Trials	Current participation in another trial excluded	Not specified (investigator discretion)	Not specified

Source: Jefferies Analysis, Company Publications

At ASH’25 we got updates for multiple MOAs in wAIHA, but cross trial comparison is difficult due to different BL populations, different endpoints, different refractory status – to us, there are multiple tx being developed, which crowds the field for Pove, but pricing for the class still makes wAIHA an attractive opportunity

We’ll flag there are many therapies currently seeking label expansion for wAIHA. This includes mAbs, B-cell receptor pathways inhibitors, IgG-mediated phagocytosis inhibitors, complement inhibitors. However, currently **there is no targeted therapy that is FDA approved for wAIHA**

First line therapy is corticosteroids, and second line therapy is splenectomy therefore this is a disease of unmet need.

At ASH’25 we got updated data for a variety of drugs that are being repurposed for wAIHA. We’ll flag that the MOAs, population / line of tx, endpoints are all different btw the trials, therefore making cross trial comparison difficult.

We’ll highlight that the ORR for Sirolimus and Iptacopan were numerically similar (77-81% vs 86.7% resp.). As a reminder, the primary endpoint in RUBY-4 is safety. That said, we know the tested population will be “History of failure or relapse to at least 2 treatment regimens for wAIHA” – suggesting the Sirolimus, Rilza and Zanu tx will be the main read across.

Drug/Regimen	MOA	Population & Line	Study Design / n=	Primary Endpoint	Response Rate	Time to Response	Durability / Relapse
Rituximab	Anti-CD20	Adult wAIHA; 1L/2L; real-world	Retrospective n=403	Durable Hb response ≥2 g/dL x28d	65% (1st); 52% (2nd)	90d (1st) / 160d (2nd)	Relapse 34-47% by 12 mo
Pred + MMF	Steroid	Newly dx wAIHA; 1L	Prospective n=20	CR/PR at early timepoints	95–100% (2–12 wks)	Median 4 wks	Relapse 10%
Sirolimus	mTOR Inhibitor	R/R wAIHA; 2L+	Prospective n=78	ORR/CRR at 3-12 mo	ORR 77–81%; CRR 40–52%	OR: 26d; CR: 76d	Relapse 12.7%
Iptacopan combo	Complement Inhibitor	Severe refractory flares	Cohort n=15	Hb increase >20 g/L + TI	ORR 73.3% at 1mo, 80% at 2mo; 86.7% achieved Hb increase >20g/L	Median 29d	C3d decrease in 87%; relapse NR
Rilzabrutinib	BTK Inhibitor	Prior-treated wAIHA	Phase 2b n=22	Hb ≥2 g/dL at wk24	63.6% wk24; 54.5% durable	By wk24	Durable to wk50; relapse NR
Zanubrutinib	BTK Inhibitor	Heavily pretreated wAIHA	Phase 2 n=13	ORR at wk12	44.4% (CR2/PR2)	By wk12	Responders 7-20 mo

Source: Jefferies Analysis, Company Publications

Given that Rituximab is being used as 2L tx, we think Pove would need to show greater efficacy to be favored by docs – we’re seeing various studies of ritux in wAIHA show 84.7% ORR and 48.5% CR on average (caveats w/ study BL pop)

We’ll flag there are numerous studies looking at Rituximab as tx in wAIHA, so we compiled a significant portion and found that the **average ORR is 84.7% and CR is 48.5% (Caveat: BL severity was different across populations)**. To us, rituximab is the key benchmark for Pove to surpass given that tx sequence has it as 2L.

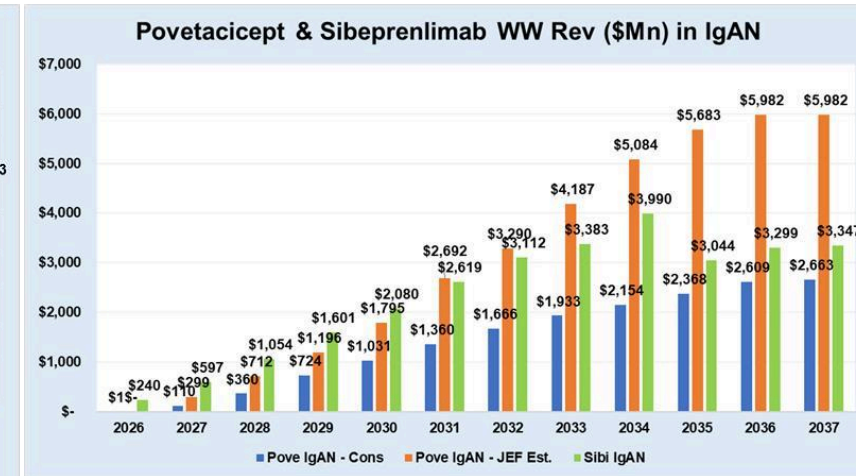
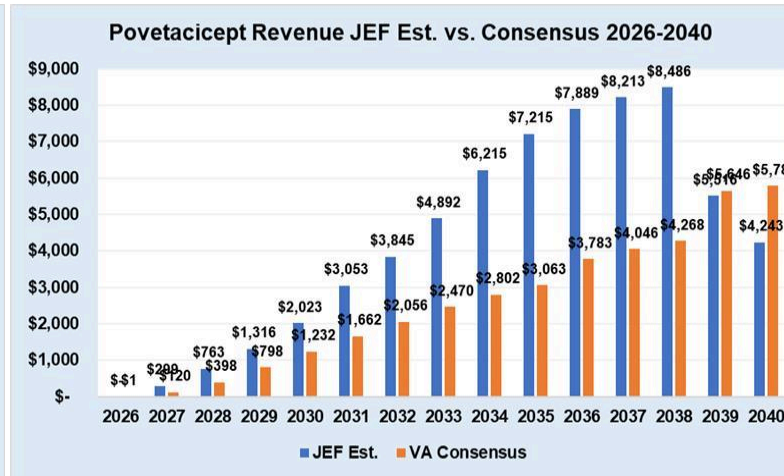
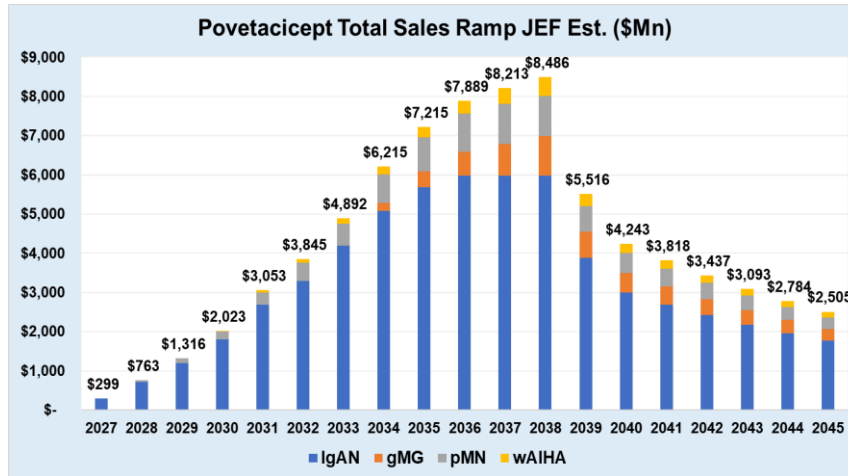
Reference	Number of Patients	Splenectomy (N)	ORR (%)	CR (%)	Duration of Response (mo)	BL population comments
Narat (2005)	11	NR	64	27	NR	6 pts had chronic ITP
D’Arena (2007)	11	1	100	73	~95mo	Steroid-refractory warm idiopathic AIHA adults; weekly rituximab x4
Bussone (2009)	27	6	93	30	NR	5 (18% relapses) after median f/u 20.9 mo, 3 retreated with R and responded
Dierickx (2009)	53	10	79	47	1yr PFS 72%, 2yr PFS 56%	
Penalver (2010)	36.00	13	77	61	Duration of response > 6mo in patients in CR	
Maung (2013)	34	3	71	27	Median time to next tx 16.5mo	50% relapse; 28.5% maintained response at 3 years
Roumier (2014)	25	NR	80	NR	50% relapse after mean of 14 ± 8	62% secondary wAIHA
Barcellini (2012)	23.00	NR	83 (100% in wAIHA subgroup)	60	NR	Primary AIHA; low dose Ritux
Jaime-Pérez (2019) - First-line	18	N/A	100 (initial)	NR	Mean MR was 81.93mo	Low dose Ritux + high dose dexamethasone (40 mg/day) for 4 days
Jaime-Pérez (2019) - Relapsed	8	NR	100	63	Mean maintained response 82 ± 18	Low dose Ritux
			Average = 84.7%	Average = 48.5%		

While Rituximab has reported efficacy in-terms of ORR and CR in the past, these days, most studies reported increase in Hb as the main efficacy endpoint. Therefore, we won’t be surprised if pove reports Hb increase as the efficacy endpoint for RUBY-4

Source: Jefferies Analysis, Company Publications

Taken together, we model \$8.5Bn total risk-adj peak sales for pove vs \$5.8Bn for cons (90% POS for IgAN, 60% POS gMG, 55% POS for pMN, 50% POS wAIHA) + we're seeing Sibe is under modeled past 2034

For IgAN alone, we model risk adj. ~\$8.5Bn - with 90% PoS given pove's positive Ph.3 (we model PS of \$6Bn for IgAN). For gMG, we model risk adj. peak sales of \$1Bn total – with 60% PoS in gMG given Telitacipt's data has helped derisk the MOA. To us pMN and wAIHA can work given the \$390K/yr price, hence we model risk adj. ~\$1Bn peak sales (PoS 55%) & for wAIHA, we model risk adj. \$475M (PoS 50%)

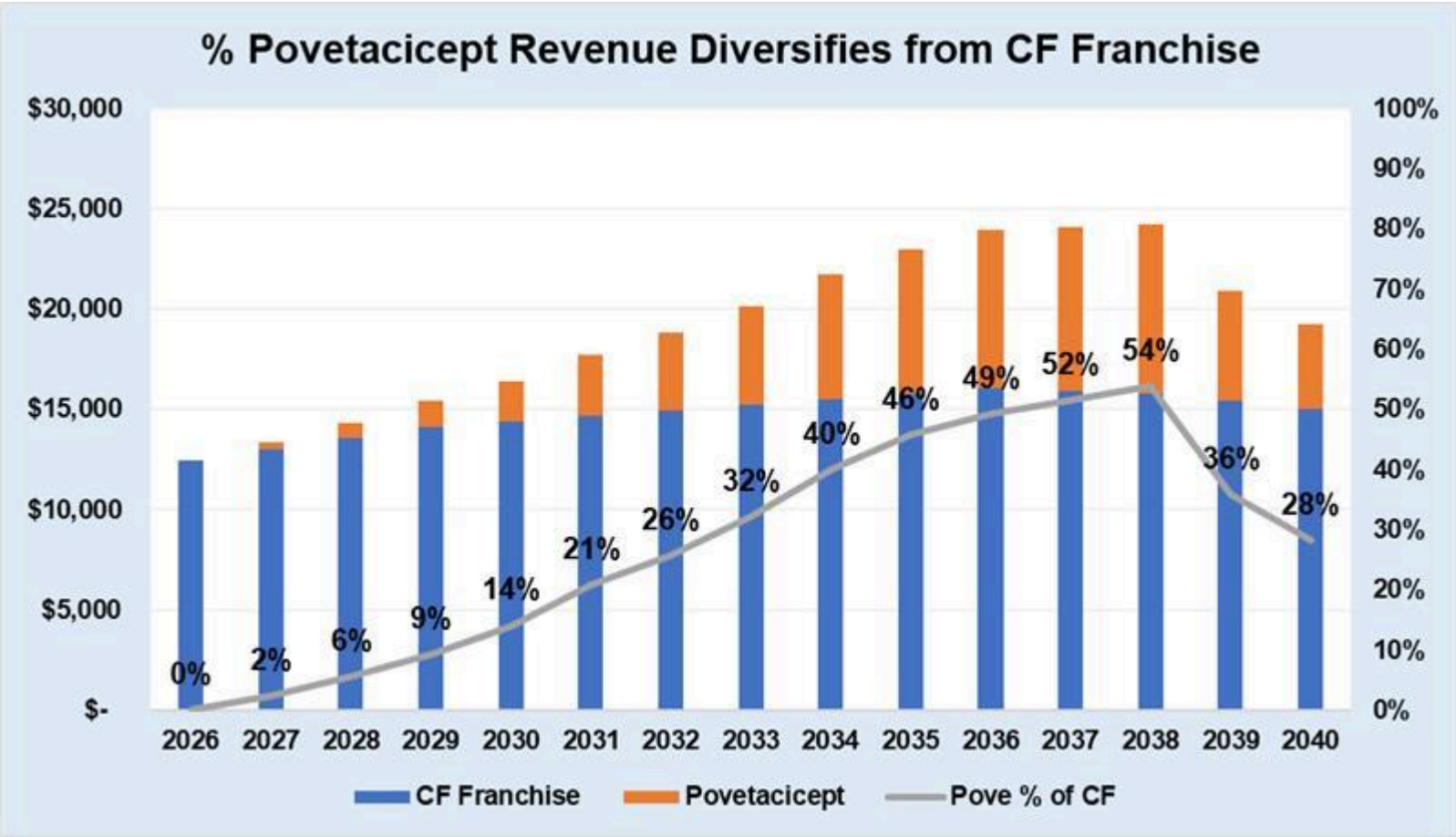


Mgmt previously commented at Q4 earnings that the renal franchise could be as large as CF. *“povetacipt is the first in a series of potentially transformative medicines that tackle the underlying cause of several renal diseases: IgAN, pMN, AMKD, and ADPKD. We anticipate that the renal franchise will ultimately rival the scale of our CF business and we're seeking to bring the best elements of our success in CF to these kidney disease areas” – VRTX mgmt. 12 Feb 2026..*

Based on our analysis, we're not quite there yet, but what we will say is we think cons. under models Sibe in IgAN (peak sales ~\$4Bn in 2034). As a reminder, we model Pove in IgAN to be ~\$8.5Bn peak sales by 2037 with the discrepancy coming from too few analysts modeling past 2043 + we think the IgAN market is larger

Source: Jefferies Analysis, Visible Alpha

Overall, we think pove has the potential to materially diversify VRTX beyond CF, with pove revenue reaching ~50% of CF revenues longer term



Source: Jefferies Analysis

Next, we'll highlight VRTX's other kidney programs – AMKD & ADPKD

Cystic Fibrosis

We expect VRTX's CF franchise to remain durable

- We're expecting CF to remain a durable revenue base for VRTX and expect gradual switch & new starts for pts onto Alyftrek
- We expect upcoming comps will work as add-on tx to VRTX, but we're not seeing evidence as a mono-tx they could produce greater efficacy
- We expect minimal pt adds from VRTX mRNA CF tx, but limited ability for generics to gain market share

IgAN & autoimmune

We see versatility for APRIL/BAFF across autoimmune indications

In IgAN, we saw 49.8% Pbo adj. UPCR reduction at 9mo (clinically in line w/ Sibe):

- To us, the impt point is: a) IgAN is a bigger market with favorable pricing (\$390K/yr), b) Pove has a convenience advantage with 1.5sec admin autoinjector
- We're optimistic on the indication expansion opportunity into gMG, pMN, wAIHA



AMKD & ADPKD

We see LT upside from Inaxaplin becoming first/best in class

- We're expecting Inaxaplin to be first & best in class in the AMKD market (currently no tx)
- We expect 64-70% UPCR reduction at 48wks + we think VRTX has the more favorable BL population vs comps
- We think the AMKD market could be bigger than initially expected given a) an approved tx will likely increase dx rates, b) our KOL expects spillover dx from the IgAN market growing from Povetacicept
- In ADPKD, we think the data is too early for VX-407 but could see pot'l reduced LFTs based on Trikafta read across (similar MOA)

Pain

We're mixed on Ph.3 DPN hitting, but are more confident in the acute pain opportunity

- We're mixed on whether chronic DPN can hit for Suze, given the high variability of previous data and Pbo (combo approach is more likely here).
- In acute pain, we expect 3x growth for Journavx based on recent access expansion

Rest of Pipeline

Rest of pipeline is interesting but we're skeptical on cons valuation

- We're skeptical on expansion opportunity for rest of pipeline:
- For Casgevy, we've seen precedence that viable tx's in SCD launch slower / underpenetrate the market due to patient/physician concerns & access barriers
- Zimislecel appears limited to severe T1D population based on immunosuppression requirement & clinical trial delays

Source: Jefferies analysis

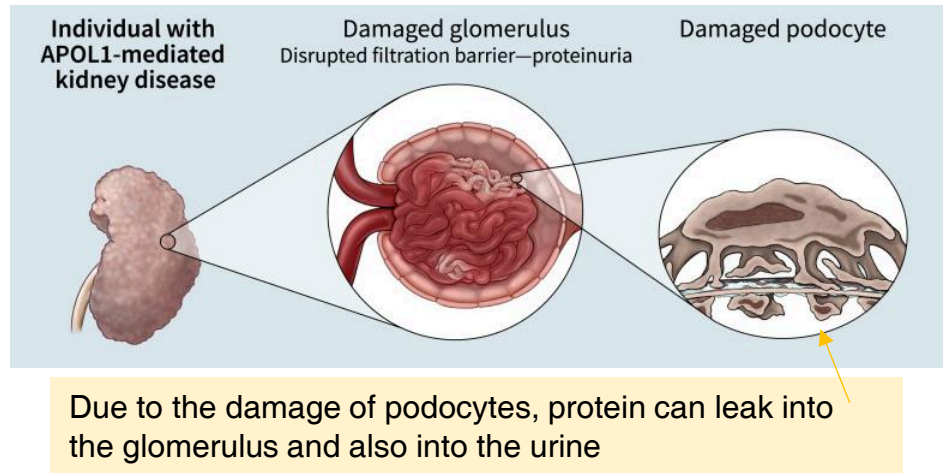
Starting with AMKD, we think VRTX could be first-in-class/best-in-class:
VRTX is running AMPLITUDE Ph.3 for VX-147/inaxaplin in primary AMKD with interim data exp YE’26/early’27 + AMPLIFIED Ph.2 PoC trial AMKD w/ moderate proteinuria or diabetes (data exp 2H’26)

Attribute	VRTX VX-147 Ph2a	VRTX AMPLIFIED (Ph2)	VRTX AMPLITUDE (Ph2/3)	MAZE MZE829 Ph2
Study ID	VX19-147-101	VX24-147-202	VX21-147-301	MZE829-201
Phase	Phase 2	Phase 2	Phase 2 / Phase 3	Phase 2
Study Design	Open-label, Single-arm	Open-label	Adaptive, Double-blind, Placebo-controlled	Open-label
Enrollment (n)	16	45	466	56
Start Date	6/8/2020	2/4/2025	3/30/2022	2/5/2025
Primary Completion (Est.)	11/11/2021	12/30/2026	5/31/2026	2026-09
Study Completion (Est.)	12/9/2021	12/30/2026	6/30/2026	2026-09
Population	APOL1-mediated FSGS	Proteinuric AMKD ± comorbidities	Adult & pediatric AMKD	CKD with proteinuria & APOL1 high-risk
Treatment	VX-147	Inaxaplin	VX-147	MZE829
Comparator	None	None	Placebo	None
Key Inclusion Criteria	G1/G1, G2/G2, or G1/G2; biopsy-proven	G1/G1, G2/G2, or G1/G2; eGFR ≥25	G1/G1, G2/G2, or G1/G2; proteinuric	G1/G1, G2/G2, or G1/G2; persistent
Key Exclusion Criteria	Non-APOL1 FSGS; sickle cell; transplant	FSGS w/ known non-APOL1 cause; uncontrolled HTN	Transplant; uncontrolled HTN; diabetes; sickle cell; non-APOL1 kidney disease	Transplant; cancer past 2y (exc. NMSC/stage 0-1); bariatric surgery; T1DM; pregnant/nursing

- VRTX’s AMPLITUDE is an adaptive Ph.2/3, double-blind, placebo-controlled trial (N~466 adult and pediatric AMKD pts; excludes diabetes, transplants, sickle cell) designed to support registration (interim data exp YE’26/early’27)
- AMPLIFIED is a single-arm, open-label Ph.2b (N~45) of inaxaplin in proteinuric AMKD ± comorbid CKD drivers incl. diabetes (data exp 2H’26);
- Maze’s MZE829-201 is an open-label Phase 2 (N~56 adult APOL1 high-risk CKD pts), using albuminuria as the primary endpoint and excluding recent cancer, bariatric surgery, T1DM and transplant to minimize PK variability. Data from the MAZE trial is exp Q1’26.

Source: Jefferies Analysis, Company Publications

As a background, APOL1 Mediated Kidney Disease (AMKD) is a genetic condition where 2 copies of the risk variant leads to accelerated kidney function decline & damage to key kidney cells (podocytes) - Inaxaplin (VX-147) is an APO-L1 channel blocker that is furthest along in development (Ph.2/3; PCD May 2026)



- AMPLITUDE is VRTX’s Global adaptive Ph.2/3 study in adults with generically confirmed biallelic APOL1 variants and proteinuria kidney disease. Study aims to have evaluate UPCR reduction & eGFR slope at 48wks (interim filing data)
- By contrast, AMPLIFIED (Ph.2b OL), is for AMKD pts w/ proteinuric APOL1 with or without comorbidities, to assess safety, tolerability, PK/PD, and is testing 13wk daily dosing.

APOL1-mediated kidney disease (AMKD)	Inaxaplin – Primary AMKD	~150K	AMPLITUDE	IA cohort enrollment complete
	Inaxaplin – AMKD with moderate proteinuria or diabetes	~100K		
			AMPLIFIED	On track to complete enrollment by YE '25

AMKD is a rapidly progressive genetic kidney disease affecting people of West/Central African ancestry who inherit two APOL1 risk variants, presenting with proteinuria and steep eGFR loss, with patients reaching dialysis ~9-12 years earlier than those without the genotype. Diagnosis requires APOL1 genetic testing (and sometimes biopsy), and with no approved therapies targeting the root cause, patients often progress to end-stage disease requiring dialysis or transplant.

Inaxaplin (VX-147) is a first-in-class APOL1 channel inhibitor for patients with two APOL1 risk variants, blocking abnormal ion flux that drives podocyte injury; it has FDA breakthrough designation and is in Phase 3 (PCD May 2026). While Inaxaplin is furthest ahead in development, we’ll flag Baricitinib (JAK-STAT pathway inhibitor), AZD2372 (ASO), MZE-829, MZ-301/2 are also being tested in AMKD (*we’ll go through each one more later...*)

APOL1 variant:
0 copies = lowest risk
1 copy = carrier
2 copies = highest risk

Drug	Sponsor	Mechanism / Indication	Stage	NCT Number	PCD
Inaxaplin (VX-147)	Vertex Pharmaceuticals	APOL1 channel-blocker	Phase 2/3 (currently in Ph.3)	NCT05312879	May-26
VX-840	Vertex Pharmaceuticals	APOL1 channel blocker	Phase 1 completed	NCT05324410	Nov-22
MZE829	Maze Therapeutics	APOL1 small-molecule inhibitor	Phase 2	NCT06830629	Sep-26
MZ-301	Maze Therapeutics	APOL1 small-molecule inhibitor	Preclinical	-	
Baricitinib	Lilly (collaborator)	JAK-STAT pathway inhibitor (APOL1-CKD)	Phase 2 (exploratory)	NCT05237388	Mar-26
AZD2373	AstraZeneca	APOL1-targeted antisense oligonucleotide	Phase 2b (dose-ranging)	NCT06824987	Aug-27

Source: Jefferies Analysis, Company Publications

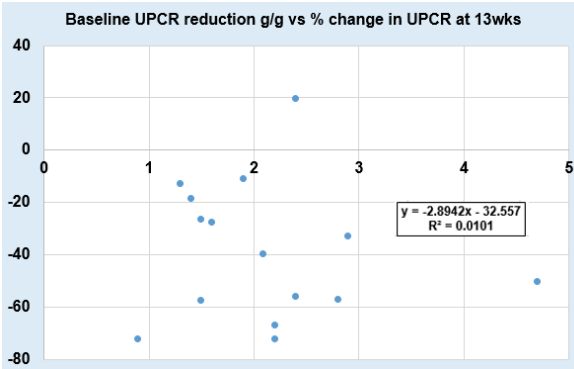
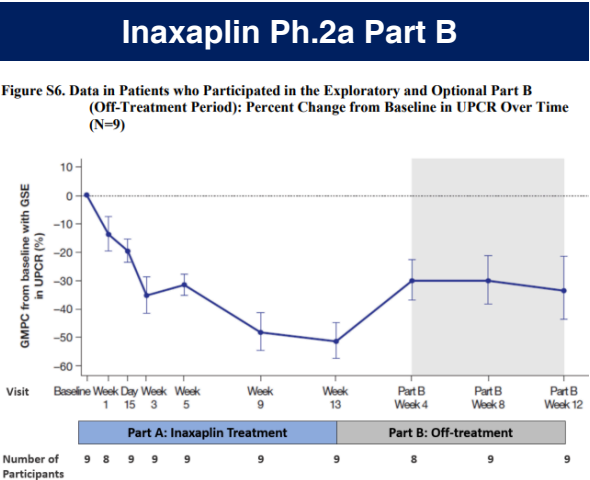
On efficacy, inaxaplin showed ~48% UPCR reduction at 13wks in Ph.2 – we think VRTX can continue to show similar or better UPCR decline in Ph.3 because a) 13wks may not capture full effect of APOL1 blockade + we expect a slightly less severe population which could optically improve UPCR

VRTX mgmt. has mentioned Ph.2/3 could either continue to show decline in UPCR at 48wks, or plateau: “We had a 47.6% reduction in proteinuria in 13 weeks. And that’s quite remarkable if you look at other proteinuria conditions. And so we’re in Phase III. I’m super excited to see what those results look like because we don’t know for certain, but the **benefit could stay stable at 13 weeks. That could be the maximal effect.** Obviously, there’s always the theoretical, so they could not be as much as you want. **But there’s other reasons to think it could continue to fall.** in the long period of time. And so that, I think, has - is a potentially really transformative medicine as well”. – mgmt. comments Dec 3 2025

- Here are reasons we think the UPCR will continue to decline:
- 1) We’ve seen in Zimmerman et al. 2025 **preclinical data with APOL1 inhibitors show reversal of established glomerular damage and partial restoration of the filter structure.**
 - 2) Ph.2 13wk effects may have only captured the functional blockade of APOL1 but not the partial reversal of cumulative glomerular injury
 - 3) In pMN (Ruggenti et al. 2015- read across to APOL1 due to upstream targeting of Anti-PLA2R using Ritux.) there was continued decline in UPCR past 13wks + Pts in Ph.2 Inaxaplin only showed ~20% regression in UPCR when stopping dosing at wk13 – suggesting remodeling had started but still had room to continue further

Source: Jefferies Analysis, Company Publications

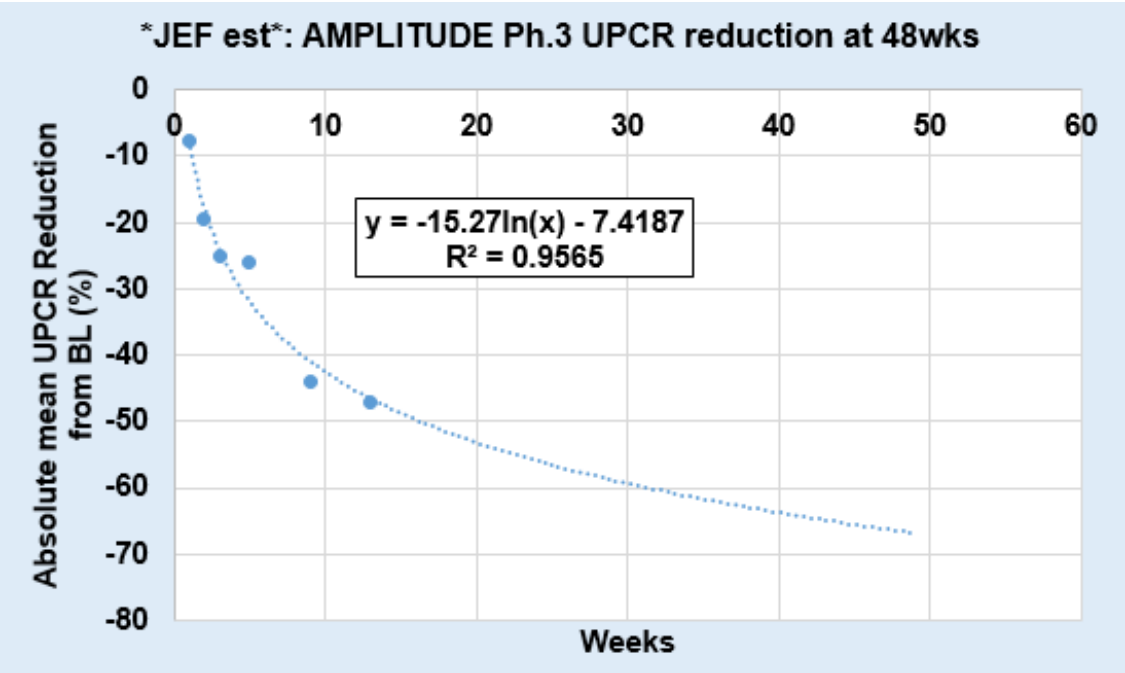
We’ll also note when Inaxaplin was stopped at 13wks, there was a regression in UPCR from -50% to -30%, but it stabilized at 30% (caveat: n=9 in part B follow-up). Given that it did not return to BL UPCR, we suspect this points to kidney remodeling occurring under Inaxaplin that could mean with greater duration of sustain APOL1 efficacy, there could be further UPCR decline.



We expect having younger pts (<18) and lowering the proteinuria threshold (2-3 g/g vs 0.7 g/g) could yield more reversible BL disease, but allowing eGFR 25-45 mL/min/1.73m² inc. more severe pts. Overall, we expect Ph.2/3 baseline disease to be more reversible than Ph.2a, making a longer 48-week follow-up better for capturing full UPCR decline.

Feature	Phase 2a	AMPLITUDE Phase 2/3
Population	Biopsy-proven FSGS only	Broad APOL1-mediated proteinuric kidney disease
Age	18-60	Ph.2: 18-65; Ph.3: 10-65
Proteinuria threshold	Cohort 1: UPCR ratio of ≥3 g/g and <10 g/g in the first morning void; Cohort 2: UPCR ratio of ≥ 2 g/g and <10 g/g in the first morning void	UPCR≥0.7 to <10g/g
eGFR floor	≥45 ml/min/1.73m2	≥25 to <75mL/min/1.73m2
Disease severity	Narrow high-proteinuric cohort	Broader spectrum including advanced CKD
Background therapy	Strictly stable RAAS/SGLT2	More flexible SOC

b) We estimate inaxaplin can show ~64-70% UPCR reduction at 48wks & eGFR stabilization at ~48ml/min/1.73m², which we believe would set a competitive bar for a first-in-class tx in AMKD

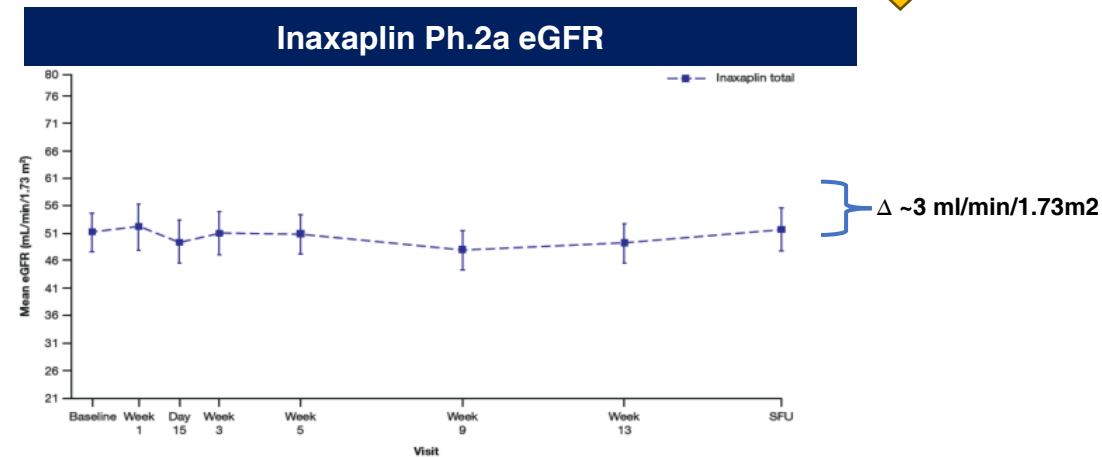


We'll flag the Ph.2/3 (NCT05312879) is the registrational trial for **accelerated approval w/ PCD May 2026 & n=466 (~233 per arm)**

Method:

- We digitized the UPCR reduction reported in Ph.2a Egbuna et al. 2023. and plotted the values over time to fit a natural log regression ($R^2=0.96$). From this we est. **~67% UPCR reduction 48wks** (*caveats: this is extrapolating further out that the data exists*).
- We took the 95% CI from Ph.2a (-31.1 to -60) and adjusted it to reflect the SE for a single arm of Ph.2/3 (n=233). From this we calculated Ph.2/3 SE of 1.73 which translates to a 95% CI range of 6.77. **This gave us a range of ~64-70% UPCR reduction at 48wks.**

In Egbuna et al. 2023, there was sustained eGFR till safety follow-up past 13wks. Given the small size of the error bars and consistency of this trend, **we're expecting this to continue into 48wks for Inaxaplin given our extrapolation showed only a ~3ml/min/1.73m² absolute reduction in eGFR (resulting in ~48ml/min/1.73m² by 48wks).** To us this is clinically stabilizing the eGFR.



Source: Jefferies Analysis, Company Publications

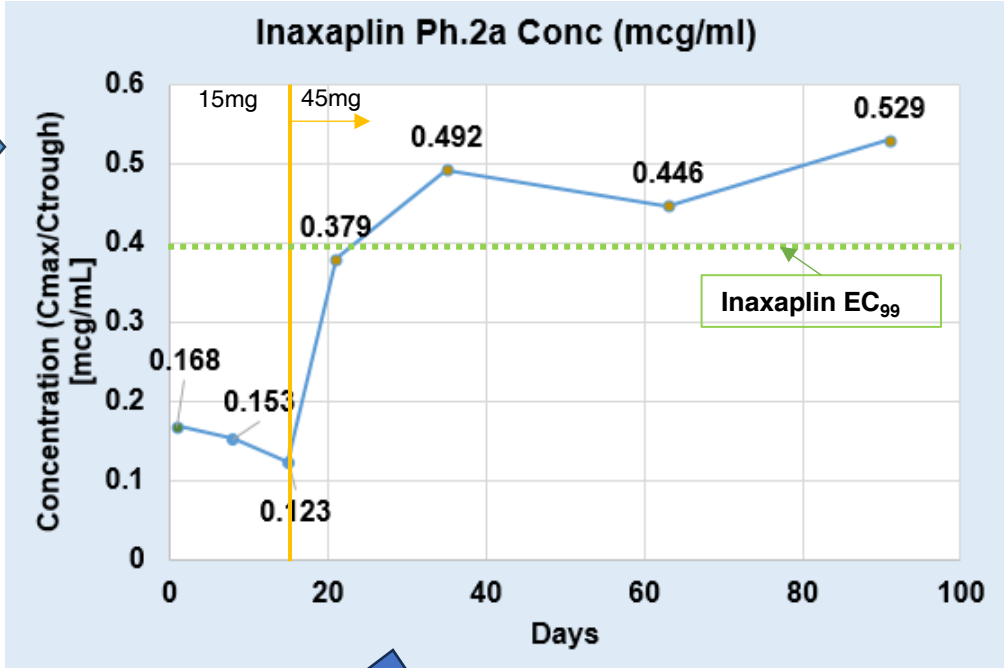
On safety, we're not seeing any dose-limiting signals as >90% of AEs were mild / moderate & none lead to d/c. We est. the go fwd dose can maintain C_{trough} above EC₉₉, thus we think efficacy can carry forward into Ph3 w/o being limited by safety

When considering that the Ph.1 had n=134 and the Ph.2 had n=16 total – we view the AE rates that can be directly compared btw both trials as similar in proportion (e.g. headache was the highest for both & on an absolute basis the numbers were similar). **Therefore, we're expecting a largely conserved safety signal from Ph.2a into AMPLITUDE.**

Adverse Event	Phase 1 Inaxaplin (n=134)	Phase 1 Placebo (n=44)	Phase 2a Inaxaplin (n=16)
Abdominal pain	1.5%	2.3%	NR
Back pain	NR	NR	18.8%
COVID-19	2.2%	0.0%	NR
Constipation	1.5%	0.0%	NR
Cough	2.2%	0.0%	NR
Decrease in blood bicarbonate	NR	NR	12.5%
Diarrhea	3.0%	0.0%	12.5%
Dizziness	NR	NR	12.5%
Dyspepsia	2.2%	0.0%	12.5%
Fatigue	1.5%	4.5%	12.5%
Headache	10.4%	2.3%	25.0%
Nausea	1.5%	0.0%	18.8%
Oropharyngeal pain	1.5%	0.0%	NR
Rhinorrhea	1.5%	2.3%	NR
Throat irritation	1.5%	0.0%	NR

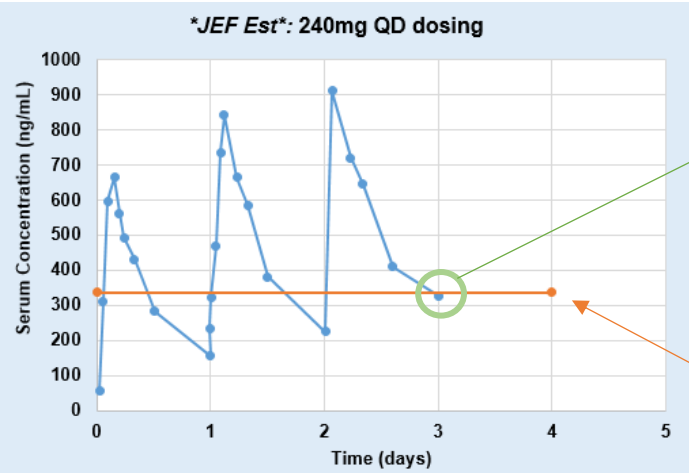
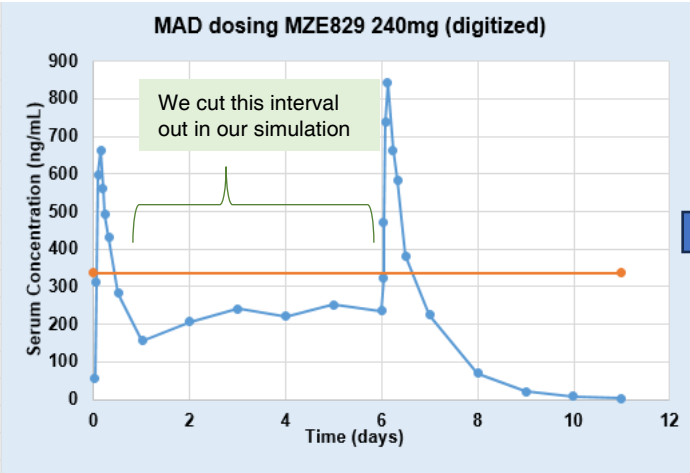
We noted that in Ph.2, 44% of AEs were mild and 50% were moderate. There were no AEs that lead to d/c. Two serious adverse events occurred in 1 participant: deep-vein thrombosis and uterine leiomyoma; neither event was considered by the investigator as related to inaxaplin. On our end, we're not seeing any concerning safety signals that could lead to dosing limitation.

The IDMC concluded that the 45mg dose (higher dose) was acceptable as the go-forward dose. We'll flag the C_{trough} mean remained above EC₉₉. To us this suggests the safety seen in Ph.2 is what we can expect into Ph.3 given the max effect is achieved



Source: Jefferies Analysis, Company Publications

We'll note competitor MAZE's Ph.2 data for MZE829 is exp 2H26, but we're not worried for VRTX – While we do think MAZE can dose above their EC99, we think VRTX is reaching similar target coverage and capping out on efficacy for the APOL1-targeted MOA

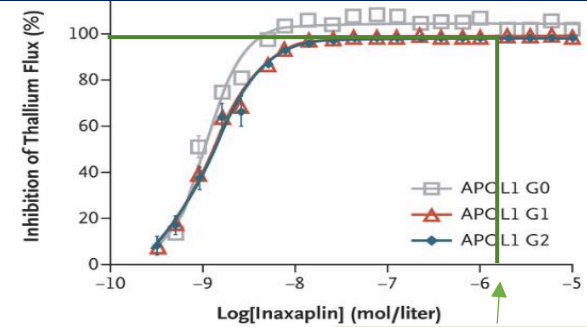


We noted that by the 3rd dose, the Ctrough was reaching the the EC99. As a reminder the Ph.2 dosing regimen is 250mg QD therefore we'd expect the Ctrough for 250mg to be slightly higher than for 240mg. To us, this suggests efficacy would be maxed out similar to the level of Inaxaplin.

We used the n=13 of tx arm and the -31.1 to -60 CI to back out a SD of 26.4 for Inaxaplin. We then adjusted this to an n=56 (based on ct.gov reported trial enrollment for Ph.2 MZE829) and **calculated a 95% CI delta of 13.82**, which would be tighter than the 95% CI from Inaxaplin Ph.2.Phase 2a (28.9).

JEF Est EC99 for MZE829

Inaxaplin PK/PD curve (channel suppression maxed-out)



At the minimum Ctrough for Inaxaplin of 0.466mcg/mL and assuming a molecular weight of 417.4g/mol, **we calculated the mol/L concentration for Inaxaplin to be 1.12uM** ($0.000466/417.4=1.12\text{uM}$). **$\log(1.12\text{uM}) = -5.95\text{mol/L}$** .

$\log(1.12\text{uM}) = -5.95$

Drug / Trial	Dose	n=	Frequency	% APOL1 channel inhibition	UPCR at 13wks	95% CI Delta
Inaxaplin (Ph.2a)	45mg	13 in tx arm	QD	~99%	47.60%	28.9
MZE829 (Ph.2) *JEF Est*	250mg	56	QD	~99%	47%	13.82

Inaxaplin showed a 47.6% UPCR reduction at 13wks and, we view this as maxed out efficacy given the ~99% channel inhibition. Therefore, we expect ~47% UPCR reduction as well for MZE829.

Source: Jefferies Analysis, Company Publications

We'll also note BL population for MZE829 Ph.2 includes diabetes pts, but VRTX excludes pts w/ a history of diabetes – while we've seen diabetes leads to faster progression, we're also expecting these pts to be more heavily pretreated thereby potl'y reducing efficacy delta from BL for MZE829

Ph.2 MZE829

- MZE829 once-daily 250 mg oral
- Open-label, basket study
 - APOL1- mediated CKD with diabetes
 - APOL1- mediated CKD without diabetes
 - Severe FSGS without diabetes*
- All patients enrolled will have two copies of the risk variants
- Patients will be on stable background CKD regimen (RAASi +/- other SoC e.g., SGLT2i, MRA, etc.) for at least 8 weeks prior to study

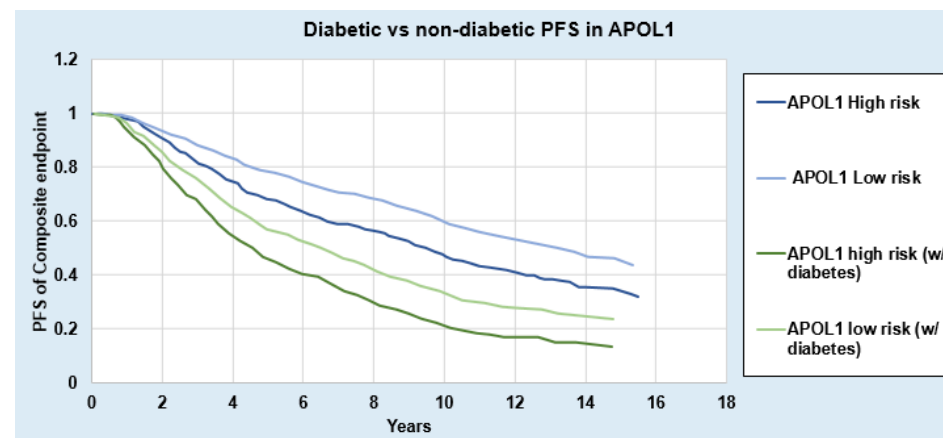
MAZE will have arms +/- diabetes. We'll flag pts with diabetes are likely already on SGLT2i/GLP-1RA/RAAS blockade.

We've seen data for both SGLT2i (Dapagliflozin, n=137) and GLP-1 (semaglutide, n=1780) that eGFR decline is slower on tx vs pbo and there was greater % change in proteinuria. We'll also flag that pts must have been on stable tx for 8wks. To us, **this could suggest the pts w/ diabetes for MZE829 might already enter with partially stabilized kidney outcomes leaving less room for MZE829 to show efficacy.**

Ph.2/3 Inaxaplin

Eligibility Criteria	
Description	Ages Eligible for Study
Key Inclusion Criteria:	10 Years to 65 Years (Child, Adult, Older Adult)
• APOL1 genotype of G1/G1, G2/G2, or G1/G2 • Proteinuric kidney disease	Sexes Eligible for Study
Key Exclusion Criteria:	All
• Solid organ or bone marrow transplant • Uncontrolled hypertension • History of diabetes mellitus • Known underlying cause of kidney disease including but not limited to sickle cell disease	Accepts Healthy Volunteers
Other protocol defined Inclusion/Exclusion criteria apply.	No

Ph.2/3 Inaxaplin has diabetes as exclusion criteria, but we'll flag the Ph.1/2a for Inaxaplin did not specify diabetes as inclusion/exclusion criteria.



We've seen from MAZE pts with diabetes had faster rate of decline for PFS composite endpoint in AMKD vs those without. On its own, this might suggest there is more disease to control, but if pts are more pretreated, this efficacy delta may be blunted.

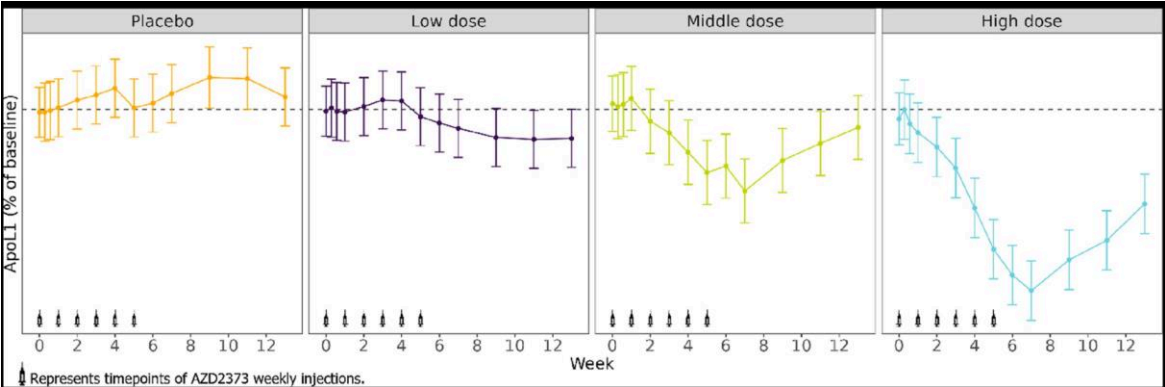
Source: Jefferies Analysis, Company Publications

We've also seen patent data for AZD2373 (APOL1 ASO) - while there was a dose response (as seen in Ph.1 data), we also noted there was 1 case of elevated liver enzymes and 3 cases of elevated CRP (suspected patent) suggesting liver tox may be a potential issue for ASO's targeting this pathway

AZD2373 is an antisense oligonucleotide (ASO) targeting APOL1 mRNA, currently in Ph.2 development (n=96; NCT06824987). In Ph.1 (n=24 total; 6x QW subQ injections), AZD2373 showed “favorable safety and tolerability profile”, w/ no major concerns reported at any dose. The most frequent adverse events were mild to moderate injection site reactions, which were dose-dependent but did not result in treatment d/c. PK analysis revealed rapid distribution and prolonged elimination. Importantly, AZD2373 produced a dose- and time-dependent reduction in plasma APOL1 concentrations. We saw a dose response for AZD2373, however they reported APOL1 decline from BL but not UPCR decline. APPREICATE trial PCD: Aug 2027

We found in a suspected patent for an ASO jointly filed by AZN and IONS (WO 2019/226611) that there was a dose dependent % inhibition of APOL1 (as we saw in the Ph.1 results). Interestingly there was a wide variability in the response where some mice did not response to the low dose and the highest response was 65%

However, we noted that in 1 cynomolgus monkey, there was a raised ALT/AST and there was an elevated CRP in 3 monkeys. While this is the unconfirmed patent for AZD2373, we think there is points to pot'l liver related AEs that we're not seeing to the same extent for inaxaplin.



AZN has established 45% reduction in uACR as a threshold for secondary endpoints

Table 100

Liver function markers in cynomolgus monkey plasma

	ALT (IU/L)	AST (IU/L)	Bilirubin (mg/dL)	Albumin (g/dL)
PBS	37	57	0.3	4.3
793406	59	55	0.3	4.3
904763	50	48	0.3	4.4
905469	54	68	0.2	3.9
905505	46	54	0.2	4.1
905634	566	417	0.5	4.1
905665	57	80	0.3	4.3
972163	58	81	0.2	4.1
972190	47	46	0.2	4.2

Table 105

C-reactive protein levels (mg/L) in cynomolgus monkey plasma

	CRP
Saline	1.7
793406	1.4
904763	2.4
905469	8.4
905505	4.1
905634	6.7
905665	10.7
972163	4.7
972190	9.5

(10) International Publication Number
WO 2019/226611 A1

Source: Jefferies Analysis, Company Patent

To get a better sense of the AMKD oppty, we spoke to the Director of Kidney Genetics at Harvard Medical School (BIDMC) – here are our takeaways:

KOL Background: Dr. Peter Czarnecki is a nephrologist at BWH/HMS and Director of the PKD/Kidney Genetics Clinic at BIDMC. Sees ~25 new AMKD dx/yr; clinic functions largely as a 2nd-opinion referral hub, often w/o clear initial dx.

- On AMKD diagnosis, our KOL emphasized APOL1 high-risk alleles are not deterministic; a **“second hit” (often infection/cytokine surge) can precipitate overt AMKD**. Canonical phenotype is FSGS w/ collapsing features, though genotype can worsen other kidney injuries (diabetic, lupus, etc.). **Our KOL estimates ~75% ultimately receive correct dx (timing heterogeneous), while ~15–25% have unfavorable courses partly driven by delay/misdirection, some progressing to ESRD pre-dx.**
- On diagnostic workup (genetics + biopsy), our KOL views AMKD as genetically actionable (2 high-risk genotypes), but sees histopathology as complementary SOC in advanced proteinuria. He favors genetics + biopsy together. Underdx driven by delayed lab recognition and referral; ideal pathway (~3-6 mo) vs real-world delays often spanning years, w/ irreversible damage accruing.
- On Inaxaplin data, our KOL called it “very encouraging,” highlighting its mechanistic specificity (inhibiting pathologic APOL1 ion flux). Proteinuria reduction signals viable substrate (potentially reversible podocyte injury), but irreversible structural damage remains the key unknown. **He believes earlier identification is critical; views AMPLITUDE enrichment as favorably tilted and Ph2a as representative of RW AMKD.**
- On clinical outcomes/endpoints, **our KOL stated UPCR is important, but clinically meaningful endpoint is preserving/slowing eGFR decline.** He expects eGFR to lag proteinuria w/ uncertain timing, influenced by baseline histopathology severity.
- On whether UPCR approval alone would drive adoption, our KOL noted no approved alternatives, so limited academic resistance; any proteinuria reduction signals biologic activity. Magnitude of benefit likely scales w/ earlier tx, but he would still consider later-stage use despite attenuated efficacy.
- On launch dynamics, **our KOL expects academic centers to adopt first, w/ community uptake driven by education diffusion. Cited tolvaptan in ADPKD as example of slower diffusion w/ REMS/monitoring burden; if inaxaplin remains oral + well-tolerated, friction should be lower.**
- On prescribing headwinds, **our KOL emphasized community time constraints (~15 min visits) make extra labs/screening a deterrent; economics/workflow (CKD clinic + dialysis rounding) can shape intensity of management. Low-friction therapy more likely to penetrate broadly.**
- On pricing/access and duration, our KOL expects tx reserved for established AMKD (not genotype+ alone) and, if effective/tolerable, likely lifelong. He believes VRTX unlikely to price unreasonably given duration, volume, and health equity considerations.
- On market sizing, our KOL framed two layers: 1) large genetically vulnerable pool, 2) smaller immediate treatable market (~e.g., ~250k US/EU). Upside from underdx pts + incident cases over time; increasing kidney disease awareness (e.g., recent IgAN approvals) could lift dx rates.
- On competitors (MAZE, LLY, AZN), our KOL sees MAZE/VRTX small-molecule channel inhibitors as conceptually similar, w/ VRTX advantaged by first-mover + visible data (NEJM). Most skeptical of AZN’s ASO/siRNA if primarily targeting liver-derived APOL1; believes podocyte biology is key and kidney delivery must be proven.
- On expected prior auth criteria, our KOL expects **payers to require 1) high-risk genotype, 2) histopathology consistent w/ AMKD-spectrum lesions, and 3) clinical activity (proteinuria and/or early eGFR loss).** Genotype alone unlikely sufficient; expects MAZE to mirror VRTX access strategy in-market.

Source: Jefferies Analysis

Additionally, we think the market could be bigger than expected - VRTX estimates the market at ~250K, but our illustrative TAM analysis suggests ~1.2M pts could develop AMKD

VRTX estimated ~250k TAM in US & EU. That said, we've seen from Pollak et al. 2023 that 13% of the Black/African American population in the US has the APOL1 high risk genotype. We suspect with the advent of the first tx, there could be an increase in diagnosis rates beyond 250k (our illustrative TAM suggests this could be ~1.2M pts that could benefit from Inaxaplin at some point). That said, socio-economic factors could reduce penetration, but we still think this is directionally +ve for market to be larger.

We see increased dx rates coming about from two key levers: **1)** our KOL suspects there could be a spillover for AMKD dx given Pove will enter the market for IgAN first and raise kidney disease awareness, **2)** VRTX currently has a no-cost APOL1 genetic testing (partnerships with Arkana, Labcorp, and Natera) for eligible pts that we expect to expand once there would be a proven tx for the disease.

US MARKET SIZING			
Metric	Value	Units	Source / Notes
Bottom-Up Epidemiology Approach			
US Black/African American Population	48,300,000	people	US Census 2023
APOL1 High-Risk Genotype Prevalence	13.0%	%	Pollak 2023
US High-Risk APOL1 Population	6,279,000	people	Calculated
Disease Penetrance (% developing kidney disease)	20.0%	%	Pollak 2023: ~20% develop overt disease
Total AMKD Patient Pool (US) - Bottom-Up	1,255,800	patients	Calculated
VRTX Company Estimates			
VRTX Total AMKD (US+EU)	250,000	patients	VRTX ASN Presentation 2025
Estimated US Share of Total	85%	%	US has ~83% of US+EU Black pop
VRTX Implied US AMKD Patients	212,500	patients	Calculated
US Patient Estimate (Average)			
	734,150	patients	Average of bottom-up & VRTX estimate
Clinically Relevant US AMKD Population			
VRTX US AMKD Estimate (base)	212,500	patients	VRTX-based estimate
Currently Diagnosed (%)	30%	%	VRTX: 'majority not diagnosed'
Currently Diagnosed Patients (US)	63,750	patients	Calculated
Future Diagnosis Rate (improved awareness)	60%	%	With genetic testing adoption
Future Diagnosed Patients (US)	127,500	patients	At market maturity
Target Patient Segments			
Primary AMKD (AMPLITUDE trial target)	127,500	patients	Est. 60% primary AMKD
AMKD w/ Moderate Proteinuria or Diabetes	85,000	patients	VRTX AMPLIFIED trial target
Total Addressable US Patients	212,500	patients	Sum of segments

VRTX no-cost APOL1 testing program

INTERESTED IN A NO-COST APOL1 GENETIC TEST FOR ELIGIBLE PATIENTS?*

There are different types of tests for APOL1, such as single-gene tests⁴ and panels that include APOL1.⁷ The diagnostic companies below offer no-cost APOL1 genetic testing and counseling options for eligible patients, sponsored by Vertex Pharmaceuticals.



Arkana
Laboratories

Offers a single-gene test for APOL1 and free genetic counseling before and after.

LEARN MORE >



labcorp

Offers a single-gene test for APOL1 and free genetic counseling for patients with confirmed APOL1 risk variants.

LEARN MORE >



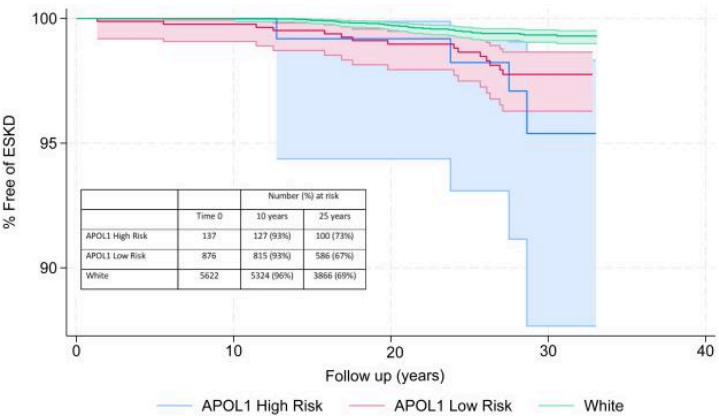
natera™

Offers a renal panel test for 397 genes, including APOL1, and free genetic counseling before and after.

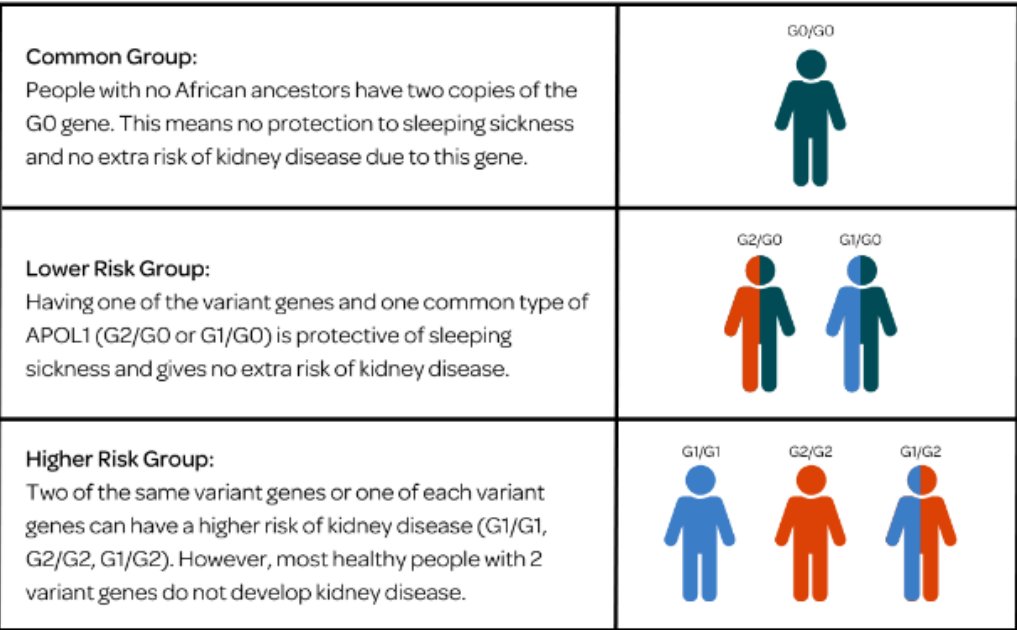
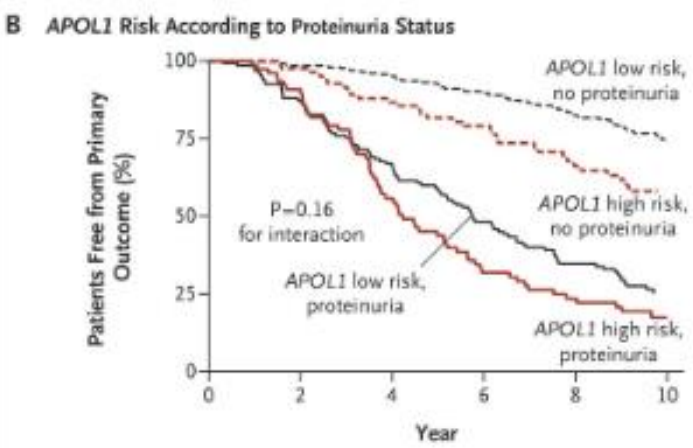
LEARN MORE >

Source: Jefferies Analysis, Company Presentation

We're also expecting Inaxaplin to show durability given a) it reduces disease progression, but does not correct the underlying genetic modification, b) we see a read across to Alport syndrome (another genetic kidney disease with lifelong treatment)



We've seen that AMKD is a progressive renal disease (e.g. % free ESKD white control clearly split from low-risk/high risk at ~20yrs of follow-up) . Furthermore, Parsa et al. reported that the risk variant & proteinuria status affect the rate of disease progression. We've also seen with other genetic kidney diseases (e.g. Alport's syndrome) that treatment is life-long given that they don't resolve the underlying genetic mutation. Likewise, if Inaxaplin were approved, it would be the first targeted therapy for AMKD, but also not be genetically modifying, hence we'd expect pts to stay on tx lifelong.



Source: Jefferies Analysis, Company Presentation

We'll quickly touch on Autosomal Dominant Polycystic Kidney Disease (ADPKD):

ADPKD is a genetic disorder (cysts on kidney) – VX-407 aims to correct protein folding for the PKD1 mutation. The current bar for efficacy in ADPKD is Tolvaptan (only approved targeted tx), which slowed eGFR change by 1.27ml/min/1.73m² vs Pbo

As a background, ADPKD is the most common inherited cause of end-stage kidney disease. The disease is characterized by patients having **fluid-filled cysts on their kidney that lead to progressive kidney function decline**. The disease is associated with PKD1 (85%) of cases and PKD2 (15%) mutation.

VRTX is advancing **VX-407**, a **first-in-class small molecule corrector that aims to restore function to the defective PC1 protein**. This is a protein critical for kidney cell regulation. VX-407 aims to restore PC1 function by correcting its folding, thereby halting cyst growth and reducing kidney volume to prevent progression to kidney failure.

VRTX commented that, since there are no approved therapies that address the underlying cause of ADPKD, it is an indication of significant unmet need. Mgmt estimates that VX-407 could reach ~30k pts. Ph.1 VX-407 completed Jun 2025 and Ph.2 PCD is July 2027 – we noted Ph.1 had n=159 but Ph.2a n=24

We're watching 3 comps in ADPKD:

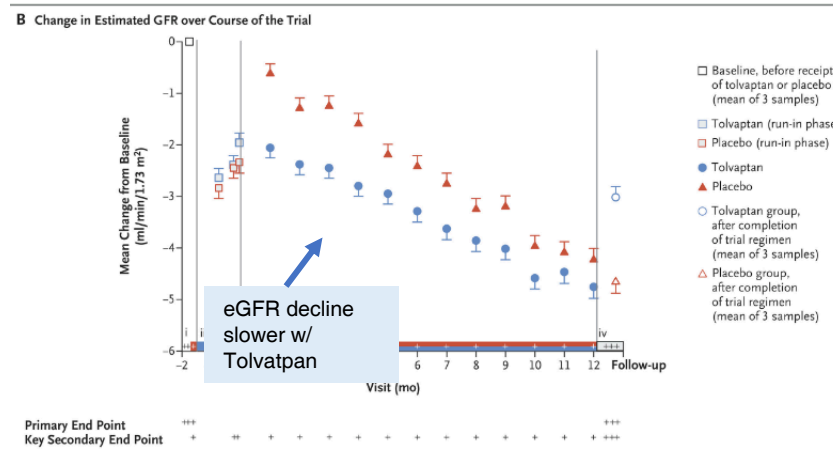
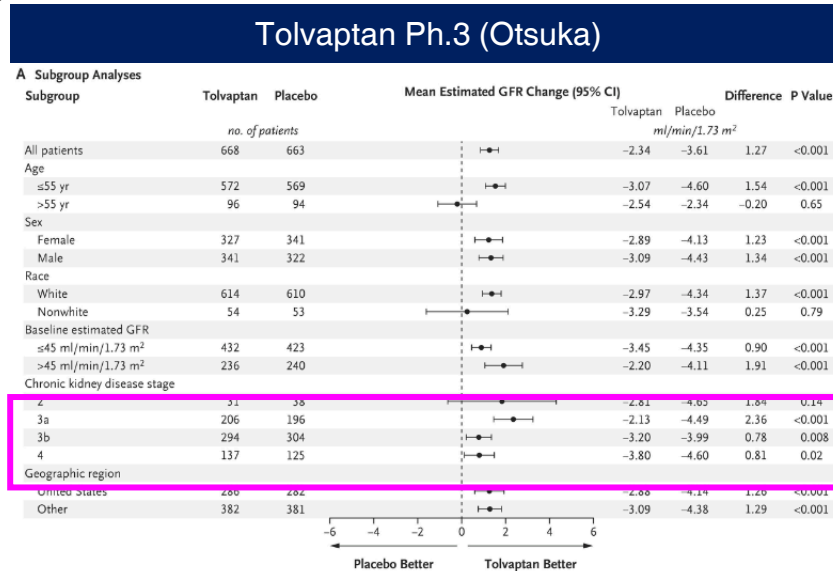
Name	Mechanism of Action
ABBV-CLS-628	mAb targeting PAPP-A to reduce IGF-1 bioavailability and slow cyst growth
Farabursen	Anti-sense Oligonucleotide inhibiting microRNA-17 to restore polycystin expression and reduce cyst formation
Lixivaptan	Selective vasopressin V2 receptor antagonist lowering intracellular cAMP to slow cyst proliferation

Source: Jefferies Analysis, Mayo Clinic

Source: Jelle
Akash Tewari

Healthcare Equity Research

Email: atewari@jefferies.com Tel. +1-212-284-3416



As a background, Tolvaptan is a V2 receptor antagonist that blocks vasopressin signaling (key driver of cyst growth on kidneys).

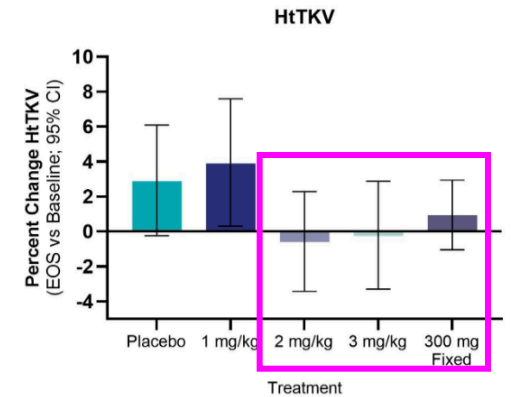
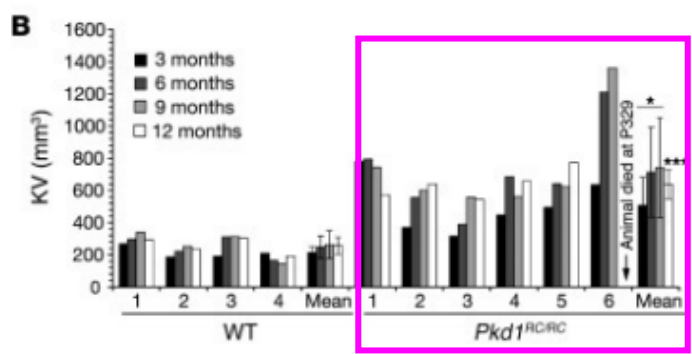
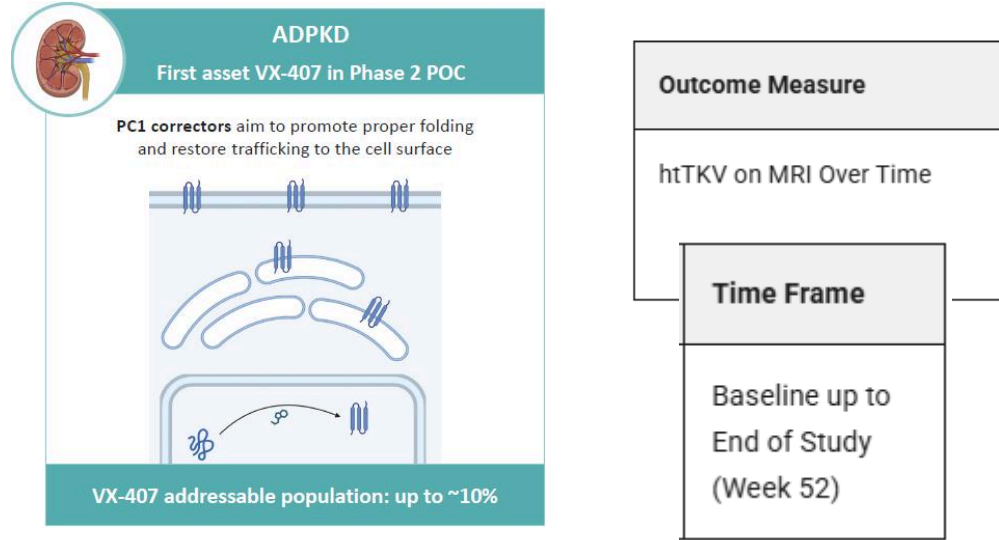
The REPRISE trial was the pivotal trial that showed efficacy for Tolvaptan in ADPKD. Tolvaptan showed an **eGFR drop of 2.34 ml/min/1.73 m² vs 3.61 ml/min/1.73 m² in Pbo (p < 0.001)**. In subgroup analysis Tolvaptan showed efficacy in CKD stage 3+, but not in stage 2. Also, no efficacy was shown in pts above 55y/o..

While data for VX-407 is still premature, we're seeing 3 datapoints suggesting it can show efficacy: 1) small increases in PC1 can lead to TKV stability, 2) Farabursen's efficacy shows PC1 is a valid target, 3) VRTX chose htTKV as an endpoint in Ph.2 suggesting it has been derisked

1) VX-407 is a small molecule designed to help misfolded PKD1 protein (PC1) work better by correcting its folding and improving its movement and signaling in the cell. Research shows that even partially restoring PC1 function (not completely fixing it) can slow cyst growth in polycystic kidney disease. Because of this, VX-407 does not need to fully correct PKD1 to have an effect on TKV. We've seen VRTX chose htTKV as the main proof-of-concept outcome to directly test if improving PC1 folding and function can slow disease progression, similar to the effects seen with tolvaptan (which lowers cAMP) and farabursen (which increases PC1/PC2), even though VX-407 works through a different mechanism.

2) Multiple genetic and preclinical models show PKD1 behaves as a dosage-sensitive gene, with graded phenotypes rather than binary loss-of-function. In hypomorphic and conditional PKD1 mouse models, restoring ~20-30% of normal PC1 expression is sufficient to reduce cyst burden and slow kidney volume increase, but complete loss produced rapid TKV increase. Importantly, heterozygous/mosaic PKD1 expression states generated slower cyst growth rates vs null models, supporting the concept that even modest increases in functional PC1 can shift the slope of TKV progression. Taken together, it suggests htTKV is a continuous, polycystin-responsive endpoint, rather than one requiring full pathway suppression.

3) Farabursen (anti-miR-17) derisks VX-407 by showing upregulating endogenous PC1/PC2 translated to stabilized TKV. In 12wk dosing, this magnitude and directionality are notable given that historical untreated ADPKD cohorts typically show ~3-5% TKV growth/year. To us this reinforces turning up polycystin alone (without targeting cAMP) can lead to similar efficacy as Tolvaptan.



Source: Jefferies Analysis, Company Publications

We looked to Trikafta as a read across to VX-407 and found it had elevated liver markers (e.g. ALT 3x >ULN 8% vs 5%) – while we’re expecting LFTs to be raised in VX-407, it could pot’ly avoid REMS if MOA is indeed similar to Trikafta

Since we’ve not seen the Ph.1 safety data for VX-407, we looked at Trikafta given the similarity in MOA (both are small molecules proteostasis modulators that improve protein folding/trafficking). In the FDA review, we saw that Trikafta had elevated ALT/AST (ALT >3x ULN 8% vs 5% pbo), **suggesting there could also be liver tox for VX-407 as well. That said, the MOA for VX-407 is indeed similar to Trikafta it could pot’ly avoid REMS – theoretically making it safer vs Tolvaptan.**

Trikafta

WARNING: DRUG-INDUCED LIVER INJURY AND LIVER FAILURE
See full prescribing information for complete boxed warning.

- TRIKAFTA can cause serious and potentially fatal drug-induced liver injury. Liver failure leading to transplantation and death has been reported. (5.1, 6)
- Assess liver function tests (ALT, AST, alkaline phosphatase, bilirubin) in all patients prior to initiating TRIKAFTA. (2.1, 5.1)
- Monitor liver function tests (ALT, AST, alkaline phosphatase, bilirubin) every month for the first 6 months of treatment, then every 3 months for the next 12 months, then at least annually. (2.1, 5.1)
- Interrupt TRIKAFTA for significant elevations in liver function tests or signs or symptoms of liver injury. Follow patients closely with clinical and laboratory monitoring until abnormalities resolve. (5.1)
- Resume TRIKAFTA if abnormalities resolve and only if the benefit is expected to outweigh the risk. (5.1)
- TRIKAFTA should not be used in patients with severe hepatic impairment (Child-Pugh Class C). TRIKAFTA is not recommended in patients with moderate hepatic impairment (Child-Pugh Class B). (2.3, 5.1, 8.7, 12.3)

Tolvaptan

WARNING: RISK OF SERIOUS LIVER INJURY

JYNARQUE (tolvaptan) can cause serious and potentially fatal liver injury. Acute liver failure requiring liver transplantation has been reported [see Warnings and Precautions (5.1)].

Measure ALT, AST and bilirubin before initiating treatment, at 2 weeks and 4 weeks after initiation, then monthly for the first 18 months and every 3 months thereafter [see Warnings and Precautions (5.1)]. Prompt action in response to laboratory abnormalities, signs, or symptoms indicative of hepatic injury can mitigate, but not eliminate, the risk of serious hepatotoxicity.

Because of the risks of serious liver injury, JYNARQUE is available only through a restricted distribution program under a Risk Evaluation and Mitigation Strategy (REMS) called the Tolvaptan for ADPKD Shared System REMS [see Warnings and Precautions (5.2)].

For Tolvaptan, while the exact cause of liver tox is unknown a key hypothesis is the inhibition of bile acid transport.

- Tolvaptan has a REMS because clinical trials and post-marketing data showed a significant risk of serious and potentially fatal drug-induced liver injury, inc. **Hy’s Law cases and a liver transplant** (est. risk of about 1 in 3,000 pts). This risk requires a boxed warning, mandatory monthly liver function monitoring, and a REMS to ensure early detection and intervention.
- In contrast, Trikafta showed mild to moderate, **reversible liver enzyme elevations**, but no **Hy’s Law cases or irreversible liver injury were seen in trials**, so the FDA determined that routine monitoring and labeling were sufficient, and a REMS was not needed.

Trikafta and Tolvaptan had higher rates of ALT>3x ULN vs Pbo.

Parameter / Event	Trikafta vs Pbo	Tolvaptan vs Pbo
ALT >3x ULN (vs. placebo)	8% vs 5%	4.1% vs 1.0% (TEMPO); 5.6% vs 1.2% (REPRISE)
ALT >5x ULN	2% vs 1%	NR
ALT >8x ULN	1% vs 1%	NR
Total Bilirubin >2x ULN	3% vs <1%	NR
Liver-related discontinuations	2% vs 1%	5.4% vs 0.6% (REPRISE)

“The potential risks of LFT elevation, rash, CK elevation, cataracts, increased blood pressure, and drug-drug interactions can be managed through labeling and routine pharmacovigilance; **a REMS is not required. No substantial safety findings were identified in the clinical development program that outweigh the potential benefit.**” – FDA Review Trikafta

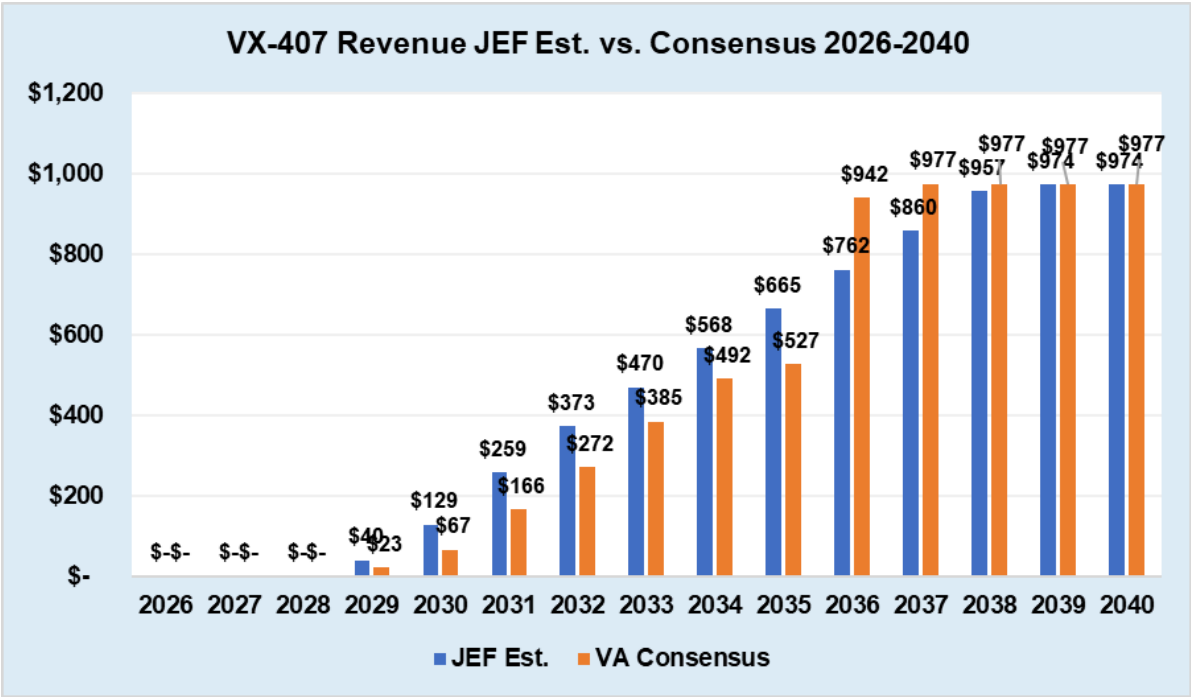
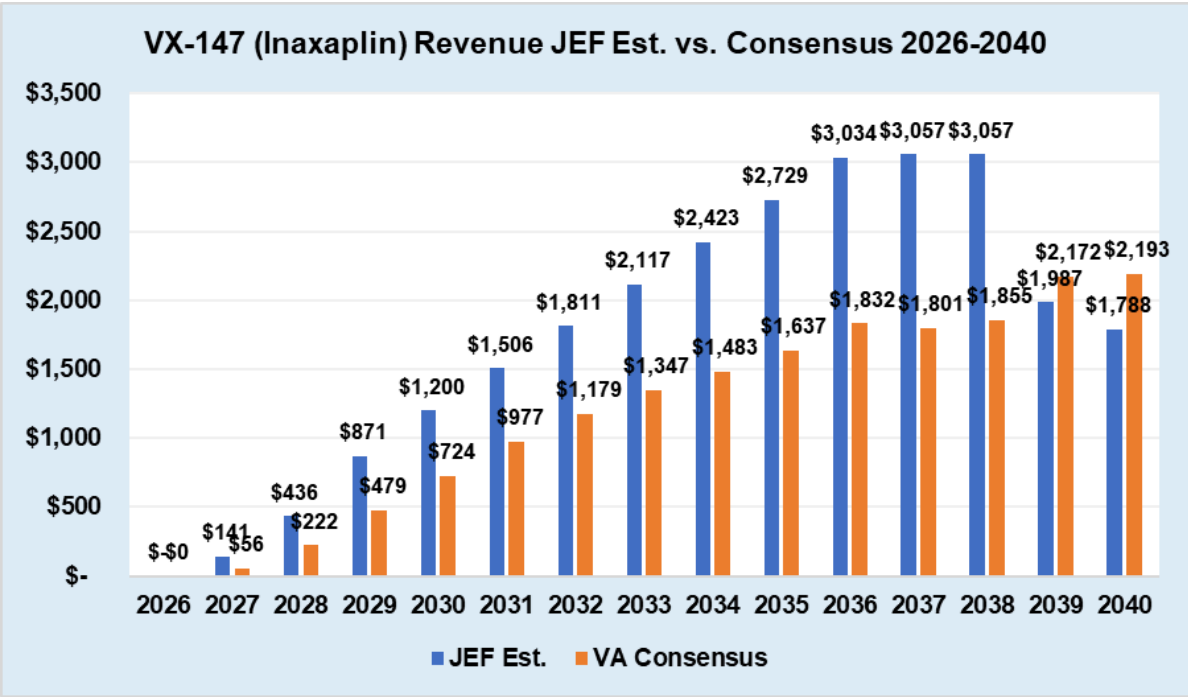
Source: Jefferies Analysis, FDA

We model ~\$3Bn peak sales for Inaxaplin in AMKD (above cons at \$2.2Bn, but we suspect TAM could be even larger w/ improved genetic screening/effective tx) and ~\$974M for VX-407 in ADPKD (in line with cons)

We model ~\$3Bn risk adjusted peak sales for Inaxaplin (VX-147) and our ramp is above consensus based on the early mover advantage for VRTX being first to market. We'll flag a) VRTX's entrance will improve genetic screening for a clearly defined genotype leading to a bigger market size (we've seen could be ~1.2M), b) our KOL believes the entrance of Pove could lead to spillover awareness of kidney disease that leads to higher dx rates.

We model ~\$974M peak sales by 2037 for VX-407 which is approx in line with consensus (few analysts model past 2035)

We'll also flag VRTX mgmt. has commented that the kidney franchise could become as large as the CF franchise long-term: *"We anticipate that the renal franchise will ultimately rival the scale of our CF business and we're seeking to bring the best elements of our success in CF to these kidney disease areas."* – VRTX mgmt. 12 Feb 2026



Source: Jefferies Analysis, Visible Alpha

Next, let's go over VRTX's pain program

Cystic Fibrosis

We expect VRTX's CF franchise to remain durable

- We're expecting CF to remain a durable revenue base for VRTX and expect gradual switch & new starts for pts onto Alyftrek
- We expect upcoming comps will work as add-on tx to VRTX, but we're not seeing evidence as a mono-tx they could produce greater efficacy
- We expect minimal pt adds from VRTX mRNA CF tx, but limited ability for generics to gain market share

IgAN & autoimmune

We see versatility for APRIL/BAFF across autoimmune indications

In IgAN, we saw 49.8% Pbo adj. UPCR reduction at 9mo (clinically in line w/ Sibe):

- To us, the impt point is: a) IgAN is a bigger market with favorable pricing (\$390K/yr), b) Pove has a convenience advantage with 1.5sec admin autoinjector
- We're optimistic on the indication expansion opportunity into gMG, pMN, wAIHA

AMKD & ADPKD

We see LT upside from Inaxaplin becoming first/best in class

- We're expecting Inaxaplin to be first & best in class in the AMKD market (currently no tx)
- We expect 64-70% UPCR reduction at 48wks + we think VRTX has the more favorable BL population vs comps
- We think the AMKD market could be bigger than initially expected given a) an approved tx will likely increase dx rates, b) our KOL expects spillover dx from the IgAN market growing from Povetacicept
- In ADPKD, we think the data is too early for VX-407 but could see pot'l reduced LFTs based on Trikafta read across (similar MOA)



Pain

We're mixed on Ph.3 DPN hitting, but are more confident in the acute pain opportunity

- We're mixed on whether chronic DPN can hit for Suze, given the high variability of previous data and Pbo (combo approach is more likely here).
- In acute pain, we expect 3x growth for Journavx based on recent access expansion

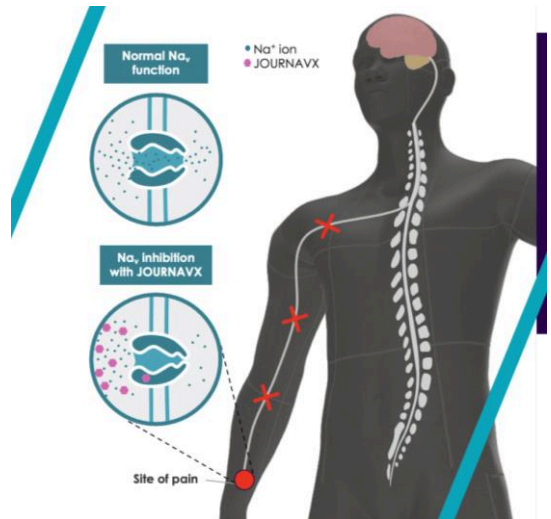
Rest of Pipeline

Rest of pipeline is interesting but we're skeptical on cons valuation

- We're skeptical on expansion opportunity for rest of pipeline:
- For Casgevy, we've seen precedence that viable tx's in SCD launch slower / underpenetrate the market due to patient/physician concerns & access barriers
- Zimislecel appears limited to severe T1D population based on immunosuppression requirement & clinical trial delays

Source: Jefferies analysis

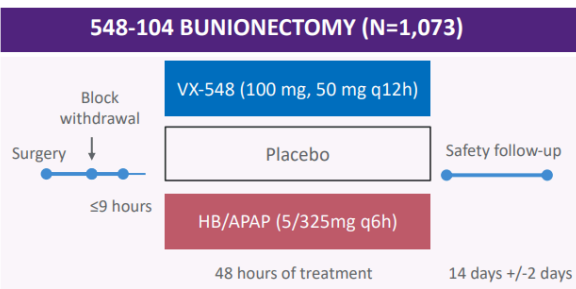
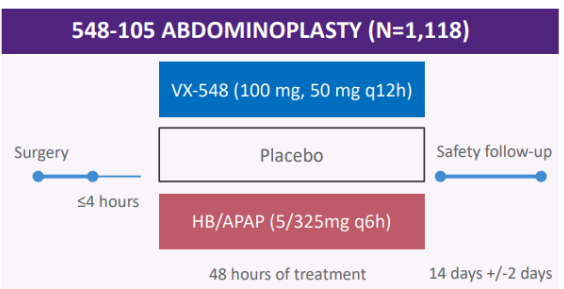
Starting off, Journavx (suzetrigine) is a selective NaV 1.8 inhibitor that was approved in 2025 for **acute pain** following 2 Ph.3 studies testing primary endpoint of SPID48 pain score – the trials were mixed, with suzetrigine hitting stat sig vs pbo but missing vs active comparator (hydrocodone/acetaminophen)



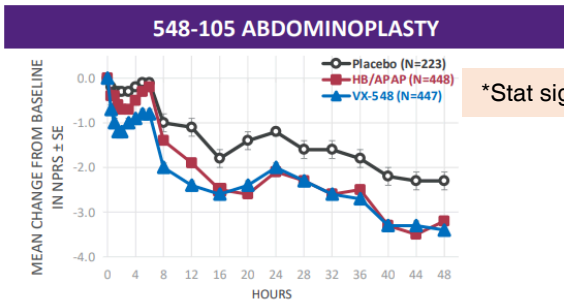
Journavx (suzetrigine) is a selective NaV 1.8 Inhibitor selectively targeting peripheral neurons to prevent the transmission of pain signal to the brain. In acute pain trials, it had a maintenance dose of oral 50mg BID.



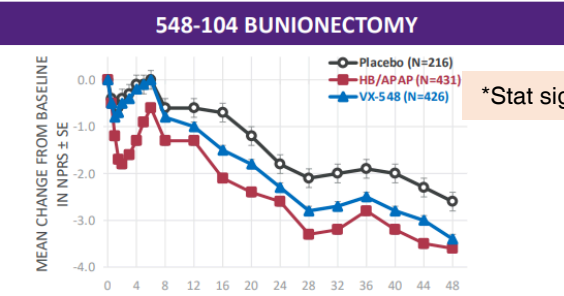
VRTX ran 2 Ph.3 trials in acute pain (delivery after abdominoplasty & bunionectomy surgery). Both tested Journavx vs Pbo and also vs Hydrocodone/Acetaminophen



On primary endpoint (SPID48 Score vs Pbo), Journavx showed a stat sig reduction in Ph.3 Abdominoplasty & Bunionectomy (P<0.001) vs Pbo, but was not supported to be stat sig superior to Hydrocodone/Acetaminophen for either acute pain trials. That said, the NPRS secondary endpoint was stat sig for both trials.



	PLACEBO N=223	HB/APAP 5 mg / 325 mg q6h N=448	VX-548 100 mg, 50 mg q12h N=447
Baseline NPRS, mean	7.5	7.4	7.3
Change from baseline in NPRS, mean	-2.3	-3.2	-3.4
% reduction from baseline in mean NPRS	31%	43%	47%

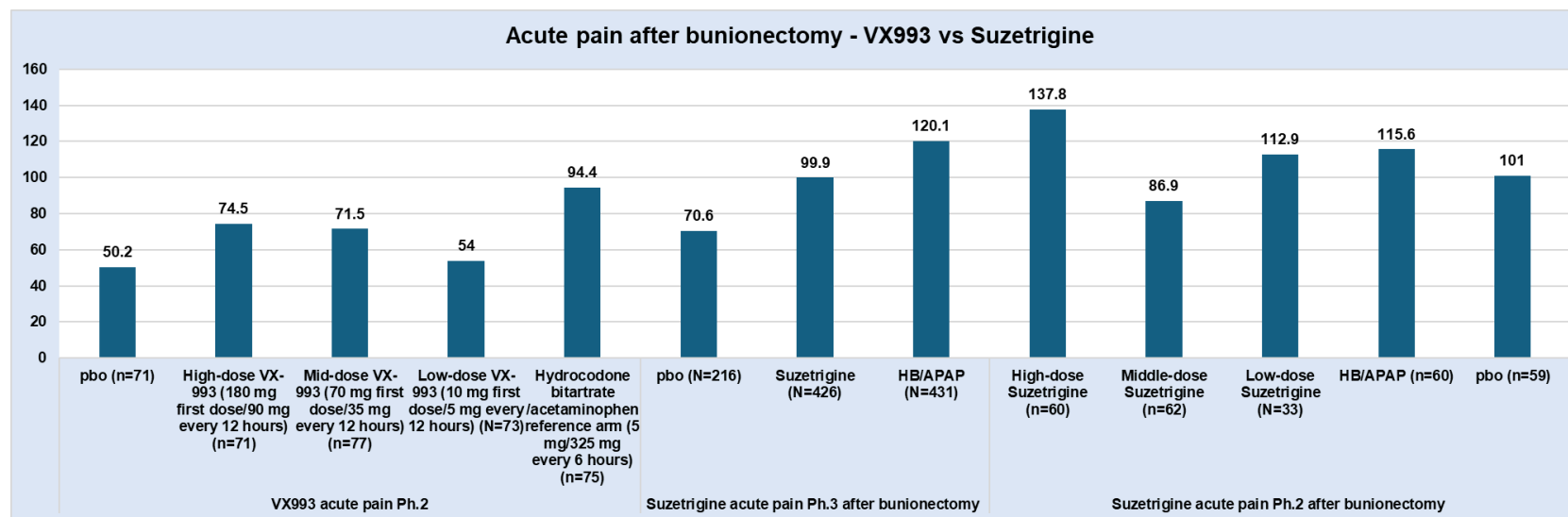


	PLACEBO N=216	HB/APAP 5 mg / 325 mg q6h N=431	VX-548 100 mg, 50 mg q12h N=426
Baseline NPRS, mean	6.8	6.8	6.7
Change from baseline in NPRS, mean	-2.6	-3.6	-3.4
% reduction from baseline in mean NPRS	38%	53%	51%

Source: Jefferies Analysis, Company Publications

Interestingly, while next gen NaV1.8 inhibitor VX993 aimed for greater target inhibition, it failed to show a stat sig benefit in Ph.2 acute pain – to us, this suggests suzetrigine may already cap out on NaV1.8 inhibition efficacy in acute setting

- In VRTX's Ph.2 bunionectomy trial (N=367, August 2025), VX-993 - a next-generation NaV1.8 inhibitor designed to test whether greater target inhibition yields superior efficacy - failed to demonstrate statistically significant pain reduction versus placebo across all three doses tested. The high-dose arm (180mg loading/90mg Q12H, n=71) achieved a SPID48 difference of 24.3 points versus placebo (95% CI: -6.3 to 54.9; p=0.119), while the mid-dose (70mg/35mg Q12H, n=77) showed a 21.2-point difference (p=0.164) and low-dose (10mg/5mg Q12H, n=73) showed only 3.7 points (p=0.809).
- In contrast, suzetrigine's Phase 2 bunionectomy trial (n ≈ 274) demonstrated a statistically significant 36.8-point SPID48 improvement versus placebo (95% CI: 4.6 to 69.0; p<0.05). This efficacy gap widened in suzetrigine's confirmatory Phase 3 program: the bunionectomy pivotal (N=1,073) achieved a 29.3-point SPID48 advantage (95% CI: 14.0 to 44.6; p=0.0002), while the abdominoplasty trial (N=1,118) showed a 48.4-point difference (95% CI: 33.6 to 63.1; p<0.0001). Both compounds exhibited favorable safety profiles versus the opioid comparator (hydrocodone/acetaminophen), with VX-993 showing numerically lower rates of nausea (4.1% vs. 19.0% for suzetrigine) and comparable CNS-related adverse events.



“Sure. So, I think that what and perhaps why there was what felt to us like a disproportionate sell-off, I think it was primarily related to concerns about the ultimate peripheral neuropathic pain potential. But then, in VX-993, a next-gen that was being studied in bunionectomy and acute pain, we did see a similar size treatment effect, but had hoped for more essentially based on what we had seen in terms of the potency and efficacy.

*But what we learned, and I think probably in hindsight on IR we should have framed better, is that **going higher up the curve did not get you greater treatment back. Essentially, we are getting all we can hope to get out of suzetrigine out of JOURNAVX by inhibiting the NaV1.8 channel.**” – VRTX Mgmt, Sep’25*

Source: Jefferies Analysis, Company Publications

Suzetrigine is also being tested for **chronic pain**, in two Diabetic Peripheral Neuropathy (DPN) Ph.3's – the trials have primary endpoint of pain score (NPRS) vs either active comparator (pregabalin) or pbo control

On Q2'25 call, VRTX mgmt. indicated that they will initiate a second Ph.3 to pursue PNP approval based on FDA feedback from EOP2 after deciding not to initiate another LSR trial: “Now shifting to pain. First, in peripheral neuropathic pain or PNP, we had a productive end of Phase II meeting with the FDA on our PNP program. While a broad PNP label remains our goal, at this time, the FDA does not see a path to a broad indication. As such, we will not be initiating an LSR trial at present. Instead, in light of the discussions and clear agreement with the FDA regarding approval requirements for diabetic peripheral neuropathy or DPN, we will begin a second DPN Phase III study shortly in order to secure DPN as our first PNP indication for Suzetrigine.” –VRTX Mgmt, Q2'25 EPS call

VRTX DPN trial comparison

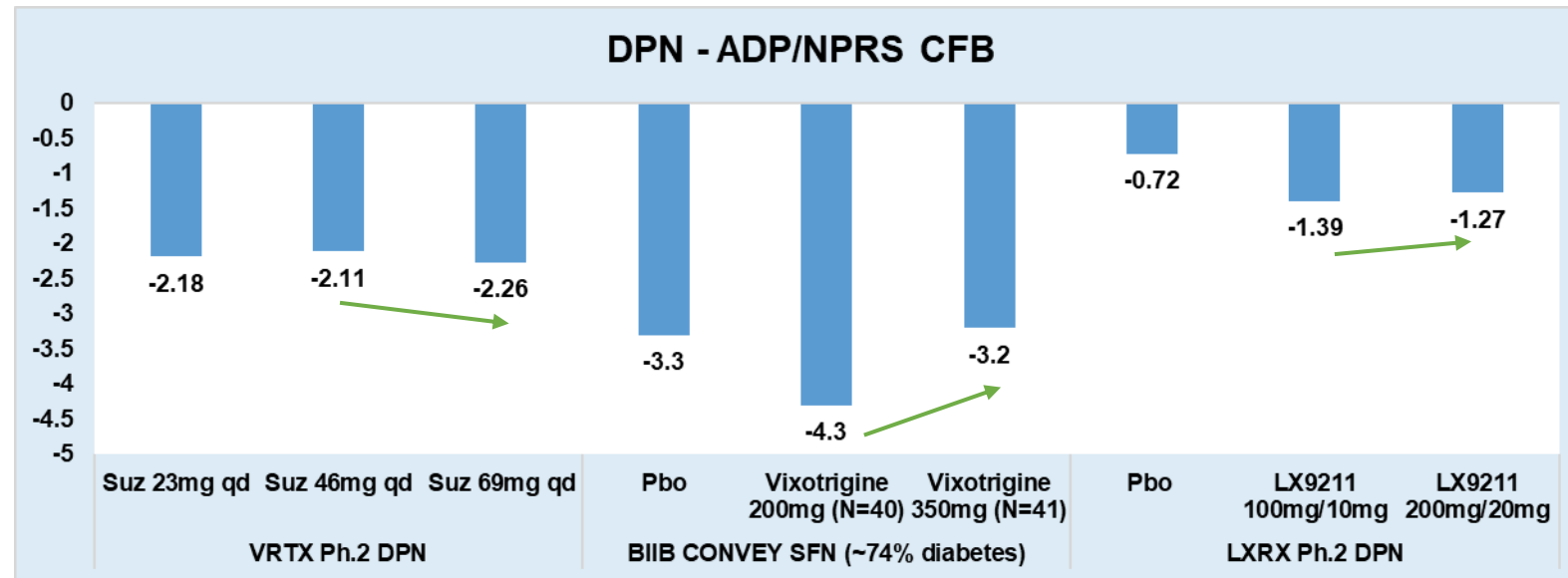
Attribute	VX-993 Ph2 (DPN)	VX-548 Ph3 (DPN, NCT06628908)	VX-548 Ph3 (DPN, NCT07231419)
Compound	VX-993	Suzetrigine (VX-548)	Suzetrigine (VX-548)
Study ID	VX24-993-103	VX24-548-110	VX24-548-111
Indication	Diabetic Peripheral Neuropathy (DPN)	DPN	DPN
Study Phase	Phase 2	Phase 3	Phase 3
Design	Randomized, double-blind, active-controlled, dose-ranging, parallel	Randomized, double-blind, placebo- and active-controlled	Randomized, double-blind, placebo-controlled
Active Comparator	Pregabalin	Pregabalin	None
Placebo Arms	Two placebos (matched to each active)	Two placebos (matched to each active)	Placebo (matched to SUZ)
Estimated Enrollment	300	1100	734
Study Start	Oct-24	Oct-24	Nov-25
Primary Completion	Apr 2026 (est.)	May 2027 (est.)	Apr 2027 (est.)
Primary Endpoint	Change in weekly avg. daily pain (NPRS) at Week 12	Change in weekly avg. daily pain (NPRS) at Week 12	Change in weekly avg. daily pain (NPRS) at Week 12 vs placebo
Duration of Pain Required	≥1 year of bilateral DPN pain	Not specified	≥1 year of bilateral DPN pain
BMI Criteria	Not specified	≥18.0 and <40.0 kg/m ²	≥18.0 and <40.0 kg/m ²
Baseline NPRS Requirement	Not specified	Avg. NPRS 4–9	Avg. NPRS 4–9
Key Exclusions	Non-DPN neuropathy, recent CV events, arrhythmias	Sensory abnormalities (non-DPN), >3 missing NPRS days	Journavx exposure, sensory abnormalities (non-DPN), >3 missing NPRS days

Source: Jefferies Analysis, Company Publications

VRTX is running two Ph.3's for suzetrigine (VX-548) in diabetic peripheral neuropathy (DPN) evaluating 12-week NPRS pain reduction but with different randomization/comparator schemes. The first (VX24-548-110; n=1100) is a placebo- and active-controlled study looking at both pregabalin and placebo, more closely following VX-993's Ph.2 structure which was run vs pregabalin. The second (VX24-548-111; n=734) is pbo-controlled only, and could provide confirmatory efficacy. Both trials share similar inclusion criteria (BMI 18–40, NPRS 4–9, HbA1c ≤9%, ≥ 1 yr bilateral DPN pain), but we noticed the second trial explicitly excludes recent Journavx exposure, pot'l to manage cross-trial interference from previous Nav1.8 studies.

Mechanistically, we're not seeing strong evidence that non-opioids (including Suzetrigine's NaV1.8 inhibition) can show large or dose-dependent effects on pain reduction in DPN

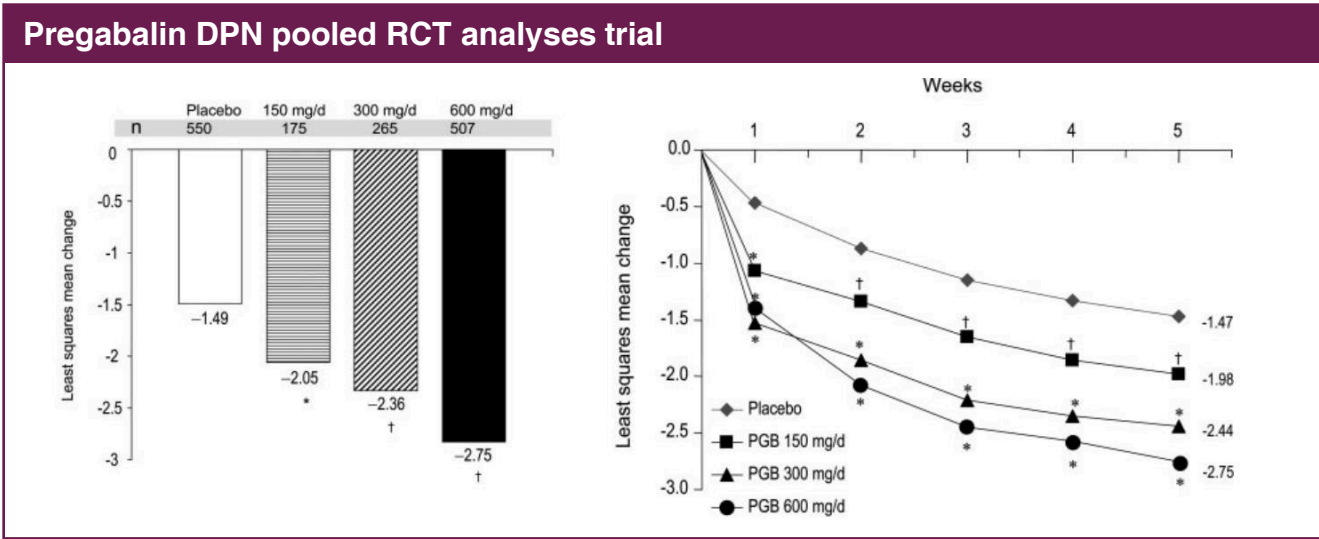
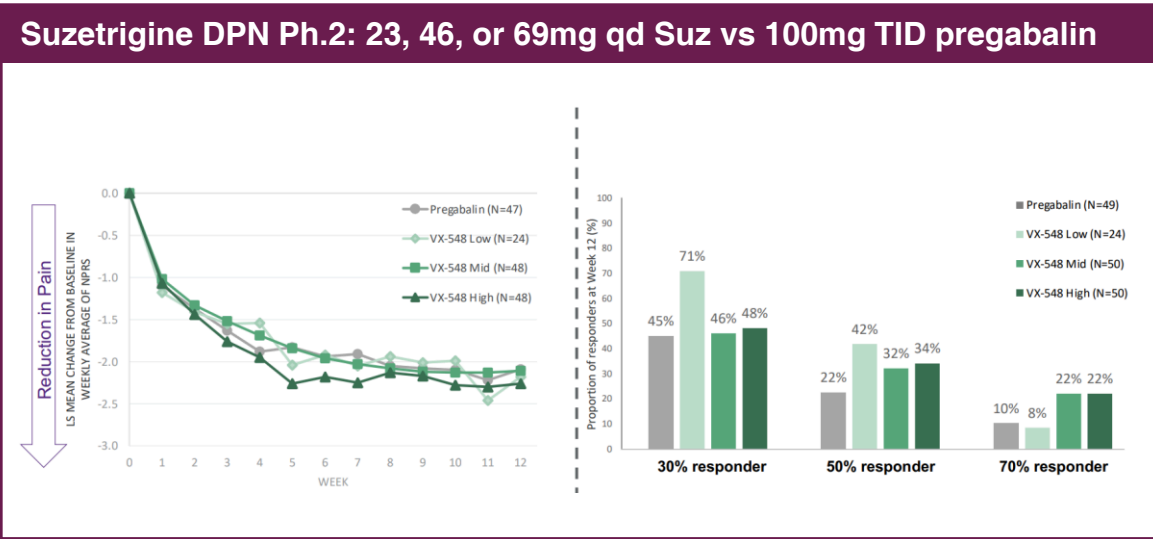
- We don't have a large dataset for NaV1.8 inhibitors like suzetrigine in chronic neuropathic pain indications yet, however, **we've noticed that across three non-opioid programs, more drug hasn't translated into more pain relief.** E.g. VRTX's suzetrigine DPN Ph.2 showed -2.1 to -2.3 across 23-69mg qd, BIIB's vixotrigine Ph.2 in SFN (~74% diabetic pts) also showed that the lower 200 mg BID dose was numerically better than the higher 350 mg BID (LS mean vs placebo: -0.85 and -0.17 at wk12, respectively), and LXR's AAK1 inhibitor LX9211 in Ph.2 RELIEF-DPN1 showed that both 100 mg/10 mg and 200 mg/20 mg regimens were similar improvements (LS mean chg: -1.39 and -1.27 points vs -0.72 with placebo).
- Mechanistically, suzetrigine and vixotrigine are highly selective Nav1.8 sodium-channel blockers, while LX9211 inhibits AAK1 to modulate clathrin-mediated endocytosis and downstream nociceptive signaling. All three aim to dampen hyper-excitable neurons, yet their dose-response curves flatten early, suggesting that once a threshold of target engagement is reached, further channel blockade or AAK1 suppression provides little incremental benefit. Theoretically, this could be because downstream network plasticity, phenotype heterogeneity, or central sensitization, rather than target occupancy become efficacy bottlenecks.



Source: Jefferies Analysis, Company Publications

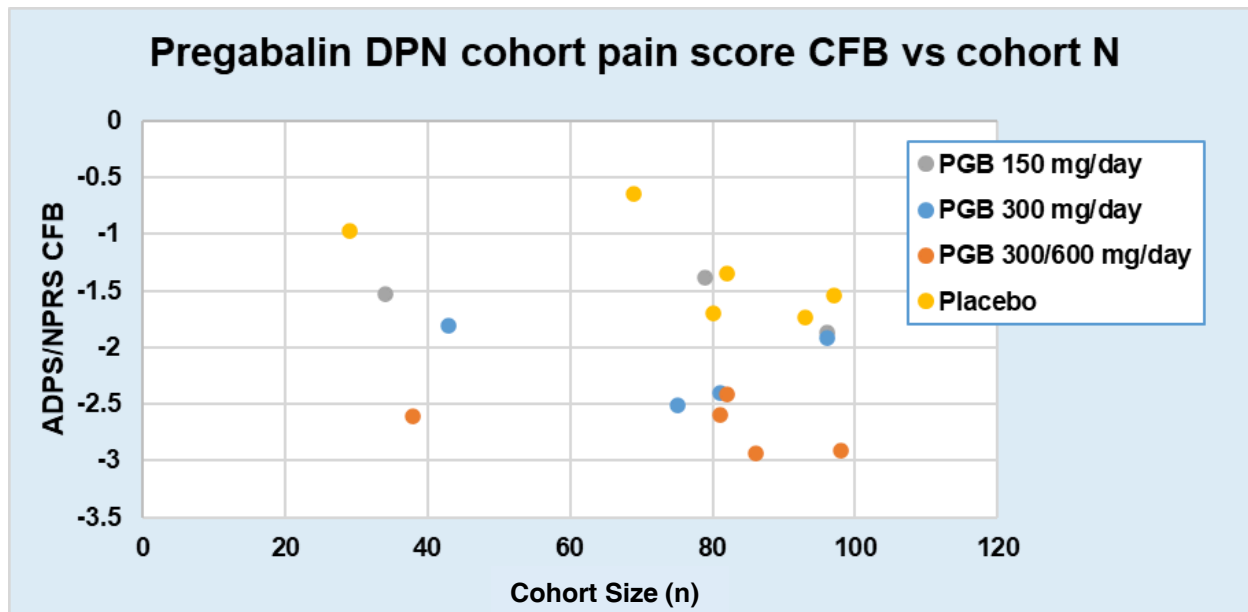
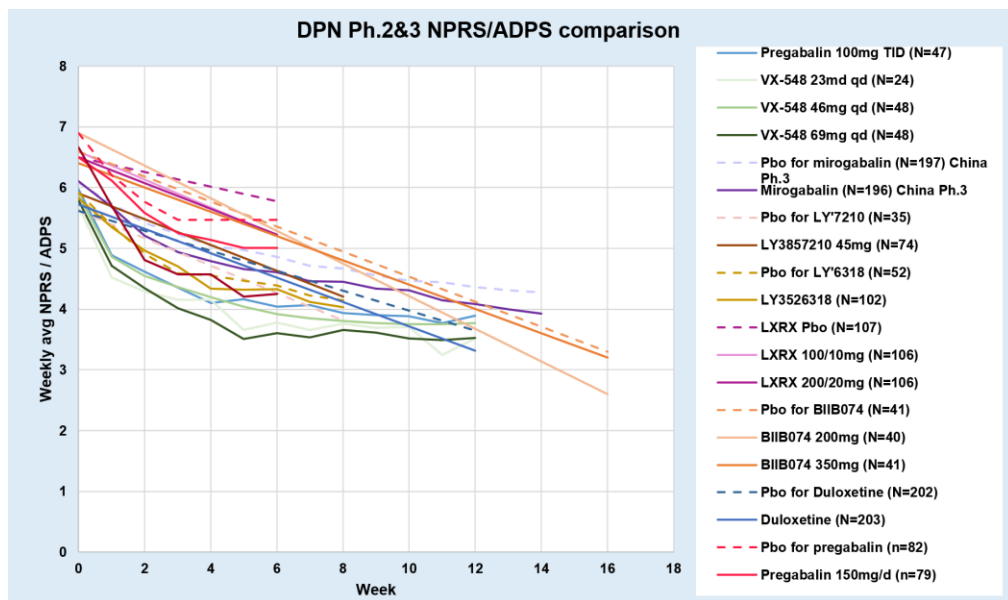
Overall, we're mixed on both chronic pain DPN trials & we model 30% POS: In Ph.2, suzetrigine 23-69mg QD did not show dose dependent reductions on NPRS & numerical pain reductions of -2.18 to -2.26 points looked worse than higher doses of pregabalin (-2.44 for 300mg/d, -2.75 for 600 mg/d)

- VRTX's 12wk DPN Ph.2 study showed that all three once-daily doses of suzetrigine produced numerical pain reductions of -2.18 to -2.26 points on the 11-point NPRS that were not clearly dose dependent, vs pregabalin 100 mg TID fell -2.09 points over the same period, suggesting similar efficacy.
- Notably, there wasn't a clear dose response for VRTX, but for pregabalin, we've historically seen that pregabalin at higher doses separates pbo separation – e.g. Richter et al. showed in a 6wk RCT that pregabalin 600mg/d was stat sig vs pbo (-1.264; p=0.0002), but 150mg/d was not clearly stat sig. (-0.440; p=0.1763)
- To us, pregabalin 100 mg TID (300 mg/d) seems like a relatively softer benchmark as suzetrigine wasn't compared to an optimally effective dose. That said, under N~300-350 pts per arm, the -0.440 pbo adj delta that pregabalin 150mg/d showed in Richter et al would likely be stat sig. based on our illustrative stat analysis.



For Ph.3, we expect similar efficacy to Ph.2 of around 2.11~2.26 point absolute change from baseline, as we've generally seen efficacy stay flat or get better as trial N increases in DPN studies

- When we caught up w. mgmt., they mentioned using specific experienced centers with well-trained investigators that know how to instruct patients on how to use the pain scale. On our end, we noticed that when pregabalin went into larger N cohorts (N~30-40 to ≥ 80), efficacy tended to stay flat or even improve slightly vs smaller cohorts – directionally, we think this could mean Suzetrigine could replicate its Ph.2 efficacy in the DPN Ph.3.
- Putting it together, our base case is that the Ph.2 efficacy shown so far (~2.11-2.26pts at 23-69mg qd) will carry over into the Ph.3 which is looking at 69mg qd based on the schema in VRTX's '953 patent, but we're not necessarily expecting the 70mg qd dose to get to the higher end of the range as there wasn't a clear dose response at wk12.



Source: Jefferies Analysis, Company Publications

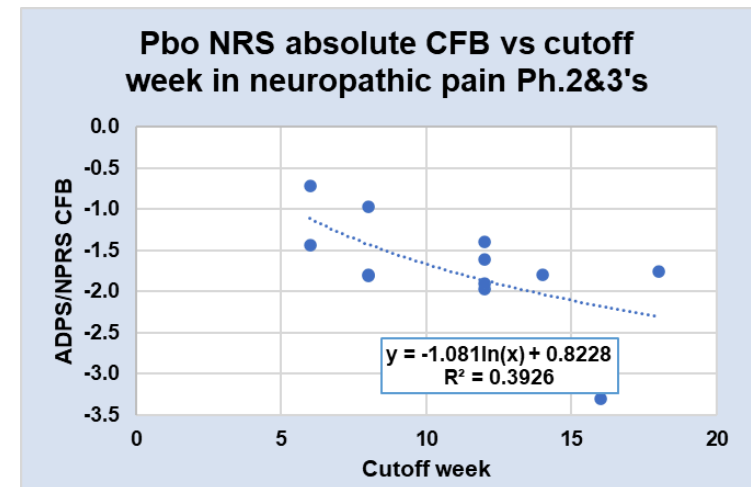
To us, the real risk is pbo variability (note that VRTX's DPN Ph.2 wasn't pbo controlled) – we've seen avg pbo reductions of 1.7~1.9 on pain scores in DPN Ph.2&3's + longer cutoff time generally results in steeper pbo response (1.4~2.0 at wk12 vs ~0.7-1.8 at wk6-8)

- We performed a meta analysis of 12 modern neuropathic-pain RCTs (6-18 wk) and see placebo CFB ranges from –1.6 to –2.0 on NRS whenever exposure is ≥ 8 weeks, with mean of ~1.7. Imptly, weighting by patient counts leaves that figure essentially unchanged. However, a relatively important covariate seems to be the cutoff weeks – as longer cutoff week weakly correlates to a steeper pbo response with $R^2 \sim 0.39$ based on our illustrative log regression fit.
- Variability around pbo response SE/SD also looks generally consistent as 9 of 12 trials report (or imply) SDs between 1.8-2.3, averaging ~2.0 pts.
- **Taken together, we'd expect a ~1.7-1.9pt placebo reduction the primary pain score endpoint at wk12 with SD~2.0 VRTX's DPN Ph.3's based on comps and our regression analysis**, taking into account duration-matched experience from other trials.
- It seems to us for chronic pain, a combo approach may ultimately be required

Placebo response comparison from neuropathic pain Ph.2&3's

Trial	Mechanism/Drug	N (Pbo)	Duration (weeks)	Baseline NRS	Endpoint NRS	Wk 12 NRS CFB (actual or LOCF)	SD (actual or implied)
MEDI7352 Ph.2a	NGF/TNF α Bispecific	54	18	6.61	4.85	-1.2	1.7
BIIB074 CONVEY SFN Ph.2 (~71% diabetes)	Nav1.7 Inhibitor	41	16	6.6	3.3	-3.3	2.0
Mirogabalin China Ph.3	$\alpha 2\delta$ Ligand	197	14	6.08	4.28	-1.7	1.8
Duloxetine China Ph.3	SNRI	173	12	5.62	3.65	-2.0	1.8
Duloxetine Japan Ph.3	SNRI	167	12	5.78	4.17	-1.6	2.3
Duloxetine Study HMAW (US, PR)	SNRI	111	12	5.7	3.8	-1.9	2.3
Duloxetine HMAV (US, PR, Canada, Argentina)	SNRI	106	12	5.8	4.4	-1.4	2.4
LY3526318 DPN Ph.2	TRPA1 Antagonist	52	8	6	4.2	-1.8	2.0
LY3857210 DPN Ph.2	P2X7 Antagonist	35	8	5.62	3.81	-1.8	1.8
Tanezumab DPN Ph.2	NGF antibody	35	8	6.9	5.14	-1.0	1.8
LX9211 RELIEF-DPN1 Ph.2	T-type Ca Channel	107	6	6.5	5.78	-0.7	1.9
Richter 2004 (Pregabalin)	$\alpha 2\delta$ Ligand	82	6	6.9	5.47	-1.4	2.2

Source: Jefferies Analysis, Company Publications



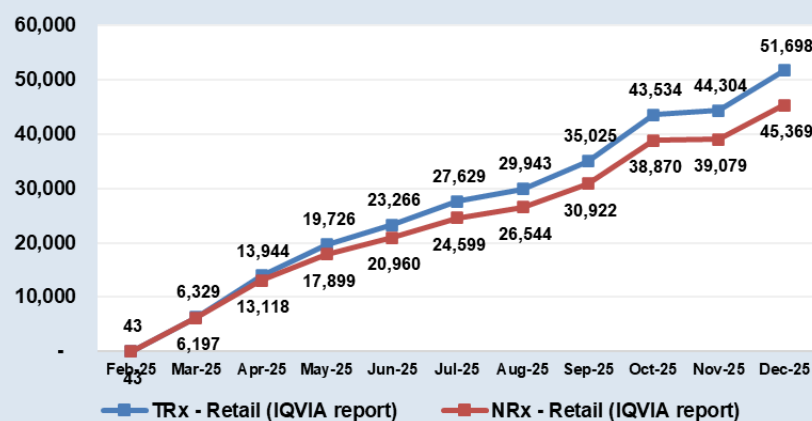
That said, we still think Journavx will get used in acute pain – we built a tracker to illustrate launch dynamics: 1) we model script capture rate of 50%-65% in line w/ mgmt. comments + utilization rate of PSP decrease overtime for new pt starts

Journavx US Revenue Build	Feb-25	Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Q1'26	Q2'26	Q3'26	Q4'26
TRx - Retail (IQVIA report)	43	6,329	13,944	19,726	23,266	27,629	29,943	35,025	43,534	44,304	51,698	158,713	183,913	209,113	234,313
% Total Scripts	65%	65%	65%	65%	65%	55%	55%	55%	55%	53%	50%	50%	50%	50%	50%
Total TRx	66	9,737	21,452	30,348	35,794	49,872	54,049	63,222	78,581	83,750	102,575	314,907	364,907	414,907	464,907
NRx - Retail (IQVIA report)	43	6,197	13,118	17,899	20,960	24,599	26,544	30,922	38,870	39,079	45,369	139,283	161,398	183,513	205,628
% Total Scripts	65%	65%	65%	65%	65%	55%	55%	55%	55%	53%	50%	50%	50%	50%	50%
Total NRx	66	9,534	20,182	27,537	32,246	44,403	47,913	55,816	70,162	73,873	90,018	276,355	320,234	364,113	407,992
NRx/TRx %	100%	98%	94%	91%	90%	89%	89%	88%	89%	88%	88%	88%	88%	88%	88%
% on PSP	55%	55%	57%	58%	60%	64%	68%	71%	73%	74%	74%	66%	59%	51%	44%
Patient Start w/ Coverage	30	4,290	8,779	11,566	13,060	16,207	15,572	16,187	19,295	19,207	23,855	93,961	132,897	178,415	230,515
Rollover Patients		203	1,271	2,811	3,548	5,469	6,135	7,406	8,419	9,877	12,558	38,552	44,673	50,794	56,915
Total Pts -> Revenue	30	4,493	10,050	14,376	16,607	21,676	21,707	23,593	27,713	29,084	36,412	132,513	177,570	229,209	287,431

Journavx Summary	Q1'25	Q2'25	Q3'25	Q4'25	FY'25
Total Scripts	9,803	87,594	167,143	264,907	529,447
New Pt Adds		77,791	79,549	97,764	

Our estimated scripts for 2025 is generally inline w/ VRTX's guidance: Q1 2025: ~10k prescriptions; Q2 2025: ~90k prescriptions; Q3 2025: ~170k prescriptions; FY 2025: ~550k total scripts; and as many rx Q4'25 as there were in the prior 3Qs combined

Journavx IQVIA TRx/NRx Feb'25 - Dec'25



Source: Jefferies Analysis, IQVIA

According to VRTX, IQVIA only reports the script data in the retail/discharge setting, which is ~65% as of Q2 and ~50% towards the end of '25 and beginning of '26 (we assume TRx and NRx has the same capture rate overtime).

"...think about where those scripts are coming from, **about half are coming from the hospital setting and about half are coming from discharge or in retail...**we estimate over 400,000 Americans have already benefited from JOURNAVX." -- VRTX, Jan'26

We assumed 55%-74% patients starts treatment w/ PSP and backed out the rev reported each Q. Additionally, VRTX recently guided that the company plan to extend the PSP to end of 2026 or into early 2027. We assume the % patients initiate on PSP to decrease over the next a few Qs as the company further sunset the program. We assume the PSP program only applicable to a portion of new patients (NRx).

"...while we will decrease and eventually sunset towards the end of this year, beginning of next, our patient support program, or PSP, we will continue to offer the PSP to those patients who are not yet covered by insurance because we believe the most important factors for long-term success are threefold..." -- VRTX, Jan'26

2) We estimate implied PSP utilization by backing out VRTX revenue & IQVIA scripts (only uninsured NRx are eligible) + insured pts and rollover pts contribute to the quarterly rev + assuming each patient has script duration of 14-days bottle per VRTX guidance

Journavx US Revenue Build	Feb-25	Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Q1'26	Q2'26	Q3'26	Q4'26
TRx - Retail (IQVIA report)	43	6,329	13,944	19,726	23,266	27,629	29,943	35,025	43,534	44,304	51,698	158,713	183,913	209,113	234,313
% Total Scripts	65%	65%	65%	65%	65%	55%	55%	55%	55%	53%	50%	50%	50%	50%	50%
Total TRx	66	9,737	21,452	30,348	35,794	49,872	54,049	63,222	78,581	83,750	102,575	314,907	364,907	414,907	464,907
NRx - Retail (IQVIA report)	43	6,197	13,118	17,899	20,960	24,599	26,544	30,922	38,870	39,079	45,369	139,283	161,398	183,513	205,628
% Total Scripts	65%	65%	65%	65%	65%	55%	55%	55%	55%	53%	50%	50%	50%	50%	50%
Total NRx	66	9,534	20,182	27,537	32,246	44,403	47,913	55,816	70,162	73,873	90,018	276,355	320,234	364,113	407,992
NRx/TRx %	100%	98%	94%	91%	90%	89%	89%	88%	89%	88%	88%	88%	88%	88%	88%
% on PSP	55%	55%	57%	58%	60%	64%	68%	71%	73%	74%	74%	66%	59%	51%	44%
Patient Start w/ Coverage	30	4,290	8,779	11,566	13,060	16,207	15,572	16,187	19,295	19,207	23,855	93,961	132,897	178,415	230,515
Rollover Patients		203	1,271	2,811	3,548	5,469	6,135	7,406	8,419	9,877	12,558	38,552	44,673	50,794	56,915
Total Pts -> Revenue	30	4,493	10,050	14,376	16,607	21,676	21,707	23,593	27,713	29,084	36,412	132,513	177,570	229,209	287,431
WAC/Patient	\$ 465	\$ 465	\$ 465	\$ 465	\$ 465	\$ 465	\$ 465	\$ 465	\$ 465	\$ 465	\$ 465	\$ 465	\$ 465	\$ 465	\$ 465
GTN (w/ coverage)	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%	30%
Compliance	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%	90%
Net Price	\$ 293	\$ 293	\$ 293	\$ 293	\$ 293	\$ 293	\$ 293	\$ 293	\$ 293	\$ 293	\$ 293	\$ 293	\$ 293	\$ 293	\$ 293
JEF Est. Rev	\$ 0.0	\$ 1.3	\$ 2.9	\$ 4.2	\$ 4.9	\$ 6.4	\$ 6.4	\$ 6.9	\$ 8.1	\$ 8.5	\$ 10.7	\$ 38.8	\$ 52.0	\$ 67.1	\$ 84.2

In our illustrative model, we backed out the implied PSP utilization rate using revenue reported by VRTX and scripts by IQVIA. We assume that only new patients (NRx) without insurance coverage are eligible for the program. These eligible patients, together w/ rollover patients from prior Q, contribute to the revenue in the current Q.

In terms of pricing, VRTX has guided that the average script duration is ~14 days (consistent with the broader market) and there are patients w/ sig. longer duration. In our analysis, we conservatively assumed 14 days duration (1 bottle) for each patient.

"And for that, **while we will decrease and eventually sunset towards the end of this year, beginning of next, our patient support program, or PSP**, we will continue to offer the PSP to those patients who are not yet covered by insurance because we believe the most important factors for long-term success are threefold: one, transform the standard of care and the management of acute pain from opioids to away from opioids." -- VRTX, Jan'26

Source: Jefferies Analysis

3) Using current IQVIA script trends and assuming ~50k patient adds per Q (in line with VRTX's expectation to 3x prescriptions this year), we estimate FY'26 US sales of ~\$242M (vs. ~\$238M cons)

Journavx Summary	Q1'25	Q2'25	Q3'25	Q4'25	FY'25	Q1'26	Q2'26	Q3'26	Q4'26	FY'26
Total Scripts	9,803	87,594	167,143	264,907	529,447	314,907	364,907	414,907	464,907	1,559,628
New Pt Adds		77,791	79,549	97,764		50,000	50,000	50,000	50,000	
JEF Est. Rev	\$ 1	\$ 12	\$ 20	\$ 27	\$ 60	\$ 39	\$ 52	\$ 67	\$ 84	\$ 242
VA Consensus	\$ 1	\$ 12	\$ 20	\$ 27	\$ 60	\$ 36	\$ 46	\$ 62	\$ 91	\$ 234
Beat (Miss) vs. Consensus						3	6	6	(7)	8

% PSP
Decrease
in 2026

Prescriptions Adds Per Q in 2026					
	30,000	40,000	50,000	60,000	70,000
-20%	\$ 187.0	\$ 201.3	\$ 215.5	\$ 229.8	\$ 244.0
-25%	\$ 198.4	\$ 213.6	\$ 228.9	\$ 244.1	\$ 259.3
-30%	\$ 209.8	\$ 226.0	\$ 242.2	\$ 258.4	\$ 274.6
-35%	\$ 221.2	\$ 238.4	\$ 255.5	\$ 272.7	\$ 289.8
-40%	\$ 232.6	\$ 250.7	\$ 268.9	\$ 287.0	\$ 305.1

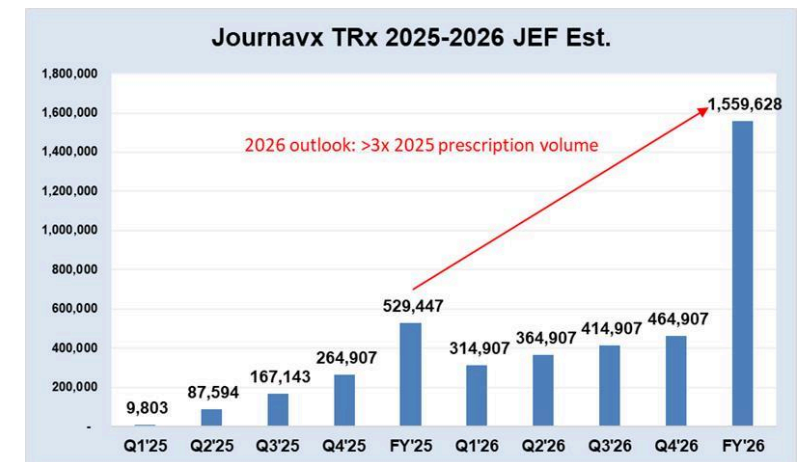
GTN
With
Coverage

Patient Compliance					
	80%	85%	90%	95%	100%
40%	\$ 184.5	\$ 196.1	\$ 207.6	\$ 219.1	\$ 230.7
35%	\$ 199.9	\$ 212.4	\$ 224.9	\$ 237.4	\$ 249.9
30%	\$ 215.3	\$ 228.7	\$ 242.2	\$ 255.6	\$ 269.1
25%	\$ 230.7	\$ 245.1	\$ 259.5	\$ 273.9	\$ 288.3
20%	\$ 246.0	\$ 261.4	\$ 276.8	\$ 292.2	\$ 307.5

In our sensitivity analysis, assuming a 35% (vs. 30% for our base case) reduction in PSP utilization, a ~40k net patient adds/Q would still be slightly higher/in line w/ consensus for a FY'26 rev of ~\$234M.

Looking at 2026, we think ~50k net patient adds/quarter is doable, (in line w/ VRTX's guidance for ~3x prescription growth in 2026 vs. 2025) based on 78k–98k/Q in 2025. We also assume a ~30% stepdown in PSP utilization rate, ending in 2026 at ~44% in our base case. Taken together, these assumptions result in FY26 revenue of ~\$242M in the US (vs. cons of \$234M).

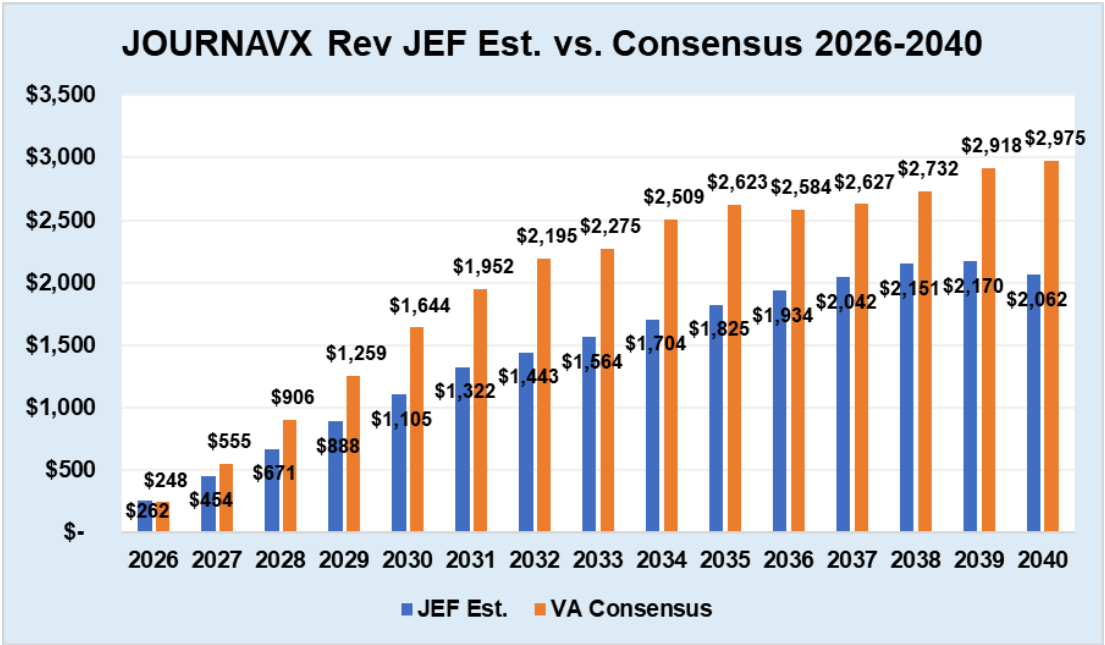
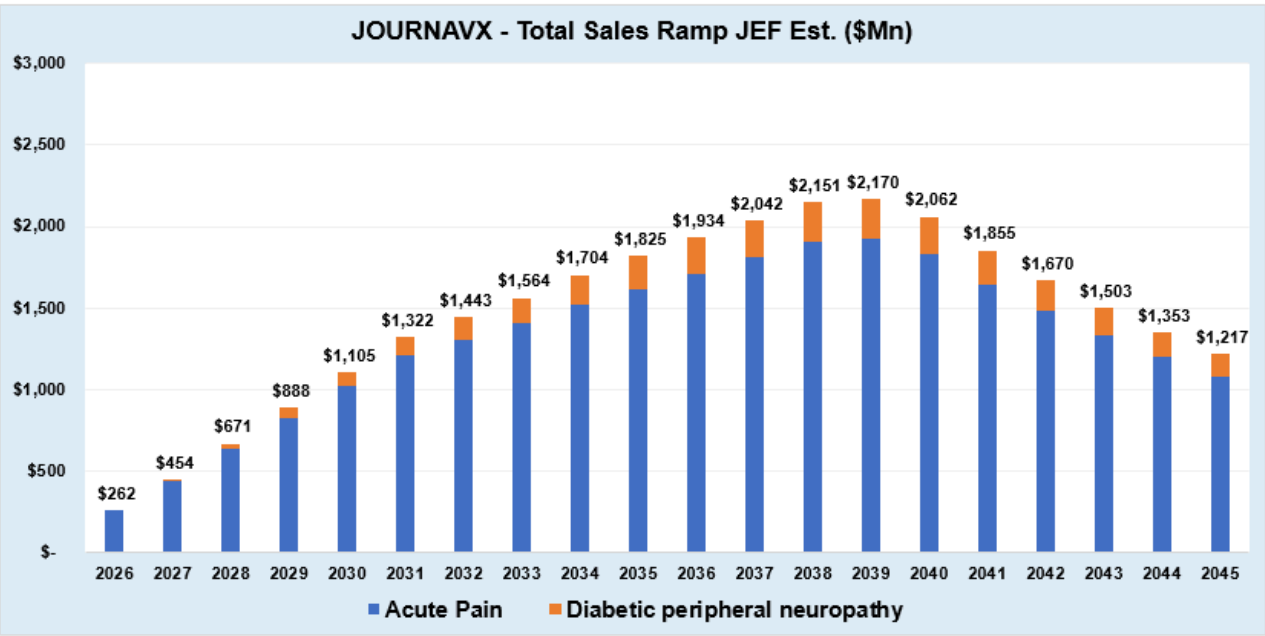
“Consistent with that movement and additional investments that we are making in JOURNAVX, this is why we see moving from the 500,000 scripts or so in '25 to 3x that number in '26. One very important point to make on JOURNAVX is that we are building a strong, durable and long-term franchise in pain, and we seek a leadership position here just like we did in CF” – VRTX, Jan'26



Source: Jefferies Analysis

While we're below cons on pain (we assume 30% POS for DPN chronic pain trial) & model ~\$2.2B risk-adj total PS vs \$3B cons, we still estimate the program can add ~\$25/share in NPV

Overall, in pain, we model ~\$2.2Bn peak sales (vs ~\$3Bn cons), but we still assign an NPV of \$25. We're in line with acute pain growth of ~3x in 2026 based on mgmt. guidance at JPM'26.



Source: Jefferies Analysis, VisibleAlpha

Finally, we'll quickly go over the rest of VRTX's pipeline

Cystic Fibrosis

We expect VRTX's CF franchise to remain durable

- We're expecting CF to remain a durable revenue base for VRTX and expect gradual switch & new starts for pts onto Alyftrek
- We expect upcoming comps will work as add-on tx to VRTX, but we're not seeing evidence as a mono-tx they could produce greater efficacy
- We expect minimal pt adds from VRTX mRNA CF tx, but limited ability for generics to gain market share

IgAN & autoimmune

We see versatility for APRIL/BAFF across autoimmune indications

In IgAN, we saw 49.8% Pbo adj. UPCR reduction at 9mo (clinically in line w/ Sibe):

- To us, the impt point is: a) IgAN is a bigger market with favorable pricing (\$390K/yr), b) Pove has a convenience advantage with 1.5sec admin autoinjector
- We're optimistic on the indication expansion opportunity into gMG, pMN, wAIHA

AMKD & ADPKD

We see LT upside from Inaxaplin becoming first/best in class

- We're expecting Inaxaplin to be first & best in class in the AMKD market (currently no tx)
- We expect 64-70% UPCR reduction at 48wks + we think VRTX has the more favorable BL population vs comps
- We think the AMKD market could be bigger than initially expected given a) an approved tx will likely increase dx rates, b) our KOL expects spillover dx from the IgAN market growing from Povetacicept
- In ADPKD, we think the data is too early for VX-407 but could see pot'l reduced LFTs based on Trikafta read across (similar MOA)

Pain

We're mixed on Ph.3 DPN hitting, but are more confident in the acute pain opportunity

- We're mixed on whether chronic DPN can hit for Suze, given the high variability of previous data and Pbo (combo approach is more likely here).
- In acute pain, we expect 3x growth for Journavx based on recent access expansion

Rest of Pipeline

Rest of pipeline is interesting but we're skeptical on cons valuation

- We're skeptical on expansion opportunity for rest of pipeline:
- For Casgevy, we've seen precedence that viable tx's in SCD launch slower / underpenetrate the market due to patient/physician concerns & access barriers
- Zimislecel appears limited to severe T1D population based on immunosuppression requirement & clinical trial delays

Source: Jefferies analysis

#1) Casgevy was approved for Sickle Cell Disease and Transfusion Dependent Beta Thalassemia in Dec 2023 – we’ve seen positive efficacy data in 5-11y/o pts & expect label expansion post 1H26 submission

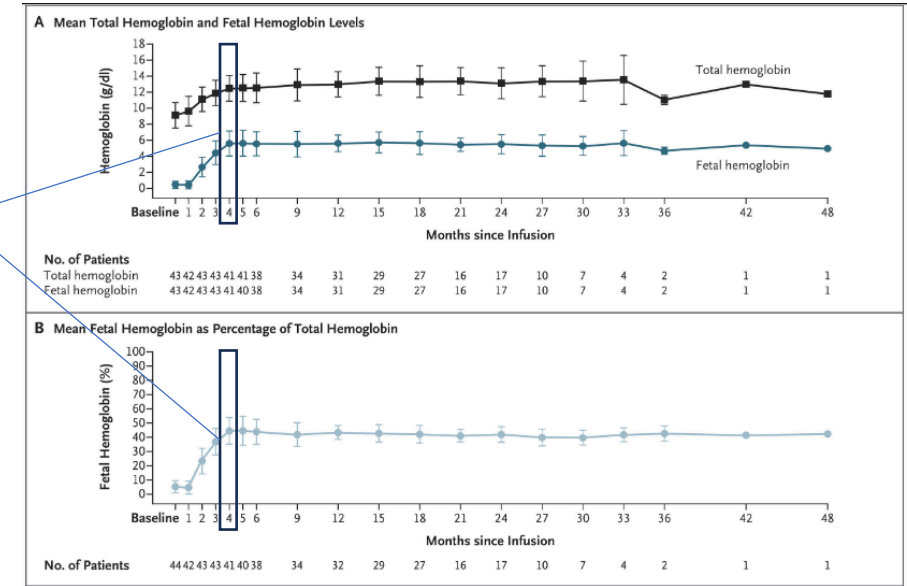
Casgevy was approved by the FDA in Dec 2023 as a CRISPR editing tx targeting BCL11A tx for sickle cell disease and transfusion dependent b-thalassemia. In Ph.3, both total and fetal hemoglobin stabilized at ~4mo since infusion. While the FDA acknowledged risks associated with gene editing BCL11A given its association with multiple blood cell associated roles & proto-oncogene status, it was approved given only ~20% of pts have a matched donor for curative SCD and **Casgevy aims to be a 1-time curative option**. In 2025, 300 pts has initiated Casgevy (inc. 150 had first cell collections and 60 were infused). As of Aug 2025 there are 75 tx-centers globally.

We’ve recently seen 3 datapoints at ASH 2025 for Casgevy we’ll flag:

1) The first FAERS- based safety analysis suggested Casgevy was associated with early post-infusion GI/mucosal bleeding (4/9 FAERS events).

2) In Ph.3 CLIMB SCD-121 (n=46; mean age 21.4; BL 4.2 VOCs/yr), Casgevy achieved VOC elimination in 41/45 pts (91.1%) and hospitalization freedom in 44/45 (97.8%). Across SCD-121 + CLIMB-131, 45/45 (100%) achieved both endpoints, with mean VOC-free duration 35.3 mo and median follow-up 41.3 mo. HbF rose durably $\geq 40\%$ with ~95% pancellular distribution and sustained hematologic/QOL gains. Safety matched myeloablative conditioning, with 1 non-tx-related COVID-19 death and no malignancies.

3) The first 5-11y Casgevy data showed 4/4 SCD pts were VOC-free for ≥ 12 mo (longest 2 yrs) and 6/6 TDT pts achieved ≥ 12 mo transfusion independence with Hb > 9 g/dL (one pneumonia-related death occurred). In the broader 5-11y cohort (13 TDT, 10 SCD), TDT pts had median FU 12.6 mo with 9/13 transfusion-free and 5/5 achieving TI12 (longest 19.1 mo), with Hb ≥ 11.8 g/dL and HbF ≥ 11.0 g/dL by 6 mo. In SCD (median FU 8.3 mo), no VOCs occurred, 2/2 achieved VF12/HF12, and HbF rose $> 40\%$ with durable pan cellular expression. **To us, this data shows efficacy in a population of high unmet need therefore we’re expecting the label expansion to be approved (+ mgmt. commented their national priority review vouchers gives them an expected shorter review time (mgmt. guided submission in 1H26).**



Source: Jefferies Analysis, Company Publications

Casgevy has 2 comps (Lyfgenia & Zynteglo) - while efficacy looked directionally similar btwn the two drugs, we've seen Casgevy show lower Gr3+ Thrombocytopenia (25% vs 66-96%) + Neutropenia (23% vs 54-78%) making safety the key differentiator to us

We'll flag Casgevy has competition from Genetix drugs Lyfgenia in SCD and Zynteglo in transfusion-dependent beta-thalassaemia. As a background, **Lyfgenia and Zynteglo use lentiviral vectors vs Casgevy that uses CRISPR.**

Therapy	CASGEVY (exa-cel)	LYFGENIA (lovo-cel)	ZYNTEGLO (beti-cel)
Mechanism of Action (MOA)	CRISPR-Cas9 editing of BCL11A enhancer to reactivate fetal hemoglobin (HbF)	Lentiviral addition of β A-T87Q anti-sickling β -globin gene	Lentiviral addition of functional β A-T87Q β -globin gene
Disease Indication	Severe sickle cell disease with recurrent VOCs	Sickle cell disease (β S/ β S, β S/ β 0, β S/ β + genotypes)	Transfusion-dependent β -thalassemia (non- β 0/ β 0 genotypes)

While cross-trial comparison is difficult due to differences in the primary/secondary endpoints, from the FDA clinical reviews, **we've seen Casgevy and Lyfgenia show directionally similar data (e.g. 93.5% achieved VF12 for Casgevy vs 87.5% achieved VOE-CR for Lyfgenia)**

Feature	CASGEVY (exa-cel)	Lyfgenia (lovo-cel)
Primary Endpoint	VF12: Freedom from severe VOCs $\geq$ 12 months	VOE-CR: Resolution of all VOE (6-18 mo)
Primary Endpoint Result	29/31 (93.5%) achieved VF12	28/32 (87.5%) achieved VOE-CR
Key Secondary Result	30/30 (100%) achieved HF12	30/32 (93.8%) achieved sVOE-CR
Durability (24-mo)	Majority remained sVOC-free; follow-up ongoing	VOE-CR24: 84.4%; sVOE-CR24: 90.6%
Hemoglobin / PD Effects	HbF $\geq$ 40% from Month 6; stable editing $\geq$ 70%	HbA T87Q ~45% of total Hb at 24 mo
Study Population Size	44 infused; 30 efficacy-evaluable	36 infused; 32 efficacy-evaluable

Source: Jefferies Analysis, Company Publications, FDA

On safety, we saw Casgevy had numerically lower Gr3+ Thrombocytopenia (25% vs 66-96%), Neutropenia (23% vs 54-78%), but had greater decrease appetite (41% vs 9-13%). Net-net we think Casgevy has a slightly more favorable safety profile vs Lyfgenia and Zynteglo which we expect to help with prescriber preference over competition long-term.

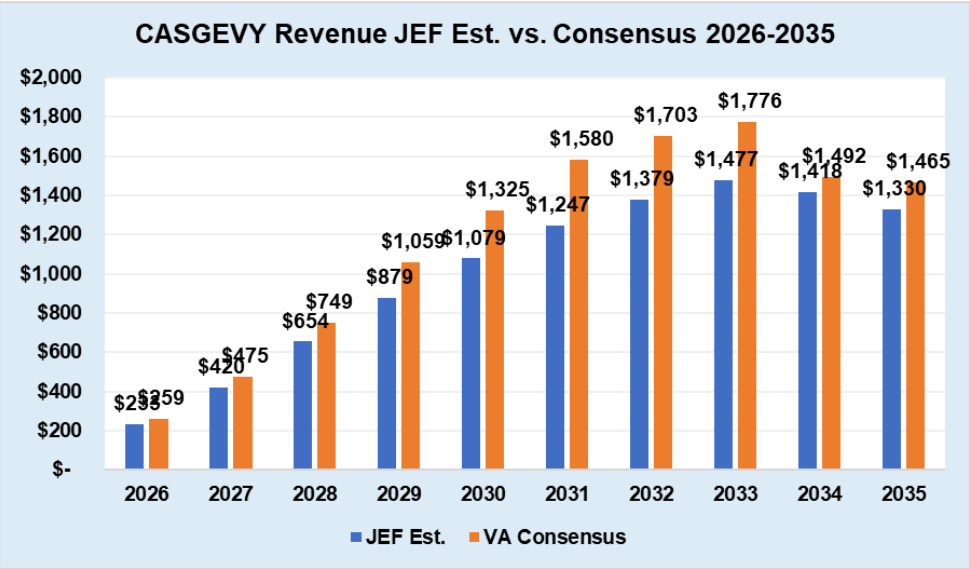
AE	Casgevy (exa-cel)	Lyfgenia (lovo-cel)	Zynteglo (beti-cel)
n =	44	35	23
Stomatitis (Grade $\geq$ 3)	24 (55%)	24 (69%)	14 (61%)
Thrombocytopenia (Grade $\geq$ 3)	11 (25%)	23 (66%)	22 (96%)
Neutropenia (Grade $\geq$ 3)	10 (23%)	19 (54%)	18 (78%)
Febrile neutropenia (Grade $\geq$ 3)	21 (48%)	15 (43%)	8 (35%)
Anemia (Grade $\geq$ 3)	11 (25%)	13 (37%)	14 (61%)
Leukopenia / WBC $\downarrow$ (Grade $\geq$ 3)	6 (14%)	11 (31%)	13 (57%)
Decreased appetite (Grade $\geq$ 3)	18 (41%)	3 (9%)	3 (13%)
Nausea (Grade $\geq$ 3)	4 (9%)	4 (11%)	2 (9%)
Abdominal pain (Grade $\geq$ 3)	5 (11%)	2 (6%)	
Hepatic veno-occlusive disease (VOD)	1 (2%)	0 (0%)	3 (13%)

That said, the TAM for Casgevy is small (we est ~31K for SCD and ~12K for TDT in US and EU) + we've seen precedence of slow uptake / under penetration from prior-tx – hence we model peak sales \$1.5Bn by 2033 vs \$1.8Bn for cons

In 2024 VRTX signed an agreement to have Casgevy produced by Lonza in the Netherlands with expansion to Portsmouth, NH.

“The good news about that is that **we have very good line of sight into how many infusions we expect in 2026**. And so, as we give revenue guidance in February at our earnings call, that's going to be very high certainty because the – for lack of a better term, that **patient funnel is full and moving forward**.” – Jan '26, VRTX

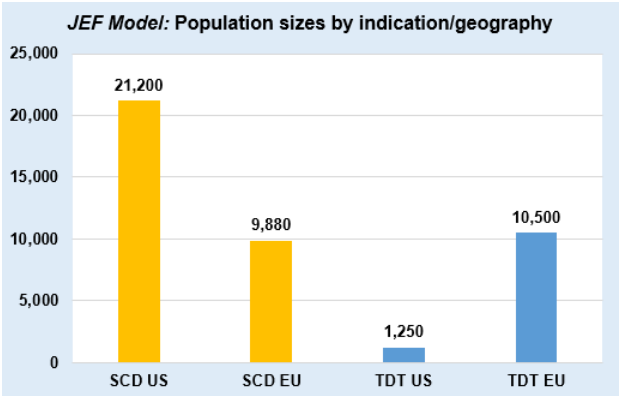
We model \$1.5Bn Risk adj. peak sales by 2033 vs \$1.8Bn for cons.



Source: Jefferies Analysis, Company Publications

We've seen precedence of slow RW uptake of SCD therapies e.g hydroxyurea showed strong clinical benefit but was adopted gradually in practice. McGann et al. 2015 noted that “the accumulated body of evidence over 30 years demonstrates that hydroxyurea is a safe and effective therapy for sickle cell anemia, but hydroxyurea remains underutilized.” The authors attribute this to several barriers including patient fears of AEs, the need for laboratory monitoring, and adherence challenges. Similarly, Sinha et al. 2018 report that “HU remain underutilized” highlighting additional factors such as limited provider familiarity and access barriers. Together, these reports illustrate that even mechanistically compelling SCD therapies can still experience gradual uptake / under penetration – hence we model \$1.5Bn vs \$1.8Bn cons

VRTX estimates there are 37K pts with SCD and TDT in North American & EU. Genetix estimates ~20K pts with SCD in the US alone. Based on our analysis (see our model) we est. 31,080 for SCD in US and EU and 11,750 total for TDT EU.

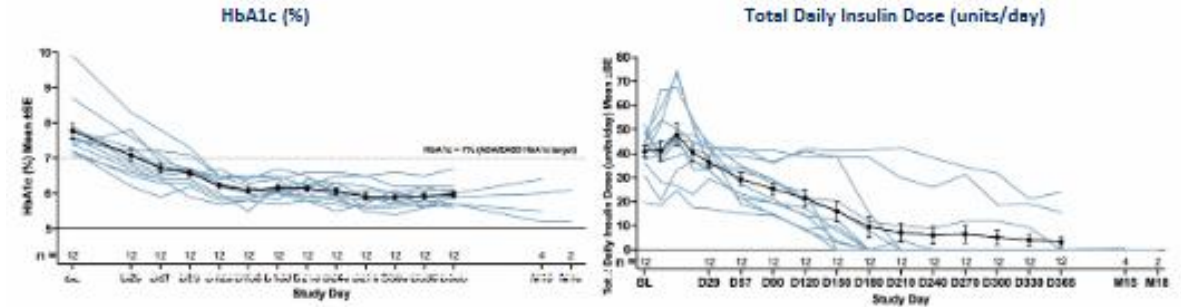
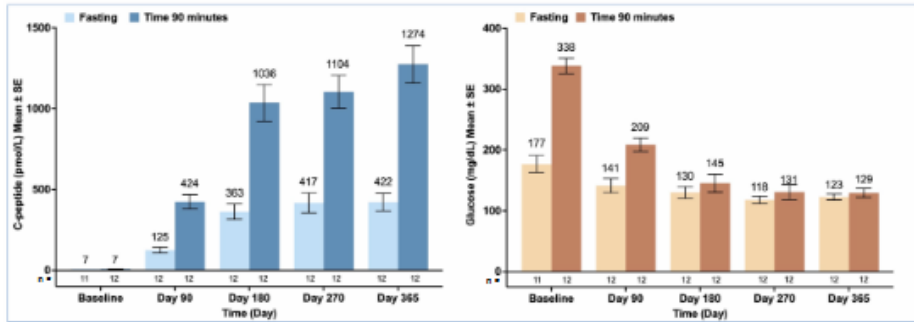


We'll flag VRTX is participating in the CMMI pilot 'Cell and Gene Therapy Access Model' to increase Medicare Access to gene-tx. We think this will help increase uptake for Casgevy.

#2) Zimislecel is a stem-cell tx for T1D being tested in pts with 2+ severe hypoglycemia in a year – Ph.1/2 showed efficacy from increase in insulin biomarker (C-peptide) and reduction in HbA1c

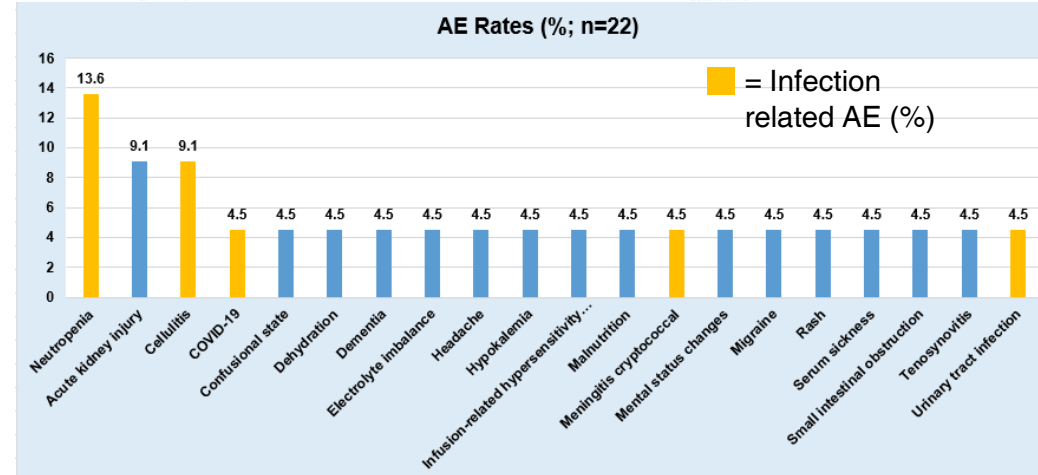
As a background, Zimislecel cell therapy for T1D delivering insulin producing islet cells to the pancreas. In an n=12 population, we've seen **C-peptide (marker of insulin) increases organically post-Zimislecel from ~7pmol/L to over 1274pmol/L at 1yr.** Furthermore, **glucose levels decreases from 338 mg/dL to 129mg/dL after 90min post-food.** To us, this is a clear efficacy signal given these pts cannot produce insulin on their own.

We've also seen this translate into longer-term markers – HbA1C decreased from ~7.8% to ~6.5%.



VRTX is positioning Zimislecel in a population with T1D & recurrent severe hypoglycemic events (mgmt. estimates ~60K), but longer term they aim to target the full 125K pt population.

On safety, in Ph.1/2 the most common AE was neutropenia (13.6%) and two deaths (one due to cryptococcal meningitis, the other due to severe dementia with agitation due to exacerbation of preexisting cognitive impairment). That said, these deaths were not attributed to Zimislecel. Of the AEs that had a causal relationship to the study drug, most were attributed to immunosuppressive tx (tacrolimus/sirolimus). Regardless, to us infection risk remains a limiting factor in expanding to more benign forms of T1D.



Source: Jefferies Analysis, Company Publications

While we’re expecting fileable Ph.3 data in 2026, we see infection risk as a limiting factor to increase TAM beyond severe T1D population

The transition to Ph.3 was supported by RMAT and **fast track designations in the U.S., PRIME designation in the EU, and an innovation passport in the UK**. Ph.1/2 data is used to establish proof of concept the Ph.3 trial is intended to serve as the basis for regulatory approvals. **VRTX aims to have global regulator submission for Zimislecel in 2026.**

While VRTX is further along in development, we’ll flag Sana Biotech is developing SC451 that uses CRISPR to gene edit donor islet cells so they don’t induce an immune response. So far, we’ve only seen a single case report of this being done successfully, therefore, it is too early to tell whether there is holds into clinical trials yet.

Lantidra is the first FDA approved tx using donor islets cells for T12D. **We’ll flag infection risk was a key concern highlighted by the FDA clinical review, hence the risk was only deemed acceptable in a tiny subset of T1D pts with life-threatening hypoglycemia.** Furthermore, they found that it was difficult to predict which pts would benefit. **Zimislecel has also produced neutropenia in Ph.1/2 therefore we’re anticipating infection risk to remain in Ph.3 and limit pt uptake to those that are have confirmed repeated severe hypoglycemia.**

Program / Company	Therapy Name / Type	Cell Source	Immune Strategy	Stage / Data	Key Notes
VRTX	Zimislecel (VX-880)	Stem cell-derived, fully differentiated islet cells	Systemic immunosuppression	Phase 1/2 reported; Phase 3 ongoing	High insulin independence rates (~83% at 1yr in P1/2); pivotal ~50 pt study ongoing
Sana Biotechnology	UP421 / SC451 (hypoimmune islet cells)	Stem cell-derived, gene-edited islets	Gene-edited immune evasion (no systemic IS intended)	Early human study ongoing	6-mo data showed graft survival & C-peptide production without chronic IS
Seraxis	SR-02 / SRE Clusters	Stem cell-derived pancreatic clusters	Systemic immunosuppression	IND cleared; early clinical entry	Aims to transplant pancreatic clusters with immune shielding strategy
CellTrans	Lantidra (donislecel)	Cadaveric donor islets	Systemic immunosuppression	FDA Approved (US)	First approved islet cell therapy; donor-dependent supply constraints

Source: Jefferies Analysis, Company Publications

We’ll call out the following key inclusion criteria:

- T1D 5yr+ with insulin dependence
- 2 episodes of documented severe hypoglycemia in the past 12mo prior to enrollment.

The key differences btw Ph.1/2 and Ph.3 are: 1) Ph.1/2 had n=12 while Ph.3 enrolled n=~50. 2) Ph.1/2 primary endpoint was safety & tolerability and freedom from severe hypoglycemic events, but Ph.3 also includes proportion of participants who are insulin independent.

Primary Outcome Measures	
Outcome Measure	
Part A: Safety and Tolerability as Assessed by Number of Participants With Adverse Events (AEs)	
Parts B and C: Proportion of Participants who are Insulin Independent with Absence of Severe Hypoglycemic Events (SHEs)	
Secondary Outcome Measures	
Outcome Measure	
Parts B and C: Proportion of Participants who are Insulin Independent	
Parts B and C: Proportion of Participants Free from SHEs with a Glycosylated Hemoglobin (HbA1c) less than (<) 7.0%	
Parts B and C: Change From Baseline in Glycosylated Hemoglobin (HbA1c)	
Parts B and C: Proportion of Participants Who Maintain Insulin Independence for at least 1 year	
Parts B and C: Safety and Tolerability as Assessed by Number of Participants With Adverse Events (AEs)	

Long-term, we've seen islet replacement tx show durable response at 5yrs

Hypo scores decreased to 0 and Beta scores rose and stabilized (n=10) – that said, this doesn't help increase TAM beyond severe T1D population

We saw in Qi et al. 2014, there was a 5-year Ph.1/2 allogeneic islet. **All patients achieved insulin independence after transplantation, and 6/10 remained insulin-independent at 5yrs.** Most maintained +ve C-peptide, HbA1c <6.0%, and improved β -scores. The study shows long-term islet graft function without T-cell-depleting induction, with 60% insulin independence at 5 years.

In particular, we'll flag there was a decrease for all pts in Hypo score to 0 by 5yrs. Also, the beta score increased, then stabilized at 5yrs. To us, this is +ve for the durability of Zimislecel, however this does not circumvent the infection risk.

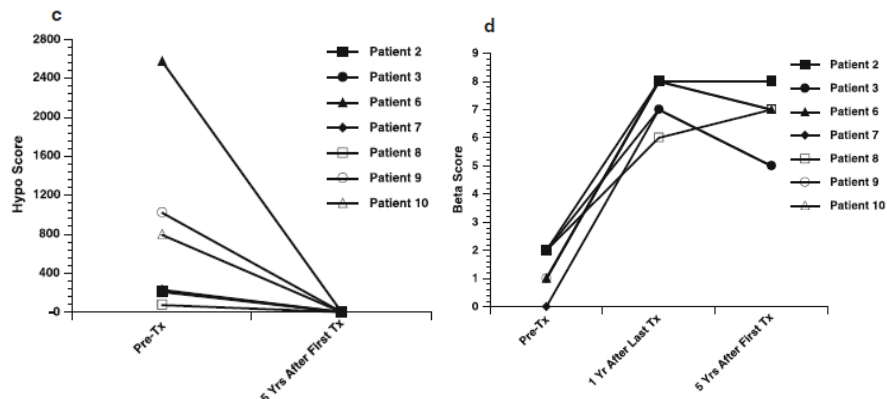
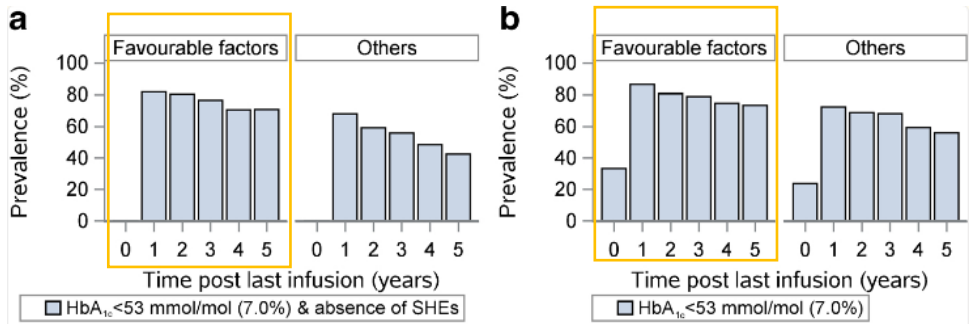


Table 2 Antibody profiles before and after transplantation

Test	Subjects									
	Patient 1	Patient 2	Patient 3	Patient 4	Patient 5	Patient 6	Patient 7	Patient 8	Patient 9	Patient 10
%PRA antibodies HLA Class I/Class II										
Before Tx	0/0	0/0	0/0	0/0	0/0	0/0	0/0	33/0	0/0	0/0
1 year after first Tx	0/0	0/0	0/0	0/0	0/0	0/0	0/0	51/0	0/0	0/0
2 years after first Tx	NA	0/0	0/0	16/0	0/0	0/0	0/0	45/0	0/0	0/0
3 years after first Tx	NA	0/0	0/0	NA	NA	0/0	0/0	45/0	0/0	0/0
4 years after first Tx	NA	0/0	0/0	NA	NA	0/0	0/0	55/52	0/0	0/0
5 years after first Tx	NA	0/0	0/12	NA	NA	0/0	0/0	54/0	0/0	0/0

This paper also highlights that autoimmune reaction is key to control for Zimislecel, given that 1 pt developed ADAs. **Given that Zimislecel is co-administered with Tacrolimus/sirolimus, we suspect this is a key driver of the infection risk that may limit TAM growth.**

We also saw from Hering et al. 2022 there were 4 factors associated with the highest rates of favorable outcomes: **1) recipient age ≥ 35 years, 2) total infused islets $\geq 325,000$ islet equivalents 3) induction immunosuppression with T cell depletion and/or TNF- α inhibition 4) maintenance with both mechanistic target of rapamycin (mTOR) and calcineurin inhibitors.**



The authors also showed that pts who met all four favorable factors were much more likely to achieve good glycemic control (HbA_{1c} < 7.0%) and avoid severe hypoglycemic events over five years after their last islet infusion, with success rates remaining higher vs pts who did not meet all four factors. We also saw that the 4CFF group maintained significantly higher rates of HbA_{1c} < 7.0% across all five years compared with the <4CFF group.

Source: Jefferies Analysis, Company Publications

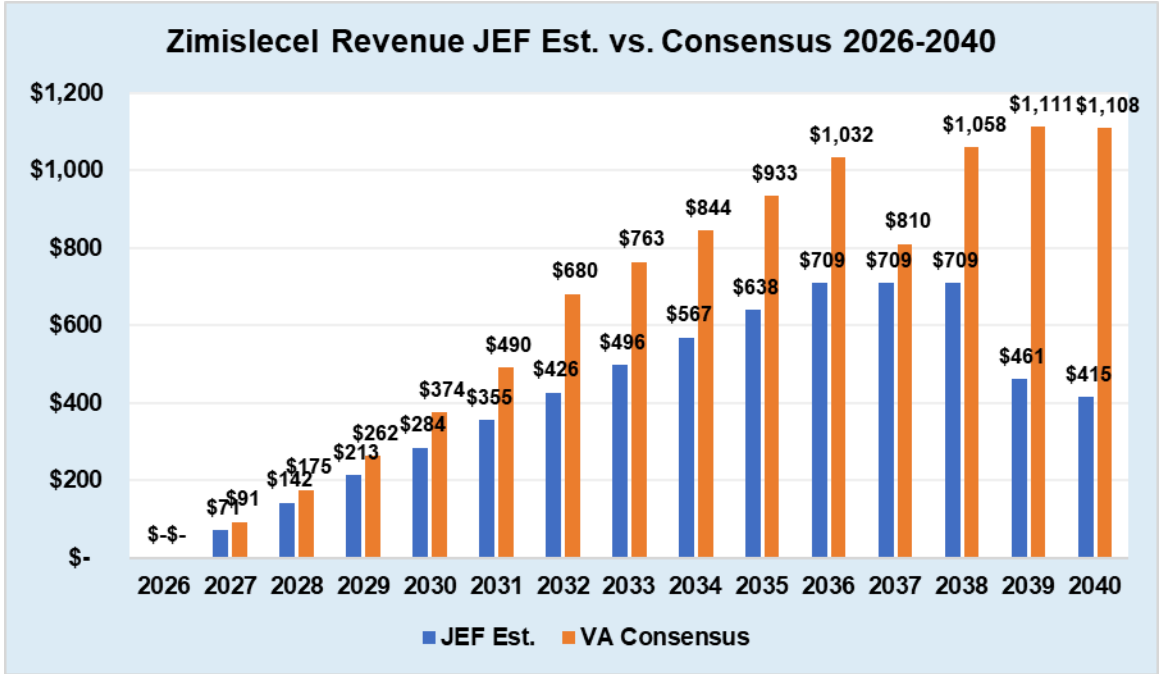
Recall, mgmt has halted Ph.3 for internal manufacturing analysis – given we don’t have visibility on resolution + immunosuppression risk overhang, we model \$709M risk adj. peak sales by 2036 vs ~\$1Bn for cons

At Q3 2025 earnings, mgmt. announced that temporarily postponed completion of dosing while they work through an internal manufacturing analysis, but did not comment any further. For 2026, their stated goal is to complete the manufacturing review but did not guide whether they were still on track to hitting their 2026 regulatory submission goal. We’ll also flag there is an ongoing partnership with Lonza in New Hampshire to build manufacturing capacity and Treefrog for scalability.

At Q4’25 earnings, the review was still characterized as “ongoing”. Given this delay, we model \$709M (vs ~\$1Bn for cons) risk adj peak sales.

“While we have completed enrollment in the pivotal trial for T1D, we have temporarily postponed completion of dosing while we work through an internal manufacturing analysis” – VRTX mgmt. Nov 3 2025 Q3 earnings

In total, VRTX believes there are 125,000 pts that have severe T1D out of 3.8M in the US.



Source: Jefferies Analysis, VisibleAlpha

Here are the main catalysts we're watching for in 2026/27 for VRTX

Drug	Co.	Event	Timing
Inaxaplin	VRTX	Ph.3 Interim readout (48wks)	1H26
Suzetrigine	VRTX	Ph.3 DPN (Trial 1)	1H27
Suzetrigine	VRTX	Ph.3 DPN (Trial 2)	1H27
VX-522	VRTX	Ph.1/2 Safety/efficacy data	2H26
Alyftrek	VRTX	Ph.3 label expansion into children 2-5	1H26
VX-828	VRTX	Completion of dosing and data sharing	2H26
VX-880	VRTX	T1D Ph.3 readout	2H27
MZE829	MAZE	MZE829 Ph.2 readout	2H26
SION-719	SION	SION-719 Ph.2 readout	2H26
SION-451 & SION-2222/SION-109	SION	Ph.1 SION-451 & SION-2222/SION-109 readout	1H26

Source: Jefferies Analysis

Vertex Pharmaceuticals Management Biographies



Reshma Kewalramani, M.D., FASN, President and Chief Executive Officer

Reshma Kewalramani has served as President and Chief Executive Officer of Vertex Pharmaceuticals since 2020. Prior to becoming CEO, she held roles of increasing responsibility at Vertex, including Chief Medical Officer and Executive Vice President, Global Medicines Development and Medical Affairs, where she led the advancement of Vertex's cystic fibrosis portfolio and expansion into new disease areas. Before joining Vertex, she held senior clinical development roles at Amgen and Biogen. Dr. Kewalramani is a practicing nephrologist and has authored numerous peer-reviewed publications. She received her M.D. from Boston University School of Medicine and completed her residency and fellowship training at Massachusetts General Hospital.



Carmen Bozic, M.D., Executive Vice President, Global Medicines Development and Medical Affairs, and Chief Medical Officer

Carmen Bozic serves as Executive Vice President, Global Medicines Development and Medical Affairs, and Chief Medical Officer at Vertex. She oversees clinical development, medical affairs, pharmacovigilance, and regulatory strategy across the company's portfolio. Prior to joining Vertex, Dr. Bozic held senior leadership roles at Biogen, where she led global development programs across neurology and immunology. She received her M.D. from Harvard Medical School and completed residency training in internal medicine and neurology at Massachusetts General Hospital and Brigham and Women's Hospital.



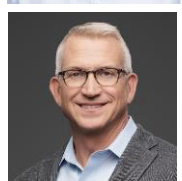
Charles F. Wagner, Executive Vice President and Chief Operating & Financial Officer

Charles Wagner is Executive Vice President and Chief Operating and Financial Officer of Vertex, with responsibility for finance, corporate strategy, business development, information technology, and operational execution. He joined Vertex in 2018 and has played a key role in scaling the company's global infrastructure and financial performance. Prior to Vertex, Mr. Wagner held senior finance and operational leadership roles at Alexion Pharmaceuticals and Pfizer. He holds a bachelor's degree in economics from the University of Connecticut and an MBA from Harvard Business School.



Mark Bunnage, D.Phil., FRSC, FRSB, Executive Vice President and Chief Scientific Officer

Mark Bunnage serves as Executive Vice President and Chief Scientific Officer, leading Vertex's global research organization and discovery strategy. Prior to joining Vertex, he was Chief Scientific Officer at BenevolentAI and previously spent more than two decades at AstraZeneca, where he held multiple senior leadership roles in medicinal chemistry and early discovery research. Dr. Bunnage earned his D.Phil. in organic chemistry from the University of Oxford and is a Fellow of the Royal Society of Chemistry & Royal Society of Biology.



E. Morrey Atkinson, Ph.D., Executive Vice President and Chief Technical Operations Officer

E. Morrey Atkinson is Executive Vice President and Chief Technical Operations Officer at Vertex, overseeing biopharmaceutical sciences, manufacturing operations, supply chain, and quality. He brings extensive experience in biologics development and large-scale manufacturing. Prior to Vertex, Dr. Atkinson held senior technical and operations leadership roles at Biogen, including responsibility for global manufacturing and process development. He received his Ph.D. in chemical engineering and has led multiple commercial product launches across complex biologic platforms.



Duncan McKechnie, Executive Vice President and Chief Commercial Officer

Duncan McKechnie serves as Executive Vice President and Chief Commercial Officer, responsible for global commercial strategy, market access, and product commercialization. He joined Vertex after holding senior commercial leadership roles at Biogen, where he led global franchises across multiple therapeutic areas. At Vertex, he has overseen the continued growth of the cystic fibrosis franchise and preparation for launches in new disease areas. Mr. McKechnie holds a bachelor's degree from the University of Edinburgh.

Source: Jefferies analysis, company data

Investment risks to our BUY rating include but are not limited to the following:

- Competitor CF programs take share from VRTX or Alyftrek switch rate is minimal
- Povetacicept has poor market uptake in IgAN or high rates of chronic safety issues emerge (eg, infection, hypogammaglobulinemia)
- AMKD fails in its Ph.3 trial or competitor program shows better data

Source: Jefferies analysis

Company Description

Vertex

Vertex Pharmaceuticals is a global biotechnology company developing therapies for serious diseases with high unmet need. It is a leader in cystic fibrosis (CF), with Trikafta/Alyftrek treating most eligible patients and driving a durable, cash-generative franchise. Beyond CF, Vertex is expanding into pain with suzetrigine (NaV1.8 inhibitor), kidney disease with inaxaplin & povetacept, as well as autoimmune disease. In type 1 diabetes, zimislecel is a stem cell-derived islet therapy designed to restore insulin production. Vertex also co-developed the CRISPR/Cas9 gene-edited therapy Casgevy for sickle cell disease and beta thalassemia, reflecting its strategy to build a diversified, fully integrated biopharmaceutical company spanning discovery through global commercialization.

Company Valuation/Risks

Vertex

We see upside pot'l from pipeline diversification (pove + AMKD + ADPKD; ~\$131/sh), which in our view the market currently attributes minimal value. Our SOTP analysis implies ~\$580/sh, including cystic fibrosis (~\$376/sh), pain (~\$25/sh + we model below consensus), and cash (~\$48/sh).

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Investment Recommendation Record

(Article 3(1)e and Article 7 of MAR)

Recommendation Published	March 10, 2026 5:32 A.M.
Recommendation Distributed	March 10, 2026 5:32 A.M.

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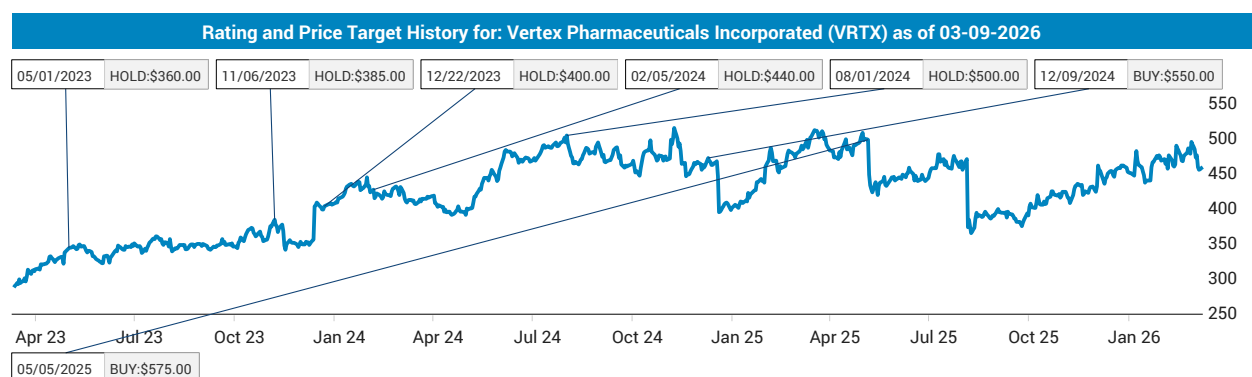
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- Argenx SE (ARGX: \$740.40, BUY)
- AstraZeneca PLC (AZN LN: p14,442.00, BUY)
- Biogen Inc. (BIIB: \$188.24, BUY)
- Bristol-Myers Squibb Co (BMY: \$60.63, BUY)
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- Cigna Corp. (CI: \$269.66, BUY)
- Eli Lilly & Co (LLY: \$1,008.39, BUY)
- GSK Plc (GSK: \$55.51, BUY)
- GSK Plc (GSK LN: p2,046.00, BUY)
- IQVIA (IQV: \$178.55, BUY)
- Ionis Pharmaceuticals, Inc. (IONS: \$75.36, BUY)

- Jade (JBIO: \$14.43, BUY)
- Krystal Biotech Inc (KRYS: \$258.65, BUY)
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- Lonza Group AG (LONN SW: CHF498.70, BUY)
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- Merck & Co Inc (MRK: \$117.11, BUY)
- Moderna, Inc. (MRNA: \$55.74, HOLD)
- Natera Inc (NTRA: \$205.22, BUY)
- Novartis AG (NVS: \$162.79, HOLD)
- Novartis AG (NOVN SW: CHF125.30, HOLD)
- Novo Nordisk (NVO: \$39.78, HOLD)
- Novo Nordisk (NOVOB DC: DKK254.50, HOLD)
- Otsuka Holdings (4578 JP: ¥10,160, BUY)
- Pfizer Inc (PFE: \$26.81, BUY)
- Prime Medicine, Inc. (PRME: \$4.11, BUY)
- Regeneron Pharmaceuticals, Inc. (REGN: \$781.60, BUY)
- Roche (ROG SW: CHF332.40, UNDERPERFORM)
- Sana Biotechnology Inc (SANA: \$3.33, BUY)
- Sanofi (SAN FP: €75.81, BUY)
- Sanofi (SNY: \$44.35, BUY)
- Signet Jewelers Ltd (SIG: \$91.61, BUY)
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- Vera Therapeutics, Inc (VERA: \$38.97, BUY)



Madeleine Lee or a member of her household holds an equity long position in Vertex Pharmaceuticals Inc Katherine Wang holds an equity long position in Vertex Pharmaceuticals Inc

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	IB Serv./Past12 Mos.		JIL Mkt Serv./Past12 Mos.	
	Count	Percent	Count	Percent
BUY	2168	61.54%	362	16.70%
HOLD	1194	33.89%	104	8.71%
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